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Résumé en français

Les déformations du fond marin, en particulier celles induites par les séismes sous-marins, constituent l'un des principaux mécanismes responsables de la génération des tsunamis. Ces phénomènes, bien que rares, sont potentiellement dévastateurs et soulèvent des enjeux scientifiques, environnementaux et sociétaux majeurs. Initialement, au-dessus de l'épicentre du séisme, les tsunamis se manifestent sous la forme d'ondes de très faible amplitude pouvant passer inaperçues sans instruments de mesure précis. Toutefois, les tsunamis possèdent des longueurs d'onde très importantes, souvent supérieures à la profondeur des océans, ce qui provoque une amplification de leur amplitude à mesure que la profondeur diminue.

D'un point de vue physique, la génération d'un tsunami résulte d'un transfert d'énergie entre une déformation rapide du plancher océanique et la colonne d'eau sus-jacente. Cette interaction met en jeu, dans certains régimes, des ondes acoustiques se propageant dans le fluide à une vitesse $c = 1500\text{m/s}$, lesquelles constituent un signal précurseur annonçant l'arrivée d'un tsunami sans représenter elles-mêmes un danger majeur. Plus important encore, elle met en jeu des ondes de gravité se propageant en surface et beaucoup plus lentement $c_g = \sqrt{gH} \approx 173\text{m/s}$ où H correspond à la profondeur de l'océan, correspondant à l'onde visible du tsunami et constituant la composante véritablement dangereuse du phénomène.

La prévision d'un tsunami et de son évolution est essentielle pour la protection des populations côtières ainsi que des infrastructures situées dans des zones exposées aux submersions. Le terme de « prévision » est ici quelque peu trompeur, dans la mesure où les séismes à l'origine des tsunamis sont des phénomènes hautement aléatoires. L'objectif consiste plutôt à prédire a posteriori, aussi rapidement que possible, le comportement du tsunami avant qu'il n'atteigne les côtes. Le premier défi réside donc dans la détection des signaux précurseurs du tsunami à travers la mesure des ondes acoustiques rapides, par exemple au moyen de capteurs acoustiques déployés dans les grandes profondeurs océaniques. Cependant, si ces capteurs permettent d'identifier les signaux précurseurs, ils ne permettent pas de déduire les caractéristiques physiques de l'événement, à savoir la déformation initiale du fond marin responsable du tsunami, et donc son évolution future.

Afin de prédire la propagation d'un tsunami, il est nécessaire de mesurer la surface libre de l'océan au niveau du front d'onde des ondes gravitaires de surface afin de reconstruire, à partir de mesures de surface bruitées, la déformation initiale avec précision et rapidité, et ainsi prévoir l'évolution du tsunami. Dans cette perspective, au-delà des acquisitions d'images satellitaires ou des données fournies par les bouées DART stratégiquement positionnées à proximité des dorsales océaniques, physiciens et mathématiciens se sont attachés à modéliser ce phénomène.

Il convient néanmoins de souligner que la cartographie des océans, en particulier à proximité des dorsales et des rifts océaniques où se produisent la majorité des séismes, constitue une tâche difficile et extrêmement coûteuse (une seule bouée DART coûte approximativement 250,000 dollars). De plus, la couverture spatiale offerte actuellement par les bouées DART semble insuffisante pour disposer d'un jeu de données assez vaste permettant la mise en œuvre de méthodes de détection avancées. Dans cette thèse, nous supposerons que l'acquisition des données est réalisée par imagerie satellitaire, permettant une précision supérieure à l'échantillonnage spatial discret fourni par les bouées DART.

Cette thèse s'interesse ainsi au problème inverse suivant : comment retrouver les déformations du fond marin à partir de la déformation mesurée sur la surface libre?

La littérature, tant physique que mathématique, s'est intéressée à la propagation des tsunamis provoqués par des déformations du fond marin. De nombreux travaux développent, de manière très

approfondie, une large gamme de modèles directs permettant de décrire ce phénomène, depuis la connaissance de la déformation du fond marin jusqu'à la propagation du tsunami. À cet égard, on peut citer, par exemple, [101, 69, 109], qui compilent plusieurs travaux consacrés à la modélisation de la propagation des ondes de surface sur cette base. Un modèle particulièrement notable est le modèle de Saint-Venant, également appelé modèle en eaux peu profondes (*shallow-water model*). Ce modèle fait intervenir la hauteur d'eau ainsi que la composante horizontale moyenne de la vitesse sur la hauteur d'eau. Cette modélisation non linéaire apparaît dans de nombreux travaux de la littérature ; voir, par exemple, [6, 30], qui utilisent tous deux le modèle en eaux peu profondes pour effectuer des calculs numériques dans le cadre de problèmes océanographiques. Si l'etude du problème direct est expansive dans la littérature, tant mathématique que physique, l'étude, quant à lui du problème inverse est plus restreinte dans un cadre mathématique rigoureux.

Chapter 1

Introduction

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1.1 About tsunami detection

Seabed deformations, particularly those induced by underwater earthquakes, are among the main mechanisms responsible for the generation of tsunamis. These phenomena, although rare, are potentially devastating and pose major scientific, environmental, and societal challenges. Immediately above the epicentre of the earthquake, tsunamis are waves of very small amplitude that can pass undetected without precise instruments. However, tsunamis have very large wavelengths, often greater than ocean depths, which causes an amplification of their amplitude as the depth of the ocean decreases.

Predicting a tsunami and its evolution is essential for protecting coastal populations as well as infrastructures built in flood-prone areas. The term prediction is somewhat misleading here, since the earthquakes responsible for tsunamis are highly random phenomena. The objective is rather to predict the behaviour of the tsunami a posteriori as quickly as possible before it reaches the coast. The first challenge is therefore to detect the precursor signals of the tsunami through the measurement of fast acoustic waves, for example by deploying acoustic sensors deep in the ocean. However, while these acoustic sensors make it possible to identify precursor tsunami signals, they do not allow one to infer the physical characteristics of the event, namely the initial seabed deformation responsible for the tsunami and therefore its future evolution. From a physical point of view, the generation of a tsunami results from an energy transfer between a rapid deformation of the ocean floor and the overlying water column. This interaction involves, in certain regimes, acoustic waves propagating through the fluid at a speed $c = 1500$ m/s, which constitute a precursor signal announcing the arrival of a tsunami but do not themselves represent a major danger. More importantly, it involves surface gravity waves propagating much more slowly, corresponding to the visible tsunami wave and representing the truly hazardous component of the phenomenon. To predict the propagation of a tsunami, the free surface of the ocean must be measured at the level of the wavefront of the surface gravity wave in order to reconstruct, from noisy surface measurements, the initial deformation accurately and rapidly, and thus predict the tsunami evolution. With this objective in mind, beyond satellite image acquisition 1.1b or the data provided by DART buoys strategically positioned near oceanic ridges 1.1a, physicists and mathematicians have sought to model the phenomenon. It is nevertheless important to note that mapping the oceans, particularly near oceanic ridges or oceanic rifts where most earthquakes occur, is a difficult and extremely expensive task (a single DART buoy costs approximately 250,000 dollars). Moreover, the spatial coverage of the ocean provided by DART buoys currently appears insufficient to obtain a dataset large enough for the implementation of advanced detection methods. In this thesis, we shall assume that data acquisition is performed using satellite imaging, which allows for greater precision than the discrete spatial sampling provided by DART buoys.



(a) DART buoy measuring the free surface..



(b) The SWOT satellite from CNES studying the ocean topography..

Both the physical and mathematical literature have studied the propagation of a tsunami caused by bottom deformations. In a very expansive manner, the literature develops a wide range of direct modelling (from the knowledge of the bottom deformation to the tsunami propagation) of this phenomenon. In this regard, one may cite for example [101, 69, 109], which compiles several works devoted to modelling the propagation of surface waves on this basis. One model worth noting is the Saint-Venant, or shallow-water, model. This model involves the water height and the average

horizontal component of the velocity over the height. This nonlinear modelling appears in many works in the literature; see for example [6, 30], which both use the shallow-water model for numerical computations in oceanographic problems. Another study [52] describes the different behaviours of a tsunami at each stage of its lifetime and notably shows that, except during its generation, tsunami propagation is a highly nonlinear phenomenon. This earlier work shows that near the coast the Saint-Venant model is more appropriate, while further offshore it is the Euler equations that provide the best modelling of the phenomenon. From this Euler equation, a linearisation is applied as in [51], leading to two main problems: an asymptotic one that takes into account only gravity waves, and the wave equation coupling both gravity and acoustic waves. One may also cite [51], which rigorously derives different propagation models under physically justified assumptions. A more geology-oriented approach [49] highlights, through laser strainmeter measurements, the link between crustal deformation anomalies and the occurrence of tsunamis. This work is based on a modelling approach using the sine-Gordon equation. One may further mention [5], which, under the assumption of linear propagation in the water wave model, considers the propagation of surface tsunamis caused by seabed deformation in the presence of a viscous layer deposited on the bottom. The literature concerning the modelling of the direct problem is extensive, and many models are derived under various assumptions; in this perspective one may cite [75, 99]. In [70], a 3D coupled acoustic-gravity wave equation is used to model the forward problem, while a Bayesian digital twin is constructed using pressure data from seafloor sensors. This model enables the effect of the earthquake on the seafloor, and its real-time impact on tsunami propagation, to be taken into account.

The purpose of this thesis is to investigate the following inverse problem: given measurements of the free surface, how can one reconstruct the shape of the seabed deformation responsible for the detected tsunami? One may notably cite [42] which, in the context of a flat-bottom tank, studies this inverse problem through Fourier–Laplace transforms. Starting from a controlled excitation of the tank bottom by a piston, [42] seeks numerically to reconstruct this deformation by measuring the induced deformation at the surface of the tank. In an analogous manner, [11] uses a Fourier–Hankel transform to solve the same inverse problem. Another method widely used in the geophysical community is the four-dimensional variational data assimilation (4D-Var) approach, employed to solve data assimilation inverse problems in oceanography or climate forecasting. This method minimises a least-squares functional and then follows a discretise-then-optimise strategy; see, for example, [8, 86, 76]. The study of inverse problems for tsunami propagation has been conducted within the shallow-water model using an optimal control approach; see for example [79], which considers both satellite and DART buoy data. One can also mention [61], which proposes a direct solution of the inverse problem using the Saint-Venant equations. Finally, by measuring both the average speed and the free surface and using an optimal control approach in a slightly modified model compared with the Saint-Venant equations, called the Boussinesq–Peregrine system, [93] proposes an efficient reconstruction method, verified numerically, while proving the well-posedness of the forward problem used. We can finally cite [15], which, in the case of the Saint-Venant equations, develops an approach using a Luenberger observer to monitor the evolution of the tsunami. Here the approach is slightly different from those studied in the other papers: the goal is not to reconstruct the bottom deformation but to converge asymptotically in time towards the correct trajectory associated with the actual deformation. Concerning an approach based on observers, one can cite [74] that uses observers to solve an initial data inverse problem.

The purpose of the work we have carried out has been to develop mathematically rigorous methods that allow the resolution of the inverse problem described above. This thesis therefore tackle the following question : what theoretical guarantees can be established for the solution of the inverse tsunami detection problem, while ideally preserving computational efficiency? Its intuitive difficulty becomes immediately apparent: one seeks to reconstruct a source located at the bottom of the ocean, nearly 4000 meters deep, solely from measurements at the free surface. Can one even assume that a given free surface uniquely determines a seabed deformation?

With this objective in mind, the work conducted has been organised around two different inverse problems. First, there is the study of a nonlinear inverse problem of active imaging, referred to as bathymetry, treated in a simplified setting in Chapter 2. This problem differs from tsunami detection in that the objective here is to map the shape of the ocean floor while it is at rest. To do

so, one actively generates a source in the ocean – micro-explosions or controlled wave emissions – in order to perturb the medium and infer information about the nature of the seabed. Then, and this is the problem most extensively developed throughout the thesis, we consider a passive imaging problem for tsunami detection using a linear model describing the early stage of tsunami propagation, the model being derived in [51]: from measurements of the free surface s, how can one reconstruct the shape of the seabed deformation χ responsible for the detected tsunami? In [52], it is stated that the short-time behaviour of a tsunami can, without introducing excessive errors, be approximated by a linear model. It is within this simplified framework that we shall reconstruct the source χ at the ocean floor. Consequently, throughout this thesis, the observation times considered must be understood as sufficiently short to ensure that the model used for inversion remains physically relevant with respect to tsunami dynamics. In the following, within the context introduced above, we present the mathematical framework chosen to study our inverse problem.

1.2 Problem setting for an early-stage tsunami propagation problem

As described earlier, the propagation of a tsunami is a highly non linear phenomenon. The closer you get from the coast, the more turbulent it gets and the less linear it becomes. Since our goal is to detect the tsunami in its early stage of life, the approximation of a linear problem is reasonable [51]. The goal of the following section is to present the linear direct problems that will describe the propagation of the tsunami at its early stage of life. The description of the second problem : the bathymetry problem, will be made in Chapter 2 only in the harmonic framework.

In this section, we introduce the so-called "complete" model considered in this thesis. This model is rigorously obtained by linearizing the Euler equations in [51]. The following problem corresponds to the "barotropic case" in [51] making the assumption that the density of the fluid remains constant with depth. However, we will continue to call this model the "complete" model as it couples both effects : the acoustic propagation at a speed c in the fluid and the gravity waves propagation only occurring at the surface at a slower pace of $c_g = \sqrt{gH}$ where H being the depth of the ocean. This model establishes a relationship between the velocity potential φ, the free surface perturbation s, and the normal displacement of the seabed ζ. Furthermore, the source term ζ, which in our case is caused by an underwater earthquake, is related to the free surface s, i.e., the resulting tsunami, through the complete model. The ocean is assumed to be at rest for negative times $t \leq 0$, with the earthquake occurring only for $t > 0$.

In our model, the ocean at rest is a two-dimensional domain Ω_θ, infinite in the longitudinal direction, defined for $X = (x, z)$ by:

$$\Omega_\theta = \left\{ X \in \mathbb{R}^2,\ -H + \theta(x) < z < 0 \right\},$$

where θ is a continuous function with compact support corresponding to a perturbation of the flat bottom described by the equation $z = -H$. The undisturbed free surface is given by the equation $z = 0$. The domain Ω_θ is thus bounded by two boundaries

$$\Gamma_0 = \left\{ X \in \mathbb{R}^2,\ z = 0 \right\}, \quad \Gamma_{-H+\theta} = \left\{ X \in \mathbb{R}^2,\ z = -H + \theta(x) \right\}.$$

We denote g the gravitational acceleration and c the celerity of acoustic waves. Let $\zeta(x, t)$ be the normal displacement of the seabed and $s(x, t)$ the perturbation of the free surface. The velocity potential $\varphi(X, t)$, the free surface $s(x, t)$, and the seabed displacement $\zeta(x, t)$ can then be related

through the equations of the complete model in the domain Ω_θ:

$$\begin{cases} \frac{1}{c^2}\partial_t^2\varphi - \Delta\varphi = 0 & \text{in } \Omega_\theta \times (0,T), \\ \partial_t\varphi + gs = 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_z\varphi - \partial_t s = 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_\nu\varphi = \chi & \text{on } \Gamma_{-H+\theta} \times (0,T), \\ \varphi = 0 & \text{in } \Omega_\theta \times \{0\}, \\ \partial_t\varphi = 0 & \text{in } \Omega_\theta \times \{0\}, \\ s = 0 & \text{on } \Gamma_0 \times \{0\}. \end{cases} \tag{1.1}$$

with $T > 0$, $\chi = -\partial_t\zeta$, and ν the outward normal to the domain. The regularity of the source term χ will be specified later. It is assumed that there is no tsunami for negative times, and in particular, $\chi(0) = \partial_t\chi(0) = 0$. Note that the two equations on Γ_0 can be merged into a single one:

$$\partial_t^2\varphi + g\partial_\nu\varphi = 0.$$

The model (1.1) is designated "complete" because it accounts for both gravity waves and acoustic waves. Two asymptotic models can then be easily derived from it: the so-called gravity model, corresponding to the case $c \to \infty$, and the so-called acoustic model, in which gravity effects are neglected, i.e., $g = 0$. Since the water waves model will be studied in detail, let us write the relationship between $\varphi(X,t)$ and $\zeta(x,t)$ in this asymptotic framework in the domain Ω_θ:

$$\begin{cases} \Delta\varphi = 0 & \text{in } \Omega_\theta \times (0,T), \\ \partial_t^2\varphi + g\partial_\nu\varphi = 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_\nu\varphi = \chi & \text{on } \Gamma_{-H+\theta} \times (0,T), \\ \varphi = 0 & \text{in } \Gamma_0 \times \{0\}, \\ \partial_t\varphi = 0 & \text{in } \Gamma_0 \times \{0\}. \end{cases} \tag{1.2}$$

The difference between problem (1.1) and (1.2) is the behaviour of the fluid. While the complete model accounts for both acoustic and gravity waves propagation, in the asymptotic case we only consider the gravity waves. In this thesis, the problem (1.2) will refer to the "asymptotic" model.

Remark 1.2.1. *The initial conditions on s, φ, and $\partial_t\varphi$ are set to zero, since the ocean is assumed to be at rest at $t = 0$. This assumption greatly simplifies the subsequent analysis without compromising the modeling of the phenomenon.*

Now we are going to introduce important results and classical methods to study both the direct and the inverse problems. A key component when it comes to build a functional framework for a system are some energy estimate. These identities help us to understand in which spaces we can expect a control of the solutions, in order to get existence and uniqueness results. We formally derive some of these results that will be proven later on. The first energy estimate is obtained in a semi infinite domain. However, in practice, we wish to implement a resolution scheme on a bounded domain. The question is : what is the best choice of boundary conditions for this problem? To help answering this question, we study the velocity of the propagation of the energy which is closely related to the former question. Only then we will briefly present two classes of methods to solve such inverse problems : a one-shot mixed formulation approach and an optimal control method. To give an idea on the ill-posedness of the inverse problems, we finally derive a stability inequality for the asymptotic model which illustrates the degree of ill-posedness of the problem.

1.3 Study of the direct problem

1.3.1 Energy estimate

In this section, we formally derive some energy identities for the problem (1.1). These inequalities are essential when it comes to construct a functional framework for the system and to get a proper control on the solutions. In particular, these inequalities will be essential in the construction of strong semi-group solutions and variational solutions in Chapter 3. We first recall a classical Grönwall Lemma, which will be useful in the following.

Lemma 1.3.1 (Grönwall Lemma)**.** *Let φ and ψ be two continuous functions with ψ positive, satisfying:*

$$\forall t \geq t_0, \quad \varphi(t) \leq K + \int_{t_0}^{t} \psi(s)\varphi(s)\ ds$$

where K is a constant. Then

$$\forall t \geq t_0, \quad \varphi(t) \leq K exp\left(\int_{t_0}^{t} \psi(s)\ ds \right).$$

Proposition 1.3.1. *Let us introduce the following energy functional:*

$$\mathcal{E}(t) = \frac{1}{c^2}\int_{\Omega_\theta} |\partial_t\varphi|^2\ dX + \int_{\Omega_\theta} |\nabla\varphi|^2\ dX + g\int_{\Gamma_0} |s|^2\ dx, \tag{1.3}$$

with $(\varphi, s)^T$ satisfying (1.1). Assuming enough regularity we have for all t in $[0,T]$:

$$\frac{1}{2}\frac{d}{dt}\mathcal{E}(t) = \int_{\Gamma_{-H+\theta}} \chi\partial_t\varphi\ dx.$$

Proof. We multiply the first equation of (1.1) by $\partial_t\varphi$ and integrate over Ω_θ. The result of the Proposition then follows naturally from Green's formula. □

Proposition 1.3.2. *For $\chi \in H^1(0,T;L^2(\Gamma_{-H+\theta}))$, there exists $K > 0$ such that for all t in $[0,T]$:*

$$\frac{1}{2c^2}\int_{\Omega_\theta} |\partial_t\varphi|^2\ dX + \frac{1}{2}\int_{\Omega_\theta} |\nabla\varphi|^2\ dX + \frac{1}{2}\int_{\Omega_\theta} \varphi^2\ dX + \frac{1}{2g}\int_{\Gamma_0} |\partial_t\varphi|^2\ dx \leq K\|\chi\|^2_{H^1(0,T;L^2(\Gamma_{-H+\theta}))}.$$

with φ a solution of (1.1) with enough regularity.

Proof. The regularity of χ allows us to assert that there exists a constant $C' > 0$ such that:

$$\sup_{t\in[0,T]} \|\chi(t)\|^2_{L^2(\Gamma_{-H+\theta})} \leq C'\|\chi\|^2_{H^1(0,T;L^2(\Gamma_{-H+\theta}))}.$$

In the following, we denote $G = \|\chi\|_{H^1(0,T;L^2(\Gamma_{-H+\theta}))}$. We also denote

$$\tilde{\mathcal{E}}(t) = \frac{1}{2c^2}\int_{\Omega_\theta} |\partial_t\varphi|^2\ dX + \frac{1}{2}\int_{\Omega_\theta} |\nabla\varphi|^2\ dX + \frac{1}{2}\int_{\Omega_\theta} \varphi^2\ dX + \frac{1}{2g}\int_{\Gamma_0} |\partial_t\varphi|^2\ dx.$$

Noticing that

$$\tilde{\mathcal{E}}(t) = \frac{1}{2}\mathcal{E}(t) + \frac{1}{2}\int_{\Omega_\theta} \varphi^2\ dX,$$

and using the result of Proposition 1.3.1, it follows that

$$\frac{d}{dt}\tilde{\mathcal{E}}(t) = \int_{\Gamma_{-H+\theta}} \chi\partial_t\varphi\ dx + \frac{1}{2}\frac{d}{dt}\int_{\Omega_\theta} \varphi^2\ dX, \quad \forall t \in [0,T].$$

Hence, by integrating in time, we obtain:

$$\tilde{\mathcal{E}}(t) = \int_0^t \int_{\Gamma_{-H+\theta}} \chi \partial_t \varphi \, dx ds + \frac{1}{2} \int_{\Omega_\theta} \varphi^2 \, dX, \quad \forall t \in [0,T].$$

We treat each of the two terms of $\tilde{\mathcal{E}}$ separately, starting with the second one:

$$\begin{aligned} \frac{1}{2} \int_{\Omega_\theta} \varphi^2 \, dX &= \frac{1}{2} \int_{\Omega_\theta} \left(\int_0^t \partial_t \varphi \, ds \right)^2 dX \\ \text{(By Cauchy-Schwarz inequality)} \quad &\leq \frac{T}{2} \int_{\Omega_\theta} \int_0^t |\partial_t \varphi|^2 \, ds dX \\ &\leq c^2 T \int_0^t \tilde{\mathcal{E}}(s) \, ds, \quad \forall t \in [0,T]. \end{aligned}$$

We now handle the second term of $\tilde{\mathcal{E}}$ by integrating by parts in time:

$$\begin{aligned} \int_0^t \int_{\Gamma_{-H+\theta}} \chi \partial_t \varphi \, dx ds &= \int_{\Gamma_{-H+\theta}} [\chi \varphi]_{s=0}^{s=t} \, dx - \int_{\Gamma_{-H+\theta}} \int_0^t \frac{\partial \chi}{\partial t} \varphi \, ds dx \\ &= \int_{\Gamma_{-H+\theta}} \chi(t) \varphi(t) \, dx - \int_0^t \left(\int_{\Gamma_{-H+\theta}} \frac{\partial \chi}{\partial t} \varphi \, dx \right) ds \\ &\leq \frac{1}{2\epsilon} \|\chi(t)\|^2_{L^2(\Gamma_{-H+\theta})} + \frac{\epsilon}{2} \|\varphi(t)\|^2_{L^2(\Gamma_{-H+\theta})} \\ &+ \frac{1}{2} \int_0^t \left(\left\| \frac{\partial \chi}{\partial t} \right\|^2_{L^2(\Gamma_{-H+\theta})} + \|\varphi\|^2_{L^2(\Gamma_{-H+\theta})} \right) ds. \end{aligned}$$

By combining the two obtained inequalities, we then have:

$$\tilde{\mathcal{E}}(t) \leq \frac{C' G^2}{2\epsilon} + \frac{\tilde{C}\epsilon}{2} \|\varphi\|^2_{H^1(\Omega_\theta)} + \frac{1}{2} G^2 + \tilde{C} \int_0^t \tilde{\mathcal{E}}(s) \, ds + c^2 T \int_0^t \tilde{\mathcal{E}}(s) \, ds, \quad \forall t \in [0,T],$$

with $\tilde{C}$ the trace constant on $\Gamma_{-H+\theta}$ such that:

$$\forall u \in H^1(\Omega_\theta), \quad \|u\|_{L^2(\Gamma_{-H+\theta})} \leq \tilde{C} \|u\|_{H^1(\Omega_\theta)}. \tag{1.4}$$

Hence, for ϵ small enough, we induce that there exists $C = \frac{C' G^2}{2\epsilon} + \frac{1}{2} G^2 > 0$ and $D = \tilde{C} + c^2 T > 0$ such that:

$$\tilde{\mathcal{E}}(t) \leq C + D \int_0^t \tilde{\mathcal{E}}(s) \, ds, \quad \forall t \in [0,T].$$

Taking $\psi = D$ in Lemma 1.3.1 we conclude that

$$\tilde{\mathcal{E}}(t) \leq C e^{Dt} \quad \forall t \in [0,T].$$

So that

$$\tilde{\mathcal{E}}(t) \leq C e^{DT} = K \|\chi\|^2_{H^1(0,T;L^2(\Gamma_{-H+\theta}))},$$

with $K = e^{DT} \left(\frac{C'}{2\epsilon} + \frac{1}{2} \right)$. □

Remark 1.3.1. *The performed calculations are, for now, formal and in fact requires a certain regularity of the function φ solving (1.1). We note that, in the following, strong solutions of the problem (1.1), constructed in Chapter 3, possess this regularity.*

Remark 1.3.2. *This result is fundamental for the construction of variational solutions, which will be detailed in Appendix A. Indeed, the boundedness of this energy $\tilde{\mathcal{E}}$ allows for the extraction of Cauchy subsequences and the construction of solutions with lower regularity.*

Remark 1.3.3. *The result of Proposition 1.3.2 extends naturally to $\chi \in H^1(0,T;H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))$.*

These energy identities are essential when it comes to construct a proper functional framework for solutions of problems (1.1)(1.2). Let us now present existence and uniqueness results for both of these problems.

1.3.2 Existence, uniqueness and regularity of solutions

Here, we briefly present one of the main component of the study conducted in this thesis. Which regularity can we expect for the solution of the direct problem? And more precisely, in which sense should we consider the solutions? This section will present results that will be detailed and proved in the different chapters of the thesis. We put here the emphasis on the particular difficulty our problem creates. Because of the boundary source χ, it is particularly subtle to define a solution in the space-time framework. Let us present the different frameworks we introduced for both of the model.

The complete model

This section presents existence and uniqueness results for the complete model (1.1). Let us start by introducing strong solutions in the semigroup sense, a result that is proven in Chapter 3.

Theorem 1.3.1. *Let us assume that $\chi \in H^2(0,T;H^{-1/2}(\Gamma_{-H+\theta}))$ with $\chi(\cdot,0)=0$ on $\Gamma_{-H+\theta}$. There exists a unique strong solution φ to problem (1.1), that is*

$$\varphi \in C^2([0,T];L^2(\Omega)) \cap C^1([0,T];H^1(\Omega)) \cap C^0([0,T];X),$$

where

$$X = \{\varphi \in H^1(\Omega),\ \Delta\varphi \in L^2(\Omega),\ \partial_z\varphi|_{\Gamma_0} \in L^2(\Gamma_0)\}.$$

The semigroup framework is quite classical when it comes to find strong solutions for time dependent evolution problem. We introduce two unbounded operators A and B such that $(\varphi,s)^T$ are solutions of (1.1) if and only if $z=(\varphi,\psi,s)^T$ is solution of

$$\begin{cases} \dot{z} + Az = B\chi, \\ z(0) = 0. \end{cases}$$

Theorem 1.3.1 is based on the analysis of A and B. The main difficulties relies on the non admissibility of the lifting operator B in the sense of [107]. This is related to the fact that χ is a source living only on the bottom surface, this causes difficulties which are dealt with in Chapter 3. While this framework is handful as it gives us strong solutions, the solutions in themselves are not very useful when it comes to numerical analysis. To prevent these issues, a type of solutions is considered which are the variational solutions, see for example [88] which deals with the necessity of weakening the notion of solutions when it comes to control.

We first define the Hilbert space

$$\widetilde{W}_0(0,T) = \Big\{\varphi \in L^2(0,T;H^1(\Omega)),\ \partial_t\varphi \in L^2(0,T;L^2(\Omega)),\ \partial_t(\varphi|_{\Gamma_0}) \in L^2(0,T;L^2(\Gamma_0)),\ \text{and } \varphi(0)=0\Big\}, \quad (1.5)$$

equipped with the norm $\|\cdot\|_{\widetilde{W}_0(0,T)}$ defined by

$$\|\varphi\|^2_{\widetilde{W}_0(0,T)} = \int_0^T \|\varphi\|^2_{H^1(\Omega)}\,dt + \int_0^T \|\partial_t\varphi\|^2_{L^2(\Omega)}\,dt + \int_0^T \|\partial_t(\varphi|_{\Gamma_0})\|^2_{L^2(\Gamma_0)}\,dt.$$

Definition 1.3.1. *Let $\chi \in H^1(0,T;H^{-1/2}(\Gamma_{-H}))$, we say that $\varphi \in \widetilde{W}_0(0,T)$ is a variational solution to Problem (3.1), if*

$$\partial_t^2\varphi \in L^2(0,T;H^1(\Omega)') \text{ and } \partial_t^2(\varphi|_{\Gamma_0}) \in L^2(0,T;H^{-1/2}(\Gamma_0)),$$
$$\text{while } \partial_t\varphi(0) = 0 \text{ and } \partial_t(\varphi|_{\Gamma_0})(0) = 0,$$

and for all $\tilde{\varphi} \in L^2(0,T;H^1(\Omega))$,

$$\int_0^T \frac{1}{c^2}\langle\partial_t^2\varphi,\tilde{\varphi}\rangle_{H^1(\Omega)',H^1(\Omega)} + \frac{1}{g}\left\langle\partial_t^2(\varphi|_{\Gamma_0}),\tilde{\varphi}|_{\Gamma_0}\right\rangle_{H^{-1/2}(\Gamma_0),H^{1/2}(\Gamma_0)} + (\nabla\varphi,\nabla\tilde{\varphi})_{L^2(\Omega)}\,dt = \int_0^T \langle\chi,\tilde{\varphi}\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})}\,dt. \quad (1.6)$$

Theorem 1.3.2. *Suppose that $\chi \in H^1(0,T;H^{-1/2}(\Gamma_{-H}))$, with $\chi(\cdot,0) = 0$ on Γ_{-H}. Then, there exists a unique variational solution $\varphi \in \widetilde{W}_0(0,T)$ to Problem (3.1).*

Here we have defined second order variational solutions. The proof of this Theorem is made in Chapter 3. The main idea is that, the variational solutions are limits of Cauchy sequences of strong solutions. The rigorous construction is also based on the energy estimate proved in Proposition 1.3.2. We see that we can manage to construct weaker solutions for a weakened assumption on the regularity of the source χ. An equivalent construction is made for first order variational solutions in Appendix A. Although these solutions seem to be well-fitted for optimal control, we still see some technical difficulties when it comes to handling them in practice. The main difficulties being that in the same $\tilde{W}_0(0,T)$ the derivative in time and the trace of φ on Γ_0 do not commute. The other one being that we are unable to get an inf-sup condition within this framework. The importance of such a condition will be made clear in Chapter 4.
The difficulty we face is thus to construct a workable framework compatible with an inf-sup condition. Such a framework is detailed in Chapter 4 thanks to the addition of a parabolic regularisation of system (1.1) which corresponds to a viscosity term in the system [88].

The asymptotic model

In this section, we recall some existence and uniqueness results for the asymptotic direct problem in the case where $\theta = 0$. These results will be properly shown in Chapter 3. Since we need those results in the following, let us recall then and let us prove some additional regularity results.

Theorem 1.3.3. *Suppose $\chi \in \mathcal{C}^2([0,T];L^2(\Gamma_{-H}))$ with $\chi(0) = \partial_t\chi(0) = 0$ on Γ_{-H}. Then, there exists a unique solution in a strong sense φ of (1.2) such that*

$$\varphi \in \mathcal{C}^2([0,T],H^{\frac{1}{2}}(\Omega_0)) \cap \mathcal{C}^1([0,T];H^1(\Omega_0)) \cap \mathcal{C}^0([0,T];H^{\frac{3}{2}}(\Omega_0)).$$

Remark 1.3.4. *Here, we consider strong solutions in the semigroup sense which are equivalent to solutions of Problem (1.2)*

Remark 1.3.5. *Following the proof of the previous Theorem made in Chapter 3, one can show that with $\chi \in \mathcal{C}^1([0,T];L^2(\Gamma_{-H}))$ with $\chi(0) = \partial_t\chi(0) = 0$ on Γ_{-H}, there is a unique strong solution φ of (1.2) such that*

$$\varphi \in \mathcal{C}^1([0,T];H^1(\Omega_0)) \cap \mathcal{C}^0([0,T];H^{\frac{3}{2}}(\Omega_0)).$$

The two following corollaries gives additional regularity to the solution to (1.2) to the cost of additional regularity for the source χ. These results will become essential in the analysis made in the following chapters.

Corollary 1.3.1. *Suppose* $\chi \in \mathcal{C}^2([0,T];L^2(\Gamma_{-H}))$ *with* $\chi(0) = \partial_t\chi(0) = \partial_t^2\chi(0) = 0$ *on* Γ_{-H}. *Then there exists a unique solution* φ *of (1.2) such that :*

$$\varphi \in \mathcal{C}^2([0,T];H^1(\Omega_0)) \cap \mathcal{C}^1([0,T];H^{\frac{3}{2}}(\Omega_0)).$$

Proof. By taking the derivative in time of (1.2), we can show that $\eta = \partial_t\varphi$ with φ satisfying (1.2), verifies

$$\begin{cases} \Delta\eta = 0 & \text{in } \Omega_\theta \times (0,T), \\ \partial_t^2\eta + g\partial_\nu\eta = 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_\nu\eta = \partial_t\chi & \text{on } \Gamma_{-H} \times (0,T) \\ \eta = 0 & \text{in } \Gamma_0 \times \{0\}, \\ \partial_t\eta = 0 & \text{in } \Gamma_0 \times \{0\}, \end{cases} \tag{1.7}$$

The initial condition $\eta(0) = 0$ on Γ_0 comes from the fact that $\partial_t\varphi(0) = 0$ on Γ_0. Let us check that $\partial_t\eta(0) = 0$ on Γ_0 On the one hand, we have that $\partial_t\eta + g\partial_\nu\varphi = 0$ on $\Gamma_0 \times (0,T)$, on the other hand considering (1.2) at initial time gives

$$\begin{cases} \Delta\varphi = 0 & \text{in } \Omega_0 \times \{0\}, \\ \partial_\nu\varphi = 0 & \text{on } \Gamma_{-H} \times \{0\}, \\ \varphi = 0 & \text{on } \Gamma_0 \times \{0\}, \end{cases}$$

which ensures that $\varphi = 0$ in $\Omega_0 \times \{0\}$ and then finally $\partial_\nu\varphi = 0$ on $\Gamma_0 \times \{0\}$. Hence, $\partial_t\eta(0) = 0$ on Γ_0. Finally, applying the result of Theorem 1.3.3 (more particularly the Remark 1.3.5) to (1.7) gives us the desired result. □

Corollary 1.3.2. *Suppose* $\chi \in \mathcal{C}^2([0,T];H^1(\Gamma_{-H}))$ *with* $\chi(0) = \partial_t\chi(0) = \partial_t^2\chi(0) = 0$ *on* Γ_{-H}. *Then there exists a unique solution* φ *of (1.2) such that :*

$$\partial_x\varphi \in \mathcal{C}^2([0,T];H^1(\Omega_0)) \cap \mathcal{C}^1([0,T];H^{\frac{3}{2}}(\Omega_0)),$$

and more particularly

$$s = -\frac{1}{g}\partial_t\varphi|_{\Gamma_0} \in H^1(0,T;H^1(\Gamma_0)).$$

Proof. By taking the derivative with respect to x of (1.2), we can show that $\eta = \partial_x\varphi$ with φ satisfying (1.2), verifies

$$\begin{cases} \Delta\eta = 0 & \text{in } \Omega_\theta \times (0,T), \\ \partial_t^2\eta + g\partial_\nu\eta = 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_\nu\eta = \partial_x\chi & \text{on } \Gamma_{-H} \times (0,T) \\ \eta = 0 & \text{in } \Gamma_0 \times \{0\}, \\ \partial_t\eta = 0 & \text{in } \Gamma_0 \times \{0\}, \end{cases} \tag{1.8}$$

Now applying Theorem 1.3.3 to (1.8) gives us the result. Furthermore, since $\partial_x\varphi \in \mathcal{C}^2([0,T];H^1(\Omega_0))$ then $\partial_x\partial_t\varphi|_{\Gamma_0} \in \mathcal{C}^1([0,T];L^2(\Gamma_0))$. This comes from the fact that the tangential trace and the derivative in time commute with the trace operator on Γ_0. Finally, we see that

$$s := -\frac{1}{g}\partial_t\varphi|_{\Gamma_0} \in \mathcal{C}^1([0,T];H^1(\Gamma_0)) \subset H^1([0,T];H^1(\Gamma_0)).$$

□

Theorem 1.3.4. *If $\chi \in H^3_{0,l}(0,T;H^1(\Gamma_{-H}))$, then there exists a unique solution φ solution of (1.2) such that :*

$$\varphi \in L^2([0,T];H^2(\Omega_0)),$$

with

$$H^3_{0,l}(0,T;H^1(\Gamma_{-H})) = \left\{\chi \in H^3(0,T;H^1(\Gamma_{-H})), \chi(0) = \partial_t\chi(0) = \partial_t^2\chi(0) = 0\right\},$$

Proof. Let us assume that χ belongs to $H^3_{0,l}(0,T;H^1(\Gamma_{-H}))$. It then follows that χ belongs to $\mathcal{C}^2([0,T];H^1(\Gamma_{-H}))$ such that $\chi(0) = \partial_t\chi(0) = \partial_t^2\chi(0) = 0$. Then, using Corollary 1.3.1, it follows that there exists a unique solution φ of (1.2) such that $\varphi \in \mathcal{C}^2([0,T];H^1(\Omega_0)) \cap \mathcal{C}^1([0,T];H^{\frac{3}{2}}(\Omega_0))$. In this proof, we denote $\eta = \partial_x\varphi$ and $\mu = \partial_y\varphi$. These two quantities then satisfy:

$$\begin{cases} \Delta\eta = 0 & \text{in } \Omega_0 \times (0,T), \\ \partial_t\eta + g\partial_x s = 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_y\eta = \partial_x\partial_t s & \text{on } \Gamma_0 \times (0,T) \\ \partial_\nu\eta = \partial_x\chi & \text{on } \Gamma_{-H} \times (0,T), \\ \eta = 0 & \text{in } \Omega_0 \times \{0\}, \\ \partial_t\eta = 0 & \text{in } \Omega_0 \times \{0\}. \end{cases} \qquad \begin{cases} \Delta\mu = 0 & \text{in } \Omega_0 \times (0,T), \\ \mu = -\dfrac{1}{g}\partial_t^2\varphi & \text{on } \Gamma_0 \times (0,T), \\ \mu = -\chi & \text{on } \Gamma_{-H} \times (0,T), \\ \mu = 0 & \text{in } \Omega_0 \times \{0\}, \\ \partial_t\mu = 0 & \text{in } \Omega_0 \times \{0\}. \end{cases}$$

Since $\chi \in \mathcal{C}^2([0,T];H^1(\Gamma_{-H}))$, it follows that $\partial_x\chi \in \mathcal{C}^1([0,T];L^2(\Gamma_{-H}))$. According to Theorem 1.3.3, we have $\eta \in \mathcal{C}^1([0,T];H^1(\Omega_0))$. Moreover, since $\chi \in \mathcal{C}^0([0,T];H^{\frac{1}{2}}(\Gamma_{-H}))$ and $\partial_t^2\varphi|_{\Gamma_0} \in \mathcal{C}^0([0,T];H^{\frac{1}{2}}(\Gamma_0))$, it follows that $\mu \in \mathcal{C}^0([0,T];H^1(\Omega_0))$.

Hence, we have that $\varphi \in \mathcal{C}^0([0,T];H^2(\Omega_0)) \subset L^2([0,T];H^2(\Omega_0))$. □

1.3.3 Propagation speed and artificial boundary conditions

With the goal of bounding the volumetric domain in order to solve the problem numerically, this part aims to determine the energy propagation velocity. More precisely, we ask whether the information (or energy) propagates at a finite or infinite speed. Indeed, if the information propagates at infinite speed, it will be necessary to bound the domain using an exact artificial condition involving a Dirichlet-to-Neumann operator or PML layers. Otherwise, any boundary condition on a sufficiently large domain will provide a correct numerical solution over a given time interval. To study this propagation speed, we introduce the following domain moving at speed $V > 0$ in the positive x-direction:

$$\Omega_t = \left\{X \in \mathbb{R}^2,\ x > R + Vt,\ -H < z < 0\right\}.$$

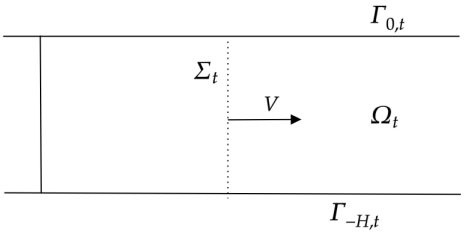
with $V > 0, R > 0$ fixed. The domain Ω_t is thus bounded by three edges defined by:

$$\Gamma_{0,t} = \left\{X \in \mathbb{R}^2,\ x > R + Vt,\ z = 0\right\}, \quad \Gamma_{-H,t} = \left\{X \in \mathbb{R}^2,\ x > R + Vt,\ z = -H\right\}$$

$$\Sigma_t = \left\{X \in \mathbb{R}^2,\ x = R + Vt,\ -H < z < 0\right\}.$$

We aim to determine, as a function of V and for sufficiently regular u, the sign of the derivative of $\mathcal{E}(\varphi,\Omega_t,t)$, defined by

$$\mathcal{E}(\varphi,\Omega_t,t) = \frac{1}{c^2}\int_{\Omega_t} |\partial_t\varphi|^2\ dxdz + \int_{\Omega_t} |\nabla\varphi|^2\ dxdz + \frac{1}{g}\int_{\Gamma_{0,t}} |\partial_t\varphi|^2\ dx.$$



with φ being the solution of the following problem

$$\begin{cases} \dfrac{1}{c^2}\partial_t^2\varphi - \Delta\varphi = 0 & \text{in } \Omega_0 \times (0,T), \\ \partial_t\varphi + gs = 0 & \text{on } \Gamma_{0,0} \times (0,T), \\ \partial_y\varphi - \partial_t s = 0 & \text{on } \Gamma_{0,0} \times (0,T), \\ \partial_\nu\varphi = 0 & \text{on } \Gamma_{-H,t} \times (0,T), \\ \varphi = f_0 & \text{in } \Omega_0 \times \{0\}, \\ \partial_t\varphi = f_1 & \text{in } \Omega_0 \times \{0\}, \\ s = f_2 & \text{on } \Gamma_{0,t} \times \{0\}. \end{cases}$$

with the $(f_i)_{0\leq i\leq 2}$ being nonzero initial conditions in $[-R,R]\times[-H,0]$. We then seek to determine the sign of $\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t)$. To this end, we need to differentiate the energy with respect to the time-dependent domain Ω_t. The following lemma presents a result that is useful in this regard.

Lemma 1.3.2. *Let $f(x,t)\in\mathcal{C}^1([-H,0]\times\mathbb{R}_+)$, it follows that for all $t\geq 0$*

$$\frac{d}{dt}\left(\int_{\Omega_t} f(X,t)\ dX\right) = \int_{\Omega_t}\frac{\partial f}{\partial t}(X,t)\ dX - V\int_{\Sigma_t} f(X,t)\ d\sigma. \tag{1.9}$$

Proof. From the definition of the domain Ω_t, we can write

$$X=(x,y)\in\Omega_t \Longrightarrow x > R+Vt.$$

It then follows that for all $t\geq 0$

$$\int_{\Omega_t} f(X,t)\ dX = \int_{-H}^{0}\left(\int_{R+Vt}^{+\infty} f(x,y;t)\ dx\right)\ dy.$$

We now compute the derivative of the foregoing quantity

$$\begin{aligned}\frac{d}{dt}\left(\int_{\Omega_t} f(X,t)\ dX\right) &= \int_{-H}^{0}\left(\frac{d}{dt}\int_{R+Vt}^{+\infty} f(x,y;t)\ dx\right)\ dy \\ &= \int_{-H}^{0}\int_{R+Vt}^{+\infty}\frac{\partial f}{\partial t}(x,y;t)\ dxdy - V\int_{-H}^{0} f(R+Vt,y;t)\ dy \\ &= \int_{\Omega_t}\frac{\partial f}{\partial t}(X,t)\ dX - V\int_{\Sigma_t} f(X,t)\ d\sigma\end{aligned}$$

□

In the following, we suppose we have enough regularity on the solution φ since the point of this section is to get an intuitive idea of how the energy propagates in the fluid. We now study the sign of $\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t)$ using Lemma 1.3.2. It then follows that

$$\begin{aligned}\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t) =& \frac{2}{c^2}\int_{\Omega_t}\partial_t^2\varphi\partial_t\varphi\ dxdy + 2\int_{\Omega_t}\nabla(\partial_t\varphi).\nabla\varphi\ dxdy + \frac{2}{g}\int_{\Gamma_{0,t}}\partial_t^2\varphi\partial_t\varphi\ dx\\ &- V\int_{\Sigma_t}\left(\frac{1}{c^2}|\partial_t\varphi|^2+|\nabla\varphi|^2\right)dy - \frac{V}{g}|\partial_t\varphi|^2(R+Vt,t).\end{aligned}$$

Now applying a Green's formula to the term $\nabla(\partial_t\varphi)\cdot\nabla\varphi$ gives

$$\begin{aligned}\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t) =& -V\int_{\Sigma_t}\left(\frac{1}{c^2}|\partial_t\varphi|^2+|\nabla\varphi|^2\right)dy + 2\int_{\partial\Omega_t}\partial_t\varphi\partial_\nu\varphi\ dy - \frac{V}{g}|\partial_t\varphi|^2(R+Vt,t)\\ &+\frac{2}{g}\int_{\Gamma_{0,t}}\partial_t^2\varphi\partial_t\varphi\ dx + \frac{2}{c^2}\int_{\Omega_t}\partial_t^2\varphi\partial_t\varphi\ dxdy - 2\int_{\Omega_t}\partial_t\varphi\Delta\varphi\ dxdy.\end{aligned}$$

Using the facts that $\frac{1}{c^2}\partial_t^2\varphi-\Delta\varphi=0$ in Ω_t, $\partial_\nu\varphi=0$ on $\Gamma_{-H,t}$, and $\partial_\nu\varphi+\frac{1}{g}\partial_t^2\varphi=0$ on $\Gamma_{0,t}$, it follows that

$$\begin{aligned}\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t) &\le -V\int_{\Sigma_t}\left(\frac{1}{c^2}|\partial_t\varphi|^2+|\partial_x\varphi|^2\right)dy - 2\int_{\Sigma_t}\partial_t\varphi\partial_x\varphi\ dy\\ &\le -V\int_{\Sigma_t}\left(\frac{1}{c^2}|\partial_t\varphi|^2+|\partial_x\varphi|^2+\frac{2}{V}\partial_t\varphi\partial_x\varphi\right)dy.\end{aligned}$$

The integrand of the last inequality can then be rewritten as follows

$$\frac{1}{c^2}|\partial_t\varphi|^2+|\partial_x\varphi|^2+\frac{2}{V}\partial_t\varphi\partial_x\varphi = \left(\frac{1}{c}\partial_t\varphi+\frac{c}{V}\partial_x\varphi\right)^2+|\partial_x\varphi|^2\left(1-\frac{c^2}{V^2}\right),$$

hence:

$$\begin{aligned}\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t) &\le -V\int_{\Sigma_t}\left(\frac{1}{c}\partial_t\varphi+\frac{c}{V}\partial_x\varphi\right)^2 dy - V\int_{\Sigma_t}|\partial_x\varphi|^2\left(1-\frac{c^2}{V^2}\right)dy\\ &\le -V\int_{\Sigma_t}|\partial_x\varphi|^2\left(1-\frac{c^2}{V^2}\right)dy.\end{aligned}$$

We thus observe that if $V\ge c$, then $\frac{d}{dt}\mathcal{E}(\varphi,\Omega_t,t)\le 0$. We can therefore conclude that the propagation occurs at a speed lower than c. Therefore we can consider an ad hoc condition to bound the domain, provided that the domain is sufficiently large with respect to the final time T. In the following, we will not emphasize on the correlation between the choice of the boundary condition and the size of the domain. In other words, even if it creates some reflection on the boundaries of the domain we will still choose a domain that is not too big in order to efficiently solve the problem numerically. A similar proof made for the asymptotic gravity case would have shown that the energy propagates at an infinite velocity. It follows that, to bound the domain in the x direction, we need for the asymptotic problem an exact boundary condition such as a DtN operator or PML layers. However, because obtaining such an exact condition reveals to be quite technical we chose for the asymptotic model an order 1 boundary condition that has been developed in [78]. The choice of these boundary will be make clear in Chapter 3.

1.4 Inverse problems setting

After having introduced the direct -or forward- problems, let us now present the inverse problems related to deformation bed reconstruction. The bathymetry inverse problem will be detailed in

Chapter 2. As described earlier, the goal is to retrieve the deformation of the sea-bottom χ from the measurements of the free surface s, so that the problem amounts to solve for the complete model : *For any s, find $\chi = \partial_\nu \varphi$ on $\Gamma_{-H+\theta}$ with φ satisfying*

$$\begin{cases} \frac{1}{c^2}\partial_t^2 \varphi - \Delta \varphi = 0 & \text{in } \Omega_\theta \times (0,T), \\ \partial_z \varphi = \partial_t s & \text{on } \Gamma_0 \times (0,T), \\ \partial_t \varphi = -gs & \text{on } \Gamma_0 \times (0,T), \\ \varphi = 0 & \text{in } \Omega_\theta \times \{0\}, \\ \partial_t \varphi = 0 & \text{in } \Omega_\theta \times \{0\}. \end{cases}$$

In fact, a measurement y of the free surface s will be polluted by noise. Of course, for the asymptotic problem the inverse problem amounts to solve the following problem : *For any s, find $\chi = \partial_\nu \varphi$ on $\Gamma_{-H+\theta}$ with φ satisfying*

$$\begin{cases} \Delta \varphi = 0 & \text{in } \Omega_\theta \times (0,T), \\ \partial_z \varphi = \partial_t s & \text{on } \Gamma_0 \times (0,T), \\ \partial_t \varphi = -gs & \text{on } \Gamma_0 \times (0,T), \\ \varphi = 0 & \text{in } \Gamma_0 \times \{0\}, \\ \partial_t \varphi = 0 & \text{in } \Gamma_0 \times \{0\}. \end{cases}$$

Mathematically, these inverse problems can be recast as Cauchy inverse problems with lateral conditions, the Cauchy data being $(\varphi, \partial_\nu \varphi)$ on $\Gamma_0 \times (0,T)$. Such problems are notoriously ill-posed. More intuitively, the ill-posedness comes from the fact that we solve a system where we have on the one hand too much conditions on the surface Γ_0 and on the other hand no condition on the bottom surface $\Gamma_{-H+\theta}$. Before studying these inverse problems it is important to know whether or not there is at most one solution χ to it. This uniqueness result will be shown later on Chapter 2 in the harmonic framework and in the space-time framework in Chapter 3. While the asymptotic inverse problem will always guarantee uniqueness of a solution, for the complete model this uniqueness will only be satisfied in a subdomain of the domain $\Omega_\theta \times (0,T)$ we will characterize. Our goal will be to retrieve χ in such subdomain.

The study of these inverse problems is the main goal of this thesis : we will indeed explore existence, uniqueness, methods of reconstruction and numerical analysis. Different approaches will be detailed in the following chapters. Let us present two classes of inverse problems resolution method that will be studied in detail in the thesis. The question is: how well these two types of approach manage to reconstruct the source χ at the bottom of the ocean? What are the pros and the cons of each approach?

1.4.1 One-shot mixed formulation

Consider V and H to be two real Hilbert spaces, and let $A : V \to H$ be a continuous non-invertible operator. Denoting by $F \in H$, let us consider the following inverse problem: *Find $u \in V$ such that*

$$Au = F. \tag{1.10}$$

This canonical form typically represents the type of inverse problem that we will encounter in this thesis. The problem is ill-posed since A is not invertible. To regularize this problem, one classically

introduces [72, 104] the Tikhonov regularization of the operator A as follows, for $\epsilon > 0$: *Find* $u_\epsilon \in V$ *such that*

$$A^*Au_\epsilon + \epsilon u_\epsilon = A^*F. \tag{1.11}$$

This problem is now well-posed in V for every $\epsilon > 0$. The fundamental result of Tikhonov regularization theory states that assuming A is injective and that there exists $u \in V$ such that $Au = F$, then $u_\epsilon \to u$ as $\epsilon \to 0$. This result therefore makes it possible to design a method for solving the inverse problem (1.10). It is sufficient to solve problem (1.11) for ϵ sufficiently small, and one may then reasonably approximate u_ϵ by u.
This approach becomes more technical and less direct [24] when noise contaminates the measurement F, since the solution of the Tikhonov regularization no longer converges to the exact solution as $\epsilon \to 0$. The method developed in the presence of measurement noise will be presented in greater detail in Chapters 2 and 3. Moreover, the operator A^*A appearing in the Tikhonov regularization may be difficult to compute in practice; to overcome this difficulty, a mixed formulation approach will be presented in specific cases of operators A.
The particular case that will be studied is the following. Let us introduce M and O two real Hilbert spaces. Let us consider two operators : $B : V \to M$ which roughly speaking accounts for the physics of the problem and $C : V \to O$ which corresponds the measurement operator, with $f \in O$ the measurement. Denoting $H = M \times O$, the non invertible operator A is:

$$\begin{cases} A : V \to H \\ \qquad u \mapsto (Bu, Cu) \end{cases}.$$

The inverse problem considered is then : *Find* $u \in V$ *such that*

$$Au = F,$$

with $F = (L, f) \in H$. Because of the definition of A, we understand the reason why it is hard to implement the operator A^*A while computing the Tikhonov regularisation. For example, if B represents the Helmholtz equation, the operator A^*A corresponds to a bilaplacian which numerically forces the use of Hermite's polynomial interpolation. To prevent such difficulties, we introduce an auxiliary variable $\lambda_\epsilon = Bu_\epsilon - L \in M$ such that solving for u_ϵ in (1.11) amounts to look for $(u_\epsilon, \lambda_\epsilon) \in V \times M$ solution of

$$\begin{cases} \varepsilon(u_\varepsilon, v)_V + b(v, \lambda_\varepsilon) + (Cu_\varepsilon, Cv)_O &= (f, Cv)_O, \quad \forall v \in V, \\ b(u_\varepsilon, \mu) - (\lambda_\varepsilon, \mu)_M &= (L, \mu)_M, \quad \forall \mu \in M, \end{cases}$$

where b represents the bilinear form associated with B. This form is much easier to implement as it only introduces second order differential operators in the case of the Helmholtz equation. This method has been introduced and rigorously described in [29]. One can also mention [44] that also uses mixed formulations to solve an inverse problems.

1.4.2 Optimal control method

Chapter 4 presents an optimal control method for solving our inverse problem. The objective is to reconstruct the source χ, representing the deformation of the seabed, as the minimizer of a least-squares functional. This approach was developed from the 1970s within the data assimilation community [88, 90] interested in weather forecasting. Consider the following dynamics

$$\begin{cases} \dot{z}(t) = Az(t) + B\chi(t) \\ z(0) = 0 \\ y(t) = Cz(t), \end{cases}$$

where C denotes an observation operator and y a measurement of the state z. Here, we choose not to precise the functional spaces used as they are quite technical to introduce. The functional framework will be rigorously described in Chapter 4. The central question of the control theory,

developed in [107] is observability: given the measurement y, can one recover the state of the system z and, consequently, the source χ? One is thus led to solve for χ as the minimizer of the following functional, for $\epsilon > 0$

$$\inf_{\chi} \mathcal{J}(\chi) := \int_0^T \frac{1}{2}\|y(t) - Cz|_\chi(t)\|^2 + \frac{\epsilon}{2}\|\chi(t)\|^2 \, dt, \tag{1.12}$$

with $z|_\chi$ satisfying

$$\begin{cases} \dot{z}(t) = Az(t) + B\chi(t) \\ z(0) = 0 \end{cases} \tag{1.13}$$

The challenge here is to chose the right norms in the criterion. The functional $\mathcal{J}$ may also be viewed as a Tikhonov regularization of an operator and can therefore be studied within the same framework as the approach introduced previously. In the optimal control problems that will be considered later, we will not address the problem in the same manner, since the Tikhonov-regularized operator associated with the functional $\mathcal{J}$ is far more complicated than those introduced in in the previous one-shot method. Although Chapter 4 will not focus on this issue, a major problem in control theory is obtaining observability inequalities for this problem of the form

$$\|\chi\|^2 \leq K \int_0^T \|Cz|_\chi\|^2 \, dt.$$

It should be noted that the notion of observability is dual to that of controllability of the operator B [107]. It is essential to notice that having such an equality gives us the well-posedness of the inverse problem. In the following section, we will derive for the asymptotic model a stability inequality. These stability inequalities are different from the observability inequalities we just introduced in the sense that the control between the source and the observation is not Lipschitz but rather logarithmic. In this sense, it does not gives us the well-posedness of the problem but allows the analysis of the degree of the ill-posedness of the inverse problem. Such an inequality is detailed in the following section through a Carleman inequality and highlights the severe ill-posedness of our inverse problem. Such inequality are for instance, developed in [7]. Chapter 4 will instead focus more on the numerical analysis aspects of such an approach. Such an analysis is particularly difficult when it comes to inverse problems. To conclude this section, we focus on two major aspects of optimal control. First, we propose an observability inequality for the asymptotic model. Finally, we present a classical engineer approach to solve such problems and show why it is mathematically challenging to study this approach.

1.5 Stability analysis of a reconstruction via optimal control

This section is dedicated to determining a stability inequality for an optimal control approach in the asymptotic case. This will illuminate the degree of the ill-posedness of the problem which we can see through the inequality we obtain. To simplify the discussion, we assume in this section that the seabed is flat, that is, $\theta = 0$. Accordingly, we denote the seabed by Γ_{-H} and the domain by Ω_0 in what follows. The study of the general case, where $\theta \neq 0$, proceeds in a similar manner but requires greater technical care in the proof of Theorem 1.3.4. Indeed, in that case, one can no longer differentiate the equations with respect to x and y directly, and it becomes necessary to work within local coordinate charts.

With the aim of establishing a convergence rate for the regularization method presented later, the following Lemma provides a Carleman inequality which will then be used to derive a logarithmic stability inequality. Let us first recall the following Carleman inequality.

Lemma 1.5.1. *Let $v \in H^2(\Omega_0)$ such that $v|_{\Gamma_{-H}} = \partial_\nu v|_{\Gamma_{-H}} = 0$. Then, there exists α, K, λ_0 such that for all $\lambda \geq \lambda_0$:*

$$\begin{aligned}\lambda^3 \int_{\Omega_0} v^2 e^{2\lambda\phi}\, dX + \lambda \int_{\Omega_0} |\nabla v|^2 e^{2\lambda\phi}\, dX \leq K \int_{\Omega_0} |\Delta v|^2 e^{2\lambda\phi}\, dX \\ + K\lambda \int_{\Gamma_0} |\nabla v|^2 e^{2\lambda\phi}\, dx + K\lambda^3 \int_{\Gamma_0} v^2 e^{2\lambda\phi}\, dx.\end{aligned} \tag{1.14}$$

with $\phi(x,y) = e^{\alpha(y+H)}$.

Proof. The proof is a slight adaptation of the one of [17]. □

Proposition 1.5.1. *For all $r \in (0,1)$, there exists $C > 0$ such that for all $v \in L^2(0,T;H^2(\Omega_0))$ such that $\|v\|^2_{L^2(0,T;H^2(\Omega_0))} \leq M$ where $M > 0$*

$$\|\partial_\nu v\|_{L^2(0,T;L^2(\Gamma_{-H}))} \leq C \frac{M}{\log^{\frac{r}{2}}\left(2 + \frac{M}{D(v)}\right)} + C\|\Delta v\|_{L^2(0,T;L^2(\Omega_0))},$$

with $D(v) = \|\Delta v\|_{L^2(0,T;L^2(\Omega_0))} + \|v\|_{L^2(0,T;H^1(\Gamma_0))} + \|\partial_z v\|_{L^2(0,T;L^2(\Gamma_0))}$.

Proof. The following proof is inspired by [17, 96]. Let us introduce the following weight function

$$g(z) = e^{\alpha z},$$

and $\psi(z) = z + H$ and $\phi(z) = e^{\alpha\psi(z)}$. We then introduce the following set

$$\Omega_\epsilon = \{X \in \Omega_0, z + H > \epsilon\},$$

and the following cut-off function

$$\chi_\epsilon(z) = \begin{cases} 1 & \text{if } z + H > 2\epsilon, \\ 0 & \text{if } z + H < \epsilon, \end{cases}$$

such that $\chi_\epsilon \in \mathcal{C}^2([-H, 0])$. Finally, we set $u = \chi_\epsilon v$. We then have that u satisfies the hypotheses of the Carleman estimate (1.14) so that

$$\begin{aligned}\lambda^3 \int_{\Omega_0} u^2 e^{2\lambda\phi}\, dX + \lambda \int_{\Omega_0} |\nabla u|^2 e^{2\lambda\phi}\, dX \leq K \int_{\Omega_0} |\Delta u|^2 e^{2\lambda\phi}\, dX \\ + K\lambda \int_{\Gamma_0} |\nabla u|^2 e^{2\lambda\phi}\, dx + K\lambda^3 \int_{\Gamma_0} u^2 e^{2\lambda\phi}\, dx.\end{aligned}$$

Considering $\lambda > 1$, it follows that

$$\lambda^3 \int_{\Omega_0} u^2 e^{2\lambda\phi}\, dX + \lambda \int_{\Omega_0} |\nabla u|^2 e^{2\lambda\phi}\, dX \geq \lambda e^{2\lambda g(3\epsilon)} \|v\|^2_{H^1(\Omega_{3\epsilon})}.$$

Let us now estimate the term on the right-hand side. We can choose χ_ϵ such that

$$\|\nabla \chi_\epsilon\|_{L^2(\Omega_0)} \leq \frac{C}{\epsilon}, \quad \|\Delta \chi_\epsilon\|_{L^2(\Omega_0)} \leq \frac{C}{\epsilon^2},$$

we get that

$$\begin{aligned}K \int_{\Omega_0} |\Delta u|^2 e^{2\lambda\phi}\, dX + K\lambda \int_{\Gamma_0} |\nabla u|^2 e^{2\lambda\phi}\, dX + K\lambda^3 \int_{\Gamma_0} u^2 e^{2\lambda\phi}\, dx \leq K e^{2\lambda g(H)} \int_{\Omega_0} |\Delta v|^2\, dX \\ + K\epsilon^{-4} e^{2\lambda g(2\epsilon)} \int_{\Omega_\epsilon \setminus \Omega_{2\epsilon}} (|v|^2 + |\nabla v|^2)\, dX \\ + K\lambda^3 e^{2\lambda g(H)} (\|v\|^2_{H^1(\Gamma_0)} + \|\partial_z v\|^2_{L^2(\Gamma_0)}),\end{aligned}$$

so that for $\lambda > 1$

$$\|v\|^2_{H^1(\Omega_{3\epsilon})} \leq C\lambda^2 e^{2\lambda(g(H)-g(3\epsilon))}(\|\Delta v\|^2_{L^2(\Omega_0)} + \|v\|^2_{H^1(\Gamma_0)} + \|\partial_z v\|^2_{L^2(\Gamma_0)}) + \frac{C}{\lambda\epsilon^4} e^{-2\lambda(g(3\epsilon)-g(2\epsilon))}\|v\|^2_{H^1(\Omega_0)}.$$

Taking $\epsilon < H/3$, we get that there exists $C > 0$ such that $g(3\epsilon) \leq C < g(H)$, so that, there exists $C_1 > 0$ such that

$$g(H) - g(3\epsilon) > C_1.$$

It is easy to check that

$$g(3\epsilon) - g(2\epsilon) \leq \alpha\epsilon,$$

so that

$$\|v\|^2_{H^1(\Omega_{3\epsilon})} \leq C\lambda^2 e^{2C_1\lambda}(\|\Delta v\|^2_{L^2(\Omega_0)} + \|v\|^2_{H^1(\Gamma_0)} + \|\partial_z v\|^2_{L^2(\Gamma_0)}) + \frac{C}{\lambda\epsilon^4} e^{-2\alpha\lambda\epsilon}\|v\|^2_{H^1(\Omega_0)}.$$

After possibly rescaling ϵ, we get that there exists $C, C_1, C_2 > 0$ such that

$$\|v\|_{H^1(\Omega_\epsilon)} \leq Ce^{C_1\lambda}(\|\Delta v\|_{L^2(\Omega_0)} + \|v\|_{H^1(\Gamma_0)} + \|\partial_z v\|_{L^2(\Gamma_0)}) + \frac{C}{\epsilon^2} e^{-C_2\lambda\epsilon}\|v\|_{H^1(\Omega_0)}.$$

Let us now choose λ with respect to ϵ such that for all $r \in (0,1)$

$$\epsilon^{-2}e^{-C_2\lambda\epsilon} = \epsilon^r.$$

It amounts to choose

$$\lambda = -\frac{r+2}{C_2}\frac{\ln\epsilon}{\epsilon} > 1,$$

so that

$$e^{C_1\lambda} = e^{-C\frac{\ln\epsilon}{\epsilon}}$$

Finally:

$$\|v\|_{H^1(\Omega_\epsilon)} \leq e^{-C\frac{\ln\epsilon}{\epsilon}}(\|\Delta v\|_{L^2(\Omega_0)} + \|v\|_{H^1(\Gamma_0)} + \|\partial_z v\|_{L^2(\Gamma_0)}) + \epsilon^r\|v\|_{H^1(\Omega_0)}. \tag{1.15}$$

Let us introduce $R_\epsilon = \Omega_0 \setminus \Omega_\epsilon$ and let us notice that on R_ϵ : $z + H \leq \epsilon$. Applying the Hardy inequality recalled here:

$$\left\| \frac{v}{(z+H)^{\frac{r}{2}}} \right\|_{L^2(\Omega_0)} \leq C\|v\|_{H^{\frac{r}{2}}(\Omega_0)}, \forall r \in (0,1), \forall v \in H^{\frac{r}{2}}(\Omega_0),$$

to v we get that

$$\begin{aligned}
\|v\|^2_{L^2(R_\epsilon)} &\leq C\epsilon^r\|v\|^2_{H^{\frac{r}{2}}(\Omega_0)} \\
&\leq C\epsilon^r\|v\|^2_{H^{\frac{1}{2}}(\Omega_0)} \\
&\leq C\epsilon^r\|v\|_{L^2(\Omega_0)}\|v\|_{H^1(\Omega_0)} \\
\text{(By Young inequality)} &\leq C\eta\|v\|^2_{L^2(\Omega_0)} + \frac{\epsilon^{2r}}{\eta}\|v\|^2_{H^1(\Omega_0)}.
\end{aligned}$$

With the same process applied on the partial derivatives of v, we get that

$$\|v\|^2_{H^1(R_\epsilon)} \leq C\eta\|v\|^2_{H^1(\Omega_0)} + \frac{\epsilon^{2r}}{\eta}\|v\|^2_{H^2(\Omega_0)}. \tag{1.16}$$

Finally, adding (1.15) and (1.16) for a η small enough, we get that for all $r \in (0,1)$, there exists $C > 0$, for all $\epsilon > 0$ small enough

$$\|v\|^2_{H^1(\Omega_0)} \leq e^{\frac{2C}{\epsilon}}(\|\Delta v\|^2_{L^2(\Omega_0)} + \|v\|^2_{H^1(\Gamma_0)} + \|\partial_z v\|^2_{L^2(\Gamma_0)}) + \epsilon^{2r}\|v\|^2_{H^2(\Omega_0)}.$$

Integrating over time, we then get that

$$\begin{aligned}\|v\|^2_{L^2(0,T;H^1(\Omega_0))} \leq e^{\frac{2C}{\epsilon}}(\|\Delta v\|^2_{L^2(0,T;L^2(\Omega_0))} &+ \|v\|^2_{L^2(0,T;H^1(\Gamma_0))} + \|\partial_z v\|^2_{L^2(0,T;L^2(\Gamma_0))}) \\ &+ \epsilon^{2r}\|v\|^2_{L^2(0,T;H^2(\Omega_0))}.\end{aligned}$$

Using the fact that v is bounded by M, we get that

$$\|v\|^2_{L^2(0,T;H^1(\Omega_0))} \leq e^{\frac{2C}{\epsilon}}(\|\Delta v\|^2_{L^2(0,T;L^2(\Omega_0))} + \|v\|^2_{L^2(0,T;H^1(\Gamma_0))} + \|\partial_z v\|^2_{L^2(0,T;L^2(\Gamma_0))}) + \epsilon^{2r}M.$$

By optimising in ϵ we finally get that for all $r \in (0,1)$, there exists $C > 0$ such that

$$\|v\|_{L^2(0,T;H^1(\Omega_0))} \leq C\frac{M}{\log^r\left(2 + \frac{M}{D(v)}\right)}.$$

We then obtain by the continuity of the normal trace operator from the $H^{\frac{3}{2}}(\Omega_0)$ functions with $L^2(\Omega_0)$ laplacian to $L^2(\Gamma_{-H})$ (see [62, Lemma 2.4])

$$\begin{aligned}\|\partial_\nu v\|_{L^2(0,T;L^2(\Gamma_{-H}))} &\leq C\|v\|_{L^2(0,T;H^{\frac{3}{2}}(\Omega_0))} + C\|\Delta v\|_{L^2(0,T;L^2(\Omega_0))} \\ &\leq C\|v\|^{\frac{1}{2}}_{L^2(0,T;H^1(\Omega_0))}\|v\|^{\frac{1}{2}}_{L^2(0,T;H^2(\Omega_0))} + C\|\Delta v\|_{L^2(0,T;L^2(\Omega_0))} \\ &\leq C\frac{M}{\log^{\frac{r}{2}}\left(2 + \frac{M}{D(v)}\right)} + C\|\Delta v\|_{L^2(0,T;L^2(\Omega_0))}.\end{aligned}$$

□

From now on, let us explore a method for reconstructing the normal displacement of the seabed, denoted by χ from the free surface measurements. The objective is to present a Tikhonov regularization approach for recovering the unknown source term χ in the asymptotic model (1.2), and to derive the corresponding convergence rate with respect to the regularization parameter. The reconstruction principle consists in determining χ from a measurement of the free surface elevation s, which we denote by s_{obs} throughout this section. We will also suppose that s_{obs} is the free surface associated to a source $\overline{\chi}$ through equation (1.2). The idea is to find a χ that best fits the observed data, that is, to minimize $s|\chi - s_{obs}$ in a suitable norm, where $s|_\chi$ denotes the free surface elevation associated with the solution of problem (1.2) for the given source χ. However, while this principle is theoretically valid, it is numerically unstable, which motivates the introduction of a regularization on the unknown source χ. These considerations lead us to minimize, with respect to χ, the following Tikhonov-regularized functional:

$$\inf_{\chi \in H^3_{0,l}(0,T;H^1(\Gamma_{-H}))} \mathcal{J}(\chi); \quad \mathcal{J}(\chi) := \frac{\epsilon}{2}\int_0^T \|\partial_t^3\chi\|^2_{H^1(\Gamma_{-H})} + \frac{1}{2}\|s|_\chi - s_{obs}\|^2_{H^1(\Gamma_0\times(0,T))},$$

with $\epsilon > 0$ and

$$H^1(\Gamma_0 \times (0,T)) = L^2(0,T;H^1(\Gamma_0)) \cap H^1(0,T;L^2(\Gamma_0)).$$

We recall the definition of $H^3_{0,l}(0,T;H^1(\Gamma_{-H}))$ introduced in Theorem 1.3.4. In this section, we will suppose that $\overline{\chi} \in H^3_{0,l}(0,T;H^1(\Gamma_{-H}))$. We first have to make sure that this functional is well-defined. In other words, for the regularity of the χ seeked and the regularity of $\overline{\chi}$, do $s|_\chi$ and s_{obs} belong to $H^1(\Gamma_0 \times (0,T))$? This result is a direct consequence of Corollary 1.3.2, therefore the functional $\mathcal{J}$ is well-defined. The existence and uniqueness of a minimizer for the functional $\mathcal{J}$ is established in the following proposition:

Proposition 1.5.2. *The functional $\mathcal{J}$ admits a unique minimizer χ_ϵ in $H^3_{0,l}(0,T;H^1(\Gamma_{-H}))$.*

Proof. The functional $\mathcal{J}$ is continuous, strictly convex and coercive. □

Throughout the following, we denote by φ_ϵ the solution of (1.2) associated with χ_ϵ and s_ϵ the corresponding free surface. We also denote in the same way by $\overline{\varphi}$ the solution associated with $\overline{\chi}$ through system (1.2). We then aim to establish a stability inequality between χ_ϵ and $\overline{\chi}$. In other words, we seek to obtain a convergence rate of χ_ϵ towards the exact solution $\overline{\chi}$ as a function of the regularization parameter ϵ. To obtain the convergence rate of the regularized solution χ_ϵ towards the exact solution $\overline{\chi}$ as a function of the parameter ϵ, we first need to bound $\|s_{obs} - s_\epsilon\|_{H^1(\Gamma_0\times(0,T))}$ with respect to ϵ.

Proposition 1.5.3. *The following inequality is satisfied:*

$$\|s_\epsilon - s_{obs}\|_{H^1(\Gamma_0\times(0,T))} \leq \sqrt{\epsilon}\|\partial_t^3\overline{\chi}\|_{L^2(0,T;H^1(\Gamma_{-H}))}.$$

Proof. The solution χ_ϵ of the minimization problem for the functional $\mathcal{J}$ satisfies, for all $\tilde{\chi}$ in $H^3_{0,l}(0,T;H^1(\Gamma_{-H}))$,

$$\langle D\mathcal{J}(\chi_\epsilon), \tilde{\chi}\rangle = 0,$$

that is,

$$\epsilon\int_0^T (\partial_t^3\chi_\epsilon, \partial_t^3\tilde{\chi})_{H^1(\Gamma_{-H})} - (s_{obs} - s_\epsilon, s|_{\tilde{\chi}})_{H^1(\Gamma_0\times(0,T))} = 0.$$

Choosing $\tilde{\chi} = \chi_\epsilon - \overline{\chi}$, it follows that

$$\epsilon\int_0^T (\partial_t^3\chi_\epsilon, (\partial_t^3\chi_\epsilon - \partial_t^3\overline{\chi}))_{H^1(\Gamma_{-H})} + \|s_{obs} - s_\epsilon\|^2_{H^1(\Gamma_0\times(0,T))} = 0. \tag{1.17}$$

From this last equality, it follows that

$$\|\partial_t^3\chi_\epsilon - \partial_t^3\overline{\chi}\|_{L^2(0,T;H^1(\Gamma_{-H}))} \leq \|\partial_t^3\overline{\chi}\|_{L^2(0,T;H^1(\Gamma_{-H}))}.$$

Finally, the equation (1.17) gives

$$\begin{aligned}
\|s_{obs} - s_\epsilon\|^2_{H^1(\Gamma_0\times(0,T))} &\leq -\epsilon(\partial_t^3\overline{\chi}, \partial_t^3\chi_\epsilon - \partial_t^3\overline{\chi})_{L^2(0,T;H^1(\Gamma_{-H}))} \\
\text{(By Cauchy-Schwarz)} &\leq \epsilon\|\partial_t^3\overline{\chi}\|_{L^2(0,T;H^1(\Gamma_{-H}))}\|\partial_t^3\chi_\epsilon - \partial_t^3\overline{\chi}\|_{L^2(0,T;H^1(\Gamma_{-H}))} \\
&\leq \epsilon\|\partial_t^3\overline{\chi}\|^2_{L^2(0,T;H^1(\Gamma_{-H}))},
\end{aligned}$$

hence the result. □

The following Corollary shows a useful stability inequality in order to get a rate of convergence of χ_ϵ towards $\overline{\chi}$.

Corollary 1.5.1. *There exists $C, \tilde{C}, M > 0$ such that:*

$$\|\chi_\epsilon - \overline{\chi}\|_{L^2(0,T;L^2(\Gamma_{-H}))} \leq C\frac{M}{\log^{\frac{r}{2}}\left(2 + \dfrac{M}{\tilde{C}\|s_{obs} - s_\epsilon\|_{H^1(\Gamma_0\times(0,T))}}\right)}.$$

Proof. From the well-posedness of problem (1.2) and Theorem 1.3.4 it follows that there exists $C > 0$ such that

$$\|\overline{\varphi} - \varphi_\epsilon\|_{L^2(0,T;H^2(\Omega_0))} \leq C\|\chi_\epsilon - \overline{\chi}\|_{H^3_{0,l}(0,T;H^1(\Gamma_{-H}))}.$$

Using the bound proved in the proof of Proposition 1.5.3 we get that for all $\epsilon > 0$

$$\|\overline{\varphi} - \varphi_\epsilon\|_{L^2(0,T;H^2(\Omega_0))} \leq C\|\chi_\epsilon - \overline{\chi}\|_{H^3_{0,l}(0,T;H^1(\Gamma_{-H}))} \leq C\|\partial_t^3\overline{\chi}\|_{L^2(0,T;H^1(\Gamma_{-H}))} := M.$$

From Proposition 1.5.1, the previous theorem and the fact that $\overline{\varphi}$ and φ_ϵ have their Laplacian equal to 0, we get that

$$\|\chi_\epsilon - \overline{\chi}\|_{L^2(0,T;L^2(\Gamma_{-H}))} \leq C\frac{M}{\log^{\frac{r}{2}}\left(2 + \dfrac{M}{\|\overline{\varphi} - \varphi_\epsilon\|_{L^2(0,T;H^1(\Gamma_0))} + \|\partial_z\overline{\varphi} - \partial_z\varphi_\epsilon\|_{L^2(0,T;L^2(\Gamma_0))}}\right)}. \tag{1.18}$$

Furthermore, on the one hand we have that

$$\|\partial_z\overline{\varphi} - \partial_z\varphi_\epsilon\|_{L^2(0,T;L^2(\Gamma_0))} = \|\partial_t s_{obs} - \partial_t s_\epsilon\|_{L^2(0,T;L^2(\Gamma_0))} \leq \|s_{obs} - s_\epsilon\|_{H^1(\Gamma_0\times(0,T))}.$$

On the other hand, we get that

$$\begin{aligned}
\|\varphi_\epsilon - \overline{\varphi}\|^2_{L^2(0,T;H^1(\Gamma_0))} &= \int_0^T \| \int_0^t (\partial_t\varphi_\epsilon - \partial_t\overline{\varphi})(\tau)\ d\tau\|^2_{H^1(\Gamma_0)}\ dt \\
\text{(Cauchy-Schwarz)} &\leq \int_0^T t\int_0^t \|(\partial_t\varphi_\epsilon - \partial_t\overline{\varphi})(\tau)\|^2_{H^1(\Gamma_0)}\ d\tau dt \\
&\leq T^2\|\partial_t\varphi_\epsilon - \partial_t\overline{\varphi}\|^2_{L^2(0,T;H^1(\Gamma_0))}.
\end{aligned}$$

Since

$$\partial_t\varphi_\epsilon - \partial_t\overline{\varphi} = -g(s_\epsilon - s_{obs}),$$

we finally get that

$$\|\varphi_\epsilon - \overline{\varphi}\|_{L^2(0,T;H^1(\Gamma_0))} \leq gT\|s_\epsilon - s_{obs}\|_{L^2(0,T;H^1(\Gamma_0))} \leq gT\|s_{obs} - s_\epsilon\|_{H^1(\Gamma_0\times(0,T))}.$$

Using (1.18) and the two previous inequalities, we conclude. □

The following Theorem presents the final convergence result of χ_ϵ towards $\overline{\chi}$.

Theorem 1.5.1 (Rate of convergence)**.** *For all $r \in (0,1)$, there exists $C > 0$ such that:*

$$\|\chi_\epsilon - \overline{\chi}\|_{L^2(0,T;L^2(\Gamma_{-H}))} \sim \frac{C}{|\log\epsilon|^{\frac{r}{2}}}.$$

Proof. The result is a direct consequence of Proposition 1.5.3 and Corollary 1.5.1. □

The previous result therefore indicates that solving the minimization problem for $\mathcal{J}$ with a small parameter ϵ approximates the exact solution of the considered inverse problem (when it exists). In fact, the obtained convergence rate is very slow. It is commonly acknowledged that the inverse problem associated with the asymptotic model is severely ill-posed. This slow convergence provides yet another illustration of this fact. Furthermore, the reconstruction norms and the regularity required for the solution $\overline{\chi}$ are extremely restrictive. This helps us to understand the ill-posedness of the asymptotic inverse problem and how hard it is to "observe" this problem in the sense of : to get a good reconstruction you need a really good observation.

1.6 Discretisation of an optimal control approach

In this section, we focus on a classical issue when it comes to discretizing an optimal control problem such as (1.12)(1.13): what is the best choice to solve numerically such a problem? Two alternatives are in order: the optimize-then-discretize strategy and the discretize-then-optimize one.

The optimize-then-discretize strategy consists in writing the optimality condition in the continuous framework and then discretising the resulting system accordingly. Let us introduce the saddle point problem of the lagrangian $\mathcal{L}$ associated with the optimal control problem (1.12)(1.13).

$$\inf_{\chi}\inf_{z}\sup_{p}\mathcal{L}(\chi,z,p)=\frac{1}{2}\int_0^T \|y(t)-Cz(t)\|^2+\frac{\epsilon}{2}\|\chi(t)\|^2+\langle \dot{z}(t)-Az(t)-B\chi(t),p\rangle\ dt, \quad (1.19)$$

One can show, under some technical assumptions, that if χ is the solution of the optimal control problem (1.12)(1.13) (which exists and is unique since the functional is coercive, continuous and strictly convex), cancelling the partial derivatives of $\mathcal{L}$ with respect to z,χ and p gives us the following optimality conditions systems

$$\begin{cases} \dot{z}(t)=Az(t)+B\chi(t), \\ \dot{p}(t)=A^*p(t)+C^*(y(t)-Cz(t)), \\ \epsilon\chi(t)-B^*p(t)=0, \\ z(0)=p(T)=0. \end{cases}$$

To solve the optimal control problem, it only remains to discretise this system and to solve it. A first approach is to solve it as a space-time monolithic problem by inverting one big linear system associated with a discretisation by the Finite Elements method discretization for example. While this approach is quite direct and easy to implement, it suffers from a lack of efficiency when the size of the problem increases. Indeed, to numerically solve such a problem, we will need to invert a space-time matrix with three unknowns. While the resolution takes only one iteration, this iteration can really be costly as the size of the space and time meshes refines. Let us also mention the memory cost to store such a FEM matrix. Another approach is to use an iterative approach by choosing a time stepping scheme for the system, such as for instance:

$$\begin{cases} \dot{z}^k(t)=Az^k(t)+\frac{1}{\epsilon}BB^*p^k(t), \\ \dot{p}^{k+1}(t)=A^*p^{k+1}(t)+C^*(y(t)-Cz^k(t)), \\ z^k(0)=p^k(T)=0. \end{cases}$$

It is now possible, after discretization in space and in time to solve the previous scheme iteratively. The main issue concerns the convergence. Because this iterative scheme does not rely on a functional to be minimized, there is no a priori reasons for it to converge towards the correct solution. The discretization of the forward and adjoint problems are made independently from one another and can possibly lead to a lack of convergence.

To cope with this problem, the discretize-then-optimize strategy consists in writing the optimality conditions on a discretised optimal control problem, so that the scheme obtained is directly linked to a functional which ensures convergence. This approach is commonly known as the 4DVAR approach in geophysics or climate forecasting for source monitoring or source reconstruction. Let us then introduce the discretised optimal control problem. Denoting

$$\inf_{(\chi_h^n)_{n=1\cdots N}} \mathcal{J}_h^\tau((\chi_h^n)) := \sum_{n=1}^N \frac{\tau}{2}\|y_h^n-C_hz_h^n\|^2+\frac{\tau\epsilon}{2}\|\chi_h^n\|^2,$$

with z_h^n satisfying a discretisation scheme of the equation (1.13) such as the following implicit Euler

scheme:

$$\begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} = A_h z_h^n + B_h \chi_h^n, \\ z_h^0 = 0. \end{cases}$$

To solve this discretised inverse problem, we can now compute an exact discrete gradient for the functional by computing a discrete adjoint state defined through a discrete lagrangian by

$$\begin{cases} \dfrac{p_h^n - p_h^{n+1}}{\tau} = A_h^* p_h^n + C_h^*(y_h^n - C_h z_h^n), \\ p_h^N = 0, \end{cases}$$

and

$$\nabla \mathcal{J}_h^\tau(\chi_h^n) = \tau(\epsilon \chi_h^n - B_h^* p_h^n).$$

The optimize-then-discretize method is now an iterative method of optimising a discrete functional using a descent method such as the Conjugate Gradient or L-BFGS. The resolution method now consists in solving a lot of small problems and iterate over a descent method. While in the optimize-then-discretize approach, the iterative approach has no a priori reasons to converge towards the continuous solutions, in this approach since the discrete optimality conditions are directly linked to a functional, the convergence is ensured. However, two questions remains.

Firstly, when h and τ goes to 0, it is not obvious to see and prove that the discrete solution $(\chi_h^n)_{n=1\cdots N}$ converges towards the continuous one χ. Indeed, this problem boils down to a Gamma convergence problem for which it is hard to conclude or/and to compute an estimate of the speed of the convergence. The solution of the discrete optimality conditions is the minimizer of a discretized functional which approximates the continuous criterion. Is it true that the discrete solution is a good approximation of the continuous solution and how good if it is the case? Secondly, to ensure this convergence, the choice of the discretisation scheme for the equation (1.13) is essential. We see for example that for a Crank Nicolson scheme, we would not be able to get a uniform discrete energy identity on the solution z_h^n because of spurious high frequencies discrete modes (see for example [73]). A precise discretization scheme is also developed and analysed in [34] for the wave equation. These two questions are the main topic of Chapter 4 which will study a discretise-then-optimise approach adding a parabolic regularisation.

1.7 Main contributions

In this context, the major contributions of this thesis are detailed in what follows. This thesis is divided into four chapters in addition to the introductory chapter above:

- In Chapter 2, we address the different inverse problems in a simplified framework. We consider the models in a harmonic setting, thus treating the solutions as plane waves in time. In this framework, we introduce tools to solve our two inverse problems (tsunami and bathymetry), with particular attention devoted to handling the amplitude of measurement noise. We propose an approach allowing, from noisy data, the reconstruction of the seabed source via a Tikhonov regularization whose regularization parameter ϵ is constructed and computed rigorously through Morozov's Principle. This chapter contains a published article with Laurent Bourgeois and Jean-François Mercier.
- In Chapter 3, we generalize the method introduced in Chapter 2 to the time-dependent setting, restricting ourselves to the inverse problem of tsunami identification only. A delicate point in this case is the uniqueness of the solution, which only holds within a subdomain of the domain under study. Moreover, the results allowing the handling of noise amplitude are extended to account for the time variable. Particular attention is devoted to the numerical implementation of the method, which uses 3D prismatic finite elements. Because of the nature of the method, the Finite Elements matrices tends to be huge in size, so that the system we inverse takes a lot of time and a lot of space in memory. This facts prevents us from dealing with more realistic set-up or with more refined meshes. To counter this

phenomenon, we have developed a SOR iterative method to invert the matrices which helps us solving larger problems. However, we are still limited by the size of the matrices we are dealing with. This chapter also emphasizes the rigorous construction of both strong and variational solutions for the direct problem. This chapter contains a submitted article with Laurent Bourgeois and Philippe Moireau.

- In Chapter 4, we introduce another approach for solving the tsunami inverse problem based on an iterative resolution of an optimal control problem. The iterative nature of the method deals with one of the main issues of the previous method. Instead of solving one large system, we solve several smaller forward and backward problems which allow us to deal with larger problems and more refined mesh. By applying a BFGS-type descent method and adding a parabolic regularization parameter, we present numerical reconstruction results for different noise levels. The focus mainly concerns the precise numerical analysis of the method, and in particular on the convergence of the discrete solution obtained numerically toward the continuous solution of the optimal control problem. We thus present a numerical estimate result that we verify numerically. However, this method still have two main issues : firstly, the analysis we made only works by adding a parabolic regularisation parameter which makes us solve a slightly different problem; secondly compared to the previous method the handling of the noise amplitude is much more complicated within this framework and relies on the construction of a L-curve to match the regularisation parameter ϵ. This chapter contains a submitted article with Laurent Bourgeois and Philippe Moireau.
- In the concluding Chapter 5, we first present a connection between the two major classes of methods developed throughout this thesis (optimal control method and one-shot mixed formulation), thereby showing a strong similarity between the approaches despite a major difference in their underlying philosophy. Finally, we present the necessary tools to generalize the study conducted in Chapter 4, and thus introduce a functional framework adapted to treating the tsunami inverse problem in the absence of parabolic regularization. We thus present, in an abstract framework, the functional spaces used to recast our problem into a workable framework for which we derive an inf-sup condition. The chapter concludes with the presentation of numerical experiment results obtained using physical parameters. This work is a collaboration with Sébastien Impériale and Juliette Dubois.

It should be noted that all the mathematical methods employed are rigorously justified. However, the numerical applications carried out remain restricted to the study of artificially simulated surface data. Note that however the results presented in Chapter 5 are more "realistic" and try to be more in adequation with the operational constraints that engineers could face in tsunami monitoring.

Communications

Publications

- Laurent Bourgeois, Jean-François Mercier, Raphaël Terrine. *Published in 2024.* Identification of bottom deformations of the ocean from surface measurements. *Inverse Problems and Imaging*
- Laurent Bourgeois, Philippe Moireau, Raphaël Terrine. *Submitted in 2026.* Numerical analysis of an optimal control approach to solve a tsunami inverse problem.
- Laurent Bourgeois, Philippe Moireau, Raphaël Terrine. *Submitted in 2026.* An inverse tsunami problem in the time domain: a well-posedness analysis of the forward problem and an inversion strategy based on a mixed formulation of the Tikhonov regularization.

Talks

- Wave-Ned 2026, Delft - August 2026
  Invitation to a mini-symposium : Convergence of a numerical scheme for solving an inverse problem: Application to tsunami reconstruction.
- PICOF'2025, Hammamet - October 2025
  Invitation to a mini-symposium : Convergence of a numerical scheme for solving an inverse problem: Application to tsunami reconstruction.

- Biennale de la SMAI 2025, Carcans-Maubuisson - June 2025
  Talk : Identification of bottom deformations from surface measurements.
- WAVES, Berlin - July 2024
  Talk : About the identification of tsunamis from surface measurements.

Numerical simulations

The numerical results presented in this thesis were obtained using the XLiFE++ library [92]. The numerical results presented in Chapter 5 were obtained using the OndoMathX library, developed within the ANANKE team by Sébastien Impériale.

Chapter 2

Mixed formulation in the frequency domain

The objective of this work is to propose a first method to solve the tsunami and the bathymetry inverse problem in an easier harmonic framework. In this chapter, the inverse problem is solved through a Tikhonov-Morozov regularization associated with a iterative method to compute the regularisation parameter ϵ with respect to the amplitude of the noise δ. This work rigorously justifies the computation of this ϵ, we call the ϵ of Morozov, through a duality in optimization approach. The procedure finally relies on solving at each iteration a mixed formulation. In order to solve the non linear bathymetry inverse problem, we use an asymptotic expansion of the bottom of the ocean to get a problem that is solved through the same procedure as described before. We finally present some numerical results showing the efficiency of the method.
This chapter contains a published article, realized in collaboration with Laurent Bourgeois and Jean-François Mercier.

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Identification of bottom deformations of the ocean from surface measurements

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Abstract

In this paper we consider a general scheme to solve two different inverse problems related to oceanography, that is retrieving either a tsunami or the shape of the seabed from the measurement of the free surface perturbation. We consider two dimensional geometries and linear potential models in the frequency regime. Such general scheme consists in firstly recovering the potential in the whole domain and secondly compute the seeked parameter at the bottom of the ocean, which in the two inverse problems is a function involved in a more or less complicated boundary condition. The first step amounts to solve an illposed Cauchy problem for the Laplace or Helmholtz equation, which we regularize by using a mixed formulation of the Tikhonov regularization and the Morozov principle to compute the regularization parameter. The computation of such Tikhonov-Morozov solution is based on an iterative method consisting in solving a sequence of weak formulations which are discretized with the help of a simple Lagrange type Finite Element Method. In the particular case of the acoustic model, we need to solve a Laplace-type equation associated with the noisy Neumann boundary data and compute the noise amplitude of its solution. A probabilistic method is proposed to obtain such amplitude of noise. Some numerical experiments show the feasibility of our strategy.

2.1 Introduction

This paper is devoted to two different inverse problems which arise in oceanography. The first one is the identification of a tsunami from measurements of the free surface deformation, such tsunami being characterized by a brutal displacement of the bottom surface of the ocean. The second one is the bathymetry problem, which consists in recovering the underwater depth by using the same measurements as in the previous problem. In these two problems, the goal is to retrieve some sea bottom parameters from surface data, but while the loading is passive in the first problem (the tsunami is a natural phenomenon), it is active in the second one (a source artificially generated is required).

Concerning the literature on tsunamis, there is a large amount of contributions related to the forward modeling of tsunamis, for example [101, 103, 109]. In particular, a comprehensive study of the behaviour of a tsunami at all steps of its “life" and of the best models to use at each of these steps is exposed in [52]. However, the identification of tsunamis from surface measurements is far less addressed in the literature. In this direction, let us mention [79], in which a shallow-water model is handled and a least-square method is proposed, and [65], which is based on a machine-learning algorithm. For the simpler linear water wave problem in the presence of a flat bottom, in [42] a numerical approach based on integral transforms with respect to the horizontal directions and with respect to time is successfully used to invert experimental data. This paves the way to real

applications of inverse problems in oceanography. The literature on bathymetry seen as an inverse problem is also quite poor. Let us mention [106], in which a numerical reconstruction procedure is conducted, in the presence of a nonlinear water wave model and by using periodic boundary conditions in the horizontal directions. We also mention [60, 98], which address the theoretical question of uniqueness/stability for the inverse problem in the context of the nonlinear water wave problem. Note that the well-posedness of the corresponding forward problem is proved in [83].

In this contribution we address the two previous inverse problems by restricting ourselves to some two-dimensional linear models in the frequency domain. The physics of the problem is then fully governed by the scalar velocity potential. Note that such linear models are only valid during the first step of the tsunami generation and would not be appropriate during the second propagation step, as emphasized in [52]. The two inverse problems, which are strongly ill-posed, are attacked by using the same strategy, which consists in two steps. In the first step, surface data are used to compute an approximate velocity potential in the whole volumic domain by transforming the initial ill-posed problem into a well-posed regularized one. In the second step we use the partial derivatives of such potential on the bottom of the domain to compute either the time derivative of the vertical displacement which characterizes the tsunami (first inverse problem) or the geometric perturbation of the bottom shape with respect to the horizontal flat reference (second inverse problem). Note that in the bathymetry problem, we are led to make a strong assumption on the smoothness of the seabed to ensure uniqueness for the inverse problem (see Theorem 2.4.2). The novelty of our contribution is twofold. Firstly in these two steps we use weak formulations and finite element methods, which are well adapted to non-flat sea bottoms. We observe that when the bottom is flat, the forward tsunami problem is often solved in the literature by first using a Fourier transform in the horizontal directions, which is forbidden in the presence of a variable sea bottom. Secondly we wish to study the influence of the choice of the model on the quality of the reconstruction at different frequencies. In this view we start from a model which takes account of both acoustic and gravity waves, the former being dominating at high frequencies, the latter being dominating at low frequencies.

The main ingredient of the first step described above is the notion of mixed formulation of the Tikhonov regularization, which is designed to regularize ill-posed Cauchy problems and can be immediately discretized with the help of a classical Lagrange Finite Element Method. This technique was first introduced in [16] in the case of the Cauchy problem for the Laplace equation, then generalized to an abstract context in [29]. The Tikhonov formulation involves a small regularization parameter that we choose following the Morozov principle (see for example [81]) and compute numerically by using the Fenchel-Rockafellar duality in optimization (see the general theory in [53]). This technique was first proposed in the context of the Morozov principle in [18, 22] again in the case of the Cauchy problem for the Laplace equation, then generalized to an abstract setting in [24]. The duality approach leads to the problem of minimizing a convex but non-quadratic functional, which in practice turns out to be delicate. This is why, following an idea of [48], we transform such problem into a sequence of quadratic minimization problems, the solutions of which converge to the solution of the original problem. Such quadratization technique has already been used in [24].

Our paper is organized as follows. In section 2 we present the linear model of oceanography that we will consider, that is the so-called complete model, and we address the corresponding forward problems, both for the tsunami problem and the bathymetry problem. Two classical limit models deriving from the complete model, that is the gravity model and the acoustic model, are presented in section 3. Our two inverse problems, that is the tsunami identification and the bathymetry problems, are introduced in section 4. Section 5 is devoted to the abstract framework of the mixed formulation of the Tikhonov regularization method, such framework being the core of our strategy to solve these two inverse problems, whatever the chosen model is. This abstract framework is then applied to the tsunami identification problem and to the bathymetry problem in sections 6 and 7, respectively. To solve the bathymetry problem, assuming the geometric perturbation is small, we consider a second order impedance boundary condition on a flat bottom instead of the true perturbed bottom, which is an approximation based on an asymptotic analysis. Note that we assume a lower bound of the surface derivative of the approximate velocity potential at the bottom (see Assumption 2.7.1). In the case of the acoustic model, we need to convert the amplitude of

noise that contaminates a Neumann boundary data to the amplitude of noise of a volumic lifting solution of such Neumann boundary problem. To proceed, two approaches, a deterministic and a probabilistic one, are exposed in section 8. Such section is complemented by an appendix, in which we provide more theoretical results related to the probabilistic approach in a slightly simpler case from the point of view of the geometry. Lastly, numerical experiments are presented in section 9.

2.2 The selected model and the forward problems

In this article, the ocean at rest is defined as a two-dimensional domain Ω_θ, given by

$$\Omega_\theta = \{(x,z) \in \mathbb{R}^2,\ -H + \theta(x) < z < 0\},$$

where θ is a continuous and compactly supported function which can be considered as a perturbation of the flat bottom of equation $z = -H$, while the free surface at rest is characterized by $z = 0$. The domain Ω_θ is hence delimited by the two curves

$$\Gamma_0 = \{(x,z) \in \mathbb{R}^2,\ z = 0\}, \quad \Gamma_{-H+\theta} = \{(x,z) \in \mathbb{R}^2,\ z = -H + \theta(x)\}.$$

2.2.1 The complete model

We first consider the tsunami problem. Let us denote $\zeta(x,t)$ the upward normal displacement of the sea bottom, which is the only source of the problem, and $s(x,t)$ the induced perturbation of the free surface. The velocity potential $\varphi(x,z,t)$, the functions s and ζ are related to each other by the following equations and boundary conditions:

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2\varphi - \Delta\varphi &=& 0 & \text{in } \Omega_\theta \times (0,+\infty), \\ \partial_t\varphi + g\, s &=& 0 & \text{on } \Gamma_0 \times (0,+\infty), \\ \partial_z\varphi - \partial_t s &=& 0 & \text{on } \Gamma_0 \times (0,+\infty), \\ \partial_\nu\varphi &=& -\partial_t\zeta & \text{on } \Gamma_{-H+\theta} \times (0,+\infty) \end{array}\right. \tag{2.1}$$

where ν is the unit outward normal to Ω_θ, g is the gravitational acceleration and c is the celerity of the acoustic waves. The system (2.1) has to be complemented by initial conditions. Note that the two boundary conditions on Γ_0 can be merged into the single equation:

$$\partial_t^2\varphi + g\,\partial_z\varphi = 0. \tag{2.2}$$

The linear model (2.1) is rigorously derived from the Euler equations in [51] and corresponds to the so-called “barotropic case" in [51], with the additional assumption that the density is constant.

In the frequency domain, which amounts to assume that $\varphi(x,z,t) = u(x,z)e^{-i\omega t}$, $s(x,t) = \eta(x)e^{-i\omega t}$ and $\zeta(x,t) = \chi(x)e^{-i\omega t}$ for some fixed frequency $\omega > 0$, the forward scattering problem of the tsunami reads: for $\chi \in L^2(\Gamma_0)$, where χ is supposed to be compactly supported, find $u \in H^1_{\mathrm{loc}}(\Omega_\theta)$ such that

$$\left\{\begin{array}{rcll} \Delta u + (\omega^2/c^2)u &=& 0 & \text{in } \Omega_\theta, \\ \partial_z u - (\omega^2/g)u &=& 0 & \text{on } \Gamma_0, \\ \partial_\nu u &=& i\omega\chi & \text{on } \Gamma_{-H+\theta}, \\ u & \text{is} & \text{outgoing.} & \end{array}\right. \tag{2.3}$$

Here,

$$H^1_{\mathrm{loc}}(\Omega_\theta) = \{u \in \mathcal{D}'(\Omega_\theta),\ \psi(x)u \in H^1(\Omega_\theta),\ \forall \psi \in C_0^\infty(\mathbb{R})\}.$$

The last line of (2.3) corresponds to a radiation condition, which is necessary to guarantee uniqueness in a scattering problem, and which will be clarified hereafter. Note that once the potential u is known, the perturbation of the free surface is given by either of the two formulas

$$\eta = \frac{i\omega}{g}u, \quad \eta = -\frac{1}{i\omega}\partial_z u.$$

The system (2.1) is interesting in the context of tsunamis because it relies on a model which involves both acoustic and gravity waves, in other words both constants g and c appear in (2.1) (see for example [101, 103, 109]). Such model will be called the complete model in the following.

Now let us introduce the forward bathymetry problem: for some source point $N \in \Omega_\theta$, find $u \in H^1_{\mathrm{loc}}(\Omega_\theta)$ such that

$$\left\{\begin{array}{rcll} -\Delta u - (\omega^2/c^2)u &=& \delta_N & \text{in } \Omega_\theta, \\ \partial_z u - (\omega^2/g)u &=& 0 & \text{on } \Gamma_0, \\ \partial_\nu u &=& 0 & \text{on } \Gamma_{-H+\theta}, \\ u & \text{is} & \text{outgoing.} & \end{array}\right. \tag{2.4}$$

Note that in problem (2.4), the source term appears in the right-hand side of the first equation, and no more in the boundary condition on the seabed $\Gamma_{-H+\theta}$. Let us prove that the forward problems for the complete model, both in the case of the tsunami generation (2.3) and the bathymetry (2.4), are well-posed. In this view, we have to specify what we mean by "outgoing" solutions in the above forward problems, which leads us to introduce the modes.

2.2.2 Determination of the modes

Like in the pioneering work [87], we introduce the modes, that is the solutions in $\Omega_0 = \mathbb{R} \times (-H, 0)$ in the form $u(x,z) = f(z)e^{\lambda x}$ to the following problem in the absence of the source term χ or δ_N:

$$\left\{\begin{array}{rcll} \Delta u + (\omega^2/c^2)u &=& 0 & \text{in } \Omega_0, \\ \partial_z u - (\omega^2/g)u &=& 0 & \text{on } \Gamma_0, \\ \partial_\nu u &=& 0 & \text{on } \Gamma_{-H}. \end{array}\right.$$

Finding such u amounts to find $(\lambda, f) \in \mathbb{C} \times H^1(-H,0)$, $f \neq 0$, such that

$$\left\{\begin{array}{rcll} f''(z) + (\lambda^2 + \omega^2/c^2)f(z) &=& 0 & \text{in } (-H,0), \\ f'(0) - (\omega^2/g)f(0) &=& 0, & \\ f'(-H) &=& 0. & \end{array}\right.$$

Let us denote ν_0 the unique positive solution to the equation

$$\nu \tanh(\nu H) = \omega^2/g, \tag{2.5}$$

while the sequence $(\nu_n)_{n\geq 1}$ is formed by the positive and increasing solutions to the equation

$$\nu \tan(\nu H) = -\omega^2/g. \tag{2.6}$$

Let us in addition assume the following.

Assumption 2.2.1. The frequency ω is such that there are no $n \in \mathbb{N}$, $n \geq 1$, satisfying

$$\nu_n = \omega/c.$$

Such assumption implies that there exists some $P \in \mathbb{N}$, such that

$$\left\{\begin{array}{lll} \nu_n < \omega/c & \text{if} & 1 \leq n \leq P, \\ \nu_n > \omega/c & \text{if} & n > P. \end{array}\right.$$

The solutions (λ, f) to the above problem are given by:

$$\left\{\begin{array}{ll} \pm i\sqrt{\nu_0^2 + \omega^2/c^2}, & f_0(z) := A_0 \cosh(\nu_0(z+H)), \\ \pm i\sqrt{\omega^2/c^2 - \nu_n^2}, & f_n(z) := A_n \cos(\nu_n(z+H)), \quad 1 \leq n \leq P, \\ \pm\sqrt{\nu_n^2 - \omega^2/c^2}, & f_n(z) := A_n \cos(\nu_n(z+H)), \quad n > P, \end{array}\right. \tag{2.7}$$

where the A_n are arbitrary non zero constants. The modes are then given by the functions

$$u_n^\pm(x,z) = f_n(z)e^{\pm\beta_n x}, \tag{2.8}$$

where from now on the constants A_n of (2.7) are chosen such that the (f_n) form an orthonormal complete basis of $L^2(-H,0)$ (which is possible from a classical result of spectral theory for self-adjoint operators) and

$$\left\{\begin{array}{ll} \beta_0 := i\sqrt{\nu_0^2+\omega^2/c^2}, & \\ \beta_n := i\sqrt{\omega^2/c^2-\nu_n^2}, & 1\le n\le P, \\ \beta_n := -\sqrt{\nu_n^2-\omega^2/c^2}, & n>P. \end{array}\right. \tag{2.9}$$

The modes u_n^+ are either propagating (for $0\le n\le P$) or evanescent (for $n>P$) in the right direction of Ω_0, so that there are outgoing to the right. The modes u_n^- are either propagating (for $0\le n\le P$) or evanescent (for $n>P$) in the left direction of Ω_0, so that there are outgoing to the left.

We are now in a position to define an outgoing function u in Ω_θ as a function for which there exists $R>0$ such that

$$u(x,z)=\sum_{n\in\mathbb{N}} c_n^{\pm}u_n^{\pm}(x,z), \quad \pm x>R, \tag{2.10}$$

where the $c_n^{\pm}$, $n\in\mathbb{N}$, are complex numbers. More precisely, R is any sufficiently large real such that the support of θ and χ are contained in $(-R,R)$. Let us consider the problem (2.3) in the simplified case of the flat seabed, that is $\theta=0$. It will be always the case for the tsunami problem.

2.2.3 Well-posedness of the tsunami problem

In order to prove the well-posedness of (2.3), we introduce the bounded domain $\Omega_0^R:=\{(x,z)\in\Omega_0,\ -R<x<R\}$ delimited by the transverse sections $\Sigma_{\pm R}=\{\pm R\}\times(-H,0)$, assuming that the support of χ is contained in $(-R,R)$. Classically, the problem (2.3) in the unbounded domain Ω_0 is equivalent to the following problem set in the bounded domain Ω_0^R: find $u\in H^1(\Omega_0^R)$ such that

$$\left\{\begin{array}{rcll} \Delta u+(\omega^2/c^2)u &=& 0 & \text{in } \Omega_0^R, \\ \partial_z u-(\omega^2/g)u &=& 0 & \text{on } \Gamma_0^R, \\ \partial_z u &=& -i\omega\chi & \text{on } \Gamma_{-H}^R, \\ \pm\partial_x u-T_\pm^c u &=& 0 & \text{on } \Sigma_{\pm R}, \end{array}\right. \tag{2.11}$$

where Γ_0^R and Γ_{-H}^R are the subparts of Γ_0 and Γ_{-H} delimited by the transverse sections Σ_{-R} and Σ_R, respectively, while the Dirichlet-To-Neumann operators $T_\pm^c$ are defined by

$$\left\{\begin{array}{rcl} T_\pm^c : H^{1/2}(\Sigma_{\pm R}) & \to & \tilde{H}^{-1/2}(\Sigma_{\pm R}) \\ \varphi & \mapsto & \sum_{n\in\mathbb{N}}\beta_n(\varphi,f_n)_{L^2(-H,0)}f_n. \end{array}\right. \tag{2.12}$$

Here, $\tilde{H}^{-1/2}(\Sigma_{\pm R})$ is the dual space of $H^{1/2}(\Sigma_{\pm R})$, the functions f_n are given by (2.7) and the complex numbers β_n are given by (2.9). It is easy to check that in problem (2.11), the artificial boundary conditions on the transverse sections $\Sigma_{\pm R}$ are equivalent to the radiation condition (2.10). For a proof in a slightly more difficult case, the reader may refer to [14].

An equivalent weak formulation to the strong problem (2.11) is: find $u\in H^1(\Omega_0^R)$ such that for all $v\in H^1(\Omega_0^R)$,

$$a(u,v)=\ell(v), \tag{2.13}$$

where the sesquilinear form a and the antilinear form ℓ are given by

$$\begin{array}{rcl} a(u,v) &=& \displaystyle\int_{\Omega_0^R}\left(\nabla u\cdot\nabla\overline{v}-(\omega^2/c^2)u\,\overline{v}\right)dxdz \\ &-& \displaystyle(\omega^2/g)\int_{\Gamma_0^R}u\,\overline{v}\,dx-\langle T_+^c u,\overline{v}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} \\ &-& \langle T_-^c u,\overline{v}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})}, \end{array} \tag{2.14}$$

where $\langle\cdot,\cdot\rangle_{H^{-1/2}(\Sigma_{\pm R}),\tilde{H}^{1/2}(\Sigma_{\pm R})}$ denotes the duality bracket between $H^{-1/2}(\Sigma_{\pm R})$ and $\tilde{H}^{1/2}(\Sigma_{\pm R})$ and

$$\ell(v)=i\omega\int_{\Gamma_{-H}^R}\chi\,\overline{v}\,dx. \tag{2.15}$$

We have the following well-posedness result.

Theorem 2.2.2. *The weak formulation (2.13) has a unique solution.*

Proof. We introduce the decomposition $a = b + c$, where the sesquilinear forms b and c are given by

$$\begin{cases} b(u,v) &= \displaystyle\int_{\Omega_0^R} (\nabla u \cdot \nabla \overline{v} + u\,\overline{v})\,dxdz - \langle T_+^c u, \overline{v}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} \\ &- \langle T_-^c u, \overline{v}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})}, \\ c(u,v) &= -(1+(\omega^2/c^2))\displaystyle\int_{\Omega_0^R} u\,\overline{v}\,dxdz - (\omega^2/g)\int_{\Gamma_0^R} u\,\overline{v}\,dx, \end{cases}$$

and define the operators $B, C : H^1(\Omega_0^R) \to H^1(\Omega_0^R)$ such that

$$(Bu,v)_{H^1(\Omega_0^R)} = b(u,v), \quad (Cu,v)_{H^1(\Omega_0^R)} = c(u,v), \quad \forall u,v \in H^1(\Omega_0^R).$$

In view of (2.9), we have that

$$\mathcal{R}e\{b(u,u)\} = \|u\|^2_{H^1(\Omega_0^R)} + \sum_{\pm,n>P} \sqrt{\nu_n^2 - \omega^2/c^2}\,|(u|_{\Sigma_{\pm R}}, f_n)_{L^2(-H,0)}|^2 \geq \|u\|^2_{H^1(\Omega_0^R)},$$

where $\mathcal{R}e\{\cdot\}$ denotes the real part. Then B is an isomorphism while C is a compact operator, the Fredholm alternative allows us to conclude that in problem (2.13), existence is equivalent to uniqueness. In order to prove uniqueness (having in mind that $\theta = 0$), assume that u satisfies the problem (2.11) for $\chi = 0$. From the three first equations of (2.11), we obtain that u is an infinite linear combination of the modes $u_n^{\pm}$ given by (2.8). Since in addition the solution u is outgoing, that is both leftgoing and rightgoing as specified by (2.10), such linear combination is necessarily equal to 0. This complete the proof. □

2.2.4 Well-posedness of the bathymetry problem

Now let us prove well-posedness of the problem (2.4). In this view, we introduce the fundamental solution in the waveguide Ω_0, that is for some source point $N(x',z') \in \Omega_0$, the solution $\mathcal{G}^c(\cdot,N) \in L^2_{\text{loc}}(\Omega_0)$ to the problem

$$\begin{cases} -\Delta\mathcal{G}^c(\cdot,N) - (\omega^2/c^2)\mathcal{G}^c(\cdot,N) &= \delta_N & \text{in } \Omega_0, \\ \partial_z\mathcal{G}^c(\cdot,N) - (\omega^2/g)\mathcal{G}^c(\cdot,N) &= 0 & \text{on } \Gamma_0, \\ \partial_z\mathcal{G}^c(\cdot,N) &= 0 & \text{on } \Gamma_{-H}, \\ \mathcal{G}^c(\cdot,N) & \text{is outgoing.} & \end{cases}$$

A short computation shows that an explicit expression of $\mathcal{G}^c$ is given, for any $M(x,z)$, by

$$\mathcal{G}^c(M,N) = -\sum_{n\in\mathbb{N}} \frac{1}{2\beta_n} e^{\beta_n|x-x'|} f_n(z) f_n(z'), \tag{2.16}$$

where the f_n and the β_n are given by (2.7) and (2.9), respectively. In what follows we will assume that the support of θ lies between Σ_{-R} and Σ_R and that N is located such that $x' < -R$. By defining the incident field u^i in Ω_θ by

$$u^i = \begin{cases} \mathcal{G}^c(\cdot,N) & \text{if} \quad x < -R, \\ 0 & \text{if} \quad x > -R, \end{cases} \tag{2.17}$$

and by writing the problem satisfied by the corresponding outgoing scattered field $u^s = u - u^i$, we obtain that the solution u to the problem (2.4) also satisfies in Ω_θ^R:

$$\begin{cases} \Delta u + (\omega^2/c^2)u &= 0 & \text{in } \Omega_\theta^R, \\ \partial_z u - (\omega^2/g)u &= 0 & \text{on } \Gamma_0^R, \\ \partial_\nu u &= 0 & \text{on } \Gamma_{-H+\theta}^R, \\ -\partial_x u &= T_-^c u - 2\partial_x\mathcal{G}^c(\cdot,N) & \text{on } \Sigma_{-R}, \\ \partial_x u &= T_+^c u & \text{on } \Sigma_R. \end{cases} \tag{2.18}$$

Remark 2.2.1. *We note that the fundamental solution $\mathcal{G}^c(\cdot, N)$ does not satisfy a Neumann boundary condition on the domain Ω_θ (only on Ω_0). This explains why we have introduced a piecewise incident field u^i in (2.17). The fact that the incident field u^i and the scattered field u^s have a discontinuity across the section $x = -R$ may seem surprising but is mathematically correct, as long as the total field is continuous across such section.*

An equivalent weak formulation to the strong problem (2.18) is: find $u \in H^1(\Omega_\theta^R)$ such that for all $v \in H^1(\Omega_\theta^R)$,

$$a_\theta(u, v) = \ell_\theta(v), \tag{2.19}$$

where the sesquillinear form a and the antilinear form ℓ are given by

$$\begin{array}{rcl} a_\theta(u,v) & = & \displaystyle\int_{\Omega_\theta^R} \left(\nabla u \cdot \nabla \overline{v} - (\omega^2/c^2) u\, \overline{v}\right) dxdz - (\omega^2/g) \int_{\Gamma_0^R} u\, \overline{v}\, dx \\ & - & \langle T_+^c u, \overline{v}\rangle_{H^{-1/2}(\Sigma_R), \tilde{H}^{1/2}(\Sigma_R)} - \langle T_-^c u, \overline{v}\rangle_{H^{-1/2}(\Sigma_{-R}), \tilde{H}^{1/2}(\Sigma_{-R})} \end{array} \tag{2.20}$$

and

$$\ell_\theta(v) = -2 \int_{\Sigma_{-R}} \partial_x \mathcal{G}^c(\cdot, N)\, \overline{v}\, dx. \tag{2.21}$$

We have the following well-posedness result in the Fredholm sense.

Theorem 2.2.3. *Assume that the weak formulation (2.19) has at most one solution. Then such weak formulation is well-posed.*

The proof of Theorem 2.2.3 relies on a decomposition of the sesquilinear form a_θ of the same type as in the proof of Theorem 2.2.2. The difference is that uniqueness is not guarranteed. Indeed, due to the fact that the waveguide Ω_θ is not straight, some trapped solutions may exist.

2.3 Two limit cases

2.3.1 Derivation of the two limits

Two well-known limit cases can be derived from the complete model. In the case of the tsunami problem for example, starting from problem (2.3), the gravity model (also often called the water wave model) corresponds to the case when $c \to +\infty$, that is the problem (2.3) becomes: find $u \in H^1_{\text{loc}}(\Omega_\theta)$ such that

$$\left\{ \begin{array}{rcll} \Delta u & = & 0 & \text{in } \Omega_\theta, \\ \partial_z u - (\omega^2/g) u & = & 0 & \text{on } \Gamma_0, \\ \partial_\nu u & = & i\omega\chi & \text{on } \Gamma_{-H+\theta}, \\ u & \text{is} & \text{outgoing.} & \end{array} \right. \tag{2.22}$$

The acoustic model corresponds to the case when $g \to 0$, that is the problem (2.3) becomes: find $u \in H^1_{\text{loc}}(\Omega_\theta)$ such that

$$\left\{ \begin{array}{rcll} \Delta u + (\omega^2/c^2) u & = & 0 & \text{in } \Omega_\theta, \\ u & = & 0 & \text{on } \Gamma_0, \\ \partial_\nu u & = & i\omega\chi & \text{on } \Gamma_{-H+\theta}, \\ u & \text{is} & \text{outgoing.} & \end{array} \right. \tag{2.23}$$

One can wonder which model we have to choose. To decide, we can define two typical frequencies $\omega_g := \sqrt{g/H}$ and $\omega_a := c/H$, such that for typical realistic values of g, c, H we have $\omega_a \gg \omega_g$. The gravity model (2.22) is well adapted for $\omega \sim \omega_g$ while the acoustic model (2.23) is well adapted for $\omega \sim \omega_a$.

2.3.2 Analysis of the gravity case model

The results obtained for the complete model in the previous section are valid in the gravity case, except that when $c \to +\infty$ the modes have a simpler form. They are given by (2.8), where in the definition of the f_n given by (2.7) and the definition of the β_n given by (2.9) we have $P = 0$, which yields

$$\begin{cases} f_0(z) := A_0 \cosh(\nu_0(z+H)), \\ f_n(z) := A_n \cos(\nu_n(z+H)), \quad n \geq 1, \end{cases} \tag{2.24}$$

and

$$\begin{cases} \beta_0 := i\nu_0, \\ \beta_n := -\nu_n, \quad n \geq 1. \end{cases} \tag{2.25}$$

In the gravity case, to avoid ambiguity the Dirichlet-To-Neumann operators given by (2.12) will be denoted $T^g_\pm$ in the sequel.

2.3.3 Analysis of the acoustic model

Let us now address the forward acoustic problems, both in the case of the tsunami generation and the bathymetry. The analysis is slightly different from the complete model. Following the same lines as in section 2.2, we need again to compute the modes, that is the solutions in $\Omega_0 = \mathbb{R} \times (-H, 0)$ in the form $u(x,z) = g(z)e^{\mu x}$ to the problem

$$\begin{cases} \Delta u + (\omega^2/c^2)u &= 0 \quad \text{in } \Omega_0, \\ u &= 0 \quad \text{on } \Gamma_0, \\ \partial_z u &= 0 \quad \text{on } \Gamma_{-H}. \end{cases} \tag{2.26}$$

The derivation of these modes from the modes of the complete model is not obvious, this is why we compute them starting from (2.26). What follows relies on the following assumption.

Assumption 2.3.1. The frequency ω is such that there are no $n \in \mathbb{N}$ satisfying

$$\mu_n = \frac{\omega}{c}, \quad \mu_n := \left(n + \frac{1}{2}\right)\frac{\pi}{H}.$$

Such assumption implies that there exists some $P \in \mathbb{N}$, $P > 0$, such that

$$\begin{cases} \mu_n < \omega/c & \text{if} \quad 0 \leq n < P, \\ \mu_n > \omega/c & \text{if} \quad n \geq P. \end{cases}$$

If we denote

$$g_n(z) = \sqrt{2/H}\cos(\mu_n(z+H)), \quad n \in \mathbb{N}, \tag{2.27}$$

which implies that $\|g_n\|_{L^2(-H,0)} = 1$, and

$$\begin{cases} \gamma_n = i\sqrt{(\omega^2/c^2) - \mu_n^2} & \text{if} \quad 0 \leq n < P, \\ \gamma_n = -\sqrt{\mu_n^2 - (\omega^2/c^2)}, & \text{if} \quad n \geq P, \end{cases} \tag{2.28}$$

the modes are then given by the functions

$$v_n^\pm(x,z) = g_n(z)e^{\pm\gamma_n x}, \tag{2.29}$$

where the (g_n) form a complete basis of $L^2(-H,0)$. The modes v_n^+ are either propagating (for $n < P$) or evanescent (for $n \geq P$) in the right direction of Ω_0, so that there are outgoing to the right. The modes v_n^- are either propagating (for $n < P$) or evanescent (for $n \geq P$) in the left direction of Ω_0, so that there are outgoing to the left.

The problem (2.23) in the unbounded domain Ω_0 ($\theta = 0$) is equivalent to the following problem set in the bounded domain Ω_0^R: find $u \in H^1(\Omega_0^R)$ such that

$$\begin{cases} \Delta u + (\omega^2/c^2)u &= 0 & \text{in } \Omega_0^R, \\ u &= 0 & \text{on } \Gamma_0^R, \\ \partial_z u &= -i\omega\chi & \text{on } \Gamma_{-H}^R, \\ \pm\partial_x u - T_\pm^a u &= 0 & \text{on } \Sigma_{\pm R}, \end{cases} \tag{2.30}$$

where the Dirichlet-To-Neumann operators $T^a_\pm$ are defined as

$$\left\{\begin{array}{rcl} T^a_\pm : H^{1/2}(\Sigma_{\pm R}) & \to & \tilde{H}^{-1/2}(\Sigma_{\pm R}) \\ \varphi & \mapsto & \sum_{n\in\mathbb{N}} \gamma_n(\varphi, g_n)_{L^2(-H,0)} g_n, \end{array}\right. \tag{2.31}$$

the γ_n are given by (2.28) and the g_n are defined by (2.27). Introducing a similar weak formulation as (2.13), we get

Theorem 2.3.2. *The problem (2.30) has a unique solution.*

Now let us consider the problem (2.4) when $g \to 0$. In this view, we introduce the corresponding fundamental solution in the waveguide Ω_0, that is for some $N(x', z') \in \Omega_0$, the solution $\mathcal{G}^a(\cdot, N) \in L^2_{\text{loc}}(\Omega_0)$ to the problem

$$\left\{\begin{array}{rcll} -\Delta\mathcal{G}^a(\cdot,N) - (\omega^2/c^2)\mathcal{G}^a(\cdot,N) & = & \delta_N & \text{in } \Omega_0, \\ \mathcal{G}^a(\cdot,N) & = & 0 & \text{on } \Gamma_0, \\ \partial_z\mathcal{G}^a(\cdot,N) & = & 0 & \text{on } \Gamma_{-H}, \\ \mathcal{G}^a(\cdot,N) & \text{is} & \text{outgoing.} & \end{array}\right.$$

The expression of $\mathcal{G}^a$ is given, for $M(x, z)$, by

$$\mathcal{G}^a(M,N) = -\sum_{n\in\mathbb{N}} \frac{1}{2\gamma_n} e^{\gamma_n|x-x'|} g_n(z) g_n(z'), \tag{2.32}$$

where the g_n and the γ_n are given by (2.27) and (2.28), respectively. As in the case of the complete model, we obtain that the solution u to the problem (2.4) in the particular case $g \to 0$ also satisfies in Ω^R_θ the problem:

$$\left\{\begin{array}{rcll} \Delta u + (\omega^2/c^2)u & = & 0 & \text{in } \Omega^R_\theta, \\ u & = & 0 & \text{on } \Gamma^R_0, \\ \partial_\nu u & = & 0 & \text{on } \Gamma^R_{-H+\theta}, \\ -\partial_x u & = & T^a_- u - 2\partial_x\mathcal{G}^a(\cdot,N) & \text{on } \Sigma_{-R}, \\ \partial_x u & = & T^a_+ u & \text{on } \Sigma_R. \end{array}\right. \tag{2.33}$$

We have the following well-posedness result in the Fredholm sense.

Theorem 2.3.3. *Assume that the problem (2.33) has at most one solution. Then such problem is well-posed.*

Remark 2.3.1. *Comparing the gravity modes (2.25) and the acoustic modes (2.28) allows us to anticipate the influence of the model on the resolution of the inverse problems, that is the tsunami identification and the bathymetry problem. Let us restrict to the right-going modes. In the gravity model we observe that the only one propagating mode u^+_0 is exponentially decaying to the bottom direction, while the evanescent modes u^+_n, $n \geq 1$, are all oscillating in the vertical direction. We hence expect that the information related to the bottom will be hard to identify from the knowledge of the free surface. Since ν_0 is an increasing function with respect to ω, we expect that the larger is ω, the harder will be the identification. In the acoustic model, the modes v^+_n are all oscillating in the vertical direction, notably the P propagating ones. Having in mind that such number P is increasing with respect to the frequency ω, we expect that the inverse problems will be easier to solve in the acoustic case, in particular at high frequencies. Similarly, moving from the gravity case to the complete case, we observe in view of (2.9) that we increase the number of propagating modes, the first one being exponentially decreasing towards the bottom of the ocean, the other ones being oscillating in the depth direction. Of course, the resolution of the inverse problems will also benefit from this effect, in particular at high frequencies.*

2.4 Setting of the inverse problems

2.4.1 The tsunami identification problem

Assume that for some unknown function $\chi \in L^2(\Gamma^R_{-H})$, where χ is compactly supported in $(-R, R)$, the velocity potential u is the solution to the problem (2.11), which is well-posed from Theorem 2.2.2. The inverse problem consists in retrieving the solicitation χ from the measurement of the induced free surface perturbation η. In this paper, we try to approach a concrete situation where the measured data correspond to a real and thus complicated model, while we solve the inverse problem from those data with the help of a simpler model. In this sense we avoid an inverse crime. More precisely, we will try to solve the inverse problem by using either of the three models: gravity, acoustic or complete. Let us first begin with the gravity model. The free surface perturbation is given by $\eta = (i\omega/g)u$ on Γ^R_0, such measurement being noisy in practice. It is readily seen that such linear inverse problem is ill-posed. Indeed, having in mind the Robin boundary condition $\partial_z u = (\omega^2/g)u$ on Γ^R_0, the inverse problem amounts to solve a Cauchy problem for the Laplace equation in Ω^R_0, the Cauchy data being $(u, \partial_z u)$ on Γ^R_0. Such problem is well-known to be severely ill-posed. In the case of the acoustic model, we also want to retrieve the solicitation χ from the measurement of the free surface perturbation η, but since u satisfies a homogeneous Dirichlet condition on Γ^R_0, we now use the formula $\eta = -(1/i\omega)\partial_z u$ on Γ^R_0. Again we are brought back to a Cauchy problem for the Helmholtz equation, which is strongly ill-posed. If we use the complete model to solve the inverse problem, we use the relationship $\eta = (i\omega/g)u$ on Γ^R_0, like in the gravity model.

Remark 2.4.1. *It is important to note that while the data η of the inverse problem is proportional to the trace of u on Γ^R_0 (Dirichlet data) in the complete/gravity models, it is proportional to the normal derivative $\partial_z u$ on Γ^R_0 (Neumann data) in the acoustic model. It will make a significant difference in the presence of noise as we will see hereafter.*

Note that for the three models, there is at most one solution χ associated with the measurement η, this is a consequence of the uniqueness related to the Cauchy problem for the Laplace or Helmholtz equation. Here is a proof in the complete case. The other cases are similar.

Theorem 2.4.1. *Let us consider two functions $\chi_1, \chi_2 \in L^2(\Gamma^R_0)$, and let us denote u_j, $j = 1, 2$, the corresponding solutions to the problem (2.11), as well as the corresponding free surface perturbations $\eta_j = (i\omega/g)u_j$ on Γ^R_0, $j = 1, 2$. If $\eta_1 = \eta_2$, then $\chi_1 = \chi_2$.*

Proof. The fact that $\eta_1 = \eta_2$ implies that $u_1|_{\Gamma^R_0} = u_2|_{\Gamma^R_0}$. Since for $j = 1, 2$, u_j satisfies the boundary condition $\partial_z u_j - (\omega^2/g)u_j = 0$ on Γ^R_0 as well as the Helmholtz equation $\Delta u_j + (\omega^2/c^2)u_j = 0$ in Ω^R_0, uniqueness for the Cauchy problem implies that $u_1 = u_2$ in Ω. Since $\chi_j = (1/i\omega)\partial_z u_j$, we finally conclude that $\chi_1 = \chi_2$. □

2.4.2 The bathymetry problem

The goal of bathymetry is to recover some unknown function $\theta \in C^0([-R, R])$ which is compactly supported in $(-R, R)$ from the free surface perturbation η on Γ^R_0 induced by the point source $\mathcal{G}^c(\cdot, N)$. Let us denote u the solution to the problem (2.18) in the domain Ω^R_θ. Theorem 2.2.3 implies that such solution u is well-defined in the absence of trapped solutions. If we try to solve the inverse problem by using the gravity or the complete model, we exploit the boundary condition $\eta = (i\omega/g)u$ on Γ^R_0. If we use the acoustic model, the goal is the same but we use the formula $\eta = -(1/i\omega)\partial_z u$ on Γ^R_0. As a geometric inverse problem, bathymetry is a non-linear inverse problem and is of course ill-posed. Uniqueness of the bathymetry problem follows the same lines as the classical Schiffer proof (see for example [43]). For the sake of self-containment, we offer a proof in the gravity case in the case of a smooth function θ. The other cases are similar.

Theorem 2.4.2. *Let us consider two C^3 functions θ_1, θ_2 which are compactly supported in $(-R, R)$, and let us denote u_j, $j = 1, 2$, the corresponding solutions to the problem (2.18) with $c \to +\infty$, as well as the corresponding free surface perturbations $\eta_j = (i\omega/g)u_j$ on Γ^R_0, $j = 1, 2$. We assume that η_1 and η_2 are not constant functions. If $\eta_1 = \eta_2$, then $\theta_1 = \theta_2$.*

Proof. Since u_1 and u_2 have the same trace on Γ_0^R and satisfy the same Robin boundary condition on Γ_0^R, we have $(u_1|_{\Gamma_0^R}, \partial_z u_1|_{\Gamma_0^R}) = (u_2|_{\Gamma_0^R}, \partial_z u_2|_{\Gamma_0^R})$. From uniqueness for the Cauchy problem for the Laplace equation, we get that $u_1 = u_2$ in $I_{12}^R := \Omega_{\theta_1}^R \cap \Omega_{\theta_2}^R$. Assume that $\Omega_{\theta_2} \not\subset \Omega_{\theta_1}$ and let us consider $\mathcal{R} := U_{12}^R \setminus \Omega_{\theta_1}^R$, where $U_{12}^R := \Omega_{\theta_1}^R \cup \Omega_{\theta_2}^R$, which is not empty. We have $\partial\mathcal{R} \subset (\partial\Omega_{\theta_1}^R \cap \partial I_{12}^R) \cup \partial\Omega_{\theta_2}^R$. By denoting ν the outward unit normal to $\mathcal{R}$, on $\partial\Omega_{\theta_1}^R \cap \partial I_{12}^R$, we have $\partial_\nu u_1 = \partial_\nu u_2 = 0$, while on $\partial\Omega_{\theta_2}^R$, we have $\partial_\nu u_2 = 0$. As a result, the function u_2 satisfies $\Delta u_2 = 0$ in $\mathcal{R}$ with the boundary condition $\partial_\nu u_2 = 0$ on $\partial\mathcal{R} \cap \Gamma_{-H+\theta_1}^R$ and $\partial_\nu u_2 = 0$ on $\partial\mathcal{R} \cap \Gamma_{-H+\theta_2}^R$. We observe here that the domain $\mathcal{R}$ is not necessarily Lipschitz, which means that the standard Green formula in $\mathcal{R}$ does not hold. To cope with this problem we follow the argument of [4]. Since for $j = 1, 2$ the function θ_j is in $C^3([-R, R])$, the corresponding solution u_j is C^2 in a volumic neighborhood of a subset of the bottom $\Gamma_{-H+\theta_j}^R$ which does not meet the sections $\Sigma_{\pm R}$ (see [63]). Hence $u_2 \in C^2(\overline{\mathcal{R}})$. The regularity of u_2 and that the perimeter of $\mathcal{R}$ is finite imply the generalized Green formula (see [59]):

$$\int_{\mathcal{R}} \Delta u_2 \overline{u_2}\, dx = -\int_{\mathcal{R}} |\nabla u_2|^2\, dx + \int_{\partial^*\mathcal{R}} \partial_\nu u_2 \overline{u_2}\, ds,$$

where $\partial^*\mathcal{R}$ is the reduced boundary in the sense of De Georgi. We get that $\nabla u_2 = 0$ in $\mathcal{R}$, that is $u_2 = C$ in $\mathcal{R}$, where C is a constant. From unique continuation applied to the function $u_2 - C$, we get that $u_2 = C$ in $\Omega_{\theta_2}^R$, which implies that η_2 is constant on Γ_0^R. This is a contradiction. □

2.5 An abstract framework to solve ill-posed Cauchy problems

In this section we introduce an abstract framework which will be well-adapted to the regularization of the two inverse problems presented above (tsunami identification and bathymetry) and for the three different models (gravity, acoustic and complete models). This abstract framework is already presented in [24] in the case of real Hilbert spaces, and we have to adapt it to the case of complex Hilbert spaces. We then consider three complex Hilbert spaces V, M and O, equipped with the scalar products $(\cdot,\cdot)_V$, $(\cdot,\cdot)_M$ and $(\cdot,\cdot)_O$, respectively. We also consider two continuous operators $B : V \to M$ and $C : V \to O$, as well as $A : V \to M \times O$ defined with the help of B and C by $Av = (Bv, Cv)$, for all $v \in V$. We assume that A is injective and has a dense range. However, it is not supposed to be onto, which implies that the problem: for $F = (F_1, F_2) \in M \times O$, find $u \in V$ such that $Au = F$ may have no solution. For $\varepsilon > 0$, the standard Tikhonov regularization consists in the well-posed weak formulation: find $u_\varepsilon \in V$ such that

$$(Au_\varepsilon, Av)_{M\times O} + \varepsilon(u_\varepsilon, v)_V = (F, Av)_{M\times O}, \quad \forall v \in V. \tag{2.34}$$

The idea of the mixed formulation of the Tikhonov regularization consists in introducing an auxiliary unknown $\lambda_\varepsilon \in M$.

Proposition 2.5.1. *For $F = (F_1, F_2) \in M \times O$, $u_\varepsilon \in V$ is the solution to the problem (2.34) if and only if $(u_\varepsilon, \lambda_\varepsilon := Bu_\varepsilon - F_1) \in V \times M$ is the solution to the following weak formulation:*

$$\left\{ \begin{array}{rcll} \varepsilon(u_\varepsilon, v)_V + \overline{b(v, \lambda_\varepsilon)} + (Cu_\varepsilon, Cv)_O & = & (F_2, Cv)_O, & \forall v \in V, \\ b(u_\varepsilon, \mu) - (\lambda_\varepsilon, \mu)_M & = & (F_1, \mu)_M, & \forall \mu \in M, \end{array} \right. \tag{2.35}$$

where the sesquilinear form b on $V \times M$ is defined by

$$b(v, \mu) = (Bv, \mu)_M, \quad \forall (v, \mu) \in V \times M. \tag{2.36}$$

Such weak formulation is well-posed for all $\varepsilon > 0$ and if there exists $u \in V$ such that $Au = F$, then $u_\varepsilon \to u$ in V and $\lambda_\varepsilon \to 0$ in M when $\varepsilon \to 0$.

Proof. The fact that u_ε satisfies (2.34) is equivalent to the fact that

$$(Bu_\varepsilon, Bv)_M + (Cu_\varepsilon, Cv)_O + \varepsilon(u_\varepsilon, v)_V = (F_1, Bv)_M + (F_2, Cv), \quad \forall v \in V.$$

Introducing $\lambda_\varepsilon := Bu_\varepsilon - F_1$, we get that the above weak formulation is equivalent to find $(u_\varepsilon, \lambda_\varepsilon) \in V \times M$ such that

$$\left\{ \begin{array}{rcll} \varepsilon(u_\varepsilon, v)_V + \overline{(Bv, \lambda_\varepsilon)_M} + (Cu_\varepsilon, Cv)_O & = & (F_2, Cv)_O, & \forall v \in V, \\ (Bu_\varepsilon, \mu)_M - (\lambda_\varepsilon, \mu)_M & = & (F_1, \mu)_M, & \forall \mu \in M, \end{array} \right.$$

that is (2.35). Since the problem (2.34) is well-posed, this is also true for the problem (2.35). Besides, if there exists $u \in V$ such that $Au = F$, a well-known property of the Tikhonov regularization is that $u_\varepsilon \to u$ in V when $\varepsilon \to 0$. Then $\lambda_\varepsilon = Bu_\varepsilon - F_1 \to Bu - F_1 = 0$ when $\varepsilon \to 0$. □

In the inverse problems described in the previous section, the data $F = (F_1, F_2)$ is noisy in practice, so that we measure some data $F^\delta = (F_1^\delta, F_2^\delta) \in M \times O$, which satisfies the classical assumption

$$\|F^\delta - F\|_{M\times O} \leq \delta < \|F^\delta\|_{M\times O}. \tag{2.37}$$

In the applications hereafter, we will see that only one component among F_1 or F_2 is noisy, the other component being uncontaminated. In the first case

$$\|F^\delta - F\|_{M\times O} = \|F_1^\delta - F_1\|_M$$

while in the second case

$$\|F^\delta - F\|_{M\times O} = \|F_2^\delta - F_2\|_O.$$

The Morozov principle offers a rigorously defined way of choosing the regularization parameter ε in the Tikhonov formulation (2.34), or equivalently in the mixed Tikhonov regularization (2.35), as a function of the amplitude of noise δ. We have the following result.

Theorem 2.5.1. *Let us consider some data $F^\delta \in M \times O$ satisfying (2.37), and $u_\varepsilon^\delta \in V$ the solution to the problem (2.34) with data F replaced by F^δ. There exists a unique $\varepsilon > 0$ such that*

$$\|Au_\varepsilon^\delta - F^\delta\|_{M\times O} = \delta. \tag{2.38}$$

The proof of Theorem 2.5.1 is omitted since it is a slight adaptation to complex Hilbert spaces of Theorem 2.1 in [24]. We note that the equality (2.38) is equivalent to

$$\|\lambda_\varepsilon^\delta\|_M^2 + \|Cu_\varepsilon^\delta - F_2^\delta\|_O^2 = \delta^2,$$

where $(u_\varepsilon^\delta, \lambda_\varepsilon^\delta) \in V \times M$ is the solution to the problem (2.35) for data $F^\delta = (F_1^\delta, F_2^\delta) \in M \times O$ instead of $F = (F_1, F_2)$.

The solution $u_\varepsilon^\delta \in V$ to the problem (2.34) with data F^δ satisfies

$$A^*Au_\varepsilon^\delta + \varepsilon u_\varepsilon^\delta = A^*F^\delta, \tag{2.39}$$

where $A^* : M \times O \to V$ is the adjoint operator of A. It remains to find a strategy to compute the value $\varepsilon(\delta)$ defined by Theorem 2.5.1, as well as the corresponding solution $u^\delta := u_{\varepsilon(\delta)}^\delta$. One way to do that is to solve the following minimization problem in $H := M \times O$:

$$\inf_{q\in H} J^\delta(q), \quad J^\delta(q) := \frac{1}{2}\|A^*q\|_V^2 + \delta\|q\|_H - \mathcal{R}e\left\{(F^\delta, q)_H\right\}. \tag{2.40}$$

We have the following proposition.

Proposition 2.5.2. *The problem (2.40) has a unique solution $p^\delta \in H$. In addition, we have*

$$\varepsilon(\delta) = \frac{\delta}{\|p^\delta\|_H}, \quad u^\delta = A^*p^\delta.$$

Proof. Let us prove that the functional J^δ is coercive, that is $J^\delta(q) \to +\infty$ when $\|q\|_H \to +\infty$. Assume on the contrary that there exists some constant $C > 0$ and a sequence $(q_n)_{n\in\mathbb{N}}$ of elements in H such that $\alpha_n = \|q_n\|_H \to +\infty$ while $J^\delta(q_n) \leq C$. Let us define $z_n = q_n/\|q_n\|_H$. Since

$\|z_n\|_H = 1$, there exists a subsequence of $(z_n)_n$, still denoted $(z_n)_n$, such that $z_n \rightharpoonup z$ in H. We have

$$\frac{\alpha_n^2}{2}\|A^* z_n\|_V^2 + \alpha_n \delta - \alpha_n \mathcal{R}e\left\{(F^\delta, z_n)_H\right\} \leq C, \tag{2.41}$$

hence

$$\frac{1}{2}\|A^* z_n\|_V^2 \leq \frac{\|F^\delta\|_H}{\alpha_n} + \frac{C}{\alpha_n^2},$$

which implies that $A^* z_n \to 0$ in V and since $A^* z_n \rightharpoonup A^* z$ in V, we get that $A^* z = 0$. The fact that A has a dense range implies that A^* is injective, hence $z = 0$, that is $z_n \rightharpoonup 0$ in H. Another consequence of (2.41) is that

$$C \geq \alpha_n(\delta - \mathcal{R}e\left\{(F^\delta, z_n)_H\right\}) \to +\infty,$$

which is a contradiction. The functional J^δ is coercive, continuous and strictly convex, which implies that the problem (2.40) has a unique solution p^δ. Let us check that $p^\delta \neq 0$. Actually, let us take $z^\delta = F^\delta / \|F^\delta\|_H$. For $\varepsilon > 0$, we have

$$J^\delta(\varepsilon z^\delta) = \frac{\varepsilon^2}{2}\|A^* z^\delta\|_V^2 + \varepsilon(\delta - \|F^\delta\|_H).$$

For small ε, the real number $J^\delta(\varepsilon z^\delta)$ has the sign of $\delta - \|F^\delta\|_H < 0$, hence there exists $q \in H$ such that $J^\delta(q) < 0 = J^\delta(0)$, which implies that p^δ is not zero. The functional J^δ is differentiable in the sense of Fréchet at $p \neq 0$ and its derivative is given by

$$\langle DJ^\delta(p), q\rangle_{H',H} = \mathcal{R}e\left\{(AA^* p, q)_H + \frac{\delta}{\|p\|_H}(p, q)_H - (F^\delta, q)_H\right\}, \quad \forall q \in H.$$

Writing that $\langle DJ^\delta(p^\delta), q\rangle_{H',H} = 0$ for any q and iq in H, we finally get that

$$AA^* p^\delta + \frac{\delta}{\|p^\delta\|_H} p^\delta = F^\delta,$$

Denoting $w^\delta := A^* p^\delta \in V$, we obtain that

$$A^* A w^\delta + \frac{\delta}{\|p^\delta\|_H} w^\delta = A^* F^\delta, \quad \|A w^\delta - F^\delta\|_H = \delta.$$

This means that for $\varepsilon = \delta / \|p^\delta\|_H$, w^δ satisfies both (2.39) and (2.38), that is $w^\delta = u^\delta$. In other words, such value of ε corresponds to the Morozov choice in the Tikhonov regularization. □

We observe that the functional J^δ given by (2.40) is convex but is not quadratic because of the second term $\delta\|q\|_H$. In practice, the minimization of J^δ by a gradient method is likely to be more difficult than if it were quadratic. In [48], the author proposed a strategy to circumvent this issue. It consists in the following iterative algorithm, where the solution of the initial non-quadratic minimization problem (2.40) is approached by a sequence formed by the minimizers of well-defined quadratic functionals.

Algorithm 2.5.2. 1. Let $\varepsilon_n > 0$ be given

2. Compute the unique solution p_n of the minimization problem

$$\inf_{q \in H} L_n(q), \quad L_n(q) := \frac{1}{2}\|A^* q\|_V^2 + \frac{1}{2}\varepsilon_n \|q\|_H^2 - \mathcal{R}e\left\{(F^\delta, q)_H\right\} \tag{2.42}$$

3. Compute $\varepsilon_{n+1} = \delta / \|p_n\|_H$ and go to step 1 for $n \to n+1$

Comparing (2.40) and (2.42), the second non-quadratic term $\delta\|q\|_H$ was replaced by the quadratic term $\varepsilon_n \|q\|_H^2/2$. In [48], the following proposition in proved.

Proposition 2.5.3. *Let $(p_n)_{n\in\mathbb{N}}$ be the sequence of elements of H defined by the algorithm 2.5.2. Then $p_n \to p^\delta$ in H when $n \to +\infty$ and $u_n := A^* p_n \to u^\delta$ in V when $n \to +\infty$.*

We now establish that the minimization problem (2.42) is equivalent to a weak formulation satisfied by $u_n \in V$ and $p_n = (\lambda_n, h_n) \in M \times O$ for all $n \in \mathbb{N}$.

Proposition 2.5.4. *The triple $(u_n, \lambda_n, h_n) \in V \times M \times O$ is given by the well-posed weak formulation: for all $(v, \mu, h) \in V \times M \times O$,*

$$\left\{\begin{array}{rcll} (u_n, v)_V - \overline{b(v,\lambda_n)} - (h_n, Cv)_O &=& 0, & \forall v \in V, \\ b(u_n,\mu) + \varepsilon_n(\lambda_n,\mu)_M &=& (F_1^\delta,\mu)_M, & \forall \mu \in M, \\ (Cu_n, h)_O + \varepsilon_n(h_n,h)_O &=& (F_2^\delta, h)_O, & \forall h \in O. \end{array}\right. \tag{2.43}$$

Proof. Let us first notice that $A^*(\lambda, h) = A^* p = B^*\lambda + C^* h$ for all $p = (\lambda, h) \in M \times O$, hence,

$$L_n(\lambda,h) = \frac{1}{2}\|B^*\lambda + C^*h\|_V^2 + \frac{\varepsilon_n}{2}(\|\lambda\|_M^2 + \|h\|_O^2) - \mathcal{R}e\left\{(F_1^\delta,\lambda)_M + (F_2^\delta,h)_O\right\}.$$

The minimizer (λ_n, h_n) of L_n satisfies

$$\left\{\begin{array}{rcll} (B^*\lambda_n + C^*h_n, B^*\mu)_V + \varepsilon_n(\lambda_n,\mu)_M - (F_1^\delta,\mu)_M &=& 0, & \forall \mu \in M, \\ (B^*\lambda_n + C^*h_n, C^*h)_V + \varepsilon_n(h_n,h)_O - (F_2^\delta,h)_O &=& 0, & \forall h \in O. \end{array}\right.$$

Since $u_n = B^*\lambda_n + C^*h_n$, we get that

$$\left\{\begin{array}{rcll} (u_n, v)_V - (B^*\lambda_n, v)_V - (C^*h_n, v)_V &=& 0, & \forall v \in V, \\ (u_n, B^*\mu)_V + \varepsilon_n(\lambda_n,\mu)_M - (F_1^\delta,\mu)_M &=& 0, & \forall \mu \in M, \\ (u_n, C^*h)_V + \varepsilon_n(h_n,h)_O - (F_2^\delta,h)_O &=& 0, & \forall h \in O. \end{array}\right.$$

By using the definition of the operators B^*, C^* and the sesquilinear b, we obtain the system (2.43), which completes the proof of the proposition. □

2.6 Application to the tsunami identification

The goal of the section is to apply the abstract setting of the previous section to the tsunami identification problem, both for the gravity and the acoustic models. The case of the complete model is omitted, since it is very close to the gravity one.

2.6.1 The gravity model

Let us start with the gravity model. The potential u satisfies the problem

$$\left\{\begin{array}{rcll} \Delta u &=& 0 & \text{in } \Omega_0^R, \\ \partial_z u - (\omega^2/g)u &=& 0 & \text{on } \Gamma_0^R, \\ \pm\partial_x u - T_\pm^g u &=& 0 & \text{on } \Sigma_{\pm R}, \end{array}\right. \tag{2.44}$$

with additionally

$$u = F_2 := (g/i\omega)\eta \quad \text{on} \quad \Gamma_0^R, \tag{2.45}$$

where η is the measured free surface perturbation. In (2.44), we recall that the operators $T_\pm^g$ coincide with the operators $T_\pm^c$ defined by (2.12) when $c \to +\infty$, that is the functions f_n and the numbers β_n are defined by (2.24) and (2.25), respectively. The goal is to find an approximation of u in Ω_0^R from the data F_2, which is noisy in general, using the strategy described in the previous section. We specify the Hilbert spaces V, M and O as

$$V = H^1(\Omega_0^R), \quad M = \{\mu \in H^1(\Omega_0^R),\ \mu|_{\Gamma_{-H}^R} = 0\}, \quad O = L^2(\Gamma_0^R). \tag{2.46}$$

The norm of space M is chosen as the $H^1(\Omega_0^R)$ semi-norm, thanks to the Poincaré inequality. We also introduce the continuous sesquilinear form on $V \times M$:

$$\begin{aligned} b(v,\mu) &= \int_{\Omega_0^R} \nabla v \cdot \nabla \overline{\mu}\, dxdz - (\omega^2/g) \int_{\Gamma_0^R} v\, \overline{\mu}\, dx \\ &- \langle T_+^g v, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_R), \tilde{H}^{1/2}(\Sigma_R)} - \langle T_-^g v, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_{-R}), \tilde{H}^{1/2}(\Sigma_{-R})} \end{aligned} \tag{2.47}$$

and the corresponding operator $B : V \to M$ defined by (2.36) from such sesquilinear form. The operator $C : V \to O$ is chosen as the trace operator. This enables us to define the operator $A : V \to M \times O$ as $A = (B, C)$. By construction $F_1 = 0$, so that $F = (0, F_2) \in M \times O$. The following lemma establishes a connection between the tsunami identification and the abstract framework of the previous section.

Lemma 2.6.1. *The potential $u \in H^1(\Omega_0^R)$ satisfies the problem (2.44) (2.45) if and only if u satisfies $Au = F$.*

Proof. Assume that $u \in H^1(\Omega_0^R)$ satisfies $Au = F$. We hence have $Bu = 0$ and $Cu = F_2$. The fact that $Cu = F_2$ is equivalent to $u|_{\Gamma_0^R} = F_2$. The fact that $Bu = 0$ is equivalent to $b(u, \mu) = 0$ for all $\mu \in M$, that is

$$\begin{aligned} \int_{\Omega_0^R} \nabla u \cdot \nabla \overline{\mu}\, dxdz - (\omega^2/g) \int_{\Gamma_0^R} u\, \overline{\mu}\, dx - \langle T_+^g u, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_R), \tilde{H}^{1/2}(\Sigma_R)} \\ - \langle T_-^g u, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_{-R}), \tilde{H}^{1/2}(\Sigma_{-R})} = 0, \quad \forall \mu \in M. \end{aligned}$$

Let us first choose $\mu = \varphi \in C_0^\infty(\Omega_0^R)$. We obtain that $\Delta u = 0$ in Ω_0^R. By using the Green formula, we get

$$\int_{\Omega_0^R} \nabla u \cdot \nabla \overline{\mu}\, dxdz = -\int_{\Omega_0^R} \Delta u\, \overline{\mu}\, dxdz + \langle \partial_\nu u, \mu \rangle_{H^{-1/2}(\partial\Omega_0^R), H^{1/2}(\partial\Omega_0^R)}, \quad \forall \mu \in M,$$

whence

$$\begin{aligned} \langle \partial_\nu u, \overline{\mu} \rangle_{H^{-1/2}(\partial\Omega_0^R), H^{1/2}(\partial\Omega_0^R)} - (\omega^2/g) \int_{\Gamma_0^R} u\, \overline{\mu}\, dx - \langle T_+^g u, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_R), \tilde{H}^{1/2}(\Sigma_R)} \\ - \langle T_-^g u, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_{-R}), \tilde{H}^{1/2}(\Sigma_{-R})} = 0, \quad \forall \mu \in M. \end{aligned}$$

This yields $\partial_z u = (\omega^2/g) u$ on Γ_0^R and that $\pm \partial_x u = T_\pm^g u$ on $\Sigma_{\pm R}$. We eventually establish that u satisfies (2.44) and (2.45). The converse statement is obtained analogously. □

To apply the results of the previous section, we have to check that the operator A is injective and has a dense range.

Lemma 2.6.2. *The operator A is injective and has a dense range.*

Proof. Let us consider $u \in H^1(\Omega_0^R)$ such that $Au = 0$. From the proof of Lemma 2.6.1, we have that $\Delta u = 0$ in Ω_0^R, $u = 0$ on Γ_0^R and $\partial_z u = (\omega^2/g) u$ on Γ_0^R, that is $(u, \partial_z u) = (0, 0)$ on Γ_0^R. From uniqueness of the Cauchy problem for the Laplace equation, we have $u = 0$ in Ω_0^R, which shows the injectivity of the operator A.

Now let us assume that $(\mu, h) \in M \times L^2(\Gamma_0^R)$ satisfies

$$(Av, (\mu, h))_{M \times L^2(\Gamma_0^R)} = 0, \quad \forall v \in H^1(\Omega_0^R).$$

This implies that

$$\begin{aligned} \int_{\Omega_0^R} \nabla v \cdot \nabla \overline{\mu}\, dxdz - (\omega^2/g) \int_{\Gamma_0^R} v\, \overline{\mu}\, dx - \langle T_+^g v, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_R), \tilde{H}^{1/2}(\Sigma_R)} \\ - \langle T_-^g v, \overline{\mu} \rangle_{H^{-1/2}(\Sigma_{-R}), \tilde{H}^{1/2}(\Sigma_{-R})} + \int_{\Gamma_0^R} v\, \overline{h}\, dx = 0, \quad \forall v \in H^1(\Omega_0^R). \end{aligned}$$

Choosing first $v = \varphi \in C_0^\infty(\Omega_0^R)$, we get $\Delta \mu = 0$ in Ω_0^R. Using the Green formula, we obtain that $\partial_z \mu = 0$ on Γ_{-H}^R and $\partial_z \mu - (\omega^2/g)\mu + h = 0$ on Γ_0^R. Since we also have $\mu = 0$ on Γ_{-H}^R, uniqueness of the Cauchy problem for the Laplace equation implies that $\mu = 0$ in Ω_0^R, which in turn implies that $h = 0$ in $L^2(\Gamma_0^R)$. We conclude that the operator A has a dense range. □

We can then apply all the results of the abstract framework described in the previous section to that specific example. In particular, the weak formulation (2.35) becomes (note that $F_1 = 0$): find $(u_\varepsilon, \lambda_\varepsilon) \in H^1(\Omega_0^R) \times M$, with $M = \{\mu \in H^1(\Omega_0^R),\ \mu|_{\Gamma^R_{-H}} = 0\}$, such that

$$\left\{\begin{array}{c} \varepsilon(u_\varepsilon, v)_{H^1(\Omega_0^R)} + \displaystyle\int_{\Omega_0^R} \nabla\lambda_\varepsilon \cdot \nabla\overline{v}\, dxdz - (\omega^2/g) \int_{\Gamma_0^R} \lambda_\varepsilon\, \overline{v}\, dx \\ -\langle \overline{T_+^g}\lambda_\varepsilon, \overline{v}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} - \langle \overline{T_-^g}\lambda_\varepsilon, \overline{v}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})} \\ + \displaystyle\int_{\Gamma_0^R} u_\varepsilon\, \overline{v}\, dx = \int_{\Gamma_0^R} F_2\, \overline{v}\, dx, \qquad \forall v \in H^1(\Omega_0^R), \\ \displaystyle\int_{\Omega_0^R} \nabla u_\varepsilon \cdot \nabla\overline{\mu}\, dxdz - (\omega^2/g) \int_{\Gamma_0^R} u_\varepsilon\, \overline{\mu}\, dx \\ -\langle T_+^g u_\varepsilon, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} - \langle T_-^g u_\varepsilon, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})} \\ - \displaystyle\int_{\Omega_0^R} \nabla\lambda_\varepsilon \cdot \nabla\overline{\mu}\, dxdz = 0, \qquad \forall \mu \in M. \end{array}\right. \tag{2.48}$$

In the presence of a noisy data F_2^δ such that

$$\|F_2^\delta - F_2\|_{L^2(\Gamma_0^R)} \leq \delta < \|F_2^\delta\|_{L^2(\Gamma_0^R)}, \tag{2.49}$$

which, since $F_1 = 0$, amounts to (2.37), we apply the algorithm 2.5.2 to obtain the Morozov solution $u^\delta \in H^1(\Omega_0^R)$. This requires to solve for any $n \in \mathbb{N}$ the problem (2.43), which becomes here: find $(u_n, \lambda_n, h_n) \in H^1(\Omega_0^R) \times M \times L^2(\Gamma_0^R)$ such that

$$\left\{\begin{array}{c} (u_n, v)_{H^1(\Omega_0^R)} - \displaystyle\int_{\Omega_0^R} \nabla\lambda_n \cdot \nabla\overline{v}\, dxdz + (\omega^2/g) \int_{\Gamma_0^R} \lambda_n\, \overline{v}\, dx \\ +\langle \overline{T_+^g}\lambda_n, \overline{v}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} + \langle \overline{T_-^g}\lambda_n, \overline{v}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})} \\ - \displaystyle\int_{\Gamma_0^R} h_n\, \overline{v}\, dx = 0, \qquad \forall v \in H^1(\Omega_0^R), \\ \displaystyle\int_{\Omega_0^R} \nabla u_n \cdot \nabla\overline{\mu}\, dxdz - (\omega^2/g) \int_{\Gamma_0^R} u_n\, \overline{\mu}\, dx \\ -\langle T_+^g u_n, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} - \langle T_-^g u_n, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})} \\ +\varepsilon_n \displaystyle\int_{\Omega_0^R} \nabla\lambda_n \cdot \nabla\overline{\mu}\, dxdz = 0, \qquad \forall \mu \in M, \\ \displaystyle\int_{\Gamma_0^R} u_n\, \overline{h}\, dx + \varepsilon_n \int_{\Gamma_0^R} h_n\, \overline{h}\, dx = \int_{\Gamma_0^R} F_2^\delta\, \overline{h}\, dx, \qquad \forall h \in L^2(\Gamma_0^R). \end{array}\right. \tag{2.50}$$

The method that we propose to approximate the tsunami shape χ on Γ^R_{-H} is the following: starting with an initial value $\varepsilon_0 > 0$, we successively solve (2.50) and compute for $n \in \mathbb{N}$:

$$\varepsilon_{n+1} = \frac{\delta}{\|p_n\|_H}, \quad \text{with} \quad \|p_n\|_H = \sqrt{\int_{\Omega_0^R} |\nabla\lambda_n|^2\, dxdz + \int_{\Gamma_0^R} |h_n|^2\, dx}.$$

For sufficiently large n, u^δ is approximated by u_n, which we still denote u^δ for simplicity. Lastly, in view of (2.11), the function χ is approximated by

$$\chi^\delta = \frac{i}{\omega}\partial_z u^\delta.$$

2.6.2 The acoustic model

In the case of the acoustic model, the potential u satisfies the problem

$$\left\{\begin{array}{rcll} \Delta u + (\omega^2/c^2)u & = & 0 & \text{in } \Omega_0^R, \\ u & = & 0 & \text{on } \Gamma_0^R, \\ \pm\partial_x u - T_\pm^a u & = & 0 & \text{on } \Sigma_{\pm R}, \end{array}\right. \tag{2.51}$$

with additionally

$$\partial_z u = f_1 := -i\omega\,\eta \quad \text{on} \quad \Gamma_0^R, \tag{2.52}$$

where η is the measured free surface perturbation. In (2.51), the operators $T_\pm^a$ are defined by (2.31). Again, the goal is to find an approximation of u in Ω_0^R from the data f_1, which is noisy, by using the strategy described in the previous section. We specify the Hilbert spaces V, M and O as in (2.46). While the operator $C : H^1(\Omega_0^R) \to L^2(\Gamma_0^R)$ is the trace operator again, we introduce a new continuous sesquilinear form on $V \times M$:

$$\begin{array}{rcl} b(v,\mu) & = & \displaystyle\int_{\Omega_0^R} \nabla v \cdot \nabla\overline{\mu}\,dxdz - (\omega^2/c^2)\int_{\Omega_0^R} v\,\overline{\mu}\,dxdz \\ & - & \langle T_+^a u, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} - \langle T_-^a u, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})} \end{array}$$

and the corresponding operator $B : V \to M$ defined by (2.36). We again form the operator $A : V \to M \times O$ as $A = (B, C)$. Let us now denote $L_1 \in M$ the unique element of M which satisfies

$$(L_1, \mu)_M = \int_{\Gamma_0^R} f_1\,\overline{\mu}\,dx, \quad \forall \mu \in M. \tag{2.53}$$

We choose $F_1 = L_1$ and $F_2 = 0$, so that $F = (L_1, 0) \in M \times O$. The following lemma establishes a connection between the tsunami identification and the abstract framework of the previous section.

Lemma 2.6.3. *The potential $u \in H^1(\Omega_0^R)$ satisfies the problem (2.51) (2.52) if and only if u satisfies $Au = F$.*

Proof. Let us consider $u \in H^1(\Omega_0^R)$ such that $Au = F$, that is $Bu = F_1$ and $Cu = 0$. The fact that $Cu = 0$ is equivalent to $u = 0$ on Γ_0^R. The fact that $Bu = F_1$ is equivalent to

$$\begin{array}{c} \displaystyle\int_{\Omega_0^R} \nabla u \cdot \nabla\overline{\mu}\,dxdz - (\omega^2/c^2)\int_{\Omega_0^R} u\,\overline{\mu}\,dxdz - \langle T_+^a u, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_R),\tilde{H}^{1/2}(\Sigma_R)} \\ -\langle T_-^a u, \overline{\mu}\rangle_{H^{-1/2}(\Sigma_{-R}),\tilde{H}^{1/2}(\Sigma_{-R})} = \displaystyle\int_{\Gamma_0^R} f_1\,\mu\,dx, \quad \forall \mu \in M. \end{array}$$

Choosing first $\mu = \varphi \in C_0^\infty(\Omega_0^R)$ yields $\Delta u + (\omega^2/c^2)u = 0$ in Ω_0^R. Then by using the Green formula, we obtain $\partial_z u = f_1$ on Γ_0^R and $\pm\partial_x u = T_\pm^a u$ on $\Sigma_{\pm R}$, that is u solves the problem (2.51) (2.52). The converse statement is obtained the same way. □

Lemma 2.6.2 is also satisfied for our new specific operator A, so that we can again apply all the results of the abstract framework described in the previous section. The explicit forms of the systems (2.35) and (2.43) in the case of acoustics, which are not given here for brevity, are quite close to (2.48) and (2.50), respectively.

2.7 Application to bathymetry

We now apply the abstract setting of section 2.5 to the bathymetry problem, both for the gravity and the acoustic models. The case of the complete model is omitted, since it is very close to the gravity one. Contrary to the tsunami identification problem, we have to tackle a (non linear) geometric inverse problem. If we assume that the bottom perturbation θ is small, we propose to solve the bathymetry problem by approximating the boundary condition $\partial_\nu u = 0$ on the perturbed boundary $\Gamma_{-H+\theta}^R$ in (2.18) and in (2.33) by a generalized impedance boundary condition on the flat boundary Γ_{-H}^R, as proposed for example in [3] and [19] (see also the introduction of [20]).

2.7.1 The gravity model

Let us start with the gravity model. By reproducing the computations of [3] or [20], we can show that the first order approximation on Γ^R_{-H} is $\partial_z u = 0$ while the second order approximation is

$$\partial_z u = \partial_x(\theta\, \partial_x u) \quad \text{on} \quad \Gamma^R_{-H}, \tag{2.54}$$

where $\|\theta\|_{L^\infty(-R,R)}$ is supposed to be small compared to the wavelength of the wave phenomenon at stake, that is $\lambda = 2\pi\sqrt{gH}/\omega$ in the gravity model. In view of (2.18) with $c \to +\infty$, the velocity potential satisfies

$$\left\{ \begin{array}{rcll} \Delta u & = & 0 & \text{in } \Omega^R_0, \\ \partial_z u - (\omega^2/g)u & = & 0 & \text{on } \Gamma^R_0, \\ -\partial_x u & = & T^g_- u - 2\partial_x \mathcal{G}^g(\cdot, N) & \text{on } \Sigma_{-R}, \\ \partial_x u & = & T^g_+ u & \text{on } \Sigma_R \end{array} \right. \tag{2.55}$$

and (2.45), where the operators $T^g_\pm$ are defined by (2.12) for $c \to +\infty$ and the point source $\mathcal{G}^g(\cdot, N)$ is given by (2.16) for $c \to +\infty$. We reuse the Hilbert spaces V, M and O as well as the operators A, B and C specified in section 2.6.1. We also define $G^g \in M$ such that

$$(G^g, \mu)_M = -2 \int_{\Sigma_{-R}} \partial_x \mathcal{G}^g(\cdot, N) \overline{\mu}\, dy, \quad \forall \mu \in M. \tag{2.56}$$

We choose $F_1 = G^g$ and $F = (G^g, F_2) \in M \times O$, with G^g and F_2 given by (2.56) and (2.45), respectively. We easily check the following result.

Lemma 2.7.1. *The function $u \in H^1(\Omega^R_0)$ satisfies the problem (2.55) (2.45) if and only if u satisfies $Au = F$.*

Such Lemma enables us to directly apply all the results of section 2.5. In particular, when the noisy data $F^\delta_2 \in O$ is such that the $F^\delta = (G^g, F^\delta_2)$ satisfies (2.37), we again apply the algorithm 3.4.4 to obtain the Morozov solution $u^\delta \in H^1(\Omega^R_0)$.

In order to approximate θ by some function θ^δ in the presence of noisy data, let us set

$$\alpha := u^\delta|_{\Gamma^R_{-H}}, \quad \alpha_x := d\alpha/dx, \quad \alpha_z := \partial_z u^\delta|_{\Gamma^R_{-H}}, \tag{2.57}$$

where the function u^δ has been obtained in the first step, and let us compute θ^δ from the equation (2.54). From the strong equation satisfied by $u^\delta \in H^1(\Omega^R_0)$, we have $\Delta u^\delta \in L^2(\Omega^R_0)$, so that $\alpha \in H^{1/2}(\Gamma^R_{-H})$, $\alpha_x \in H^{-1/2}(\Gamma^R_{-H})$ and $\alpha_z \in H^{-1/2}(\Gamma^R_{-H})$. However, to justify this second step, we adopt the following regularity assumption:

Assumption 2.7.1.

$$\alpha_x, \alpha_z \in L^\infty(\Gamma^R_{-H}), \quad \exists\, m > 0,\ |\alpha_x(x)| \geq m \text{ for almost all } x \in \Gamma^R_{-H}.$$

In view of (2.54) and having in mind that θ^δ is compactly supported in $(-R, R)$, a first simple method consists in computing directly θ^δ as

$$\theta^\delta(x) = \frac{1}{\alpha_x} \int_{-R}^{x} \alpha_z(t)\, dt.$$

A second method consists in using a weak formulation of (2.54). Let us denote $I = (-R, R)$ and consider the problem: find $\theta \in L^2(I)$ such that

$$b(\theta, v) = \ell(v), \quad \forall v \in V_0(I), \tag{2.58}$$

where the space $V_0(I)$ is defined by

$$V_0(I) = \{v \in H^1(I),\ v(-R) = 0\}$$

and where the sesquilinear form b and the antilinear form ℓ are given by

$$b(\theta, v) := \int_I \alpha_x \, \theta \, \overline{v}' \, dx, \quad \ell(v) := -\int_I \alpha_z \overline{v} \, dx, \quad \forall (\theta, v) \in L^2(I) \times V_0(I). \tag{2.59}$$

We note that any solution to the weak formulation (2.58) satisfies the strong problem (2.54) in the sense of distributions. We now establish that the problem (2.58) is well-posed.

Theorem 2.7.2. *Under Assumption 2.7.1, the problem (2.58) has a unique solution.*

Proof. Let us apply the Brezzi-Nečas-Babuška Theorem (see for example [58]). Firstly, let us check the inf-sup condition, that is

$$\inf_{\theta \in L^2(I)} \sup_{v \in V_0(I)} \frac{|b(\theta, v)|}{\|\theta\|_{L^2(I)} \|v\|_{V_0(I)}} \geq \beta$$

for some $\beta > 0$. Indeed, we have

$$\sup_{v \in V_0(I)} \frac{|b(\theta, v)|}{\|v\|_{V_0(I)}} = \sup_{v \in V_0(I)} \frac{\left| \int_I \alpha_x \, \theta \, \overline{v}' \, dx \right|}{\|v\|_{V_0(I)}} = \|\alpha_x \theta\|_{L^2(I)} \geq m \, \|\theta\|_{L^2(I)},$$

where the second equality above is obtained by choosing v such that

$$v(x) = \int_{-R}^{x} (\alpha_x \theta)(t) \, dt.$$

We hence obtain the result with $\beta = m$. Secondly, it is easily seen that if for some $v \in V_0(I)$ we have $b(\theta, v) = 0$ for all $\theta \in L^2(I)$, then $v = 0$. Indeed, if

$$\int_I \alpha_x \, \theta \, \overline{v}' \, dx = 0, \quad \forall \theta \in L^2(I),$$

then $\alpha_x v' = 0$, and given assumption 2.7.1 we get $v' = 0$, hence $v = 0$. □

2.7.2 The acoustic model

Now let us consider the acoustic model. Again mimicking [3] or [20], we show that the first order approximation on Γ^R_{-H} is $\partial_z u = 0$ while the second order approximation is

$$\partial_z u = \partial_x (\theta \, \partial_x u) + (\omega^2/c^2) \theta \, u \quad \text{on} \quad \Gamma^R_{-H}, \tag{2.60}$$

where $\|\theta\|_{L^\infty(-R,R)}$ is supposed to be small compared to the wavelength of the acoustic waves, that is $\lambda = 2\pi c/\omega$. In view of (2.33) we have

$$\left\{ \begin{array}{rcll} \Delta u + (\omega^2/c^2) u & = & 0 & \text{in } \Omega^R_0, \\ u & = & 0 & \text{on } \Gamma^R_0, \\ -\partial_x u & = & T^a_- u - 2\partial_x \mathcal{G}^a(\cdot, N) & \text{on } \Sigma_{-R}, \\ \partial_x u & = & T^a_+ u & \text{on } \Sigma_R. \end{array} \right. \tag{2.61}$$

and (2.52), where the operators $T^a_\pm$ are defined by (2.31) while $\mathcal{G}^a(\cdot, N)$ is given by (2.32). We consider here the Hilbert spaces V, M and O as well as the operators A, B and C specified in section 2.6.2. We also define $L_1 \in M$ satisfying (2.53) and $G^a \in M$ satisfying (2.56) for $\mathcal{G}^g$ replaced by $\mathcal{G}^a$, $F_1 = L_1 + G^a$ and $F_2 = 0$, so that $F = (L_1 + G^a, 0) \in M \times O$. We easily check the following result.

Lemma 2.7.2. *The function $u \in H^1(\Omega^R_0)$ satisfies the problem (2.61) (2.52) if and only if u satisfies $Au = F$.*

The above Lemma invites us again to directly apply all the results of section 3.4. In particular, the algorithm 2.5.2 enables us to obtain the Morozov solution $u^\delta \in H^1(\Omega_0^R)$ in the presence of some noisy data $F_1^\delta = L_1^\delta + G^a \in M$ such that (2.37) is satisfied.

In order to compute an approximated θ^δ function in the presence of noise starting from the function u^δ obtained in the first step and reusing the definitions of α, α_x and α_z given by (2.57), we wish to retrieve θ^δ from the equation (2.60) in a second step. The weak formulation (2.58) is modified as follows: find $\theta \in L^2(I)$ such that

$$b(\theta, v) + c(\theta, v) = \ell(v), \quad \forall v \in V_0(I), \tag{2.62}$$

where the sesquilinear form b and the antilinear form ℓ are defined by (2.59), while the sesquilinear form c is given by

$$c(\theta, v) := -(\omega^2/c^2) \int_I \alpha\, \theta\, \overline{v}\, dx, \quad \forall (\theta, v) \in L^2(I) \times V_0(I).$$

We now establish that the problem (2.62) is well-posed in the Fredholm sense.

Theorem 2.7.3. *Under Assumption 2.7.1, if the problem (2.62) has at most one solution, such solution exists.*

Proof. With the help of the Riesz theorem, we define the operators $B, C : L^2(I) \to V_0(I)$ such that

$$(T\theta, v)_{V_0(I)} = t(\theta, v), \quad \forall (\theta, v) \in L^2(I) \times V_0(I),$$

where T is either operator B, C and t is the corresponding sesquilinear form b, c. Since the sesquilinear form b satisfies the assumptions of the Brezzi-Nečas-Babuška Theorem (see the proof of Theorem 2.7.2), the operator B is an isomorphism. Let us check that the operator C is compact. To proceed, we set $u_\theta := C\theta \in V_0(I)$ for some $\theta \in L^2(I)$. Since from Assumption 2.7.1 we have $\alpha_x \in L^\infty(I) \subset L^2(I)$, we infer $\alpha \in H^1(I) \subset L^\infty(I)$. Hence $f := -(\omega^2/c^2)\alpha\theta \in L^2(I)$. We observe that the function u_θ is the unique solution in $V_0(I)$ to the weak formulation:

$$\int_I u'_\theta\, \overline{v}'\, dx = \int_I f\, \overline{v}\, dx, \quad \forall v \in V_0(I).$$

A straightforward regularity result shows that $\theta \mapsto u_\theta$ is continuous from $L^2(I)$ to $H^2(I) \cap V_0(I)$, which implies that C is compact. The conclusion results from the Fredhom alternative. □

2.8 About the noisy data in the acoustic case

As it can be seen in the Algorithm 2.5.2 to compute the Morozov solution u^δ, the amplitude of noise δ of the data F is required in all cases. In the gravity case, such amplitude of noise coincides with the amplitude of noise on F_2, which happens to be, in view of (2.45), the amplitude of noise on the free surface perturbation η multiplied by (g/ω). In the acoustic case, δ coincides with the amplitude of noise affecting F_1, that is the amplitude of noise on L_1. Estimating such amplitude of noise is more complicated than in the gravity case. Indeed, the volumic function L_1 is defined from the Neumann data f_1 by the weak formulation (2.53). In view of (2.52), the amplitude of noise on f_1 is given by multiplying the amplitude of noise on the free surface perturbation η by ω. But "converting" the amplitude of noise on f_1 into the amplitude on L_1 is not as simple. Let us denote

$$\left\{ \begin{array}{rcl} S : L^2(\Gamma_0^R) & \to & M = \{\mu \in H^1(\Omega_0^R),\, \mu|_{\Gamma_{-H}^R} = 0\} \\ f & \mapsto & L, \end{array} \right. \tag{2.63}$$

where L is the unique solution in M of the weak formulation:

$$\int_{\Omega_0^R} \nabla L \cdot \nabla \overline{\mu}\, dxdz = \int_{\Gamma_0^R} f\, \overline{\mu}\, dx, \quad \forall \mu \in M, \tag{2.64}$$

which is equivalent to the strong problem: find $L \in M$ such that

$$\left\{\begin{array}{rcll} \Delta L & = & 0 & \text{in } \Omega_0^R, \\ \partial_z L & = & f & \text{on } \Gamma_0^R, \\ L & = & 0 & \text{on } \Gamma_{-H}^R, \\ \partial_x L & = & 0 & \text{on } \Sigma_{\pm R}. \end{array}\right. \tag{2.65}$$

If d denotes the amplitude of noise on f, a simple way of estimating the resulting amplitude of noise δ on L is

$$\delta = \|S\|\, d, \quad \|S\| := \sup_{f \in L^2(\Gamma_0^R),\, f \neq 0} \frac{\|Sf\|_M}{\|f\|_{L^2(\Gamma_0^R)}}. \tag{2.66}$$

It happens that the operator norm $\|S\|$ is easily determined.

Proposition 2.8.1. *We have $\|S\| = \sqrt{H}$ when $\theta = 0$.*

Proof. Let us introduce the operator $T : L^2(\Gamma_0^R) \to L^2(\Gamma_0^R)$ defined by $T = \gamma \circ S$, where $\gamma : M \to L^2(\Gamma_0^R)$ is the trace operator. Let us check that the operator T is self-adjoint, compact, positive and injective. Indeed, for $f, g \in L^2(\Gamma_0^R)$ we have

$$\begin{array}{rcl} (Tf, g)_{L^2(\Gamma_0^R)} & = & \displaystyle\int_{\Gamma_0^R} (Sf)\partial_z(\overline{Sg})\, dx = \int_{\partial\Omega_0^R} (Sf)\partial_\nu(\overline{Sg})\, ds \\ & = & \displaystyle\int_{\Omega_0^R} \nabla(Sf)\cdot\nabla(\overline{Sg})\, dxdz = (f, Tg)_{L^2(\Gamma_0^R)}, \end{array} \tag{2.67}$$

which proves that T is self-adjoint. The fact that S is bounded and γ is compact implies that the operator $T = \gamma \circ S$ is compact. Positivity and injectivity are straightforward consequences of (2.67). The spectral theorem implies then that the spectrum of T is formed by a sequence of real and non increasing eigenvalues $\rho_n > 0$, $n \in \mathbb{N}$, such that $\rho_n \to 0$ when $n \to +\infty$, and there exists a complete orthonormal basis of $L^2(\Gamma_0^R)$ formed with the help of the corresponding eigenfunctions e_n. This implies that

$$\|S\|^2 = \sup_{f \in L^2(\Gamma_0^R),\, f \neq 0} \frac{\|Sf\|_M^2}{\|f\|_{L^2(\Gamma_0^R)}^2} = \sup_{f \in L^2(\Gamma_0^R),\, f \neq 0} \frac{(Tf, f)_{L^2(\Gamma_0^R)}}{\|f\|_{L^2(\Gamma_0^R)}^2} = \rho_0,$$

since ρ_0 is the largest eigenvalue of T. An easy computation shows that

$$\rho_0 = H, \quad \rho_n = \frac{2R}{n\pi}\tanh\left(\frac{n\pi H}{2R}\right), \quad n \geq 1,$$

and for $x \in (-R, R)$, we can choose

$$e_0(x) = \frac{1}{\sqrt{2R}}, \quad e_n(x) = \frac{1}{\sqrt{R}}\cos\left(\frac{n\pi}{2R}(x+R)\right), \quad n \geq 1.$$

We have in particular $\rho_0 = H$, which completes the proof. □

The estimate (2.66) giving the amplitude δ of the noise on the data L in M from the amplitude of noise on the data f in $L^2(\Gamma_0^R)$, in view of the definition of $\|S\|$, is likely to overestimate δ. This is why we propose an alternative way of computing δ based on a probabilistic model of the measurements f. Let us assume that f is given on Γ_0^R by

$$f = \sum_{k=1}^{N} f_k\, \chi_k^N.$$

For $k = 1, \cdots, N$, the functions χ_k^N are indicator functions of intervals having the same length $2R/N$, which do not overlap and the union of which coincides with Γ_0^R. The complex random variables f_k, $k = 1, \cdots, N$, satisfy the following discrete Gaussian white noise assumptions:

Assumption 2.8.1. The f_k, $k = 1, \cdots, N$, are independent and identically distributed complex random variables such that their real and imaginary parts follow the normal distribution $\mathcal{N}(0, \sigma^2/2)$, where $\sigma > 0$, which implies that

$$\mathbb{E}[f_k] = 0, \quad \mathbb{E}[f_k\, \overline{f_l}] = \sigma^2 \delta_{kl}, \quad k, l = 1, \cdots, N.$$

Instead of defining δ by (2.66), we can set

$$\delta = r_N\, d, \quad r_N = \sqrt{\mathbb{E}\left[\frac{\|Sf\|_M^2}{\|f\|^2_{L^2(\Gamma_0^R)}}\right]}. \tag{2.68}$$

Such definition of δ seems more realistic: it accounts for the fact that the measurements are discrete in practice and that the error between the noisy and the exact data could reasonably satisfy Assumption 2.8.1. It is clear that $r_N \leq \|S\|$, which implies that the resulting amplitude δ is smaller in the probabilistic approach than in the deterministic one. More precisely, the number r_N is given by

Theorem 2.8.2.

$$r_N = \sqrt{\frac{1}{2R}\sum_{k=1}^{N} \|S\chi_k^N\|_M^2},$$

where we recall that $\|\cdot\|_M$ *is the semi-norm in* $H^1(\Omega_0^R)$.

The proof of Theorem 2.8.2 is omitted since it is very similar to that of Theorem 2.9.3 given in the appendix for a slightly simpler geometry. Besides, for this simpler geometry, in the appendix we also prove that $r_N \to 0$ when $N \to +\infty$. Theorem 2.8.2 offers a practical way of computing r_N: it requires to solve the weak formulation (2.64) N times for $f = \chi_k^N$, $k = 1, \cdots, N$.

2.9 Numerical experiments

2.9.1 The tsunami identification

All the numerical computations of the article are performed with the help of XLiFE++ Library [92]. We choose $c = 20$, $g = 10$, $H = 1$, which corresponds to $\omega_g \sim 3.16$ and $\omega_a = 20$. In this section, $R = 4$.

The gravity model

Let us begin with the tsunami identification using the gravity model, having in mind that the data are generated by the complete model. Starting from a function χ on Γ_{-H}^R which represents the tsunami and that will be the reference function we try to retrieve, we compute the free surface perturbation η on Γ_0^R by solving the complete forward problem (2.13) in the domain Ω_0^R with the help of a Finite Element Method. The mesh size corresponds to a subdivision of the segment Γ_0^R into intervals of size $h = 0.01$, the finite elements consisting of classical $P2$ triangles. The infinite sums which define the Dirichlet-To-Neumann operators (2.12) are truncated to a finite number of terms which include all the propagating modes and at least 5 evanescent ones. Once the free surface perturbation η has been computed, we artificially compute some noisy data η^δ on Γ_0^R such that

$$\eta^\delta = \eta + \rho\, b,$$

where b is a function defined from its values at each discretization point of Γ_0^R. Such values are random complex numbers, the real and imaginary parts of which follow a uniform distribution $\mathcal{U}(-1, 1)$. The real $\rho > 0$ is calibrated such that

$$\frac{\|\eta^\delta - \eta\|_{L^2(\Gamma_0^R)}}{\|\eta\|_{L^2(\Gamma_0^R)}} = \sigma,$$

where σ is a prescribed relative noise amplitude. Multiplying the absolute amplitude of noise $\|\eta^\delta - \eta\|_{L^2(\Gamma_0^R)}$ by g/ω provides δ. The artificial noisy data η^δ constitutes the input data of the Tikhonov-Morozov inverse method described in section 2.6.1, which lastly enables us to compute the retrieved tsunami function χ^δ. In the absence of noise, the retrieved tsunami function is given by $\chi_\varepsilon = (i/\omega)\partial_z u_\varepsilon$, where u_ε is the solution to problem (2.48) for a very small ε. In order to discretize the problems (2.50) for all $n \in \mathbb{N}$, we again use a $P2$ Finite Element Method to compute the triple (u_n, λ_n, h_n) and a mesh having the same size as for the forward problem. In the Figure 2.1, we compare the retrieved tsunami function χ^δ and the actual tsunami function χ on Γ_0^R, for $\omega = 3$ and various relative amplitudes of noise σ on η, namely $\sigma = 0$ (exact data), $\sigma = 1\%$, $\sigma = 5\%$ and $\sigma = 10\%$. Note that the value of ω is chosen close to ω_g, that is the typical frequency of gravity waves, which explains the good reconstruction results for small σ.

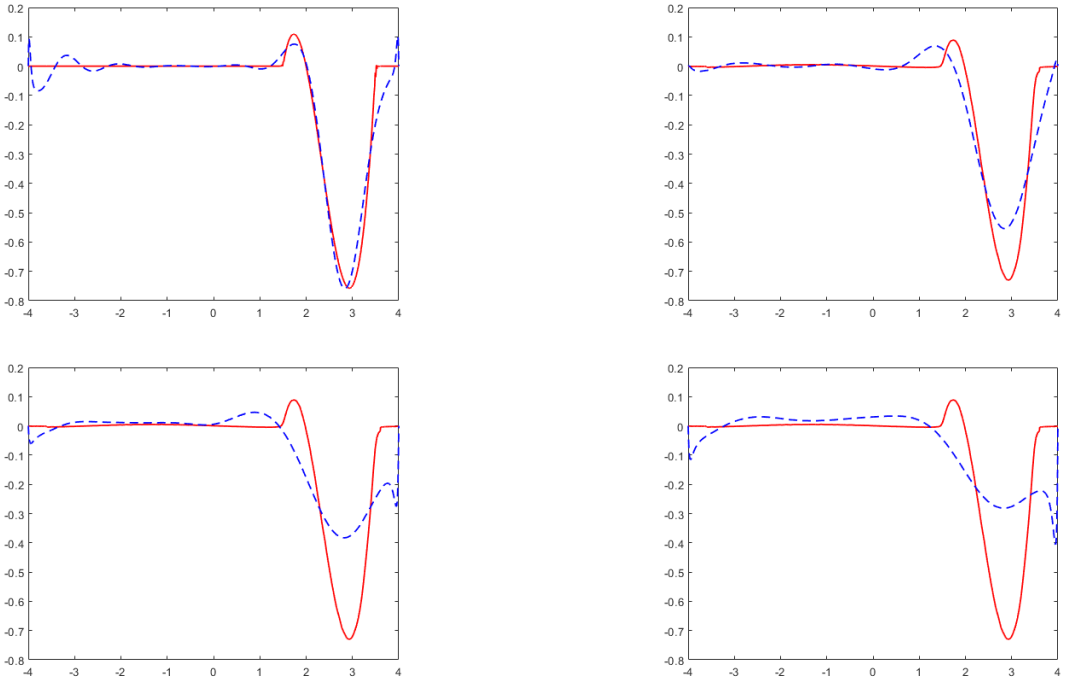


Figure 2.1: Gravity case for $\omega = 3$. Comparison between χ^δ (dashed line) and χ (continuous line) for various amplitudes of noise. Top left: $\sigma = 0$. Top right: $\sigma = 1\%$. Bottom left: $\sigma = 5\%$. Bottom right: $\sigma = 10\%$..

The acoustic model

We now consider the tsunami identification with the help of the acoustic model. In the Figure 2.2, the retrieved function χ^δ is compared to the exact function χ, for $\omega = 20$ and the different relative amplitudes of noise on η, namely $\sigma = 0$ (exact data), $\sigma = 1\%$, $\sigma = 5\%$ and $\sigma = 10\%$. The value of ω is close to ω_a, that is the typical frequency of acoustic waves. We use the same Finite Element Method and the same mesh size as in the gravity case. In the acoustic model, the amplitude of noise δ is obtained by multiplying $\|\eta^\delta - \eta\|_{L^2(\Gamma_0^R)}$ by ω and r_N, where r_N is given by (2.68) with $N = 100$.

Remark 2.9.1. *In the Figures 2.1 and 2.2, we notice that the error between the retrieved tsunami χ^δ and the exact one χ is quite large at the endpoints of Γ_{-H}^R, which correspond to the transition points between the part of the boundary $\partial\Omega^R$ on which we have data and the part Γ_{-H}^R on which we have no data. As shown by the analysis conducted in [21], we explain this phenomenon by the fact that the smoothness of the solution to the mixed formulation of the Tikhonov regularization is poorer at those transition points. Even if such error decreases when the mesh size is refined, we*

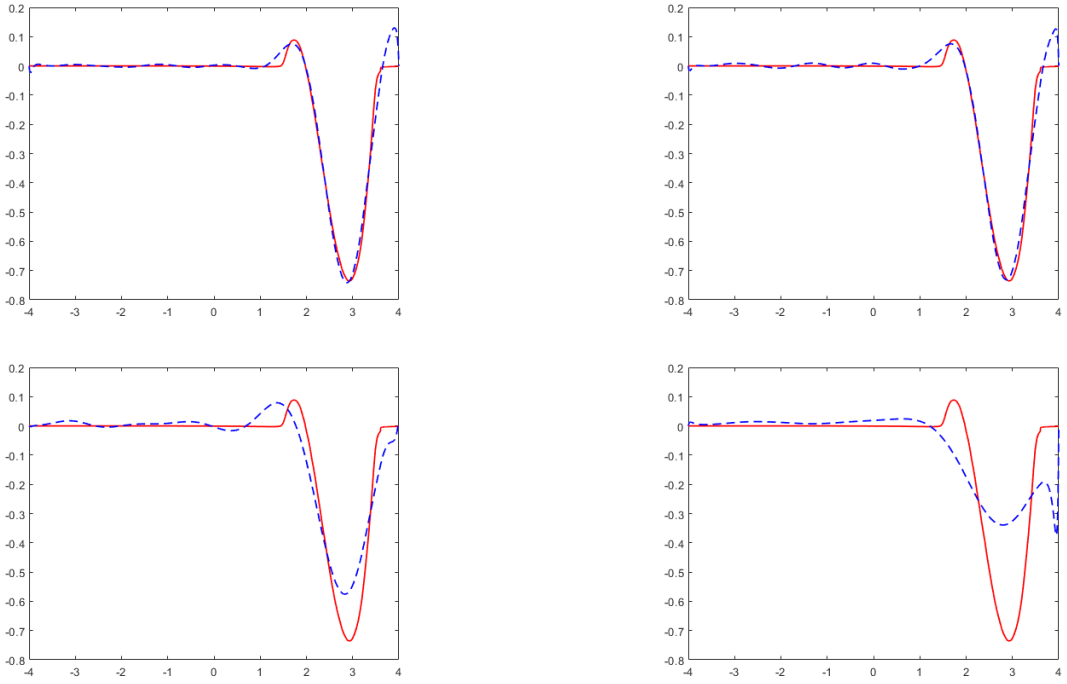


Figure 2.2: Acoustic case for $\omega = 20$. Comparison between χ^δ (dashed line) and χ (continuous line) for various amplitudes of noise. Top left: $\sigma = 0$. Top right: $\sigma = 1\%$. Bottom left: $\sigma = 5\%$. Bottom right: $\sigma = 10\%$..

have decided to apply an additional regularization to χ^δ *on* Γ^R_{-H}. *From the a priori assumption that* χ *is compactly supported in* $I = (-R, R)$, *we define* $\tilde{\chi}^\delta$ *as the solution of the one dimensional weak formulation: find* $\tilde{\chi}^\delta \in H^1_0(I)$ *such that*

$$\int_I \tilde{\chi}^\delta \overline{v}\, dx + \tilde{\varepsilon} \int_I (\tilde{\chi}^\delta)' \overline{v}'\, dx = \int_I \chi^\delta \overline{v}\, dx, \quad \forall v \in H^1_0(I).$$

Here, the regularization parameter is given by $\tilde{\varepsilon} = 0.001/k^2$, *where* $k = \omega/\sqrt{gH}$ *in the gravity case and* $k = \omega/c$ *in the acoustic case. Such regularization was already applied in Figures 2.1 and 2.2, otherwise the errors at the endpoints would have been worse. In what follows the regularized function* $\tilde{\chi}^\delta$ *will be simply denoted* χ^δ.

On the relevance of the Morozov choice

We wish now to test the relevance of the Morozov strategy to choose the regularization parameter ε as a function of the amplitude of noise δ. In the acoustic case, we have plotted in the Figure 2.3 the retrieved tsunami $\chi^\delta_\varepsilon = (i/\omega)\partial_z u^\delta_\varepsilon$ and the actual one χ, where u^δ_ε is the solution to the problem (2.35) corresponding to the acoustic case, by using a value of ε in (2.35) which seems relevant but differs from the Morozov choice, for two relative amplitudes of noise on η. Comparing the two bottom pictures of Figure 2.2 and the pictures of Figure 2.3, we realize that the Morozov choice for ε is not necessarily the best in terms of the error between the retrieved and the exact solution. To be more precise, in Figure 2.4 we have plotted the error between the retrieved tsunami χ^δ_ε and the actual one χ as a function of ε. In Figure 2.4 we also compare the deterministic and probabilistic choices of δ described in section 2.8, which are given by (2.66) and (2.68), respectively. Note that the L^2 error is limited to the subpart $\Gamma^{R'}_{-H}$ of Γ^R_{-H} with $R' = 0.9R$, since the reconstruction close to the endpoints of Γ^R_{-H} is bad and would prevent us from presenting a clear analysis (see Remark 2.9.1). Classically, we observe that in the absence of noise ($\delta = 0$) the error $\|\chi_\varepsilon - \chi\|_{L^2(\Gamma^{R'}_0)}$ increases with ε, except for very small values of ε, due to the finite element discretization. In the presence of

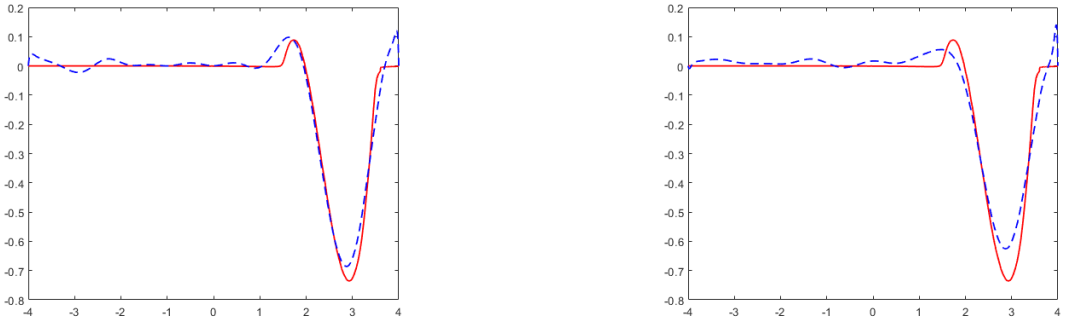


Figure 2.3: Acoustic case for $\omega = 20$. Comparison between χ_ε^δ (dashed line) and χ (continuous line) for two amplitudes of noise and well-tuned choices for ε (see Figure 2.4). Left: $\sigma = 5\%$. Right: $\sigma = 10\%$..

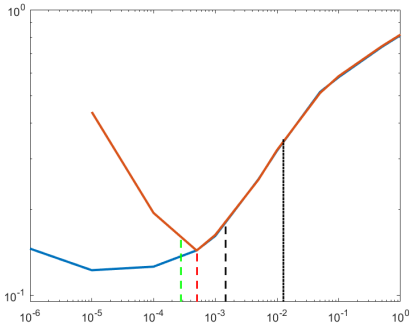


Figure 2.4: Error $\|\chi_\varepsilon^\delta - \chi\|_{L^2(\Gamma_0^{R'})}$ between the solution of problem (2.35) in the acoustic case and the exact solution, as a function of ε. The blue curve corresponds to uncontaminated data $\sigma = 0\%$ while the red one corresponds to noisy data with $\sigma = 5\%$. The vertical line on the right corresponds to the Morozov choice for δ following (2.66) while the other vertical dashed lines correspond to the Morozov choice for δ following (2.68) with $N = 100$, $N = 400$ and $N = 800$, from the right to the left..

noise, the error $\|\chi_\varepsilon^\delta - \chi\|_{L^2(\Gamma_0^{R'})}$ first decreases with ε and then increases with ε, which justifies to look for an optimal value of ε. We observe that the value of ε given by the Morozov choice is better by using the probabilistic strategy (2.68) than the deterministic strategy (2.66). More precisely the best value of N is 400, which roughly corresponds to the number of discretization points of Γ_0^R, that is number of artificial measurements, which is expected.

Influence of the chosen model

In order to compare the quality of the identification produced by the different models (gravity, acoustics and complete), in Figure 2.5 we have plotted the error between the retrieved tsunami function χ^δ using the Tikhonov-Morozov strategy and the actual one χ as a function of the frequency ω on the subpart $\Gamma_{-H}^{R'}$. When data are uncontaminated, the best model for the identification seems to be the complete one for all frequencies, which is expected since the data were produced by the complete model. The results obtained with the gravity model are good at a very low frequency but strongly deteriorates when ω increases, which was expected in view of Remark 2.3.1. In the presence of noisy data, the best model is again the complete one except at a very low frequency, for which the acoustic one is better. Note that the gravity model is competitive at frequencies around 10, strongly deteriorates at high frequencies, and more surprinsigly also at very low frequencies.

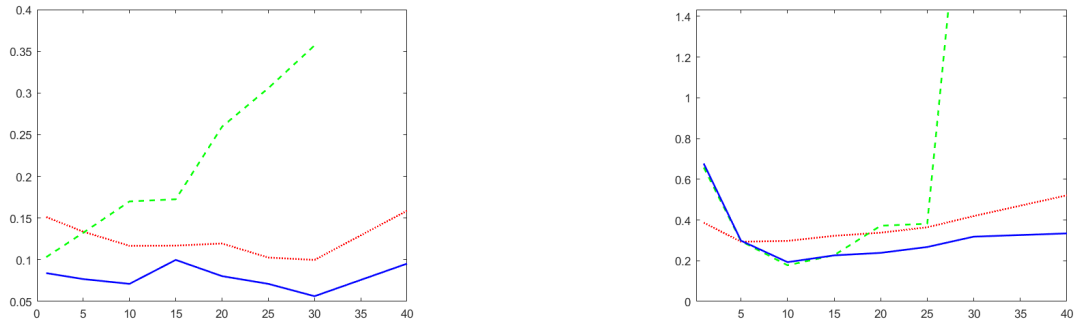


Figure 2.5: Error $\|\chi^\delta - \chi\|_{L^2(\Gamma^{R'}_{-H})}$ as a function of ω using different models: gravity (green), acoustic (red) and complete (blue). Left: uncontaminated data ($\sigma = 0\%$). Right: noisy data ($\sigma = 5\%$).

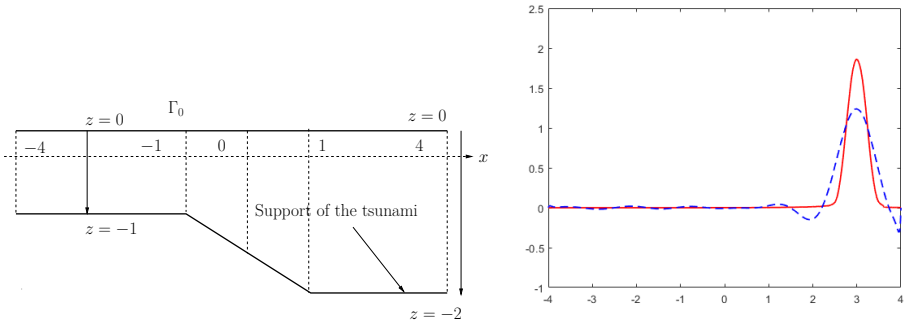


Figure 2.6: Non flat bottom, acoustic case for $\omega = 20$. Left: Geometry of the sea bottom. Right: Comparison between χ^δ (dashed line) and χ (continuous line) for $\sigma = 1\%$..

The case of a variable sea bottom

We complete this section by showing that our strategy to identify a tsunami can easily be extended to a non-flat sea bottom, contrary to the other methods mentioned in the introduction, which rely on integral transforms with respect to the horizontal variable. It suffices to use two different left and right Dirichlet-To-Neumann operators, those operators being defined with the help of the guided modes related to the left and right half-waveguides, respectively. This is illustrated by Figure 2.6 in the acoustic case, with $\omega = 20$ and $\sigma = 1\%$.

2.9.2 Bathymetry

In this section, $R = 1$ and $H = 0.5$, the other parameters being unchanged. Let us begin with the bathymetry problem using the gravity model, the data still being obtained with the complete model. Starting from a reference function θ which governs the profile $\Gamma^R_{-H+\theta}$ of the sea bottom, we compute the free surface perturbation η on Γ^R_0 by solving the forward problem (2.19) in the domain Ω^R_θ with the help of the same Finite Element Method as for the tsunami identification and the same mesh size. The source point N is chosen as $(-1.1, -0.45)$. Then we artificially perturb η to generate a noisy data η^δ on Γ^R_0 by using the same procedure as for the tsunami identification. Lastly we use the two steps strategy based on the approximate second-order boundary condition on Γ^R_{-H} and described in section 2.7.1. It is important to note that the data are obtained by solving a forward problem based on the true non-flat geometry $\Gamma^R_{-H+\theta}$ while the inverse problem is solved in a flat geometry Γ^R_{-H}. We finally obtain a function θ^δ on Γ^R_{-H} in the presence of noisy data. Note that in the second step, such θ^δ is obtained with the help of a finite element discretization of the one dimensional weak formulation (2.58). In order to discretize the spaces $L^2(I)$ and $V_0(I)$, we use the same finite dimensional space based on $P1$ finite elements. In Figure 2.7 we compare the

retrieved function θ^δ and the reference one θ, both with uncontaminated data ($\sigma = 0$) and with noisy data ($\sigma = 1\%$), for a frequency $\omega = 3$ compatible with the gravity model. Let us secondly

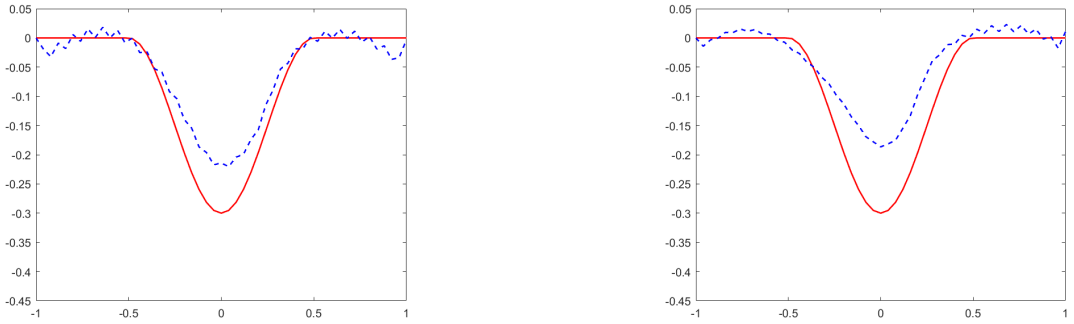


Figure 2.7: Gravity case for $\omega = 3$. Comparison between θ^δ (dashed line) and θ (continuous line). Left: uncontaminated data ($\sigma = 0$). Right: noisy data ($\sigma = 1\%$)..

consider the bathymetry problem using the acoustic model. The two step strategy is now described by section 2.7.2. In the second step, the computation of the function θ^δ consists in a finite element discretization of the weak formulation (2.62) which is the same as in the gravity case. In Figure 2.8 we compare the retrieved function θ^δ and the true one θ, both with uncontaminated data ($\sigma = 0$) and with noisy data ($\sigma = 1\%$), for frequency $\omega = 25$. Unsurprisingly, the identification results for

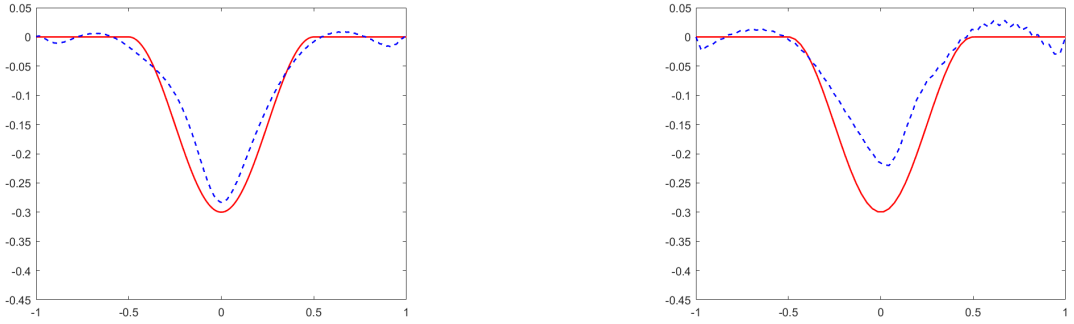


Figure 2.8: Acoustic case for $\omega = 25$. Comparison between θ^δ (dashed line) and θ (continuous line). Left: uncontaminated data ($\sigma = 0$). Right: noisy data ($\sigma = 1\%$)..

bathymetry are more sensitive with respect to σ than those obtained for the tsunami identification, which is due to the fact that the former is a geometric nonlinear inverse problem while the latter is linear. In order to compare the three different models with respect to the frequency ω, in Figure 2.9 we have plotted the error between the retrieved function θ^δ and the actual one as a function of the frequency ω. We observe that the results are better for the complete and acoustic models than for the gravity one whatever the frequency is, which can be explained by Remark 2.3.1. In the gravity case, and as for the tsunami identification problem, the results are satisfactory for small ω but strongly deteriorate when ω increases. This phenomenon also occurs for the complete/acoustic model, for ω larger than 30: this is probably due to the fact that the approximation of the true boundary condition at the bottom by a second-order impedance boundary condition becomes false when the wavelength $\lambda := 2\pi\, c/\omega$ is not large compared to the amplitude of θ. It is important to note that in the numerical results related to bathymetry, the assumption 2.7.1 seems to be satisfied.

From the numerical results obtained for the tsunami problem and for the bathymetry problem, we can conclude that these two inverse problems are approximately solvable using the gravity model only at a very low frequency (typically $\omega \simeq \sqrt{g/H}$) while they are approximately solvable at any (not too large) frequency for the acoustic/complete models. Note that the identification results

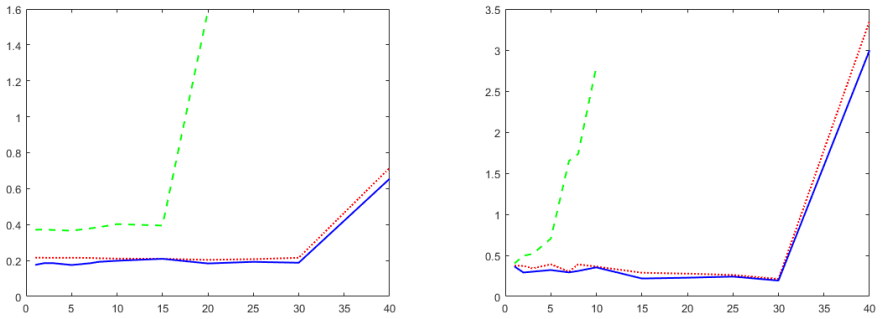


Figure 2.9: Error $\|\theta^\delta - \theta\|_{L^2(\Gamma^R_{-H})}$ as a function of ω using different models: gravity (green), acoustic (red) and complete (blue). Left: uncontaminated data ($\sigma = 0\%$). Right: noisy data ($\sigma = 1\%$)..

were slightly better using the complete model than the acoustic one, but the reader has to keep in mind that the artificial data were obtained by using the complete model in any case.

Appendix : a Laplace type problem with random Neumann boundary conditions

In order to study the limit of r_N defined by (2.68) when $N \to +\infty$, instead of considering the problem (2.65) we move to a slightly simpler problem in the two dimensional disc D centered at 0 and of radius R: for some boundary data f, find L such that

$$\left\{ \begin{array}{rcll} -\Delta L + L & = & 0 & \text{in } D, \\ \partial_\nu L & = & f & \text{on } \partial D. \end{array} \right. \tag{2.69}$$

The problem (2.69) is obviously well-posed in $H^1(D)$ for $f \in L^2(\partial D)$. We can hence define the operator

$$\left\{ \begin{array}{rcl} S : L^2(\partial D) & \to & H^1(D) \\ f & \mapsto & L, \end{array} \right. \tag{2.70}$$

such that L is the solution to problem (2.69), as well as the operator $T = \gamma \circ S : L^2(\partial D) \to L^2(\partial D)$, where $\gamma : H^1(D) \to L^2(\partial D)$ is the trace operator. A similar proposition as Proposition 2.8.1 is now obtained.

Proposition 2.9.1. *We have*

$$\|S\| = \sqrt{\frac{I_0(R)}{I_0'(R)}},$$

where for $n \in \mathbb{N}$, the $I_n(r)$ are the bounded solutions (up to a multiplicative constant) to the modified Bessel equations

$$r^2 \frac{d^2 u_n}{dr^2} + r \frac{du_n}{dr} - (n^2 + r^2) u_n = 0, \quad r \in (0, R).$$

Remark 2.9.2. *A comprehensive analysis of the functions $I_n(r)$ may be found in [1].*

Proof of Proposition 2.9.1. By using the same arguments as in the proof of Proposition 2.8.1, we show that the operator T is self-adjoint, compact, positive and injective. The spectrum of T is formed by a sequence of non increasing eigenvalues $\rho_n > 0$, $n \in \mathbb{N}$, such that $\rho_n \to 0$ when $n \to +\infty$, the corresponding eigenfunctions e_n constituting a complete orthonormal basis of $L^2(\partial D)$. As in the proof of Proposition 2.8.1, we also have $\|S\|^2 = \rho_0$. An easy computation shows that

$$\rho_n = \frac{I_n(R)}{I_n'(R)}, \quad e_n(\theta) = \frac{1}{\sqrt{2\pi R}} e^{in\theta}, \quad n \in \mathbb{N},$$

which implies the stated result. □

Remark 2.9.3. *We observe that the operator T is a Hilbert-Schmidt operator (see for example [31] for a definition), since*

$$\sum_{n=0}^{+\infty} \|Te_n\|^2_{L^2(\partial D)} = \sum_{n=0}^{+\infty} \rho_n^2 < +\infty.$$

Indeed, by using the asymptotics of the modified Bessel function given by [1], we have

$$\rho_n \underset{n\to+\infty}{\sim} \frac{R}{n},$$

which enables us to conclude.

In the following theorem, we prove that the problem (2.69) is also well-posed in $L^2(D)$ for $f \in H^{-3/2}(\partial D)$.

Theorem 2.9.1. *For $f \in H^{-3/2}(\partial D)$, the problem (2.69) has a unique solution in $L^2(D)$.*

Proof. Let us introduce the space $V := \{v \in H^2(D),\ \partial_\nu v|_{\partial D} = 0\}$ and the operator

$$\left\{ \begin{array}{rcl} A : V & \to & L^2(D) \\ v & \mapsto & -\Delta v + v. \end{array} \right.$$

From a standard regularity result for elliptic equations (see for example [31]), the continuous operator A is an isomorphism. This implies that the adjoint operator $A^* : L^2(D) \to V^*$, where V^* is the dual space of V, is also an isomorphism (see again [31]). This means that for all $L^* \in V^*$, there exists a unique $u \in L^2(D)$ such that $A^*u = L^*$. Such u satisfies

$$\langle A^*u, v\rangle_{V^*,V} = (u, Av)_{L^2(D)} = \langle L^*, v\rangle_{V^*,V}, \quad \forall v \in V.$$

If we choose $L^* \in V^*$ such that

$$\langle L^*, v\rangle_{V^*,V} = \langle f, v\rangle_{H^{-3/2}(\partial D),H^{3/2}(\partial D)}, \quad \forall v \in V,$$

we conclude that for $f \in H^{-3/2}(\partial D)$, there exists a unique $u \in L^2(D)$ such that

$$\int_D u(-\Delta v + v)\, dx = \langle f, v\rangle_{H^{-3/2}(\partial D),H^{3/2}(\partial D)}, \quad \forall v \in V. \tag{2.71}$$

Let us check that such function u satisfies problem (2.69). Taking $v = \varphi \in C_0^\infty(D)$, we first obtain that $-\Delta u + u = 0$ in $\mathcal{D}'(D)$. Hence $u, \Delta u \in L^2(D)$, so that for all $v \in V$, we have the Green formula (see [90]):

$$\begin{array}{rcl} \displaystyle\int_D u(-\Delta v + v)\, dx & = & \displaystyle\int_D (-\Delta u + u)v\, dx \\ & - & \langle u, \partial_\nu v\rangle_{H^{-1/2}(\partial D),H^{1/2}(\partial D)} + \langle \partial_\nu u, v\rangle_{H^{-3/2}(\partial D),H^{3/2}(\partial D)}. \end{array}$$

In view of (2.71), we obtain that

$$\langle \partial_\nu u - f, v\rangle_{H^{-3/2}(\partial D),H^{3/2}(\partial D)} = 0, \quad \forall v \in V.$$

We conclude that $\partial_\nu u = f$ in $H^{-3/2}(\partial D)$. □

For all $y \in \partial D$, since dimension is 2 we have $\delta_y \in H^{-3/2}(\partial D)$, so that for $f = \delta_y$, problem (2.69) has a unique solution in $L^2(D)$, which is denoted $G_y = G(\cdot, y)$.

The operator $T : L^2(\partial D) \to L^2(\partial D)$ can then be seen as an integral operator, defined for all $f \in L^2(\partial D)$, by

$$(Tf)(x) = \int_{\partial D} G(x, y) f(y)\, ds, \quad \forall x \in \partial D. \tag{2.72}$$

Remark 2.9.4. *From Remark 2.9.3, since T is a Hilbert-Schmidt operator, we have that $G \in L^2(\partial D \times \partial D)$ (see for example [31]).*

Like in section 2.8, let us assume that f is given on ∂D by

$$f = \sum_{k=1}^{N} f_k \, \chi_k^N. \tag{2.73}$$

For $k = 1, \cdots, N$, the functions χ_k^N are indicator functions of intervals having the same length $2\pi R/N$, which do not overlap and the union of which coincides with ∂D. The random variables f_k, $k = 1, \cdots, N$, satisfy the following discrete Gaussian white noise assumptions:

Assumption 2.9.2. The f_k, $k = 1, \cdots, N$, are independent and identically distributed real random variables following the normal distribution $\mathcal{N}(0, \sigma^2)$, where $\sigma > 0$, which implies

$$\mathbb{E}[f_k] = 0, \quad \mathbb{E}[f_k \, f_l] = \sigma^2 \delta_{kl}, \quad k, l = 1, \cdots, N.$$

We again introduce

$$r_N = \sqrt{\mathbb{E}\left[\frac{\|Sf\|^2_{H^1(D)}}{\|f\|^2_{L^2(\partial D)}}\right]}. \tag{2.74}$$

We now derive a simple expression for r_N.

Theorem 2.9.3. *The number r_N is given by*

$$r_N = \sqrt{\frac{N}{2\pi R}} \, \|S\chi_1^N\|_{H^1(D)}.$$

Proof. The weak formulation of problem (2.69) implies that

$$\|Sf\|^2_{H^1(D)} = \int_{\partial D} f(Tf) \, ds, \quad \forall f \in L^2(\partial D).$$

In view of (2.72), we deduce the concrete expression

$$\|Sf\|^2_{H^1(D)} = \int_{\partial D} \int_{\partial D} G(x, y) f(x) f(y) \, ds(x) ds(y).$$

The decomposition (2.73) yields

$$\|Sf\|^2_{H^1(D)} = (G^N X, X)_{\mathbb{R}^N},$$

where the vector $X \in \mathbb{R}^N$ and the matrix $G^N \in \mathbb{R}^{N \times N}$ are defined by

$$X_k := f_k, \quad G^N_{kl} = \int_{\partial D} \int_{\partial D} G(x, y) \chi_k^N(x) \chi_l^N(y) \, ds(x) ds(y), \quad k, l = 1, \cdots, N.$$

On the other hand,

$$\|f\|^2_{L^2(\partial D)} = \sum_{k,l=1}^{N} f_k \, f_l \int_{\partial D} \chi_k^N \, \chi_l^N \, ds.$$

From the definition of the χ_k^N, we obtain

$$\|f\|^2_{L^2(\partial D)} = \frac{2\pi R}{N} \sum_{k=1}^{N} f_k^2 = \frac{2\pi R}{N} \|X\|^2_{\mathbb{R}^N}.$$

We conclude that

$$r_N^2 = \frac{N}{2\pi R} \mathbb{E}\left[\frac{(G^N X, X)_{\mathbb{R}^N}}{\|X\|^2_{\mathbb{R}^N}}\right].$$

In view of Assumption 2.9.2 and Theorem 1 in [68], we have

$$\mathbb{E}\left[\frac{(G^N X, X)_{\mathbb{R}^N}}{\|X\|^2_{\mathbb{R}^N}}\right] = \frac{\mathbb{E}\left[(G^N X, X)_{\mathbb{R}^N}\right]}{\mathbb{E}\left[\|X\|^2_{\mathbb{R}^N}\right]}.$$

We easily derive, again by using Assumption 2.9.2, that

$$\mathbb{E}\left[\|X\|^2_{\mathbb{R}^N}\right] = N\sigma^2$$

and

$$\mathbb{E}\left[(G^N X, X)_{\mathbb{R}^N}\right] = \sum_{k,l=1}^{N} G^N_{kl}\,\mathbb{E}\left[f_k\, f_l\right] = \sigma^2 \sum_{k=1}^{N} G^N_{kk} = N\sigma^2\, G^N_{11}.$$

In the last equality, we have used the symetry of ∂D, which implies that G^N_{kk} does not depend on $k = 1, \cdots, N$. We conclude that

$$r_N^2 = \frac{N}{2\pi R} G^N_{11},$$

with

$$G^N_{11} = \int_{\partial D}\int_{\partial D} G(x,y)\chi^N_1(x)\chi^N_1(y)\, ds(x)ds(y) = \|S\chi^N_1\|^2_{H^1(D)},$$

which completes the proof. □

An interesting question concerns the behaviour of r_N when $N \to +\infty$. In order to study this limit, we need to specify the singularity of the kernel G (Remark 2.9.4 is not sufficient to conclude).

Lemma 2.9.1. *The function G satisfies $G \in L^p(\partial D \times \partial D)$, for all $p \geq 1$.*

Proof. Let us introduce the function $(x,y) \mapsto G^\infty(x,y) := -\ln|x-y|/\pi$ and the function $x \mapsto G^\infty_y(x) := G^\infty(x,y)$ for a given y. The latter satisfies $\Delta G^\infty_y = 0$ for $x \neq y$. We clearly have that $G^\infty \in L^p(\partial D \times \partial D)$ for all $p \geq 1$. Then it is enough to prove that $G - G^\infty$ is bounded in $\partial D \times \partial D$, which constitutes the remainder of the proof. Let us pick some $y \in \partial D$. From (2.71), the function $G_y \in L^2(D)$ satisfies:

$$\int_D G_y(-\Delta v + v)\, dx = v(y), \quad \forall v \in V. \tag{2.75}$$

On the other hand, by denoting $D_\varepsilon := D \setminus \overline{D(y,\varepsilon)}$, where $D(y,\varepsilon)$ is the disc centered at y and of radius $\varepsilon > 0$,

$$\int_D G^\infty_y(-\Delta v + v)\, dx = \lim_{\varepsilon\to 0}\int_{D_\varepsilon} G^\infty_y(-\Delta v + v)\, dx$$

$$= \lim_{\varepsilon\to 0}\left\{\int_{D_\varepsilon} v(-\Delta G^\infty_y + G^\infty_y)\, dx - \int_{\partial D_\varepsilon} G^\infty_y\, \partial_\nu v\, ds + \int_{\partial D_\varepsilon} \partial_\nu G^\infty_y\, v\, ds\right\},$$

where the unit normal ν is oriented outside D_ε. We have $\Delta G^\infty_y = 0$ in D_ε, and

$$\lim_{\varepsilon\to 0}\int_{D_\varepsilon} v(-\Delta G^\infty_y + G^\infty_y)\, dx = \int_D G^\infty_y\, v\, dx.$$

We focus now on the limit when $\varepsilon \to 0$ of the two boundary terms. Since $\partial_\nu v = 0$ on ∂D, we have

$$\int_{\partial D_\varepsilon} G^\infty_y\, \partial_\nu v\, ds = \int_{\partial D_\varepsilon \cap \partial D(y,\varepsilon)} G^\infty_y\, \partial_\nu v\, ds = -\frac{\ln\varepsilon}{\pi}\int_{\partial D_\varepsilon \cap \partial D(y,\varepsilon)} \partial_\nu v\, ds.$$

But

$$\int_{\partial D_\varepsilon \cap \partial D(y,\varepsilon)} \partial_\nu v\, ds = \int_{\partial(D\cap D(y,\varepsilon))} \partial_\nu v\, ds = \int_{D\cap D(y,\varepsilon)} \Delta v\, dx.$$

By the Cauchy-Schwarz inequality, it follows that

$$\left|\int_{\partial D_\varepsilon \cap \partial D(y,\varepsilon)} \partial_\nu v\, ds\right| \leq \|\Delta v\|_{L^2(D\cap D(y,\varepsilon))}\sqrt{\int_{D\cap D(y,\varepsilon)} 1\, dx},$$

with

$$\sqrt{\int_{D\cap D(y,\varepsilon)} 1\,dx} \le \sqrt{\frac{\pi}{2}}\,\varepsilon.$$

Gathering the previous estimates, we conclude that there exists a constant $c>0$ such that

$$\left|\int_{\partial D_\varepsilon} G_y^\infty\,\partial_\nu v\,ds\right| \le c\,\varepsilon|\ln\varepsilon|\|\Delta v\|_{L^2(D)} \underset{\varepsilon\to 0}{\to} 0.$$

Concerning the second boundary term, we have

$$\int_{\partial D_\varepsilon} \partial_\nu G_y^\infty\,v\,ds = \int_{\partial D_\varepsilon\cap\partial D} \partial_\nu G_y^\infty\,v\,ds + \int_{\partial D_\varepsilon\cap\partial D(y,\varepsilon)} \partial_\nu G_y^\infty\,v\,ds.$$

Considering the first term above, we observe that the function g_y defined by

$$g_y(x) := -\partial_\nu G_y^\infty(x) = \frac{1}{\pi}\frac{1}{|x-y|^2}(x-y)\cdot\nu_x, \quad \forall x\in\partial D\setminus\{y\},$$

is continuous on ∂D. Indeed, let us assume that the boundary ∂D is locally characterized by the equation $h(x)=0$, where h is an infinitely smooth function. For $x\in\partial D$ and x is sufficiently close to y,

$$h(y) = h(x) + \nabla h(x)\cdot(y-x) + \frac{1}{2}(y-x)^T\cdot\nabla^2 h(x)\cdot(y-x) + O(|x-y|^3),$$

where ∇h and $\nabla^2 h$ are the gradient and the Hessien of h, respectively, with $\nabla h(x)\neq 0$ for x close to y. Since $h(x)=h(y)=0$ and $\nu_x = \nabla h(x)/|\nabla h(x)|$, this yields

$$g_y(x) = \frac{1}{2\pi}\frac{(x-y)^T}{|x-y|}\cdot\frac{\nabla^2 h(x)}{|\nabla h(x)|}\cdot\frac{(x-y)}{|x-y|} + O(|x-y|) \underset{x\to y}{\to} \frac{1}{2\pi}\tau_y^T\cdot\frac{\nabla^2 h(y)}{|\nabla h(y)|}\cdot\tau_y,$$

where τ_y is the tangential vector to ∂D at point y. This proves that $g_y\in C^0(\partial D)$. Finally,

$$\int_{\partial D_\varepsilon\cap\partial D} \partial_\nu G_y^\infty\,v\,ds \underset{\varepsilon\to 0}{\to} -\int_{\partial D} g_y\,v\,ds.$$

Considering the second term, we have

$$g_y(x) = \frac{1}{\pi}\frac{1}{|x-y|^2}(x-y)\cdot\nu_x = -\frac{1}{\pi\varepsilon}, \quad \forall x\in\partial D_\varepsilon\cap\partial D(y,\varepsilon),$$

hence

$$\int_{\partial D_\varepsilon\cap\partial D(y,\varepsilon)} \partial_\nu G_y^\infty\,v\,ds \underset{\varepsilon\to 0}{\to} v(y)\frac{1}{\pi\varepsilon}\pi\varepsilon = v(y).$$

As a conclusion, the function G_y^∞ satisfies the weak formulation

$$\int_D G_y^\infty(-\Delta v + v)\,dx = \int_D G_y^\infty\,v\,dx - \int_{\partial D} g_y\,v\,ds + v(y), \quad \forall v\in V. \tag{2.76}$$

Subtracting the weak formulations (2.75) and (2.76), we obtain that the function $r_y := G_y - G_y^\infty \in L^2(D)$ satisfies the weak formulation:

$$\int_D r_y(-\Delta v + v)\,dx = -\int_D G_y^\infty\,v\,dx + \int_{\partial D} g_y\,v\,ds, \quad \forall v\in V,$$

then r_y satisfies the strong problem:

$$\left\{\begin{array}{rcll} -\Delta r_y + r_y &=& -G_y^\infty & \text{in } D,\\ \partial_\nu r_y &=& g_y & \text{on } \partial D.\end{array}\right.$$

Since $G_y^\infty\in L^2(D)$ and $g_y\in L^2(\partial D)$, by a standard regularity result we get that $r_y\in H^{3/2}(D)$, then $r_y\in C^0(\overline{D})$, in particular $r_y|_{\partial D}\in C^0(\partial D)$. Hence r_y is bounded on ∂D for all $y\in\partial D$ and the bound $\|r_y\|_{L^\infty(\partial D)}$ does not depend on y by rotation invariance. Finally, $G-G^\infty$ is bounded on $\partial D\times\partial D$, which completes the proof. □

Now let us specify the behaviour of r_N when the number N of measurements becomes large.

Theorem 2.9.4. *We have*

$$\lim_{N\to+\infty} N^{1/2-\varepsilon}\, r_N = 0, \quad \forall \varepsilon > 0.$$

Proof. We recall that

$$r_N^2 = \frac{N}{2\pi R}\int_{\partial D}\int_{\partial D} G(x,y)\chi_1^N(x)\chi_1^N(y)\, ds(x)ds(y).$$

From Lemma 2.9.1, we have in particular that $G \in L^q(\partial D \times \partial D)$, for all $q > 2$. Using the Hölder inequality, with p defined by $1/p + 1/q = 1$, we have

$$\begin{aligned} r_N^2 &\leq \frac{N}{2\pi R}\left(\int_{\partial D\times\partial D} |G(x,y)|^q\, ds(x)ds(y)\right)^{1/q} \\ &\times \left(\int_{\partial D\times\partial D} |\chi_1^N(x)\chi_1^N(y)|^p\, ds(x)ds(y)\right)^{1/p} \\ &= \frac{N}{2\pi R}\|G\|_{L^q(\partial D\times\partial D)}\left(\frac{2\pi R}{N}\times\frac{2\pi R}{N}\right)^{1/p} \\ &= \|G\|_{L^q(\partial D\times\partial D)}\left(\frac{2\pi R}{N}\right)^{1-2/q}. \end{aligned}$$

The fact that $q > 2$ enables us to conclude. □

In Figure 2.10 below, we numerically illustrate both Theorem 2.9.3 and Theorem 2.9.4. More precisely, in the left part of Figure 2.10 we compare the value of r_N defined by (2.74), which is computed with the help of Theorem 2.9.3, and its Monte-Carlo approximation given by the empirical mean

$$(r_N^M)^2 = \frac{1}{M}\sum_{m=1}^{M} \frac{\|Sf^m\|^2_{H^1(D)}}{\|f^m\|^2_{L^2(\partial D)}}, \tag{2.77}$$

where the f^m, $m = 1, \cdots, M$, form a sample of M generated functions f which satisfy Assumption 2.9.2. We can check the convergence of r_N^M to r_N for large M. In the right part of Figure 2.10, we have plotted the values of r_N with respect to N in a log-log scale, which exhibits a negative slope of 0.39 approximately. Such slope is quite close to the optimal value given by Theorem 2.9.4. Besides, the value of $\|S\|$, which majorates r_N for all N according to (2.74) and is given by Proposition 2.9.1, is such that $\log\|S\| = 0.175$. The fact that $r_N \to 0$ when $N \to +\infty$ means in some sense

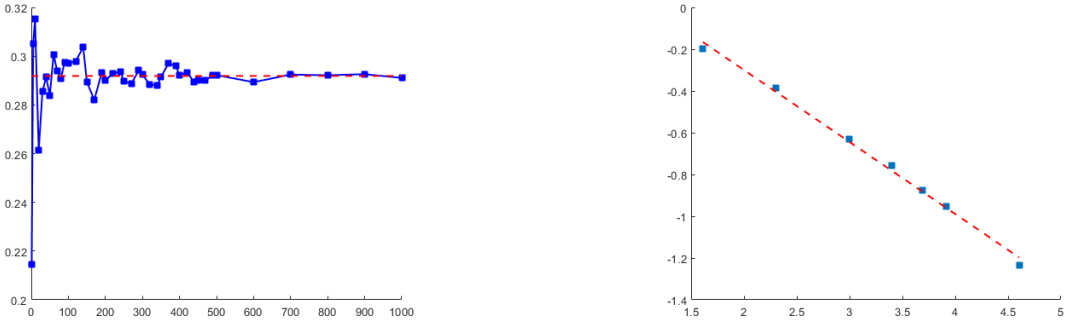


Figure 2.10: Left: Convergence of the Monte-Carlo estimates r_N^M to r_N when $M \to +\infty$. Right: Values of $\log r_N$ with respect to $\log N$ ($\log\|S\| = 0.175$)..

that noise is vanishing when the number of measurements tends to infinity. We note however that in practice N is the number of measurements, which is finite and known.

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Chapter 3

One-shot mixed space-time formulation

In this chapter and in the followings, we will study the problem in a simpler set up where we only consider flat bottom ocean. In other words, we will consider that $\theta = 0$. However, it is important to notice that this simplification does not lose any generality as the approaches detailed in the following would perfectly work with $\theta \neq 0$. We only made this choice to simplify the redaction and to consider simpler numerical cases.

This chapter generalises in a space-time framework the study conducted in the previous chapter. Here, we only focus on the tsunami problem. A big focus is also made on the rigorous study of the direct problems both complete and asymptotic. We describe the construction of strong and variational solutions for both of the models. Through the semigroup framework, we construct strong solutions for the complete and asymptotic models. In the case of the complete model, we construct second order variational solutions as limits of the strong solutions previously constructed. In Appendix A, the construction of first order variational solutions is also made. This chapter contains a submitted article, realized in collaboration with Laurent Bourgeois and Philippe Moireau.

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An inverse tsunami problem in the time domain: a well-posedness analysis of the forward problem and an inversion strategy based on a mixed formulation of the Tikhonov regularization

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Abstract

This contribution concerns an inverse problem related to a tsunami in the ocean, the tsunami being caused by a submarine earthquake. Considering the very beginning of the phenomenon, a simple linear model incorporating both gravity and acoustic waves is proposed. The main objective is to develop a strategy to solve the inverse problem of retrieving the bottom displacement from the induced free surface perturbation. Such strategy is based on a mixed formulation of the Tikhonov regularization in the space/time domain, the regularization parameter being determined by using the Morozov principle by means of duality in optimization. Some numerical experiments in 2D, which rely on a tensorized finite element method, show that our strategy is effective. A secondary objective is to prove existence and uniqueness of both strong and variational solutions to the forward problem.

3.1 Introduction

This paper concerns the identification of the seabed displacement from the measurement of the free surface perturbation of the ocean. A possible application is the following. We assume that a submarine earthquake produces a tsunami. If some specific device is likely to measure the free surface induced perturbation, then it is possible to quantify the strength of the earthquake in order to anticipate the features of the tsunami, in particular once it reaches the coast. The fluctuations of the sea level can in practice be measured by arrays of DART buoys, but the cost of such device implies that they are not densely enough distributed on the surface of the ocean. In contrast, we expect an increasing role of satellite altimetry in the future, such technique being for instance discussed in [94, 102, 66]. Satellite measurements cover a quite large surface of the sea surface and therefore could provide the kind of data we need in the present article.

Identifying the seabed displacement from the free surface measurement can be viewed as an ill-posed inverse problem. In this paper, such a problem is governed by a simple scalar linear oceanography model, more precisely a partial differential equation that is satisfied by the velocity potential and which takes both acoustic and gravity waves into account. A rigorous derivation of such linear model from the Euler equation is described in [51]. Our linear model can be considered as satisfactory only during a short period of time after the submarine earthquake. More complicated nonlinear models would have to be used to describe the propagation of the tsunami. In particular, some shallow water models are popular for describing the propagation of tsunamis, the corresponding forward problems being analyzed in [30]. An inverse problem using the Boussinesq-Peregrine model is for example analyzed in [85]. It should be noted that a similar inverse problem as ours is successfully solved in [42] with the help of real data obtained experimentally, which constitutes a strong encouragement. Throughout the article, in addition to our so-called complete model, which enables one to consider both acoustic and gravity effects, we also consider the asymptotic model we obtain when the celerity of the acoustic waves tends to $+\infty$, which boils down to the well-known linear water wave model.

As it seems that no existence/uniqueness theorem for the forward problem governed by our complete model is proposed in the literature, our paper starts with a proof of such result, both for strong solutions and for variational ones. Our contribution to the inverse problem is based on a mixed formulation of the Tikhonov regularization, which leads to a weak formulation using a simple Finite Element Method at the cost of the introduction of an additional adjoint variable. This method goes back to [16], was generalized in [29] and carried out in many situations. The Tikhonov regularization involves a small parameter that has to be calibrated as a function of the amplitude of noise on the data. In order to calibrate such small parameter, a Morozov strategy was introduced in [18] and generalized in [24]. Such strategy consists in using duality in optimization. More precisely, by solving a dual minimization problem, we immediately compute the Morozov regularization parameter as well as the corresponding Tikhonov solution. It happens that the cost function to minimize is not quadratic, which may imply slow convergence. This is why in [25] we used the strategy introduced in [48], which consists in replacing the non-quadratic minimization problem by a sequence of quadratic ones, the optimality condition of each of these quadratic minimization problems being equivalent to a simple weak formulation. In [25], as a first attempt we developed the above strategy for our tsunami inverse problem in the harmonic regime, which is a simpler case where the ill-posedness is the same. However, addressing the inverse tsunami problem in the frequency domain may seem a bit unrealistic. This is why in this paper we extend [25] to the time domain case.

When it comes to the discretization, we use tensorized finite elements, which are obtained by multiplying the basis functions of the spatial finite element space by the basis functions of the time finite element space. A drawback of our method is that we have to invert a large system involving both the velocity potential and the adjoint variable, which both depend on the spatial and the time variables. Even if we restrict to a two dimensional space, the matrix to invert may be too large, which encouraged us to introduce an iterative Successive OverRelaxation (SOR) method [105] in order to decrease the size of the matrices to invert. In [27], another strategy is proposed to solve the same tsunami inverse problem. It relies on a more classical optimal control problem in the sense of [88]. While in the present article the seabed displacement is obtained as a by-product of the approximated retrieved velocity potential, it is the control variable of the optimal control problem in [27]. In the strategy of [27], again we have to cope with the velocity potential and an adjoint variable to compute the gradient of the cost function, but these two variables satisfy a Cauchy problem, based on an initial condition for the former variable and on a final condition for the adjoint variable, these two Cauchy problems being solved alternatively. This enables us to invert smaller matrices than those involved in the present article. However, in [27] we did not develop a Morozov strategy to choose the regularization parameter, which would be more involved than in the present paper. In addition, in [27] we had to introduce a parabolic regularization of the model in order to properly define the adjoint variable. Besides, we point out that an alternative method to our mixed formulation of the Tikhonov regularization is the one proposed for example in [35, 38, 34, 37]: rather than introducing a regularization parameter at the continuous level, the authors introduce some stabilizers at the discrete level in order to ensure the well-posedness of the

discrete problem, the mesh size being considered as a regularization parameter.

The paper is organized as follows. In Section 2 we introduce the tsunami forward problem and the corresponding inverse problem we address in this contribution, both for the complete and the asymptotic models. Some existence/uniqueness results for the forward problem governed by the complete model are proved in Section 3, while the case of the asymptotic model is considered in Appendix A. In Section 4, we introduce an abstract framework for the inverse problem, which essentially gathers results already proved in [24] and in [25]. Our tsunami inverse problems are then recast in such abstract setting in Section 5, so that the convergence results are stated directly from the abstract case. Section 6 is dedicated to the SOR method used to invert our discretized systems. Lastly, numerical experiments are presented in Section 7 in order to prove the feasibility of our approach.

3.2 Description of the problem

We model the ocean by a two dimensional waveguide $\Omega = \mathbb{R} \times (-H, 0)$ for $H > 0$. The boundary $\Gamma_0 = \mathbb{R} \times \{0\}$ coincides with the free surface at rest, while $\Gamma_{-H} = \mathbb{R} \times \{-H\}$ coincides with the straight bottom of the ocean. Let us denote (x, z) the horizontal and vertical coordinates of any point $X \in \Omega$ and $t \in (0, T)$ the time variable, with $T > 0$. The tsunami problem we consider is the following: find the velocity potential $\varphi(X, t)$ such that

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2\varphi - \Delta\varphi & = & 0 & \text{in } \Omega \times (0,T), \\ \partial_t^2\varphi + g\,\partial_z\varphi & = & 0 & \text{on } \Gamma_0 \times (0,T), \\ -\partial_z\varphi & = & \chi & \text{on } \Gamma_{-H} \times (0,T), \\ \varphi = \partial_t\varphi & = & 0 & \text{on } \Omega \times \{0\}, \end{array}\right. \tag{3.1}$$

where $\chi(x,t) = -\partial_t\zeta(x,t)$, and ζ is the upward displacement of the sea bottom in the z direction, c the celerity of the acoustic waves and g the gravitational acceleration. We assume that $\chi = 0$ at negative times. Importantly, the boundary condition on Γ_0 is obtained by merging the dynamical condition

$$\partial_t\varphi + g\,s = 0 \text{ on } \Gamma_0 \times (0,T) \tag{3.2}$$

and the kinematical condition

$$\partial_z\varphi - \partial_t s = 0 \text{ on } \Gamma_0 \times (0,T) \tag{3.3}$$

on Γ_0, where $s(x,t)$ is the free surface perturbation in the z direction. Though very simple, the system (3.1) is often used in the context of tsunami prediction, as can be seen for example in [100, 103, 108], since such model enables us to take both gravity and acoustic waves into account. Passing to the limit $c \to +\infty$ in the system (3.1), the first equation degenerates into the problem: find the velocity potential $\varphi(X, t)$ such that

$$\left\{\begin{array}{rcll} \Delta\varphi & = & 0 & \text{in } \Omega \times (0,T), \\ \partial_t^2\varphi + g\,\partial_z\varphi & = & 0 & \text{on } \Gamma_0 \times (0,T), \\ -\partial_z\varphi & = & \chi & \text{on } \Gamma_{-H} \times (0,T), \\ \varphi = \partial_t\varphi & = & 0 & \text{on } \Gamma_0 \times \{0\}, \end{array}\right. \tag{3.4}$$

which correspond to the classical linear water wave model, the equations (3.2) and (3.3) remaining valid. In what follows, the system (3.1) with a finite value of c will be referred to as the complete model, while the system (3.4) with $c = +\infty$ will be referred to as the asymptotic model. Note that contrary to (3.1), it suffices to impose the initial condition on φ on Γ_0 only.

Let us now introduce the forward and the inverse problems related to the systems (3.1) and (3.4). Computing the free surface perturbation s knowing the sea bottom velocity χ will be considered as the forward problem, computing the sea bottom velocity χ from the free surface perturbation s as the inverse problem. There exist functional frameworks which guarantee well-posedness of the two above forward problems. For example, in the complete model, we have the two following results, which are proved in the next section. The first one concerns a strong solution using the theory of Hille-Yosida when the data χ lies in $H^2(0,T; H^{-1/2}(\Gamma_{-H}))$.

Theorem 3.2.1. *Let us assume that $\chi \in H^2(0,T;H^{-1/2}(\Gamma_{-H}))$ with $\chi(\cdot,0)=0$ on Γ_{-H}. There exists a unique strong solution φ to Problem (3.1), that is*

$$\varphi \in C^2([0,T];L^2(\Omega)) \cap C^1([0,T];H^1(\Omega)) \cap C^0([0,T];X),$$

where

$$X = \{\varphi \in H^1(\Omega),\ \Delta\varphi \in L^2(\Omega),\ \partial_z\varphi|_{\Gamma_0} \in L^2(\Gamma_0)\}.$$

We can now establish the existence of a weak solution to Problem (3.1) when χ is less regular; specifically, we only assume that $\chi \in H^1(0,T;H^{-1/2}(\Gamma_{-H}))$. Let us first define such a weak solution. We first define the Hilbert space

$$\widetilde{W}_0(0,T) = \Big\{\varphi \in L^2(0,T;H^1(\Omega)),\ \partial_t\varphi \in L^2(0,T;L^2(\Omega)),\ \partial_t(\varphi|_{\Gamma_0}) \in L^2(0,T;L^2(\Gamma_0)),\ \text{and } \varphi(0)=0\Big\}, \quad (3.5)$$

equipped with the norm $\|\cdot\|_{\widetilde{W}_0(0,T)}$ defined by

$$\|\varphi\|^2_{\widetilde{W}_0(0,T)} = \int_0^T \|\varphi\|^2_{H^1(\Omega)}\,dt + \int_0^T \|\partial_t\varphi\|^2_{L^2(\Omega)}\,dt + \int_0^T \|\partial_t(\varphi|_{\Gamma_0})\|^2_{L^2(\Gamma_0)}\,dt.$$

Note that, in particular, since $\varphi \in H^1(0,T;L^2(\Omega))$, $\varphi(0)$ is well defined with respect to the time variable and we can impose $\varphi(0)=0$. Moreover, since $\varphi \in L^2(0,T;H^1(\Omega))$, we have $\varphi|_{\Gamma_0} \in L^2(0,T;H^{1/2}(\Gamma_0))$, but $\partial_t\varphi \in L^2(0,T;L^2(\Omega))$ does not necessarily have a trace on Γ_0. However, we impose that for a function $\varphi \in \widetilde{W}_0(0,T)$, $\varphi|_{\Gamma_0}$ admits a weak time derivative $\partial_t(\varphi|_{\Gamma_0}) \in L^2(\Gamma_0)$.

Definition 3.2.1. *Let $\chi \in H^1(0,T;H^{-1/2}(\Gamma_{-H}))$, we say that $\varphi \in \widetilde{W}_0(0,T)$ is a variational solution to Problem (3.1), if*

$$\partial_t^2\varphi \in L^2(0,T;H^1(\Omega)') \text{ and } \partial_t^2(\varphi|_{\Gamma_0}) \in L^2(0,T;H^{-1/2}(\Gamma_0)),$$
$$\text{while } \partial_t\varphi(0)=0 \text{ and } \partial_t(\varphi|_{\Gamma_0})(0)=0,$$

and for all $\tilde\varphi \in L^2(0,T;H^1(\Omega))$,

$$\int_0^T \frac{1}{c^2}\langle\partial_t^2\varphi,\tilde\varphi\rangle_{H^1(\Omega)',H^1(\Omega)} + \frac{1}{g}\left\langle\partial_t^2(\varphi|_{\Gamma_0}),\tilde\varphi|_{\Gamma_0}\right\rangle_{H^{-1/2}(\Gamma_0),H^{1/2}(\Gamma_0)} + (\nabla\varphi,\nabla\tilde\varphi)_{L^2(\Omega)}\,dt = \int_0^T \langle\chi,\tilde\varphi\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})}\,dt. \quad (3.6)$$

We first note that this notion of weak solution is consistent with the notion of strong solution that we have already justified when χ is sufficiently regular. Indeed, we have the following proposition, the proof of which is straightforward using Green's formula and is left to the reader.

Proposition 3.2.1. *Assume that $\chi \in H^2(0,T;H^{-1/2}(\Gamma_{-H}))$ with $\chi(\cdot,0)=0$ on Γ_{-H}, and let φ be the unique strong solution to Problem (3.1) in the sense of Theorem 3.2.1. Then φ is also a variational solution to Problem (3.1).*

Theorem 3.2.2. *Suppose that $\chi \in H^1(0,T;H^{-1/2}(\Gamma_{-H}))$, with $\chi(\cdot,0)=0$ on Γ_{-H}. Then, there exists a unique variational solution $\varphi \in \widetilde{W}_0(0,T)$ to Problem (3.1).*

As concerns the asymptotic model, we have the following classical result concerning a strong solution, the proof of which is recalled in Appendix A.

Theorem 3.2.3. *Let us assume that $\chi \in C^2([0,T];L^2(\Gamma_{-H}))$. There exists a unique strong solution φ to Problem (3.4), that is*

$$\varphi \in C^2([0,T];H^{1/2}(\Omega)) \cap C^1([0,T];H^1(\Omega)) \cap C^0([0,T];H^{3/2}(\Omega)).$$

A variational solution to Problem (3.4) could probably be defined as it was done above for Problem (3.1) and an analogous theorem to Theorem 3.2.2 could probably also be established for Problem (3.4).

We observe from (3.1) and (3.4) that the inverse problem consists in solving the wave equation (complete model) or the Laplace equation (asymptotic model) from the lateral Cauchy data $(\varphi, \partial_z\varphi)$ on $\Gamma_0 \times (0,T)$ and no boundary condition on $\Gamma_{-H} \times (0,T)$, which is an ill-posed problem. In particular, existence of a solution is not ensured, neither for the complete model, nor for the asymptotic one. The question of uniqueness is the following: is there at most one solution χ on $\Gamma_{-H} \times (0,T)$ corresponding to the measurement of s on $\Gamma_0 \times (0,T)$? Such question makes a difference between the two models: there is at most one solution for the asymptotic model, while uniqueness is not satisfied for the complete model, it is only valid on a subpart of $\Gamma_{-H} \times (0,T)$. More precisely, we have the following theorem proved in [27], which is valid for both models.

Theorem 3.2.4. *Assume that φ satisfies*

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2\varphi - \Delta\varphi & = & 0 & \text{in } \Omega \times (0,T), \\ \partial_t\varphi + g\, s & = & 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_z\varphi - \partial_t s & = & 0 & \text{on } \Gamma_0 \times (0,T), \\ -\partial_z\varphi & = & \chi & \text{on } \Gamma_{-H} \times (0,T), \end{array}\right.$$

for some χ.

If $s = 0$ on $\Gamma_0 \times (0,T)$, then $\varphi = 0$ in $\{(X,t) \in \Omega \times (0,T),\, t < T + z/c\}$, hence $\chi = 0$ on $\Gamma_{-H} \times (0, T - H/c)$.

In particular, for $c = +\infty$, $\varphi = 0$ in $\Omega \times (0,T)$ and $\chi = 0$ on $\Gamma_{-H} \times (0,T)$.

3.3 Strong and weak solutions to the forward problem

This section is dedicated to the proofs of Theorems 3.2.1 and 3.2.2.

3.3.1 Proof of Theorem 3.2.1

We first recall the following result proved in [95], in [9] and mentioned in [31]. We consider $\mathcal{H}$ a Hilbert space, equipped with his scalar product $(\cdot,\cdot)_{\mathcal{H}}$, and an unbounded operator $\mathcal{A} : \mathcal{H} \to \mathcal{H}$ of domain $D(\mathcal{A})$ which is maximal accretive, that is

$$\exists \lambda_0 \geq 0, \quad (\mathcal{A}z + \lambda_0\, z, z)_{\mathcal{H}} \geq 0, \quad \forall z \in D(\mathcal{A}) \tag{3.7}$$

and

$$\forall \lambda_0 > 0, \quad \forall f \in \mathcal{H}, \quad \exists z \in D(\mathcal{A}), \quad \lambda_0\, z + \mathcal{A}z = f. \tag{3.8}$$

We have the following theorem.

Theorem 3.3.1. *For $z_0 \in D(\mathcal{A})$ and $\beta \in H^1(0,T;\mathcal{H})$ for $T > 0$, the problem*

$$\left\{\begin{array}{rcll} \dfrac{dz}{dt} + \mathcal{A}z & = & \beta & t \in (0,T), \\ z(0) & = & z_0, & \end{array}\right. \tag{3.9}$$

has a unique strong – better called strict – solution

$$z \in C^1([0,T];\mathcal{H}) \cap C^0([0,T];D(\mathcal{A})).$$

Remark 3.3.1. *The notion of strict solution is defined in [9, Definition 3.1], where Theorem 3.3.1 is proved in [9, Proposition 3.3].*

In order to apply Theorem 3.3.1 to Problem (3.1), we introduce the space

$$\mathcal{H} = H^1(\Omega) \times L^2(\Omega) \times L^2(\Gamma_0),$$

equipped with the scalar product defined, for all $z = (\varphi, \psi, s)$ and $\tilde{z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s})$ in $\mathcal{H}$, by

$$(z, \tilde{z})_{\mathcal{H}} = (\varphi, \tilde{\varphi})_{H^1(\Omega)} + \frac{1}{c^2}(\psi, \tilde{\psi})_{L^2(\Omega)} + g\,(s, \tilde{s})_{L^2(\Gamma_0)},$$

the space

$$\mathcal{V} = \{(\varphi, \psi, s) \in H^1(\Omega) \times H^1(\Omega) \times L^2(\Gamma_0),\ \psi|_{\Gamma_0} + g\,s = 0\},$$

equipped with the scalar product defined, for all z and $\tilde{z}$ in $\mathcal{V}$, by

$$(z, \tilde{z})_{\mathcal{V}} = (\varphi, \tilde{\varphi})_{H^1(\Omega)} + (\psi, \tilde{\psi})_{H^1(\Omega)} + (s, \tilde{s})_{L^2(\Gamma_0)}.$$

From [27], the space $\mathcal{V}$ is continuously embedded and dense in $\mathcal{H}$. Then, let us define the continuous bilinear form on $\mathcal{V}$ by

$$a(z, \tilde{z}) = -(\psi, \tilde{\varphi})_{H^1(\Omega)} + (\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega)}, \quad \forall z, \tilde{z} \in \mathcal{V},$$

where we have denoted

$$(\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega)} = \int_\Omega \nabla\varphi \cdot \nabla\tilde{\psi}\, dX.$$

From the above assumptions, we can define an unbounded operator $\mathcal{A}$ in $\mathcal{H}$, the domain $D(\mathcal{A})$ of which is defined by

$$D(\mathcal{A}) = \{z \in \mathcal{V},\ \exists f \in \mathcal{H}, \quad a(z, \tilde{z}) = (f, \tilde{z})_{\mathcal{H}}, \quad \forall \tilde{z} \in \mathcal{V}\},$$

and $\mathcal{A}$ is defined for $z \in D(\mathcal{A})$ by $\mathcal{A}z = f$, the uniqueness of such f resulting from the density of $\mathcal{V}$ in $\mathcal{H}$. In particular, we have the inclusions $D(\mathcal{A}) \subset \mathcal{V} \subset \mathcal{H}$.

In the following proposition, we offer a characterization of the domain of the operator $\mathcal{A}$.

Proposition 3.3.1. *We have*

$$D(\mathcal{A}) = \{z = (\varphi, \psi, s) \in \mathcal{V},\ \Delta\varphi \in L^2(\Omega),\ \partial_z\varphi|_{\Gamma_0} \in L^2(\Gamma_0),\ \partial_z\varphi|_{\Gamma_{-H}} = 0\}.$$

Proof. Assuming $z = (\varphi, \psi, s) \in D(\mathcal{A})$, there exists some $f = (f^\varphi, f^\psi, f^s) \in \mathcal{H}$ such that

$$a(z, \tilde{z}) = (f, \tilde{z})_{\mathcal{H}}, \quad \forall \tilde{z} \in \mathcal{V},$$

that is, for all $(\tilde{\varphi}, \tilde{\psi}, \tilde{s}) \in \mathcal{V}$,

$$-(\psi, \tilde{\varphi})_{H^1(\Omega)} + (\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega)} = (f^\varphi, \tilde{\varphi})_{H^1(\Omega)} + \frac{1}{c^2}(f^\psi, \tilde{\psi})_{L^2(\Omega)} + g\,(f^s, \tilde{s})_{L^2(\Gamma_0)}.$$

Choosing first $\tilde{\psi} = \tilde{s} = 0$, we get that

$$\psi = -f^\varphi,$$

which implies that for all $(\tilde{\psi}, \tilde{s}) \in H^1(\Omega) \times L^2(\Gamma_0)$ such that $\tilde{\psi}|_{\Gamma_0} + g\,\tilde{s} = 0$,

$$(\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega)} = \frac{1}{c^2}(f^\psi, \tilde{\psi})_{L^2(\Omega)} + g\,(f^s, \tilde{s})_{L^2(\Gamma_0)}.$$

Choosing now $\tilde{\psi} \in C_0^\infty(\Omega)$ and $\tilde{s} = 0$, we get that

$$-\Delta\varphi = \frac{1}{c^2} f^\psi \in L^2(\Omega).$$

Choosing then $\tilde{\psi}(x, z) = \rho(x)(1 + z/H)$, where $\rho \in C_0^\infty(\mathbb{R})$, and $\tilde{s} = -\tilde{\psi}|_{\Gamma_0}/g$, we get by using the Green formula

$$(-\Delta\varphi, \tilde{\psi})_{L^2(\Omega)} + \left\langle \partial_z\varphi, \tilde{\psi} \right\rangle_{H^{-1/2}(\Gamma_0), H^{1/2}(\Gamma_0)} = \frac{1}{c^2}(f^\psi, \tilde{\psi})_{L^2(\Omega)} + g\,(f^s, \tilde{s})_{L^2(\Gamma_0)},$$

that is

$$\langle \partial_z \varphi, \rho \rangle_{H^{-1/2}(\Gamma_0), H^{1/2}(\Gamma_0)} = -(f^s, \rho)_{L^2(\Gamma_0)}, \quad \forall \rho \in C_0^\infty(\mathbb{R}).$$

which implies that

$$\partial_z \varphi|_{\Gamma_0} = -f^s \in L^2(\Gamma_0).$$

Choosing lastly $\tilde{\psi}(x,z) = \rho(x)(z/H)$, where $\rho \in C_0^\infty(\mathbb{R})$, and $\tilde{s} = 0$, we get by using the Green formula

$$(-\Delta\varphi, \tilde{\psi})_{L^2(\Omega)} + \langle \partial_z \varphi, \tilde{\psi} \rangle_{H^{-1/2}(\Gamma_{-H}), H^{1/2}(\Gamma_{-H})} = \frac{1}{c^2}(f^\psi, \tilde{\psi})_{L^2(\Omega)},$$

which implies that $\partial_z \varphi = 0$ on Γ_{-H}. Finally,

$$D(\mathcal{A}) \subset \{z = (\varphi, \psi, s) \in \mathcal{V},\ \Delta\varphi \in L^2(\Omega),\ \partial_z\varphi|_{\Gamma_0} \in L^2(\Gamma_0),\ \partial_z\varphi|_{\Gamma_{-H}} = 0\}.$$

The converse inclusion is easily proved by inverting the above steps. □

In order to apply Theorem 3.3.1, we need to introduce the operator $R_{-H} : H^{-1/2}(\Gamma_{-H}) \to H^1(\Omega)$ defined for $\chi \in H^{-1/2}(\Gamma_{-H})$, by the solution $U = R_{-H}\chi$ to the problem

$$\left\{ \begin{array}{rcll} -\Delta U + U & = & 0 & \text{in } \Omega, \\ -\partial_z U & = & \chi & \text{on } \Gamma_{-H}, \\ \partial_z U & = & 0 & \text{on } \Gamma_0. \end{array} \right. \tag{3.10}$$

Denoting $D_{-H} := (R_{-H}, 0, 0) : H^{-1/2}(\Gamma_{-H}) \to \mathcal{H}$, the idea is to first consider the problem satisfied by $Z = z - D_{-H}\chi$. In this view, we also introduce $E : \mathcal{H} \to \mathcal{H}$ such that

$$Ez = (0, \varphi, 0), \quad \forall z = (\varphi, \psi, s) \in \mathcal{H}$$

and finally, for $\chi \in H^2(0,T; H^{-1/2}(\Gamma_{-H}))$, the space-time operator B_{-H} is given by

$$B_{-H}\chi = -D_{-H}\partial_t\chi + c^2\, E D_{-H}\chi.$$

We have the following proposition.

Proposition 3.3.2. *For $\chi \in H^2(0,T; H^{-1/2}(\Gamma_{-H}))$ and $\chi(\cdot, 0) = 0$, the problem*

$$\left\{ \begin{array}{rcll} \dfrac{dZ}{dt} + \mathcal{A}Z & = & B_{-H}\chi & t \in (0,T), \\ Z(0) & = & 0 & \end{array} \right. \tag{3.11}$$

has a unique (strict) solution

$$Z \in C^1([0,T]; \mathcal{H}) \cap C^0([0,T]; D(\mathcal{A})).$$

Proof. From the previous considerations, for all $\chi \in H^2(0,T; H^{-1/2}(\Gamma_{-H}))$, we have $B_{-H}\chi \in H^1(0,T;\mathcal{H})$. To apply Theorem 3.3.1 it suffices to prove that the operator $\mathcal{A}$ satisfies the assumptions (3.7) and (3.8).

For $z = (\varphi, \psi, s) \in D(\mathcal{A})$ and $\lambda \geq 0$, we have

$$\begin{aligned} (\mathcal{A}z + \lambda\, z, z)_{\mathcal{H}} &= a(z,z) + \lambda \|z\|_{\mathcal{H}}^2 \\ &= -(\psi, \varphi)_{H^1(\Omega)} + (\nabla\varphi, \nabla\psi)_{L^2(\Omega)} + \lambda\|\varphi\|_{H^1(\Omega)}^2 + \frac{\lambda}{c^2}\|\psi\|_{L^2(\Omega)}^2 + \lambda g\, \|s\|_{L^2(\Gamma_0)}^2 \\ &= -(\psi, \varphi)_{L^2(\Omega)} + \lambda\|\varphi\|_{H^1(\Omega)}^2 + \frac{\lambda}{c^2}\|\psi\|_{L^2(\Omega)}^2 + \lambda g\, \|s\|_{L^2(\Gamma_0)}^2 \\ &\geq -\frac{1}{2}\|\psi\|_{L^2(\Omega)}^2 - \frac{1}{2}\|\varphi\|_{L^2(\Omega)}^2 + \lambda\|\varphi\|_{L^2(\Omega)}^2 + \frac{\lambda}{c^2}\|\psi\|_{L^2(\Omega)}^2. \end{aligned}$$

To obtain (3.7), it suffices to take $\lambda \geq \lambda_0$ with $\lambda_0 = \max(1/2, c^2/2)$.

Now let us check that (3.8) is also satisfied. Let us assume that $f = (f^\varphi, f^\psi, f^s) \in \mathcal{H}$ and let us consider the weak formulation, for any $\lambda_0 > 0$: find $\varphi \in H^1(\Omega)$ such that for all $\tilde\psi \in H^1(\Omega)$,

$$\begin{aligned}&\frac{\lambda_0^2}{c^2}(\varphi,\tilde\psi)_{L^2(\Omega)} + (\nabla\varphi,\nabla\tilde\psi)_{L^2(\Omega)} + \frac{\lambda_0^2}{g}(\varphi,\tilde\psi)_{L^2(\Gamma_0)}\\ &= \frac{1}{c^2}(f^\psi,\tilde\psi)_{L^2(\Omega)} + \frac{\lambda_0}{c^2}(f^\varphi,\tilde\psi)_{L^2(\Omega)} - (f^s,\tilde\psi)_{L^2(\Gamma_0)} + \frac{\lambda_0}{g}(f^\varphi,\tilde\psi)_{L^2(\Gamma_0)}.\end{aligned} \tag{3.12}$$

A direct application of the Lax-Milgram lemma proves that such weak formulation has a unique solution. Now let us define $\psi = \lambda_0\,\varphi - f^\varphi \in H^1(\Omega)$ and $s = -\psi|_{\Gamma_0}/g \in L^2(\Gamma_0)$. From (3.12), we get

$$\frac{\lambda_0}{c^2}(\psi,\tilde\psi)_{L^2(\Omega)} + (\nabla\varphi,\nabla\tilde\psi)_{L^2(\Omega)} - \lambda_0(s,\tilde\psi)_{L^2(\Gamma_0)} = \frac{1}{c^2}(f^\psi,\tilde\psi)_{L^2(\Omega)} - (f^s,\tilde\psi)_{L^2(\Gamma_0)}.$$

The fact that $\psi = \lambda_0\,\varphi - f^\varphi$ implies that for all $\tilde\varphi \in H^1(\Omega)$,

$$\lambda_0(\varphi,\tilde\varphi)_{H^1(\Omega)} - (\psi,\tilde\varphi)_{H^1(\Omega)} = (f^\varphi,\tilde\varphi)_{H^1(\Omega)}.$$

Adding the two previous weak formulations, we obtain that for all $\tilde\varphi,\tilde\psi \in H^1(\Omega)$,

$$\begin{aligned}&\lambda_0(\varphi,\tilde\varphi)_{H^1(\Omega)} + \frac{\lambda_0}{c^2}(\psi,\tilde\psi)_{L^2(\Omega)} - \lambda_0(s,\tilde\psi)_{L^2(\Gamma_0)} - (\psi,\tilde\varphi)_{H^1(\Omega)} + (\nabla\varphi,\nabla\tilde\psi)_{L^2(\Omega)}\\ &= (f^\varphi,\tilde\varphi)_{H^1(\Omega)} + \frac{1}{c^2}(f^\psi,\tilde\psi)_{L^2(\Omega)} - (f^s,\tilde\psi)_{L^2(\Gamma_0)}.\end{aligned} \tag{3.13}$$

Introducing $\tilde s = -\tilde\psi|_{\Gamma_0}/g$, we have that $z = (\varphi,\psi,s) \in \mathcal{V}$ satisfies, for all $\tilde z = (\tilde\varphi,\tilde\psi,\tilde s) \in \mathcal{V}$,

$$\lambda_0\,(z,\tilde z)_{\mathcal{H}} + a(z,\tilde z) = (f,\tilde z)_{\mathcal{H}}.$$

We have hence found some $z \in D(\mathcal{A})$ such that

$$\lambda_0\,z + \mathcal{A}z = f,$$

which completes the proof of (3.8). □

The end of the proof of Theorem 3.2.1 consists in showing the relationship between Problem (3.11) and Problem (3.1).

Proposition 3.3.3. *Let us consider the solution Z to Problem (3.11). Then the solution $z = Z + D_{-H}\chi$ is the unique solution in $C^1([0,T];\mathcal{H}) \cap C^0([0,T];\mathcal{X})$ to the problem (3.1), where*

$$\mathcal{X} = \{z = (\varphi,\psi,s) \in \mathcal{V},\ \Delta\varphi \in L^2(\Omega),\ \partial_z\varphi|_{\Gamma_0} \in L^2(\Gamma_0)\}.$$

Proof. From Proposition 3.3.2, there exists $Z \in C^1([0,T];\mathcal{H}) \cap C^0([0,T];D(\mathcal{A}))$ satisfying (3.11), which is equivalent to $Z(0) = 0$ and

$$\left(\frac{dZ}{dt},\tilde z\right)_{\mathcal{H}} + a(Z,\tilde z) = (B_{-H}\chi,\tilde z)_{\mathcal{H}}, \quad \forall \tilde z \in \mathcal{V}. \tag{3.14}$$

For all $\tilde z = (\tilde\varphi,\tilde\psi,\tilde s) \in \mathcal{V}$, we have

$$\begin{aligned}\left(\frac{dZ}{dt},\tilde z\right)_{\mathcal{H}} + a(Z,\tilde z) &= (\partial_t\varphi,\tilde\varphi)_{H^1(\Omega)} - (\partial_t U,\tilde\varphi)_{H^1(\Omega)} + \frac{1}{c^2}(\partial_t\psi,\tilde\psi)_{L^2(\Omega)} + g\,(\partial_t s,\tilde s)_{L^2(\Gamma_0)}\\ &\quad - (\psi,\tilde\varphi)_{H^1(\Omega)} + (\nabla\varphi,\nabla\tilde\psi)_{L^2(\Omega)} - (\nabla U,\nabla\tilde\psi)_{L^2(\Omega)},\end{aligned}$$

where the function U satisfies (3.10). On the other hand, since $B_{-H}\chi = -D_{-H}\partial_t\chi + c^2\,ED_{-H}\chi$,

$$\begin{aligned}(B_{-H}\chi,\tilde z)_{\mathcal{H}} &= -((\partial_t U,0,0),(\tilde\varphi,\tilde\psi,\tilde s))_{\mathcal{H}} + c^2((0,U,0),(\tilde\varphi,\tilde\psi,\tilde s))_{\mathcal{H}}\\ &= -(\partial_t U,\tilde\varphi)_{H^1(\Omega)} + (U,\tilde\psi)_{L^2(\Omega)}.\end{aligned}$$

Then (3.14) is equivalent to, for all $(\tilde{\varphi}, \tilde{\psi}, \tilde{s}) \in \mathcal{V}$,

$$\begin{aligned}&(\partial_t\varphi, \tilde{\varphi})_{H^1(\Omega)} + \frac{1}{c^2}(\partial_t\psi, \tilde{\psi})_{L^2(\Omega)} + g\,(\partial_t s, \tilde{s})_{L^2(\Gamma_0)}\\ &- (\psi, \tilde{\varphi})_{H^1(\Omega)} + (\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega)} - (\nabla U, \nabla\tilde{\psi})_{L^2(\Omega)} = (U, \tilde{\psi})_{L^2(\Omega)}.\end{aligned} \tag{3.15}$$

Choosing $\tilde{z} = (\tilde{\varphi}, 0, 0)$ in (3.15), we get that $\partial_t\varphi = \psi$, which implies that, for all $\tilde{\psi} \in H^1(\Omega)$ and $\tilde{s} = -\tilde{\psi}|_{\Gamma_0}/g$,

$$\frac{1}{c^2}(\partial_t^2\varphi, \tilde{\psi})_{L^2(\Omega)} + g\,(\partial_t s, \tilde{s})_{L^2(\Gamma_0)} + (\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega)} - (\nabla U, \nabla\tilde{\psi})_{L^2(\Omega)} = (U, \tilde{\psi})_{L^2(\Omega)}.$$

Choosing $\tilde{\psi} \in C_0^\infty(\Omega)$ and $\tilde{s} = 0$, we obtain in view of $\Delta U = U$ in Ω that $(1/c^2)\partial_t^2\varphi - \Delta\varphi = 0$ in $\Omega \times (0, T)$, that is the first equation of (3.1).

Now for all $\tilde{\psi} \in H^1(\Omega)$ and $\tilde{s} = -\tilde{\psi}|_{\Gamma_0}/g$, using that $\Delta\varphi \in C^0([0, T]; L^2(\Omega))$ (in view of $Z \in C^0([0, T]; D(\mathcal{A}))$),

$$\begin{aligned}&\frac{1}{c^2}(\partial_t^2\varphi, \tilde{\psi})_{L^2(\Omega)} - (\partial_t s, \tilde{\psi})_{L^2(\Gamma_0)} - (\Delta\varphi, \tilde{\psi})_{L^2(\Omega)} + (\Delta U, \tilde{\psi})_{L^2(\Omega)}\\ &+ \left\langle \partial_z\varphi, \tilde{\psi}\right\rangle_{H^{-1/2}(\Gamma_0), H^{1/2}(\Gamma_0)} - \left\langle \partial_z\varphi, \tilde{\psi}\right\rangle_{H^{-1/2}(\Gamma_{-H}), H^{1/2}(\Gamma_{-H})}\\ &- \left\langle \partial_z U, \tilde{\psi}\right\rangle_{H^{-1/2}(\Gamma_0), H^{1/2}(\Gamma_0)} + \left\langle \partial_z U, \tilde{\psi}\right\rangle_{H^{-1/2}(\Gamma_{-H}), H^{1/2}(\Gamma_{-H})}\\ &= (U, \tilde{\psi})_{L^2(\Omega)}.\end{aligned}$$

Choosing $\tilde{\psi} = 0$ on Γ_{-H} and since $\psi|_{\Gamma_0} = \partial_t\varphi|_{\Gamma_0} = -gs$, in view of $z \in C^0([0, T]; D(\mathcal{A}))$, we derive the two equations (3.2) and (3.3). Choosing now $\tilde{\psi} = 0$ on Γ_0, we get that $-\partial_z\varphi = \chi$ on Γ_{-H}. Finally, that $Z(0) = 0$ implies that $z(0) = 0$ since $\chi(\cdot, 0) = 0$ on Γ_{-H}, so that $\varphi = \partial_t\varphi = 0$ on $\Omega \times \{0\}$. We conclude that φ is a solution to the Problem (3.1). Note that since $Z \in C^1([0, T], \mathcal{H})$ and $U \in H^2(0, T; H^1(\Omega)) \subset C^1([0, T], H^1(\Omega))$, we have that $\varphi \in C^1([0, T], H^1(\Omega))$ and $\psi \in C^1([0, T], L^2(\Omega))$. But in view of $\partial_t\varphi = \psi$, we obtain in turn that $\varphi \in C^2([0, T], L^2(\Omega))$.

Uniqueness if a consequence of Theorem 3.3.1 for $\beta = 0$. □

3.3.2 Proof of Theorem 3.2.2

To show the existence of a variational solution, we follow a classical approach where we approximate the data by sequences of regular functions and consider the sequence of strong solutions associated with these regular data. Then, we show that these strong solutions satisfy a variational formulation, and we pass to the limit on this formulation using an energy estimate and strong convergences. Let us first prove an energy estimate associated with strong solution of Problem (3.1).

Proposition 3.3.4. *Let $\chi \in H^2\left(0, T; H^{-1/2}\left(\Gamma_{-H}\right)\right)$ with $\chi(0) = 0$ and consider the strong solution Z to Problem (3.11). Then $z = Z + D_{-H}\chi \in C^1([0, T]; \mathcal{H}) \cap C^0([0, T]; \mathcal{X})$ satisfies: there exists $K > 0$ such that for all $t \in (0, T)$*

$$\frac{1}{2c^2}\int_\Omega \psi^2 dX + \frac{1}{2}\int_\Omega |\nabla\varphi|^2 dX + \frac{1}{2}\int_\Omega \varphi^2 dX + \frac{g}{2}\int_{\Gamma_0} s^2 dx \leq K\|\chi\|^2_{H^1\left(0,T;H^{-1/2}(\Gamma_{-H})\right)}.$$

Proof. Starting back from (3.14), we directly infer that as $Z \in C^1([0, T]; \mathcal{H}) \cap C^0([0, T]; D(\mathcal{A}))$ satisfies

$$\frac{d}{dt}\left(\frac{1}{2}\|Z\|^2_{\mathcal{H}}\right) \leq \|B_{-H}\chi\|_{\mathcal{H}}\|Z\|_{\mathcal{H}},$$

so that using a variant of Grönwall's Lemma [32, Lemma A.5] and knowing that $Z(0) = 0$, we have

$$\|Z\|_{\mathcal{H}} \leq \int_0^t \|B_{-H}\chi(t)\|_{\mathcal{H}} dt.$$

Then, we directly deduce from the definition of B_{-H} that there exists K_1 such that

$$\|Z\|_{\mathcal{H}} \leq K_1 \|\chi\|_{H^1\left(0,T;H^{-1/2}(\Gamma_{-H})\right)},$$

and as $z = Z + D_{-H}\chi$, there exists K such that

$$\|z\|_{\mathcal{H}} \leq K \|\chi\|_{H^1\left(0,T;H^{-1/2}(\Gamma_{-H})\right)},$$

which concludes the proof. □

Remark 3.3.2. *We can easily associate a natural energy functional with Problem (3.1):*

$$\mathcal{E}(t) = \frac{1}{2c^2}\int_\Omega |\partial_t\varphi|^2\, dX + \frac{1}{2}\int_\Omega |\nabla\varphi|^2 dX + \frac{1}{2g}\int_{\Gamma_0} |\partial_t\varphi|^2\, dx. \tag{3.16}$$

Indeed, if we multiply the first equation of (3.1) by $\partial_t\varphi$, integrate over Ω, and use Green's formula, we directly infer that if $\chi \in L^2(0,T;L^2(\Gamma_{-H}))$, then

$$\frac{d}{dt}\mathcal{E}(t) = \int_{\Gamma_{-H}} \chi\partial_t\varphi\, dx. \tag{3.17}$$

Normally, such an identity should typically allow us to control the energy of the solution using the source term. However, if we only assume that $\chi \in L^2(0,T;L^2(\Gamma_{-H}))$, the power induced by the external source is continuous with respect to the trace of $\partial_t\varphi$ on the boundary. Such a quantity is not controlled by the energy $\mathcal{E}$. This is the reason why we had to assume additional regularity for χ with respect to time, in order – after integration by parts in time – to control the power induced by the external source by the trace of φ on the boundary, which is well defined.

Let χ_n be a sequence of functions in $H^2(0,T;H^{-1/2}(\Gamma_{-H}))$ such that $\chi_n(0) = 0$ converging by density to χ in $H^1(0,T;H^{-1/2}(\Gamma_{-H}))$ with $\chi(0) = 0$. Then according to Theorem 3.2.1 and Proposition 3.3.3, there exists, for all $n \in \mathbb{N}$, a unique strong solution $z_n = (\phi_n, \psi_n, s_n)$ in $C^1([0,T],\mathcal{H}) \cap C^0([0,T],\mathcal{X})$ satisfying:

$$\begin{cases} \partial_t z_n + \mathcal{A} z_n = -D_{-H}\partial_t\chi_n + c^2 E D_{-H}\chi_n, \text{ in } (0,T) \\ z_n(0) = 0. \end{cases} \tag{3.18}$$

By taking the inner product of the first equation of (3.18) with $\tilde{Z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s}) \in \mathcal{V}$, we obtain that for all time $t \in (0,T)$,

$$\begin{aligned}(\partial_t\phi_n(t),\tilde{\varphi})_{H^1(\Omega)} + \frac{1}{c^2}(\partial_t\psi_n(t),\tilde{\psi})_{L^2(\Omega)} + g(\partial_t s_n(t),\tilde{s})_{L^2(\Gamma_0)} \\ - (\psi_n(t),\tilde{\varphi})_{H^1(\Omega)} + (\nabla\phi_n(t),\nabla\tilde{\psi})_{L^2(\Omega)} \\ = -(R_{-H}\partial_t\chi_n(t),\tilde{\varphi})_{H^1(\Omega)} + (R_{-H}\chi_n(t),\tilde{\psi})_{L^2(\Omega)}.\end{aligned} \tag{3.19}$$

Then introducing $\varphi_n = \phi_n + R_{-H}\chi_n$ we have

$$\begin{aligned}(\partial_t\varphi_n(t),\tilde{\varphi})_{H^1(\Omega)} + \frac{1}{c^2}(\partial_t\psi_n(t),\tilde{\psi})_{L^2(\Omega)} + g(\partial_t s_n(t),\tilde{s})_{L^2(\Gamma_0)} \\ - (\psi_n(t),\tilde{\varphi})_{H^1(\Omega)} + (\nabla\varphi_n(t),\nabla\tilde{\psi})_{L^2(\Omega)} \\ = \langle\chi_n(t),\tilde{\psi}\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})},\end{aligned} \tag{3.20}$$

which is a weak formulation of Problem (3.1).

Denoting $\varphi_{n,m} = \varphi_n - \varphi_m$ – and $\psi_{n,m}$, $s_{n,m}$ and $\chi_{n,m}$ respectively – then by linearity (3.20) remains valid by replacing φ_n with $\varphi_{n,m}$ – and $\psi_{n,m}$, $s_{n,m}$ and $\chi_{n,m}$ respectively. By choosing $\tilde{\varphi} = \varphi_{n,m}$, $\tilde{\psi} = \psi_{n,m}$ and $\tilde{s} = s_{n,m}$, from Proposition 3.3.4 the following energy estimate holds for all $t \in (0,T)$:

$$\frac{1}{2}\int_\Omega |\nabla\varphi_{n,m}(t)|^2\, dX + \frac{1}{2}\int_\Omega \varphi_{n,m}(t)^2\, dX + \frac{1}{2c^2}\int_\Omega \psi_{n,m}(t)^2\, dX + \frac{g}{2}\int_{\Gamma_0} s_{n,m}(t)^2\, dx$$

$$\leq K\|\chi_{n,m}\|^2_{H^1(0,T;H^{-1/2}(\Gamma_{-H}))}.$$

Since χ_n converges to χ in $H^1(0,T,H^{-1/2}(\Gamma_{-H}))$, it follows that $(\chi_n)_{n\in\mathbb{N}}$ is a Cauchy sequence hence, $(\varphi_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $C^0([0,T],H^1(\Omega))$, so are $(\psi_n)_{n\in\mathbb{N}}$ in $C^0([0,T],L^2(\Omega))$, and $(s_n)_{n\in\mathbb{N}}$ in $C^0([0,T],L^2(\Gamma_0))$. Hence, by completeness, there exist $\varphi \in C^0([0,T],H^1(\Omega))$, $\psi \in C^0([0,T],L^2(\Omega))$, and $s \in C^0([0,T],L^2(\Gamma_0))$ such that

$$\begin{aligned}
&\varphi_n \to \varphi \text{ in } C^0([0,T],H^1(\Omega)),\\
&\psi_n \to \psi \text{ in } C^0([0,T],L^2(\Omega)),\\
&s_n \to s \text{ in } C^0([0,T],L^2(\Gamma_0)).
\end{aligned}$$

In particular, $\varphi(0)=0$. For any $\varrho \in C_0^\infty(0,T)$, we have, for example for all $\tilde\psi \in L^2(\Omega)$,

$$\begin{aligned}
\int_0^T\int_\Omega \partial_t\psi_n(t,X)\,\tilde\psi(X)\,\varrho(t)\,dX\,dt &= -\int_0^T\int_\Omega \psi_n(t,X)\,\tilde\psi(X)\,d_t\varrho(t)\,dx\,dt\\
&\xrightarrow{n\to\infty} -\int_0^T\int_\Omega \psi(t,X)\,\tilde\psi(X)\,d_t\varrho(t)\,dX\,dt,
\end{aligned}$$

and identically for all $\tilde\varphi \in H^1(\Omega)$

$$\int_0^T\int_\Omega \partial_t\varphi_n(t,X)\,\tilde\varphi(X)\,\varrho(t)\,dX\,dt \xrightarrow{n\to\infty} -\int_0^T\int_\Omega \varphi(t,X)\,\tilde\varphi(X)\,\dot\varrho(t)\,dX\,dt,$$

while for all $\tilde s \in L^2(\Gamma_0)$

$$\int_0^T\int_{\Gamma_0} \partial_t s_n(t,x)\,\tilde s(x)\,\varrho(t)\,dx\,dt \xrightarrow{n\to\infty} -\int_0^T\int_{\Gamma_0} s(t,x)\,\tilde s(x)\,\dot\varrho(t)\,dx\,dt.$$

Therefore, by considering $\tilde\varphi = 0$ in (3.20) and passing to the limit we have that (φ,ψ,s) satisfies, for all $(\tilde\psi,\tilde s) \in H^1(\Omega)\times L^2(\Gamma_0)$ such that $\tilde\psi|_{\Gamma_0} + g\,\tilde s = 0$,

$$\begin{aligned}
&\frac{1}{c^2}\frac{d}{dt}(\psi(t),\tilde\psi)_{L^2(\Omega)} + g\frac{d}{dt}(s(t),\tilde s)_{L^2(\Gamma_0)}\\
&\qquad + (\nabla\varphi(t),\nabla\tilde\psi)_{L^2(\Omega)} = \langle\chi(t),\tilde\psi\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})} \quad \text{in } \mathcal{D}'(0,T). \qquad (3.21)
\end{aligned}$$

Then, choosing in (3.20) $\tilde\psi = 0, \tilde s = 0$ and for all $\omega \in L^2(\Omega)$, $\tilde\varphi \in H^1(\Omega)$ such that

$$\begin{cases} -\Delta\tilde\varphi + \tilde\varphi = \omega, & \text{in } \Omega,\\ \partial_z\tilde\varphi = 0, & \text{on } \Gamma_0\cup\Gamma_{-H}, \end{cases}$$

we have for all $\varrho \in C_0^\infty(0,T)$

$$-\int_0^T (\varphi_n,\omega)_{L^2(\Omega)} d_t\varrho(t)\,dt = \int_0^T (\psi_n,\omega)_{L^2(\Omega)}\varrho(t)\,dt,$$

namely from weak convergence, $\varphi \in L^2(0,T;H^1(\Omega))$ admits a weak time-derivative in $L^2(0,T;L^2(\Omega))$ and we have in $\mathcal{D}'(0,T)$ for all $\omega \in L^2(\Omega)$

$$\frac{d}{dt}(\varphi(t),\omega)_{L^2(\Omega)} = (\psi(t),\omega)_{L^2(\Omega)}. \qquad (3.22)$$

Moreover, choosing in (3.20) $\tilde\psi = 0, \tilde s = 0$ and for all $h \in L^2(\Gamma_0)$, $\tilde\varphi \in H^1(\Omega)$ such that

$$\begin{cases} -\Delta\tilde\varphi + \tilde\varphi = 0, & \text{in } \Omega,\\ \partial_z\tilde\varphi = 0, & \text{on } \Gamma_{-H},\\ \partial_z\tilde\varphi = h, & \text{on } \Gamma_0, \end{cases}$$

we have for all $\varrho \in C_0^\infty(0,T)$

$$-\int_0^T \langle \varphi_n, h\rangle_{H^{1/2}(\Gamma_0),H^{-1/2}(\Gamma_0)} d_t\varrho(t)\,dt = \int_0^T -g(s_n,h)_{L^2(\Gamma_0)}\varrho(t)\,dt,$$

namely from weak convergence, $\varphi|_{\Gamma_0} \in L^2(0,T;H^{1/2}(\Gamma_0))$ admits a weak time-derivative in $L^2(0,T;L^2(\Gamma_0))$ and we have in $\mathcal{D}'(0,T)$, for all $\tilde\psi \in H^1(\Omega)$,

$$\frac{d}{dt}(\varphi|_{\Gamma_0}(t),\tilde\psi|_{\Gamma_0})_{L^2(\Gamma_0)} = -g(s(t),\tilde\psi|_{\Gamma_0})_{L^2(\Gamma_0)}. \tag{3.23}$$

Combining (3.21), (3.22) and (3.23), we obtain that for all $\tilde\psi \in H^1(\Omega)$,

$$\begin{aligned}\frac{d^2}{dt^2}\frac{1}{c^2}(\varphi(t),\tilde\psi)_{L^2(\Omega)} + \frac{d^2}{dt^2}\frac{1}{g}(\varphi|_{\Gamma_0}(t),\tilde\psi|_{\Gamma_0})_{L^2(\Gamma_0)}\\ + (\nabla\varphi(t),\nabla\tilde\psi)_{L^2(\Omega)} = \langle\chi(t),\tilde\psi\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})}, \quad \text{in } \mathcal{D}'(0,T).\end{aligned}$$

Namely, for all $\lambda \in C_0^\infty(0,T)$ and for all $\tilde\psi \in H^1(\Omega)$, we have

$$\begin{aligned}\int_0^T \frac{1}{c^2}(\varphi(t),\tilde\psi)_{L^2(\Omega)} d_t^2\lambda(t) + \frac{1}{g}(\varphi|_{\Gamma_0}(t),\tilde\psi)_{L^2(\Gamma_0)} d_t^2\lambda(t)\,dt\\ = \int_0^T -(\nabla\varphi(t),\nabla\tilde\psi)_{L^2(\Omega)}\lambda(t) + \langle\chi(t),\tilde\psi\rangle_{H^{-1/2}(\Gamma_{-H}),H^{1/2}(\Gamma_{-H})}\lambda(t)\,dt,\end{aligned}$$

so there exists $C>0$ such that

$$\begin{aligned}\left|\int_0^T \left\langle \left(\frac{1}{c^2}\varphi(t),\ \frac{1}{g}\varphi|_{\Gamma_0}(t)\right),\ (\tilde\psi,\tilde\psi|_{\Gamma_0})\right\rangle_{L^2(\Omega)\times L^2(\Gamma_0)} d_t^2\lambda(t)\,dt\right|\\ \le C\|\lambda\|_{L^2(0,T)}\|(\tilde\psi,\tilde\psi|_{\Gamma_0})\|_{H^1(\Omega)\times H^{1/2}(\Gamma_0)}.\end{aligned}$$

This directly implies

$$\partial_t^2\varphi \in L^2(0,T;H^1(\Omega)') \quad \text{and} \quad \partial_t^2(\varphi|_{\Gamma_0}) \in L^2(0,T;H^{-1/2}(\Gamma_0)).$$

Therefore, we can consider the trace of $\partial_t\varphi$ and $\partial_t(\varphi|_{\Gamma_0})$ at $t=0$, and find that

$$\partial_t\varphi(0) = \psi(0) = 0 \text{ and } \partial_t(\varphi_{\Gamma_0})(0) = -g\,s(0) = 0.$$

By density of $C_0^\infty(0,T)$ in $L^2(0,T;H^1(\Omega))$, we have finally proven that there exists a variational solution to Problem (3.1) in the sense of Definition 3.2.1.

Let us now proceed to prove that such a solution is unique. To do this, we consider $\chi = 0$ and show that the corresponding solution φ of (3.6) is $\varphi = 0$. In this context, for all $\tau \in (0,T)$, we introduce the so-called Ladyzhenskaya test functions [82] – see their use for uniqueness results in [88, Section IV-1.3] – $\eta_\tau \in L^2(0,T;H^1(\Omega))$ defined by

$$\eta_\tau : (0,T) \ni t \mapsto \begin{cases} -\displaystyle\int_t^\tau \varphi(s)\,ds, & t<\tau,\\ 0, & t>\tau,\end{cases}$$

which indeed satisfies

$$\int_0^T \|\eta_\tau(t)\|_{H^1(\Omega)}^2\,dt \le \int_0^\tau (\tau-t)\|\varphi\|_{L^2(0,T;H^1(\Omega))}^2\,dt \le T^2\|\varphi\|_{L^2(0,T;H^1(\Omega))}^2.$$

Introducing η_τ as a test function in (3.6), we get for all $\tau \in [0,T]$

$$\int_0^\tau \left[\frac{1}{c^2}\left\langle \partial_t^2\varphi(t), -\int_t^\tau \varphi(s)\,ds\right\rangle_{H^1(\Omega)',H^1(\Omega)}\right.$$

$$+\frac{1}{g}\left\langle \partial_t^2(\varphi|_{\Gamma_0})(t), -\int_t^\tau \varphi|_{\Gamma_0}(s)\,ds\right\rangle_{H^{-1/2}(\Gamma_0),H^{1/2}(\Gamma_0)} + (\nabla(\partial_t\eta_\tau)(t), \nabla\eta_\tau(t))_{L^2(\Omega)}\Big]dt = 0.$$

Integrating by parts and recalling that $\eta_\tau(\tau) = 0$ while $\partial_t\varphi(0) = 0$ and $\partial_t(\varphi|_{\Gamma_0})(0) = 0$, we get that, for all $\tau \in [0, T]$,

$$\int_0^\tau \Big[-\frac{1}{c^2}\langle \partial_t\varphi(t), \varphi(t)\rangle_{H^1(\Omega)',H^1(\Omega)} - \frac{1}{g}\langle \partial_t(\varphi|_{\Gamma_0})(t), \varphi|_{\Gamma_0}(t)\rangle_{H^{-1/2}(\Gamma_0),H^{1/2}(\Gamma_0)}\Big]dt + \frac{1}{2}\Big[(\nabla\eta_\tau(t),\, \nabla\eta_\tau(t))_{L^2(\Omega)}\Big]_0^\tau = 0.$$

Finally, using $\varphi(0) = 0$, we get

$$\frac{1}{2c^2}\|\varphi(\tau)\|^2_{L^2(\Omega)} + \frac{1}{2g}\|\varphi|_{\Gamma_0}(\tau)\|^2_{L^2(\Gamma_0)} + \frac{1}{2}\left\|\nabla\int_0^\tau \varphi\right\|^2_{L^2(\Omega)} = 0,$$

which implies that $\varphi = 0$ and concludes the proof.

3.4 An abstract framework for the inverse problem

In this section, we recall some results already proved either in [24] or in [25]. This is why we just state them without any proof.

3.4.1 The general case

Let us consider two real Hilbert spaces V and H, equipped with their scalar products denoted $(\cdot,\cdot)_V$ and $(\cdot,\cdot)_H$, respectively. Let us consider a continuous operator $A : V \to H$. We consider that A is possibly neither injective nor onto. In addition, it has not necessarily a dense range. We recall that A has a dense range is equivalent to the fact that the adjoint operator of A, denoted A^*, is injective. Of course, for $F \in H$, finding $u \in V$ such that

$$Au = F \tag{3.24}$$

is ill-posed in general. This is why we introduce the Tikhonov regularization of such problem, that is for $\varepsilon > 0$: find $u_\varepsilon \in V$ such that for all $v \in V$,

$$(Au_\varepsilon, Av)_H + \varepsilon(u_\varepsilon, v)_V = (F, Av)_H. \tag{3.25}$$

Such problem is obviously well-posed, and we have the classical convergence result.

Theorem 3.4.1. *Assume that for $F \in H$, the Problem (3.24) has at least one solution. Then when $\varepsilon \to 0$, the solution u_ε to the Problem (3.25) converges to the unique solution of the minimization problem*

$$\inf_{v\in S}\|v\|_V \text{ with } S := \{v \in V,\, Av = F\}.$$

In the context of inverse problems, generally the data F is associated with a measurement, it is hence noisy. Considering that F is an unperturbed data, the noisy data F^δ is supposed to be such that

$$\|F^\delta - F\|_H \le \delta,$$

where δ is the amplitude of the noise. A classical way of choosing the regularization parameter ε in (3.25) when we only have access to some noisy data F^δ is the Morozov discrepancy principle. It consists in choosing ε in order to solve the inverse problem with the precision of the noise. In [24] the following result is obtained.

Theorem 3.4.2. *For some $\delta > 0$, let us assume that the data $F^\delta \in H$ is such that*

$$\|F_\perp^\delta\|_H < \delta < \|F^\delta\|_H, \tag{3.26}$$

where $F_\perp^\delta$ is the orthogonal projection of F^δ on $(\text{Range}\, A)^\perp$.
For $\varepsilon > 0$, let us denote $u_\varepsilon^\delta \in V$ the regularized solution associated with data F^δ in the sense of Tikhonov, which is defined by the weak formulation

$$(Au_\varepsilon^\delta, Av)_H + \varepsilon(u_\varepsilon^\delta, v)_V = (F^\delta, Av)_H, \quad \forall v \in V. \tag{3.27}$$

There exists a unique $\varepsilon > 0$ such that

$$\|Au_\varepsilon^\delta - F^\delta\|_H = \delta. \tag{3.28}$$

Remark 3.4.1. *It should be noted that if A is injective, Theorem 3.4.1 implies that u_ε converges to the unique solution $u \in V$ to the problem $Au = F$. If A has a dense range, Assumption 3.26 in Theorem 3.4.2 reduces to $\|F^\delta\|_H > \delta$.*

We now wish to introduce a method to compute the Morozov value of $\varepsilon(\delta)$ given by Theorem 3.4.2 as well as the corresponding solution $u_{\varepsilon(\delta)}^\delta$. To this aim we introduce the following minimization problem:

$$(\mathscr{P}^*) \quad \inf_{q \in H} G^\delta(q), \quad G^\delta(q) := \frac{1}{2}\|A^*q\|_V^2 + \delta\,\|q\|_H - (F^\delta, q)_H. \tag{3.29}$$

We have the following theorem, which is proved in [24].

Theorem 3.4.3. *If the noisy data F^δ satisfies the assumption (3.26), the problem $(\mathscr{P}^*)$ given by (3.29) has at least one solution p^δ, and $u^\delta = A^*p^\delta$ coincides with the Tikhonov solution u_ε^δ of Problem (3.27) where $\varepsilon(\delta)$ is the unique value of $\varepsilon > 0$ such that (3.28) is satisfied, according to Theorem 3.4.2. Lastly, $p^\delta \neq 0$ and*

$$\varepsilon(\delta) = \frac{\delta}{\|p^\delta\|_H}.$$

A drawback of the minimization problem $(\mathscr{P}^*)$ is that it is not a quadratic problem, so that it requires a gradient method that may happen to converge very slowly. In order to remedy this problem, following [48] we introduce an iterative method involving a quadratic cost function, which is described by the following algorithm.

Algorithm 3.4.4.

1. Let $\varepsilon_n > 0$ be given
2. Compute the unique solution p_n of the minimization problem

$$\inf_{q \in H} L_n^\delta(q), \quad L_n^\delta(q) := \frac{1}{2}\|A^*q\|_V^2 + \frac{1}{2}\varepsilon_n\|q\|_H^2 - (F^\delta, q)_H \tag{3.30}$$

3. Compute $\varepsilon_{n+1} = \delta/\|p_n\|_H$ and return to step 1 for $n \to n+1$

Such algorithm is justified by the following theorem, proved in [48].

Theorem 3.4.5. *Assume that the noisy data F^δ satisfies the assumption (3.26). Let $(p_n)_{n\in\mathbb{N}}$ be the sequence of elements of H defined by the algorithm 3.4.4. When $n \to +\infty$, we have $p_n \to p^\delta$ in H, where p^δ is a solution of problem $(\mathscr{P}^*)$ and $u_n := A^*p_n \to u^\delta$ in V.*

As a result, Algorithm (3.4.4) is an effective method to compute the Morozov solution.

Remark 3.4.2. *In [48], the proof is given in the special case when the operator A has a dense range (but not necessarily injective). In this case, the solution p^δ to the problem $(\mathscr{P}^*)$ is unique. However, it is easy to check that the proof remains true in the case when A has not a dense range.*

3.4.2 The composite case

In a view to address the inverse problems we have introduced in Section 3.2, we consider some particular case for operator A. More precisely, we introduce two continuous operators $B : V \to M$ and $C : V \to O$, where M and O are two Hilbert spaces equipped with the scalar products $(\cdot,\cdot)_M$ and $(\cdot,\cdot)_O$, so that we define the composite operator A such that $Av = (Bv, Cv)$ for all $v \in V$ and $H = M \times O$. We also introduce some exact data $f \in O$, so that, $F := (0, f)$ in an exact composite data in H and similarly, for some noisy data $f^\delta \in O$ with $\|f^\delta - f\|_O \leq \delta$, $F^\delta := (0, f^\delta)$ is a noisy data in H, with $\|F^\delta - F\|_H \leq \delta$. We wish now to specify the weak formulation (3.25) when assuming this special composite form for operator A.

We have the following theorem, which is proved in [25].

Theorem 3.4.6. *For $F = (0, f) \in M \times O$, $u_\varepsilon \in V$ is the solution to the Problem (3.25) if and only if $(u_\varepsilon, \lambda_\varepsilon := Bu_\varepsilon) \in V \times M$ is the solution to the following weak formulation:*

$$\left\{\begin{array}{rcl} \varepsilon(u_\varepsilon, v)_V + b(v, \lambda_\varepsilon) + (Cu_\varepsilon, Cv)_O &=& (f, Cv)_O, \quad \forall v \in V, \\ b(u_\varepsilon, \mu) - (\lambda_\varepsilon, \mu)_M &=& 0, \quad \forall \mu \in M, \end{array}\right. \tag{3.31}$$

where the bilinear form b on $V \times M$ is defined by

$$b(v, \mu) = (Bv, \mu)_M, \quad \forall (v, \mu) \in V \times M. \tag{3.32}$$

We also wish to derive a weak formulation satisfied for all $n \in \mathbb{N}$ by the solution $p_n = (\lambda_n, h_n) \in M \times O$ to the minimization Problem (3.30) and by $u_n = A^* p_n \in V$. The following theorem is proved in [25].

Theorem 3.4.7. *The triple $(u_n, \lambda_n, h_n) \in V \times M \times O$ satisfies the well-posed weak formulation: for all $(v, \mu, h) \in V \times M \times O$,*

$$\left\{\begin{array}{rcl} (u_n, v)_V - b(v, \lambda_n) - (h_n, Cv)_O &=& 0, \quad \forall v \in V, \\ b(u_n, \mu) + \varepsilon_n(\lambda_n, \mu)_M &=& 0, \quad \forall \mu \in M, \\ (Cu_n, h)_O + \varepsilon_n(h_n, h)_O &=& (f^\delta, h)_O, \quad \forall h \in O. \end{array}\right. \tag{3.33}$$

3.5 Application to the tsunami problem

3.5.1 Setting in a bounded domain

Having in mind the numerical implementation, we have to bound the domain in the x direction. We denote $\Omega^R = \{(x, z) \in \Omega,\ -R < x < R\}$, $\Sigma_{\pm R}$ the two transverse sections which bound Ω^R in the x direction, Γ_0^R and Γ_{-H}^R the subsets of Γ_0 and Γ_{-H}, respectively, which are delimited by the transverse sections $\Sigma_{\pm R}$. For the sake of simplicity, we impose some first-order absorbing boundary conditions on the transverse sections $\Gamma_{\pm R}$ following [54].

The tsunami Problem (3.1) for the velocity potential u in the bounded domain Ω^R becomes

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2 u - \Delta u &=& 0 & \text{in } \Omega^R \times (0, T), \\ \partial_t^2 u + g\,\partial_z u &=& 0 & \text{on } \Gamma_0^R \times (0, T), \\ \pm\partial_x u + (1/c)\partial_t u &=& 0 & \text{on } \Sigma_{\pm R} \times (0, T), \\ -\partial_z u &=& \chi & \text{on } \Gamma_{-H} \times (0, T), \\ u = \partial_t u &=& 0 & \text{on } \Omega^R \times \{0\}, \end{array}\right. \tag{3.34}$$

while the tsunami Problem (3.4) becomes

$$\left\{\begin{array}{rcll} \Delta u &=& 0 & \text{in } \Omega^R \times (0, T), \\ \partial_t^2 u + g\,\partial_z u &=& 0 & \text{on } \Gamma_0^R \times (0, T), \\ \pm\partial_x u + (1/\sqrt{gH})\partial_t u &=& 0 & \text{on } \Sigma_{\pm R} \times (0, T), \\ -\partial_z u &=& \chi & \text{on } \Gamma_{-H} \times (0, T), \\ u = \partial_t u &=& 0 & \text{on } \Gamma_0^R \times \{0\}, \end{array}\right. \tag{3.35}$$

where the choice of the celerity $\sqrt{gH}$ of gravity waves in the absorbing boundary condition is suggested and justified in [78]. In the bounded domain Ω^R, the inverse problems reads as follows: we want to retrieve χ on $\Gamma^R_{-H} \times (0,T)$ from the measurement of s on $\Gamma^R_0 \times (0,T)$, which amounts, since $\partial_t u + g\, s = 0$, to measure $f := \partial_t u$ on $\Gamma^R_0 \times (0,T)$. As a result, the inverse problem for the complete model consists, from f in $\Gamma^R_0 \times (0,T)$, in finding $u \in \Omega^R \times (0,T)$ satisfying the problem

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2 u - \Delta u &=& 0 & \text{in } \Omega^R \times (0,T), \\ \partial_t^2 u + g\,\partial_z u &=& 0 & \text{on } \Gamma^R_0 \times (0,T), \\ \partial_t u &=& f & \text{on } \Gamma^R_0 \times (0,T), \\ \pm\partial_x u + (1/c)\partial_t u &=& 0 & \text{on } \Sigma_{\pm R} \times (0,T), \\ u = \partial_t u &=& 0 & \text{on } \Omega^R \times \{0\}, \end{array}\right. \tag{3.36}$$

and then computing $\chi = -\partial_z u$ on $\Gamma^R_{-H} \times (0,T)$. In the asymptotic model, the system (3.36) is replaced by

$$\left\{\begin{array}{rcll} \Delta u &=& 0 & \text{in } \Omega^R \times (0,T), \\ \partial_t^2 u + g\,\partial_z u &=& 0 & \text{on } \Gamma^R_0 \times (0,T), \\ \partial_t u &=& f & \text{on } \Gamma^R_0 \times (0,T), \\ \pm\partial_x u + (1/\sqrt{gH})\partial_t u &=& 0 & \text{on } \Sigma_{\pm R} \times (0,T), \\ u = \partial_t u &=& 0 & \text{on } \Gamma^R_0 \times \{0\}. \end{array}\right. \tag{3.37}$$

In order to solve such inverse problems, we recast them in the abstract framework of Section 3.4, in particular Subsection 3.4.2.

3.5.2 The complete model

Concerning the regularization of the Problem (3.36), the spaces V, M, O are chosen as

$$V = \{v \in H^1(0,T;H^1(\Omega^R)),\ v(\cdot,0) = 0 \text{ in } \Omega^R\},$$

which is equipped with the natural scalar product

$$(u,v)_{H^1(0,T;H^1(\Omega^R))} = \int_0^T \left((u,v)_{H^1(\Omega^R)} + (\partial_t u, \partial_t v)_{H^1(\Omega^R)}\right)\, dt, \quad \forall u, v \in V, \tag{3.38}$$

while

$$M = \{\mu \in H^1(0,T;H^1(\Omega^R)),\ \mu(\cdot,T) = 0 \text{ in } \Omega^R,\ \mu = 0 \text{ on } \Gamma^R_{-H} \times (0,T)\},$$

which is equipped with the same scalar product as V, and lastly $O = L^2(0,T;L^2(\Gamma^R_0))$, equipped with its standard scalar product. Note that instead of (3.38), the scalar product of V may be simplified as

$$(u,v)_V = \int_0^T (\partial_t u, \partial_t v)_{H^1(\Omega^R)}\, dt, \quad \forall u, v \in V,$$

in virtue of the Poincaré inequality, as proved in the following lemma.

Lemma 3.5.1. *The norm induced by $\|\cdot\|_V$ is equivalent to the full norm $\|\cdot\|_{H^1(0,T;H^1(\Omega^R))}$.*

Proof. Denoting, for all $v \in V$,

$$f(t) = \|v(\cdot,t)\|^2_{H^1(\Omega^R)}, \quad t \in (0,T),$$

we have $f \in L^1(0,T)$. Since for all $t \in (0,T)$ and all $\eta > 0$,

$$\left|\frac{df}{dt}(t)\right| = 2|(v(\cdot,t), \partial_t v(\cdot,t))_{H^1(\Omega^R)}| \le \eta \|v(\cdot,t)\|^2_{H^1(\Omega^R)} + \frac{1}{\eta}\|\partial_t v(\cdot,t)\|^2_{H^1(\Omega^R)},$$

we obtain that $f \in W^{1,1}(0,T)$. Due to $f(0) = 0$, we have the Poincaré inequality

$$\|f\|_{L^1(0,T)} \le C \left\|\frac{df}{dt}\right\|_{L^1(0,T)},$$

$C > 0$ being a constant, that is

$$\int_0^T \|v(\cdot,t)\|^2_{H^1(\Omega^R)}\,dt \le C\eta \int_0^T \|v(\cdot,t)\|^2_{H^1(\Omega^R)}\,dt + \frac{C}{\eta}\int_0^T \|\partial_t v(\cdot,t)\|^2_{H^1(\Omega^R)}\,dt.$$

It suffices to take η sufficiently small to obtain the result. □

Let us now choose the two operators B and C. Let us define the bilinear form $b : V \times M \to \mathbb{R}$ by

$$\begin{aligned} b(v,\mu) &= -\frac{1}{c^2}\int_0^T\int_{\Omega_R} \partial_t v\,\partial_t\mu\,dX\,dt + \int_0^T\int_{\Omega^R} \nabla v\cdot\nabla\mu\,dX\,dt \\ &+ \frac{1}{c}\sum_{\pm}\int_0^T\int_{\Sigma_{\pm R}} \partial_t v\,\mu\,dz\,dt - \frac{1}{g}\int_0^T\int_{\Gamma_0^R} \partial_t v\,\partial_t\mu\,dx\,dt, \end{aligned} \tag{3.39}$$

which is continuous, and the corresponding operator $B : V \to M$ with the help of (3.32). Let us define the operator $C : V \to O$ such that

$$Cv = \partial_t v|_{\Gamma_0^R\times(0,T)}, \quad \forall v \in V.$$

In the sequel, the analysis will require the following notation: $Q^R = \Omega^R \times (0,T)$, $S_0^R = \Omega^R \times \{0\}$, $S_T^R = \Omega^R \times \{T\}$, $\Sigma_0^R = \Gamma_0^R \times (0,T)$, $\Sigma_{-H}^R = \Gamma_{-H}^R \times (0,T)$ and $W_{\pm R} = \Sigma_{\pm R} \times (0,T)$. All the sets S_0^R, S_T^R, Σ_0^R, Σ_{-H}^R and $W_{\pm R}$ are subsets of the boundary ∂Q^R of the 3-dimensional domain Q^R. The following proposition emphasizes the relationship between the inverse Problem (3.36) and the abstract setting.

Proposition 3.5.1. *If the function $u \in V$ satisfies $Au = F$, then it is a solution to Problem (3.36).*

Proof. Let us assume that $u \in V$ satisfies $Au = F$, which is equivalent to $Bu = 0$ and $Cu = f$. The fact that $u \in V$ yields $u = 0$ on $\Omega^R \times \{0\}$, which is the first initial condition of (3.36).

The fact that $Cu = f$ directly implies that $\partial_t u = f$ on $\Gamma_0^R \times (0,T)$, which is the third equation of (3.36). The fact that $Bu = 0$ is equivalent to $b(u,\mu) = 0$ for all $\mu \in M$. We first choose $\mu \in C_0^\infty(\Omega^R \times (0,T))$, which implies that

$$(1/c^2)\partial_t^2 u - \Delta u = 0$$

in the sense of distributions in $\Omega^R \times (0,T)$.

In a view to apply the Green formula in the domain Q^R, we introduce the 3-dimensional vector field

$$V_3 = \big(\nabla u, -(1/c^2)\partial_t u\big) \in (L^2(Q_R))^3.$$

We also denote, for $\mu \in H^1(Q^R)$,

$$\nabla_3\mu = (\nabla\mu, \partial_t\mu) \in (L^2(Q_R))^3.$$

Using this notation, we obtain that for all $\mu \in M$,

$$-\frac{1}{c^2}\int_0^T\int_{\Omega_R} \partial_t u\,\partial_t\mu\,dX\,dt + \int_0^T\int_{\Omega^R} \nabla v\cdot\nabla\mu\,dX\,dt = \int_{Q^R} V_3\cdot\nabla_3\mu\,dX\,dt.$$

Observing that

$$\mathrm{div}_3 V_3 := \partial_x V_3 + \partial_z V_3 + \partial_t V_3 = \Delta u - (1/c^2)\partial_t^2 u = 0$$

in Q^R, the Green formula implies that

$$\int_{Q^R} V_3\cdot\nabla_3\mu\,dX\,dt = \langle V_3\cdot\nu_3, \mu\rangle_{H^{-1/2}(\partial Q^R),H^{1/2}(\partial Q^R)}, \tag{3.40}$$

where ν_3 is the outward unit normal to Q^R and $\langle\cdot,\cdot\rangle_{H^{-1/2}(\partial Q^R),H^{1/2}(\partial Q^R)}$ means the duality product between $H^{-1/2}(\partial Q^R)$ and $H^{1/2}(\partial Q^R)$. The definition (3.39) of the bilinear form b and (3.40) implies that for all $\mu \in M$,

$$\langle V_3\cdot\nu_3,\mu\rangle_{H^{-1/2}(\partial Q^R),H^{1/2}(\partial Q^R)} + \frac{1}{c}\sum_{\pm}\int_{W_{\pm R}} \partial_t u\,\mu dz\,dt - \frac{1}{g}\int_{\Sigma_0^R} \partial_t u\,\partial_t\mu\,dx\,dt = 0. \tag{3.41}$$

Let us first choose μ in the space of functions $C^\infty(\overline{Q^R})$ satisfying $\mu = 0$ in a volumic vicinity of $\partial Q^R \setminus \overline{S_0}$, which is a subset of M. Then (3.41) becomes

$$(1/c^2)\langle \partial_t u, \mu\rangle_{\mathcal{D}'(S_0),C_0^\infty(S_0)} = 0.$$

Since the traces of the functions μ on S_0 describes the whole space $C_0^\infty(S_0)$, we have $\partial_t u = 0$ in the sense of distribution in $S_0 = \Omega^R \times \{0\}$, that is the second intial condition of (3.36). Let us secondly choose $\mu \in C^\infty(\overline{Q^R})$ with $\mu = 0$ in a volumic vicinity of $\partial Q^R \setminus \overline{W_{\pm R}}$, those μ being elements of M. Using (3.41) we obtain

$$\langle \pm\partial_x u, \mu\rangle_{\mathcal{D}'(W_{\pm R}),C_0^\infty(W_{\pm R})} + \frac{1}{c}\int_{W_{\pm R}} \partial_t u\, \mu\, dzdt = 0,$$

which implies $\pm\partial_x u + (1/c)\partial_t u = 0$ in the sense of distribution in $W_{\pm R}$, in other words the fourth equation of (3.36). Lastly, we choose $\mu \in C^\infty(\overline{Q^R})$ with $\mu = 0$ in a volumic vicinity of $\partial Q^R \setminus \overline{\Sigma_0^R}$. We obtain from (3.41) that

$$\langle \partial_z u, \mu\rangle_{\mathcal{D}'(\Sigma_0^R),C_0^\infty(\Sigma_0^R)} - \frac{1}{g}\langle \partial_t u, \partial_t\mu\rangle_{\mathcal{D}'(\Sigma_0^R),C_0^\infty(\Sigma_0^R)} = 0,$$

that is

$$\left\langle \partial_z u + \frac{1}{g}\partial_t^2 u, \mu \right\rangle_{\mathcal{D}'(\Sigma_0^R),C_0^\infty(\Sigma_0^R)} = 0,$$

hence $\partial_z u + (1/g)\partial_t^2 u = 0$ in the sense of distribution in Σ_0^R, which is the second equation of (3.36). □

Let us specify the mixed formulation of the Tikhonov regularization (3.31) in the context of the complete model: find $(u_\varepsilon, \lambda_\varepsilon) \in V \times M$ such that for all $(v, \mu) \in V \times M$,

$$\begin{cases}
\varepsilon \displaystyle\int_0^T \int_{\Omega_R} (\partial_t u_\varepsilon\, \partial_t v + \nabla(\partial_t u_\varepsilon)\cdot\nabla(\partial_t v))\, dXdt + \\
-\dfrac{1}{c^2}\displaystyle\int_0^T \int_{\Omega_R} \partial_t v\, \partial_t \lambda_\varepsilon\, dXdt + \int_0^T \int_{\Omega^R} \nabla v\cdot\nabla\lambda_\varepsilon\, dXdt \\
+\dfrac{1}{c}\displaystyle\sum_{\pm}\int_0^T \int_{\Sigma_{\pm R}} \partial_t v\, \lambda_\varepsilon\, dzdt - \frac{1}{g}\int_0^T \int_{\Gamma_0^R} \partial_t v\, \partial_t\lambda_\varepsilon\, dxdt \\
+\displaystyle\int_0^T \int_{\Gamma_0^R} \partial_t u_\varepsilon\, \partial_t v\, dxdt = \int_0^T \int_{\Gamma_0^R} f\, \partial_t v\, dxdt, \quad \forall v \in V, \\
-\dfrac{1}{c^2}\displaystyle\int_0^T \int_{\Omega_R} \partial_t u_\varepsilon\, \partial_t\mu\, dXdt + \int_0^T \int_{\Omega^R} \nabla u_\varepsilon\cdot\nabla\mu\, dXdt \\
+\dfrac{1}{c}\displaystyle\sum_{\pm}\int_0^T \int_{\Sigma_{\pm R}} \partial_t u_\varepsilon\, \mu\, dzdt - \frac{1}{g}\int_0^T \int_{\Gamma_0^R} \partial_t u_\varepsilon\, \partial_t\mu\, dxdt \\
-\displaystyle\int_0^T \int_{\Omega^R} (\partial_t\lambda_\varepsilon\, \partial_t\mu + \nabla(\partial_t\lambda_\varepsilon)\cdot\nabla(\partial_t\mu))\, dXdt, \quad \forall \mu \in M.
\end{cases} \tag{3.42}$$

If the data $f \in L^2(0,T;L^2(\Gamma_0^R))$ is such that a solution to the Problem (3.36) has a solution $u \in V$, the sequence u_ε has a limit $\tilde{u} \in V$ in virtue of Theorem 3.4.1, and we have that $\tilde{u} = u$ in

$$Q_0^R = \left\{(X,t) \in \Omega^R, \quad 0 < t < T + \frac{z}{c}\right\}$$

in virtue of Theorem 3.2.4.

Lastly, let us specify the weak formulation (3.33) in the context of the complete model: find

$(u_n, \lambda_n, h_n) \in V \times M \times O$ such that for all $(v, \mu, h) \in V \times M \times O$,

$$\left\{\begin{array}{l}
\varepsilon_n \displaystyle\int_0^T \int_{\Omega_R} (\partial_t u_n\, \partial_t v + \nabla(\partial_t u_n) \cdot \nabla(\partial_t v))\, dX dt \\
+\dfrac{1}{c^2} \displaystyle\int_0^T \int_{\Omega_R} \partial_t v\, \partial_t \lambda_n\, dX dt - \int_0^T \int_{\Omega^R} \nabla v \cdot \nabla \lambda_n\, dX dt \\
-\dfrac{1}{c} \displaystyle\sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t v\, \lambda_n\, dz dt + \dfrac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t v\, \partial_t \lambda_n\, dx dt \\
-\displaystyle\int_0^T \int_{\Gamma_0^R} h_n\, \partial_t v\, dx dt = 0, \quad \forall v \in V, \\
-\dfrac{1}{c^2} \displaystyle\int_0^T \int_{\Omega_R} \partial_t u_n\, \partial_t \mu\, dX dt + \int_0^T \int_{\Omega^R} \nabla u_n \cdot \nabla \mu\, dX dt \\
+\dfrac{1}{c} \displaystyle\sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t u_n\, \mu\, dz dt - \dfrac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t u_n\, \partial_t \mu\, dx dt \\
+\varepsilon_n \displaystyle\int_0^T \int_{\Omega^R} (\partial_t \lambda_n\, \partial_t \mu + \nabla(\partial_t \lambda_n) \cdot \nabla(\partial_t \mu))\, dX dt, \quad \forall \mu \in M, \\
\displaystyle\int_0^T \int_{\Gamma_0^R} \partial_t u_n\, h\, dx dt + \varepsilon_n \int_0^T \int_{\Gamma_0^R} h_n\, h\, dx dt = \int_0^T \int_{\Gamma_0^R} f^\delta\, h\, dx dt, \quad \forall h \in O.
\end{array}\right. \tag{3.43}$$

3.5.3 The asymptotic model

The previous choices of the spaces V, M, O and well as the operators B, C have to be adapted to the asymptotic case (3.37). The spaces V, M, O are now chosen as

$$V = \{v \in H^1(0,T; H^1(\Omega^R)),\ u(\cdot, 0) = 0 \text{ on } \Gamma_0^R\},$$

which is equipped with the scalar product (3.38),

$$M = \{\mu \in H^1(0,T; H^1(\Omega^R)),\ \mu(\cdot, T) = 0 \text{ on } \Gamma_0^R,\ \mu = 0 \text{ on } \Gamma_{-H}^R \times (0,T)\},$$

which is equipped with the scalar product

$$(\lambda, \mu)_M = \int_0^T \int_{\Omega^R} (\nabla \lambda \cdot \nabla \mu + \nabla(\partial_t \lambda) \cdot \nabla(\partial_t \mu))\, dX dt, \quad \forall \lambda, \mu \in M.$$

The space $O = L^2(0,T; L^2(\Gamma_0^R))$ is however unchanged.

Let us now choose the two operators B and C. Let us define the new bilinear form $b : V \times M \to \mathbb{R}$ by

$$\begin{array}{rcl}
b(v, \mu) & = & \displaystyle\int_0^T \int_{\Omega^R} \nabla v \cdot \nabla \mu\, dX dt \\
& + & \dfrac{1}{\sqrt{gH}} \displaystyle\sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t v\, \mu\, dz dt - \dfrac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t v\, \partial_t \mu\, dx dt,
\end{array} \tag{3.44}$$

which is continuous, and the corresponding operator $B : V \to M$ with the help of (3.32). The operator $C : V \to O$ is again defined by

$$Cv = \partial_t v|_{\Gamma_0^R \times (0,T)}, \quad \forall v \in V.$$

We have the following proposition.

Proposition 3.5.2. *If the function $u \in V$ satisfies $Au = F$, assuming the additional regularity $u \in H^3(Q^R)$, then u is a solution to Problem (3.37).*

Proof. Let us assume that $u \in V$ satisfies $Au = F$, which is equivalent to $\partial_t u = f$ on $\Gamma_0^R \times (0,T)$ and $b(u, \mu) = 0$ for all $\mu \in M$, where b is given now by (3.44). The fact that $u \in V$ yields $u = 0$

on $\Gamma_0^R \times \{0\}$, which is the first initial condition of (3.37). We first choose $\mu \in C_0^\infty(\Omega^R \times (0,T))$ in (3.45), which implies that

$$\Delta u = 0$$

in the sense of distributions in $\Omega^R \times (0,T)$. Defining the 3-dimensional vector field

$$\tilde{V}_3 = (\nabla u, 0) \in (L^2(Q_R))^3,$$

we observe that $\mathrm{div}_3 \tilde{V}_3 = 0$, hence we have

$$\int_0^T \int_{\Omega^R} \nabla v \cdot \nabla \mu \, dX dt = \int_{Q^R} \tilde{V}_3 \cdot \nabla_3 \mu \, dX dt = \langle \tilde{V}_3 \cdot \nu_3, \mu \rangle_{H^{-1/2}(\partial Q^R), H^{1/2}(\partial Q^R)}.$$

so that we obtain that for all $\mu \in M$,

$$\langle \tilde{V}_3 \cdot \nu_3, \mu \rangle_{H^{-1/2}(\partial Q^R), H^{1/2}(\partial Q^R)} + \frac{1}{\sqrt{gH}} \sum_{\pm} \int_{W_{\pm R}} \partial_t u \, \mu \, dz dt - \frac{1}{g} \int_{S_0^R} \partial_t u \, \partial_t \mu \, dx dt = 0. \tag{3.45}$$

Using the regularity assumption $u \in H^3(Q^R)$, which implies in particular that $\partial_{\nu_3} u \in L^2(\partial Q^R)$, we obtain in view of the definition of b given by (3.44) that

$$\begin{aligned} &\sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \pm \partial_x u \, \mu \, dz dt + \int_0^T \int_{\Gamma_0^R} \partial_z u \, \mu \, dx dt \\ &+ \frac{1}{\sqrt{gH}} \sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t u \, \mu \, dz dt - \frac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t u \, \partial_t \mu \, dx dt = 0. \end{aligned} \tag{3.46}$$

Choosing again $\mu \in C^\infty(\overline{Q^R})$ such that μ vanishes on some appropriate subparts of ∂Q^R like in the proof of Proposition 3.5.1, we obtain the boundary conditions $\pm \partial_x u + (1/\sqrt{gH}) \partial_t u = 0$ on $W_{\pm R}$ and $\partial_z u + (1/g) \partial_t^2 u = 0$ on Σ_0^R. Starting from (3.46) and reusing that $\pm \partial_x u + (1/\sqrt{gH}) \partial_t u = 0$ on $W_{\pm R}$, we have that for all $\mu \in C^\infty(\overline{Q^R})$,

$$\int_{\Sigma_0^R} \partial_z u \, \mu \, dx dt - \frac{1}{g} \int_{\Sigma_0^R} \partial_t u \, \partial_t \mu \, dx dt = 0.$$

The fact that $u \in H^3(Q^R)$, which implies that $u|_{\Sigma_0^R} \in H^2(\Sigma_0^R)$, we have the Green formula

$$\int_{\Sigma_0^R} \partial_t u \, \partial_t \mu \, dx dt = - \int_{\Sigma_0^R} \partial_t^2 u \, \mu \, dx dt + \int_{\Gamma_0^R \times \{T\}} \partial_t u \, \mu \, dx - \int_{\Gamma_0^R \times \{0\}} \partial_t u \, \mu \, dx.$$

Reusing that $\partial_z u + (1/g) \partial_t^2 u = 0$ on Σ_0^R, and since $\mu(\cdot, T) = 0$ on Γ_0^R, we obtain that $\partial_t u(\cdot, 0) = 0$ on Γ_0^R, which is the second initial condition in (3.37). □

Theorem 3.5.1. *The operator A is injective and has a dense range.*

Proof. Assume that $Au = 0$ for $u \in V$. The fact that $Cu = 0$ means $\partial_t u = 0$ on $\Gamma_0^R \times (0,T)$, which together with the initial condition $u = 0$ on $\Gamma_0^R \times \{0\}$ implies that $u = 0$ on $\Gamma_0^R \times (0,T)$. The fact that $b(u, \mu) = 0$ for all $\mu \in M$ implies like in the proof of (3.5.2) that $\Delta u = 0$ in $\Omega^R \times (0,T)$. Using the fact that $u = 0$ on $\Gamma_0^R \times (0,T)$, we obtain that for all $\mu \in M$

$$\langle \tilde{V}_3 \cdot \nu_3, \mu \rangle_{H^{-1/2}(\partial Q^R), H^{1/2}(\partial Q^R)} + \frac{1}{\sqrt{gH}} \sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t v \, \mu \, dz dt = 0.$$

Choosing $\mu \in C^\infty(\overline{Q^R})$ satisfying $\mu = 0$ in a volumic vicinity of $\partial Q^R \setminus \overline{\Sigma_0^R}$, we get that $\partial_z u = 0$ on $\Gamma_0^R \times (0,T)$. In conclusion, u satisfies

$$\left\{ \begin{array}{rcll} \Delta u & = & 0 & \text{in } \Omega^R \times (0,T), \\ u & = & 0 & \text{on } \Gamma_0^R \times (0,T), \\ \partial_z u & = & 0 & \text{on } \Gamma_0^R \times (0,T). \end{array} \right.$$

The fact that $u = 0$ is then a consequence of unique continuation. We conclude that A is injective.

In order to prove that A has a dense range, we assume that $(\lambda, f) \in H = M \times O$ satisfies

$$(Av, (\lambda, f))_H = 0, \quad \forall v \in V,$$

that is

$$b(v, \lambda) + (Cv, f)_O = 0, \quad \forall v \in V.$$

Choosing $v \in C_0^\infty(Q^R)$, we obtain $\Delta\lambda = 0$ in the sense of distribution in $\Omega^R \times (0,T)$. Now we choose $v \in C^\infty(\overline{Q^R})$ such that $v = 0$ in a volumic vicinity of $\partial Q^R \setminus \overline{\Sigma_{-H}^R}$, we obtain $\partial_z \lambda = 0$ on $\Gamma_{-H} \times (0,T)$, which together with $\lambda = 0$ on $\Gamma_{-H} \times (0,T)$ implies that $\lambda = 0$ in $\Omega^R \times (0,T)$. We finally conclude that $f = 0$, thus A has a dense range. □

Let us specify the mixed formulation of the Tikhonov regularization (3.31) in the context of the asymptotic model: find $(u_\varepsilon, \lambda_\varepsilon) \in V \times M$ such that for all $(v, \mu) \in V \times M$, such that

$$\left\{\begin{array}{l}
\varepsilon \displaystyle\int_0^T \int_{\Omega_R} (u_\varepsilon\, v + \nabla u_\varepsilon \cdot \nabla v + \partial_t u_\varepsilon\, \partial_t v + \nabla(\partial_t u_\varepsilon) \cdot \nabla(\partial_t v))\, dX dt \\
+ \displaystyle\int_0^T \int_{\Omega^R} \nabla v \cdot \nabla \lambda_\varepsilon\, dX dt + \frac{1}{\sqrt{gH}} \sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t v\, \lambda_\varepsilon\, dz dt - \frac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t v\, \partial_t \lambda_\varepsilon\, dx dt \\
+ \displaystyle\int_0^T \int_{\Gamma_0^R} \partial_t u_\varepsilon\, \partial_t v\, dx dt = \int_0^T \int_{\Gamma_0^R} f\, \partial_t v\, dx dt, \quad \forall v \in V, \\
\displaystyle\int_0^T \int_{\Omega^R} \nabla u_\varepsilon \cdot \nabla \mu\, dX dt + \frac{1}{\sqrt{gH}} \sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t u_\varepsilon\, \mu\, dz dt - \frac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t u_\varepsilon\, \partial_t \mu\, dx dt \\
- \displaystyle\int_0^T \int_{\Omega^R} (\nabla\lambda \cdot \nabla\mu + \nabla(\partial_t \lambda) \cdot \nabla(\partial_t \mu))\, dX dt = 0, \quad \forall \mu \in M.
\end{array}\right. \tag{3.47}$$

If the data $f \in L^2(0,T; L^2(\Gamma_0^R))$ is such that a solution to the Problem (3.37) has a solution $u \in V$, which is unique in virtue of Theorem 3.2.4, the sequence u_ε tends to u in V in virtue of Theorem 3.4.1 and Theorem 3.5.1.

We finally specify the weak formulation (3.33) in the context of the asymptotic model: find $(u_n, \lambda_n, h_n) \in V \times M \times O$ such that for all $(v, \mu, h) \in V \times M \times O$,

$$\left\{\begin{array}{l}
\displaystyle\int_0^T \int_{\Omega_R} (u_n\, v + \nabla u_n \cdot \nabla v + \partial_t u_n\, \partial_t v + \nabla(\partial_t u_n) \cdot \nabla(\partial_t v))\, dX dt \\
- \displaystyle\int_0^T \int_{\Omega^R} \nabla v \cdot \nabla \lambda_n\, dX dt - \frac{1}{\sqrt{gH}} \sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t v\, \lambda_n\, dz dt + \frac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t v\, \partial_t \lambda_n\, dx dt \\
- \displaystyle\int_0^T \int_{\Gamma_0^R} h_n\, \partial_t v\, dx dt = 0, \quad \forall v \in V, \\
\displaystyle\int_0^T \int_{\Omega^R} \nabla u_n \cdot \nabla \mu\, dX dt + \frac{1}{\sqrt{gH}} \sum_{\pm} \int_0^T \int_{\Sigma_{\pm R}} \partial_t u_n\, \mu\, dz dt - \frac{1}{g} \int_0^T \int_{\Gamma_0^R} \partial_t u_n\, \partial_t \mu\, dx dt \\
+ \varepsilon_n \displaystyle\int_0^T \int_{\Omega^R} (\nabla \lambda_n \cdot \nabla \mu + \nabla(\partial_t \lambda_n) \cdot \nabla(\partial_t \mu))\, dX dt = 0, \quad \forall \mu \in M, \\
\displaystyle\int_0^T \int_{\Gamma_0^R} \partial_t u_n\, h\, dx dt + \varepsilon_n \int_0^T \int_{\Gamma_0^R} h_n\, h\, dx dt = \int_0^T \int_{\Gamma_0^R} f^\delta\, h\, dx dt, \quad \forall h \in O.
\end{array}\right. \tag{3.48}$$

In view of Theorem 3.4.5 and Theorem 3.5.1, here $p_n = (\lambda_n, h_n)$ converges in $H = V \times O$ to the unique solution of Problem (3.29).

3.6 About an iterative inversion method

As appears in the weak formulations of the previous section, for example (3.42) or (3.43), the matrix to invert after discretization of such systems is large, since it is a space-time system involving the

two variables $(u_\varepsilon, \lambda_\varepsilon)$ in (3.42) and the three variables (u_n, λ_n, h_n) in (3.43). Let us consider the first case, in which we propose a Successive Over Relaxation method (known as the SOR method in the literature) in order to decrease the sizes of the matrices we invert. Such method is described for example in [105]. The price to pay is that those smaller matrices have to be inverted many times because of the iterations. All in all, this technique will enable us to consider smaller meshes, both in space and time. In this section, following [13] we justify the convergence of the method under the condition that the relaxation parameter is sufficiently small.

After discretization of the Problem (3.42) with the help of the Finite Element Method, the $(m+n)\times(m+n)$ system to invert is in the form

$$MX = F, \quad M = \begin{pmatrix} A & B \\ B^T & -C \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix}, \quad F = \begin{pmatrix} f \\ g \end{pmatrix}, \tag{3.49}$$

where $A \in \mathcal{M}_m(\mathbb{R})$, $C \in \mathcal{M}_n(\mathbb{R})$, $B \in \mathcal{M}_{mn}(\mathbb{R})$ are matrices, $B^T \in \mathcal{M}_{nm}(\mathbb{R})$ being the transpose of B, while $x, f \in \mathbb{R}^m$ and $y, g \in \mathbb{R}^n$ are vectors. The matrix M is invertible (but not symmetric), while the matrices A, C are symmetric positive definite matrices, hence invertible. Following [105], the Jacobi matrix associated with M is

$$J := -D^{-1}M + I, \quad D := \begin{pmatrix} A & 0 \\ 0 & -C \end{pmatrix},$$

where I is the identity matrix of size $m+n$. An easy computation shows that

$$J = \begin{pmatrix} 0 & -A^{-1}B \\ C^{-1}B^T & 0 \end{pmatrix},$$

and we use the decomposition $J = L + U$ such that

$$L := \begin{pmatrix} 0 & 0 \\ C^{-1}B^T & 0 \end{pmatrix}, \quad U := \begin{pmatrix} 0 & -A^{-1}B \\ 0 & 0 \end{pmatrix}.$$

Observing that the system (3.49) is equivalent to the system

$$(I - \omega L)X = (\omega U + (1-\omega)I)\, X + \omega D^{-1}F,$$

for all $\omega \in \mathbb{R}$, we introduce, choosing some initial value $X_0 \in \mathbb{R}^{m+n}$, the sequence of $X_p \in \mathbb{R}^{m+n}$ given by

$$(I - \omega L)X_{p+1} = (\omega U + (1-\omega)I)\, X_p + \omega D^{-1}F, \tag{3.50}$$

for all $p \in \mathbb{N}$.

The key point is of course the convergence of such sequence to X when $p \to +\infty$. To address such question, we are lead to estimate the spectral radius of

$$Ł_\omega := (I - \omega L)^{-1}\,(\omega U + (1-\omega)I)\,.$$

In this view, let us find a relationship between the spectra of matrices J and $Ł_\omega$. This relationship is established in the following theorem, which is proved in [105, see Theorem 4.3]

Theorem 3.6.1. *Assume that $\omega \neq 0$.*

If λ is an eigenvalue of L_ω such that $\lambda \neq 0$ and μ satisfies

$$(\lambda + \omega - 1)^2 = \lambda\,\omega^2\mu^2, \tag{3.51}$$

then μ is an eigenvalue of J.

Conversely, if μ is an eigenvalue of J and λ satisfies (3.51), then λ is an eigenvalue of L_ω.

In order to prove convergence, we have to check that the eigenvalues of J are purely imaginary.

Lemma 3.6.1. *The spectrum of J belongs to $i\mathbb{R}$.*

Proof. It suffices to prove that if μ is an eigenvalue of J, then $\mu^2 \in \mathbb{R}$ and $\mu^2 \leq 0$. There exists $(x,y)^T$, $(x,y) \neq 0$, such that

$$\begin{pmatrix} 0 & -A^{-1}B \\ C^{-1}B^T & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \mu \begin{pmatrix} x \\ y \end{pmatrix},$$

that is

$$-A^{-1}By = \mu\, x, \quad C^{-1}B^T x = \mu\, y.$$

Assume first that $x \neq 0$. We obtain that

$$-A^{-1}BC^{-1}B^T x = \mu^2\, x,$$

which implies that

$$\mu^2 (Ax, x)_{\mathbb{R}^m} + (C^{-1}B^T x, B^T x)_{\mathbb{R}^m} = 0.$$

Given that A and C^{-1} are both positive definite, we conclude that $\mu^2 \leq 0$.

If $x = 0$, then $y \neq 0$, hence $\mu = 0$. □

With the help of Theorem 3.6.1 and Lemma 3.6.1, we can directly apply the following theorem, proved in [13].

Theorem 3.6.2. *Let us denote ρ_{L_ω} and ρ_J the spectral radii of L_ω and J, respectively. We have $\rho_{L_\omega} < 1$ for $\omega \in (0, \omega_0)$, where $\omega_0 = 2/(1+\rho_J)$.*

In addition, the minimum of ρ_{L_ω} on $(0,\omega_0)$ is given by $1 - 2/(1+(1+\rho_J^2)^{1/2})$.

From Theorem 3.6.2, we obtain the convergence $X_p \to X$ when $p \to +\infty$ if the parameter ω is chosen sufficiently small. The optimal value of ω depends on the spectral radius of J, which is a costly computation in practice.

From a practical point of view, the system (3.50) is equivalent to

$$\begin{cases} Ax_{p+1} &= (1-\omega)Ax_p - \omega\, By_p + \omega\, f \\ \omega\, B^T x_{p+1} - Cy_{p+1} &= -(1-\omega)Cy_p + \omega\, g. \end{cases}$$

We first compute $w \in \mathbb{R}^m$ such that

$$Aw = \omega(f - By_p),$$

then

$$x_{p+1} = w + (1-\omega)x_p.$$

Next, we compute $z \in \mathbb{R}^n$ such that

$$Cz = \omega(B^T x_{p+1} - g),$$

then

$$y_{p+1} = z + (1-\omega)y_p.$$

As can be seen above, the SOR method boils down to invert the sparse smaller matrices A and C instead of the full matrix M.

3.7 Numerical experiments

3.7.1 Introduction

Throughout this section , we choose $R = 2$, $H = 0.5$, $c = 20$, $g = 10$, $T = 1$. We discretize the weak formulations (3.42), (3.43), (3.47) and (3.48) with the help of a Finite Element Method. More precisely, the discretization of the space $H^1(0,T;H^1(\Omega))$ is based on a space triangular mesh of Ω_R and a time subdivision of $(0,T)$, the approximation space being the tensor product $\mathbb{P}1(\Omega_R) \times \mathbb{P}1(0,T)$, where $\mathbb{P}1(\Omega_R)$ is the classical Lagrange two dimensional finite element space based on the space triangular mesh and $\mathbb{P}1(0,T)$ is the Lagrange one dimensional finite element space based on the time subdivision. In the weak formulations (3.43) and (3.48), we also need to discretize the space $L^2(0,T;L^2(\Gamma_0^R))$, the approximation space being $\mathbb{P}1(\Gamma_0^R) \times \mathbb{P}1(0,T)$, where $\mathbb{P}1(\Gamma_0^R)$ is formed by the traces on Γ_0^R of the elements of $\mathbb{P}1(\Omega_R)$. All our numerical results were obtained by using the Library XLIFE++ [80].

The function χ on Γ_{-H}^R which is used to obtain the artificial data and which has to be retrieved is given by

$$\chi(x,t) = \frac{1}{2}\cos^2\left(4\pi\left(x+\frac{R}{4}\right)\right)\mathbb{1}_{[-\frac{3R}{8},-\frac{R}{8}]}(x) \times \frac{d\cdot}{dt}\left(\sin(\omega t)e^{-b(t-d)^2}\right), \tag{3.52}$$

where $\mathbb{1}_I$ is the indicator function of the interval I,

$$b = \frac{10}{d^2}, \quad d = \frac{1}{2}\left(T - \frac{H}{c}\right),$$

while ω is the mean frequency of the time function above. Otherwise specified, the frequency ω is equal to 15, the time step is $\Delta t = 0.005$ while the mesh size is $h = 0.15$. The smallness of the time step, compared to the mesh size, is justified by the fact that ω is large, in other words that the time variations are very rapid to simulate a tsunami. In order to produce artificial data, we solve the forward Problems (3.1) and (3.4) with the help of finite difference schemes (with respect to time) and finite element methods (with respect to space). More precisely, to solve (3.1) we use a classical implicit leap-frog scheme and to solve (3.4) we use the theta-scheme introduced in [78]. Once the approximated free surface s has been computed, since $f = \partial_t u = -g\,s$, we define the artificial noisy data f^δ on $\Gamma_0^R \times (0,T)$ by

$$f^\delta = f + \rho\,\mathrm{rand}.$$

Here the noise function rand is given from its values at each discretization point of $\Gamma_0^R \times (0,T)$, which are random numbers satisfying a uniform distribution $\mathcal{U}(-1,1)$. The number $\rho > 0$ is then calibrated such that

$$\frac{\|f^\delta - f\|_{L^2(\Gamma_0^R\times(0,T))}}{\|f\|_{L^2(\Gamma_0^R\times(0,T))}} = \sigma,$$

where σ is the relative noise amplitude that we impose.

3.7.2 Reconstructions for the complete model

Let us present some numerical experiments in the case of the complete model, that is we regularize the ill-posed Problem (3.36) in the presence of noisy data by using Algorithm 3.4.4, which is based on the successive weak formulations (3.43) for all $n \in \mathbb{N}$. We notice that the functions (u_n, λ_n, h_n) are almost stationary for n between 10 and 15. The retrieved value of χ^δ is then given by $\chi^\delta = -\partial_z u_n$ at convergence. Note that in view of Theorem 3.2.4, the value of χ^δ in Γ_{-H}^R is correct only on the truncated time interval $(0, T - H/c)$. But given the values of T, H and c we have chosen, it is almost $(0,T)$. Besides, our strategy in the presence of noisy relies on the assumption (3.26). In practice, the right inequality $\|F^\delta\|_H > \delta$ is easy to check. On the contrary, the left inequality $\|F_\perp^\delta\|_H < \delta$, which comes from the fact that the underlying operator A does not have a dense range, is hard to check. Indeed, we didn't manage to compute the orthogonal component $F_\perp^\delta$ in our particular case. This is why we will apply our strategy assuming this condition is satisfied. In [24] we show a simpler example in which such component is actually computed. In the figures 3.1, 3.2, 3.3, we present

comparisons between the retrieved function χ and the exact one, for the relative amplitudes of noise $\sigma = 10^{-3}$, that is almost uncontaminated data, $\sigma = 5\%$ and $\sigma = 10\%$, respectively. We observe

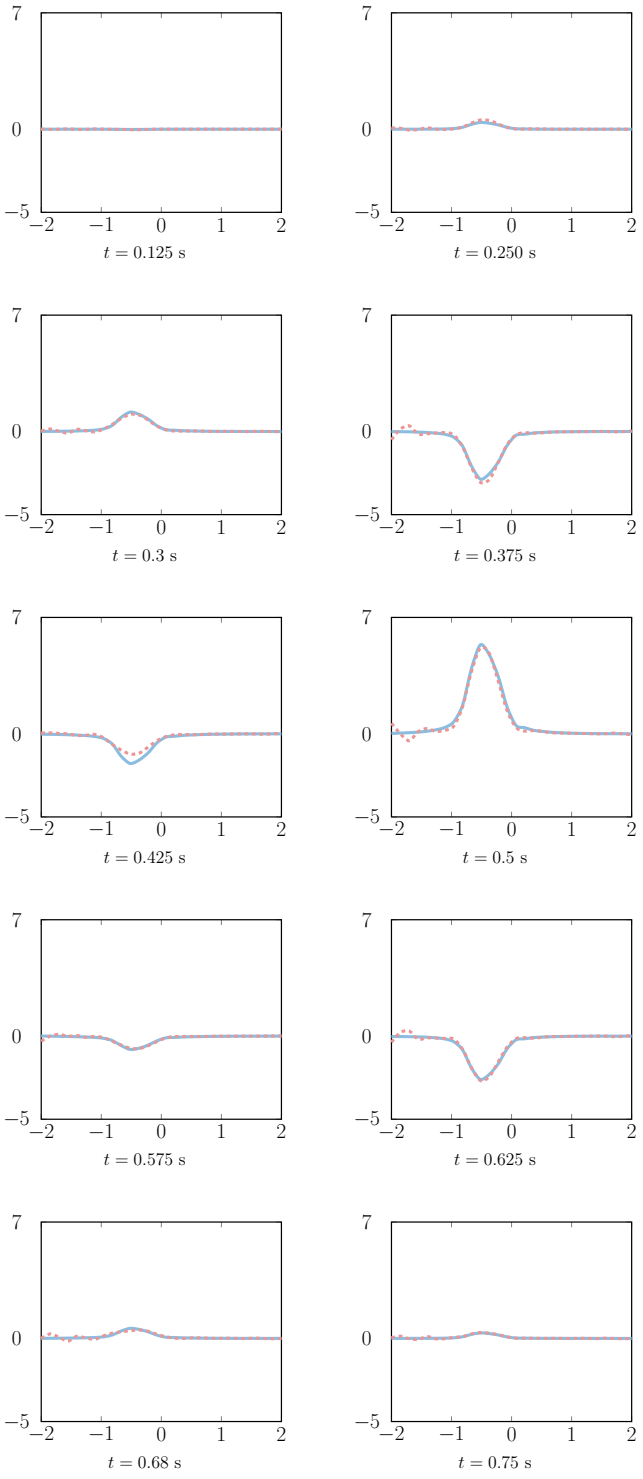


Figure 3.1: Complete model. Reconstructed χ at different times t for $\sigma = 10^{-3}$, $\Delta t = 0.005$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

that the reconstructions are robust with respect to the noise with our choice of discretization. For $\sigma = 5\%$, we now try to use a coarser time step that is $\Delta t = 0.01$ instead of 0.005, the mesh size

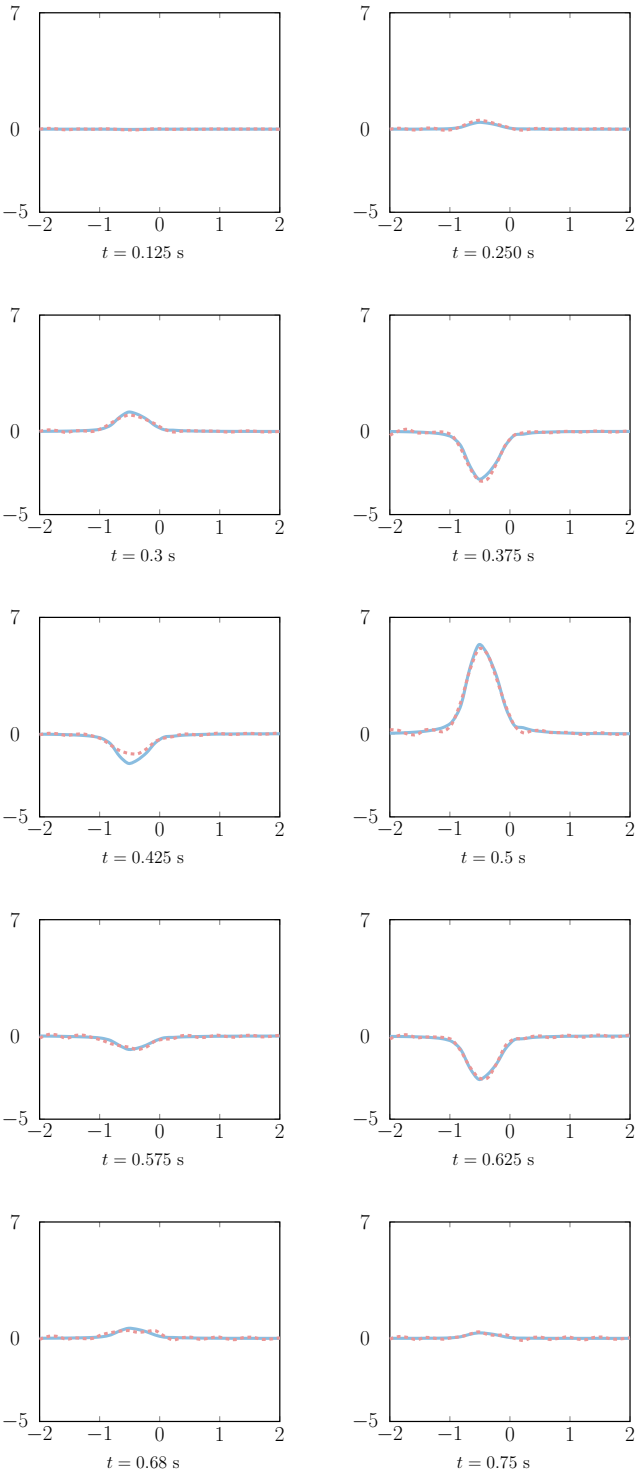


Figure 3.2: Complete model. Reconstructed χ at different times t for $\sigma = 0.05$, $\Delta t = 0.005$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

being again $h = 0.15$. We observe on figure 3.4 that the quality of the reconstruction decreases a lot (compare with figure 3.2). If now we conserve the time step $\Delta t = 0.01$ but refine the mesh, that is we choose $h = 0.075$ instead of $h = 0.15$, the matrices are too large to be inverted, this is why we use the SOR iterative method described in section 3.6. The results are presented in figure 3.5 and

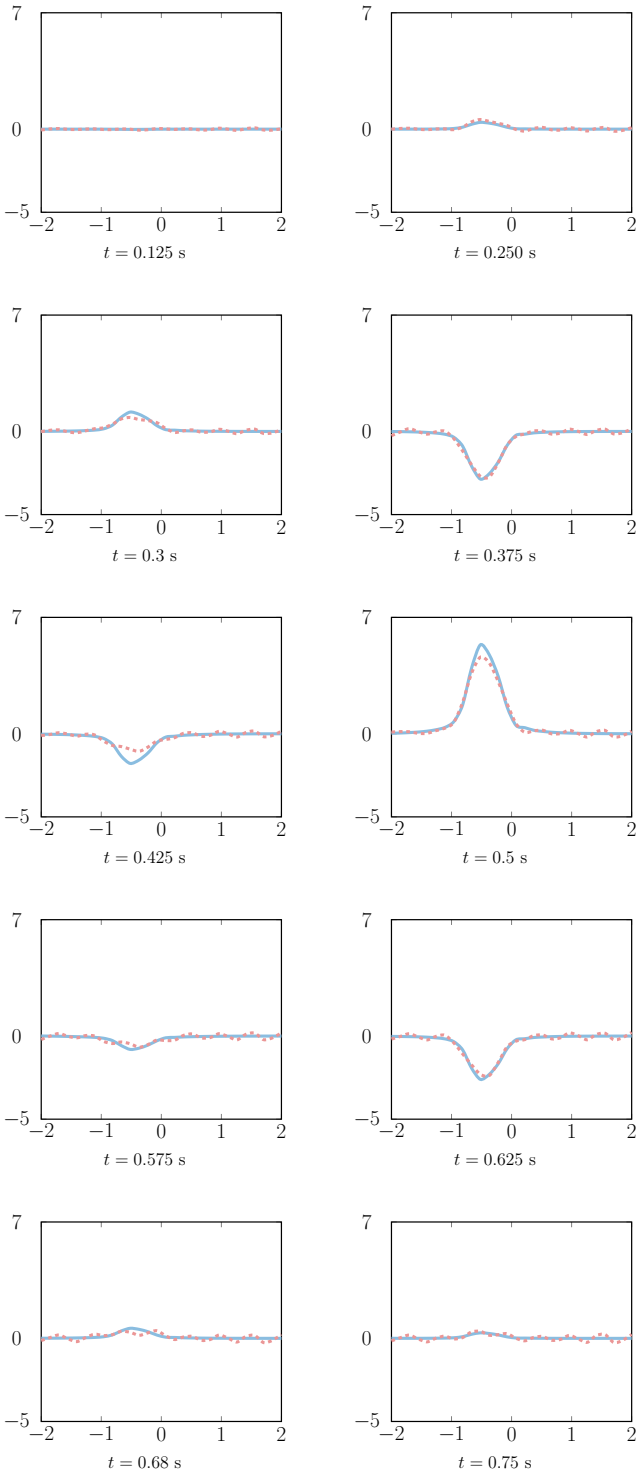


Figure 3.3: Complete model. Reconstructed χ at different times t for $\sigma = 0.1$, $\Delta t = 0.005$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

show an improvement compared to figure 3.4.

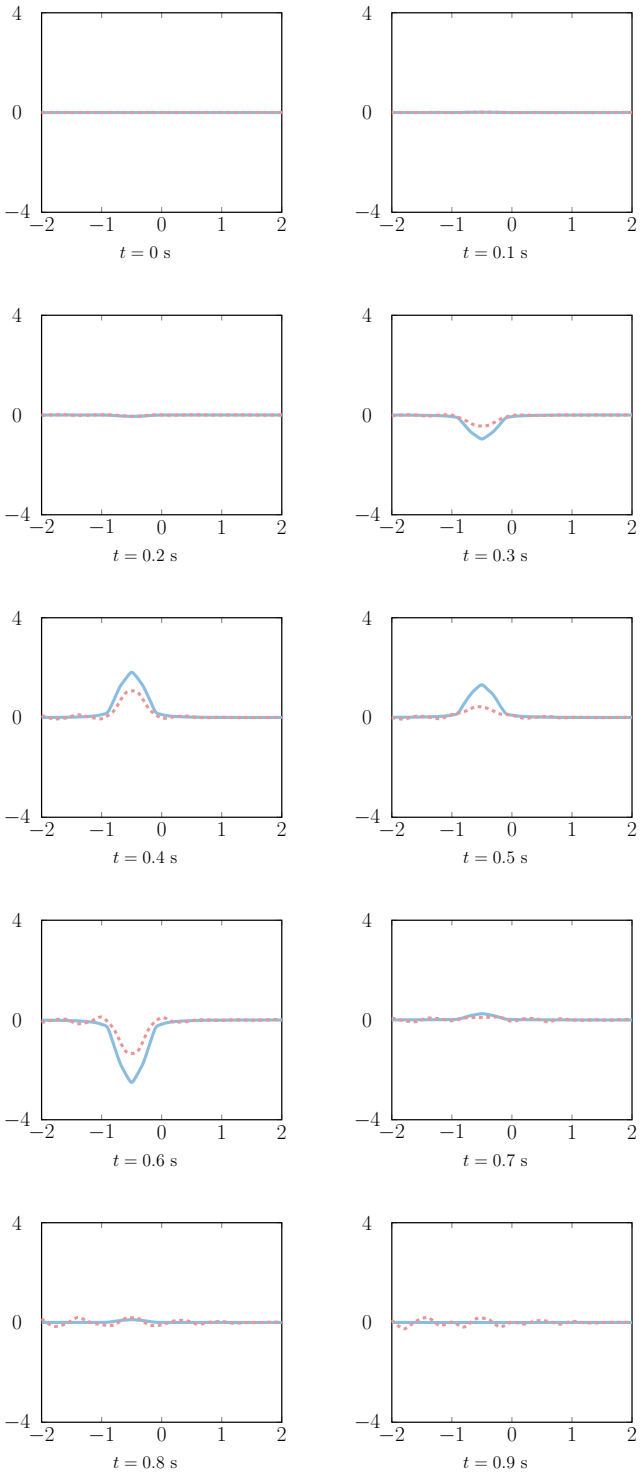


Figure 3.4: Complete model. Reconstructed χ at different times t for $\sigma = 0.05$, $\Delta t = 0.01$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

3.7.3 Reconstructions for the asymptotic model

Let us now show some numerical results in the case of the asymptotic model, that is we regularize the ill-posed Problem (3.37) in the presence of noisy data by using again Algorithm 3.4.4, which

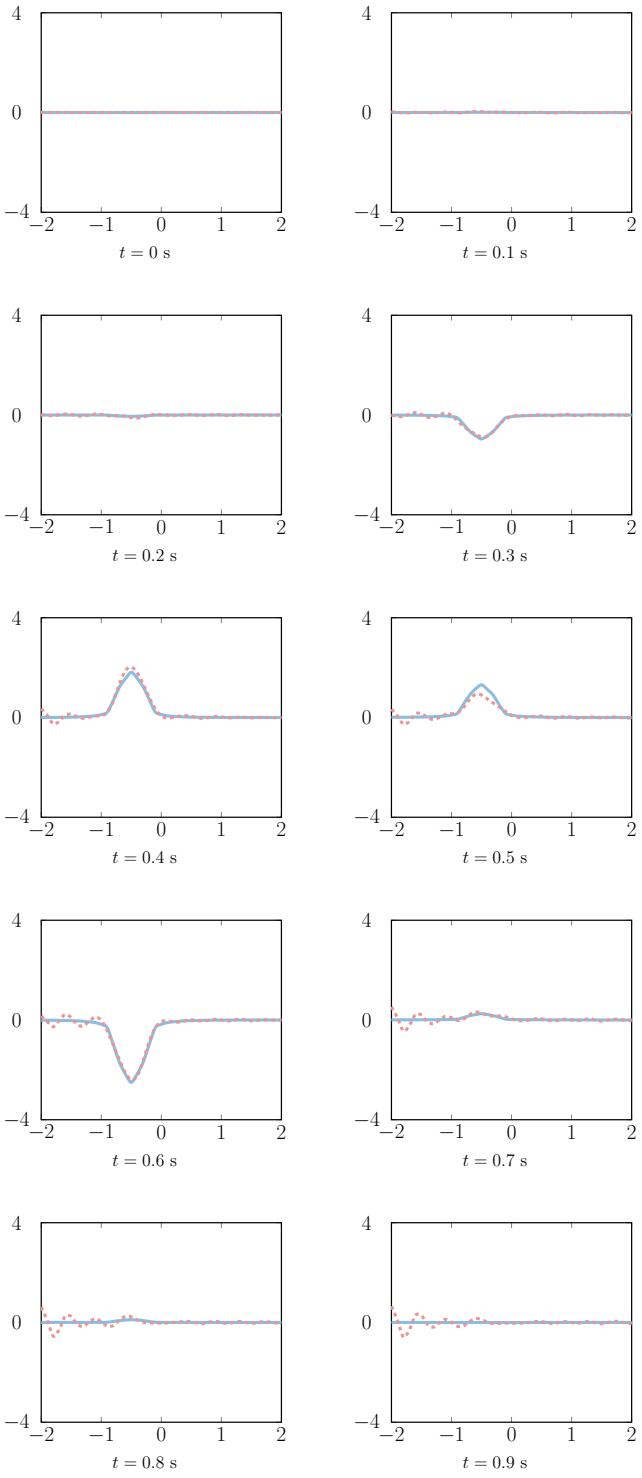


Figure 3.5: Complete model. Reconstructed χ at different times t for $\sigma = 0.05$, $\Delta t = 0.01$, $h = 0.075$ (SOR Method). Continuous line: true χ. Dashed line: reconstructed χ.

includes the weak formulations (3.48). In the figures 3.6, 3.7, 3.8, we present comparisons between the retrieved function χ and the exact one, for the relative amplitudes of noise $\sigma = 10^{-3}$, that is almost uncontaminated data, $\sigma = 5\%$ and $\sigma = 10\%$, respectively. Again, we observe a good robustness of our reconstructions even in the presence of noisy data.

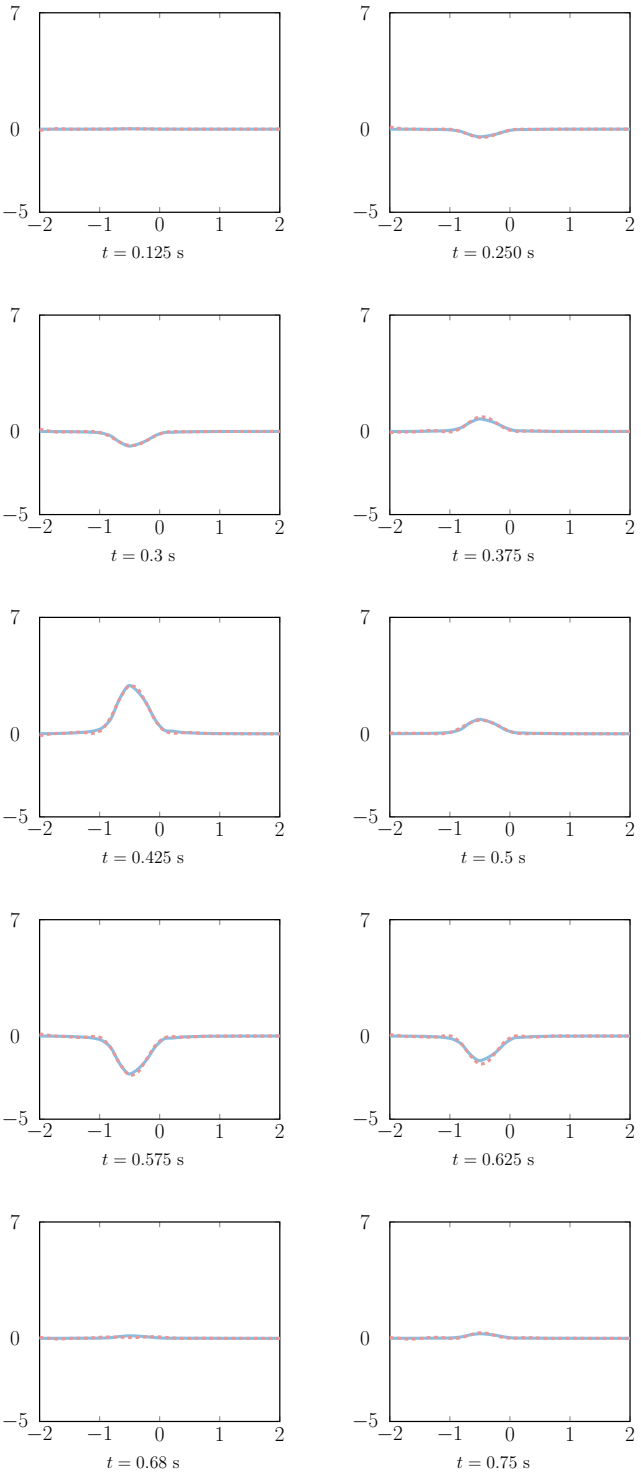


Figure 3.6: Asymptotic model. Reconstructed χ at different times t for $\sigma = 10^{-3}$, $\Delta t = 0.005$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

3.7.4 General remarks

A first interesting question is the efficiency of the Morozov choice for the regularization parameter ε in the weak formulations (3.42) and (3.47). To evaluate such efficiency, let us plot the function

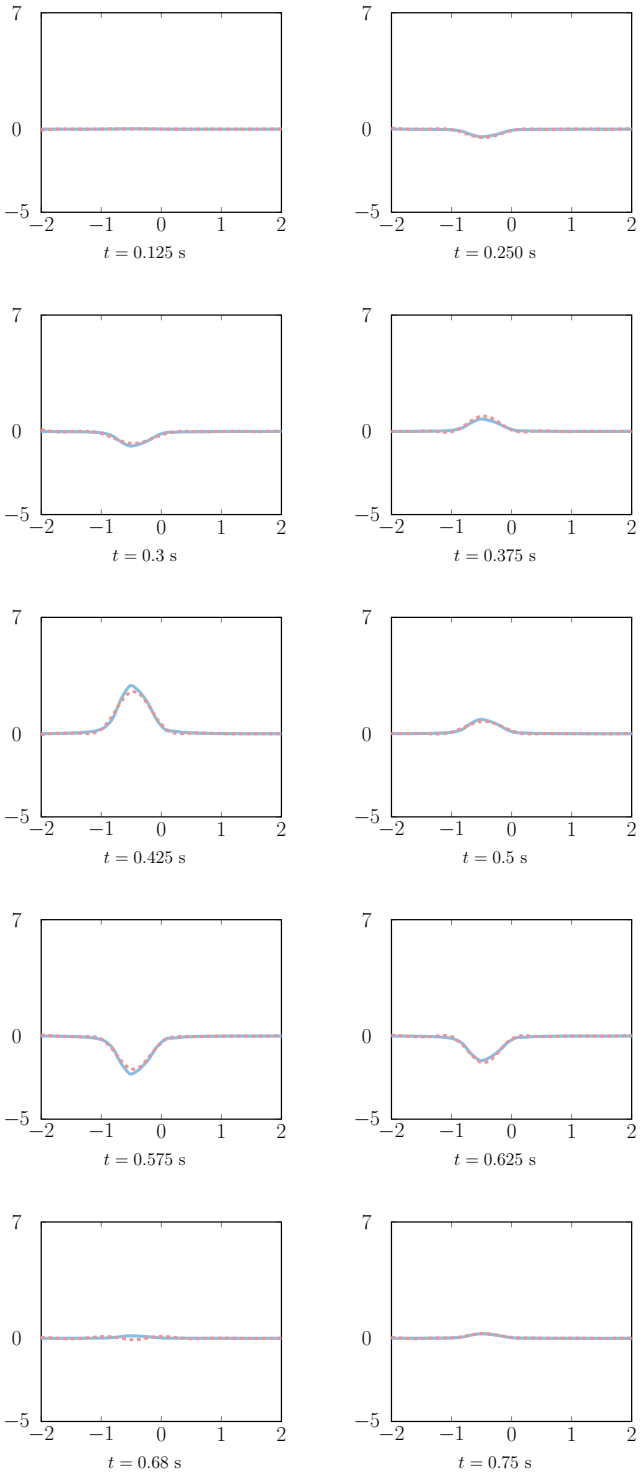


Figure 3.7: Asymptotic model. Reconstructed χ at different times t for $\sigma = 0.05$, $\Delta t = 0.005$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

$E^\delta : \varepsilon \mapsto \|\chi_\varepsilon^\delta - \chi\|_{L^2(0,T;L^2(\Gamma^R_{-H}))}$, where $\chi_\varepsilon^\delta = -\partial_z u_\varepsilon^\delta$, and u_ε^δ is the solution to either Problem (3.42) or (3.47) in the presence of a noisy data f^δ of amplitude δ. Here χ is the true solution. We wish to compare the Morozov choice $\varepsilon(\delta)$ and the minimum of the function E^δ. Such comparison is proposed in figure 3.9 for the complete and the asymptotic models. As shown in figure 3.9, the

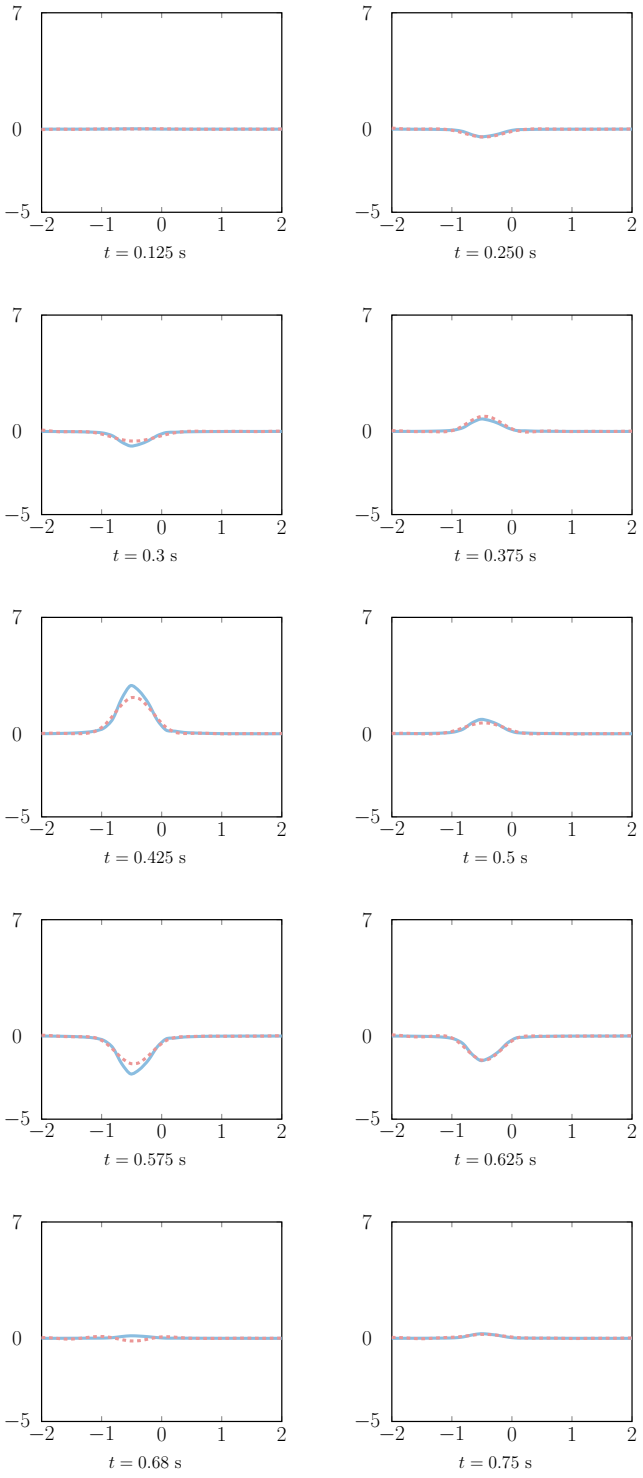


Figure 3.8: Asymptotic model. Reconstructed χ at different times t for $\sigma = 0.1$, $\Delta t = 0.005$, $h = 0.15$. Continuous line: true χ. Dashed line: reconstructed χ.

Morozov choice $\varepsilon(\delta)$, though not the optimal one, is not too far from such optimal value. Contrary to what the theory predicts in the case of uncontaminated data, the function E^δ for $\delta = 0$ is not increasing for small ε, which is due to the discretization.

A second question concerns how the quality of the reconstructions depends on the typical

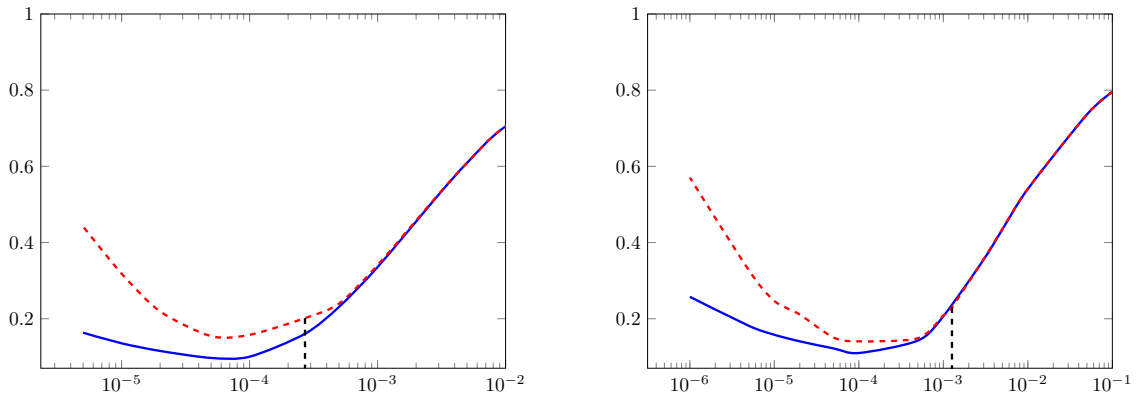


Figure 3.9: Error $\|\chi_\varepsilon^\delta - \chi\|_{L^2(0,T;L^2(\Gamma_{-H}^R))}$ as a function of ε and comparison with the Morozov choice (vertical dashed line). Left: complete model. Right: asymptotic model. Blue plain curve: uncontaminated data ($\sigma = 0\%$). Red dashed curve: noisy data ($\sigma = 10\%$).

frequency ω of the source term χ given by (3.52). To answer such question, in figure 3.10 we have plotted the error $\|\chi^\delta - \chi\|_{L^2(0,T;L^2(\Gamma_{-H}^R))}$, where χ^δ is the reconstructed χ obtained by using our Morozov strategy and χ is the true solution, as a function of the frequency ω. We are interested in three cases. Firstly, we use the complete model to obtain the artificial data and use the complete model to solve the inverse model. Secondly, we use the asymptotic model to obtain the artificial data and use the asymptotic model to solve the inverse model. Lastly, we use the complete model to obtain the artificial data and use the asymptotic model to solve the inverse model, as an attempt to avoid an inverse crime: data are generated by a more complicated model than the one used in the inverse strategy. We observe on figure 3.10 that in the case of the complete model for data and

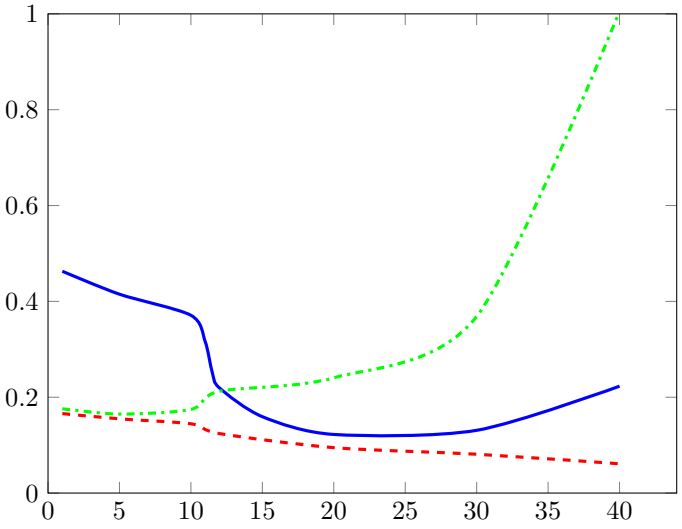


Figure 3.10: Reconstruction error as a function of the frequency for $\sigma = 5\%$. Blue plain curve: complete model for data and for inversion. Red dashed line: asymptotic model for data and for inversion. Green dashed line: complete model for data and asymptotic model for inversion. .

inversion, provided that the frequency is not too large, the quality of the reconstructions increases a lot with respect to ω, which is less the case for the asymptotic model for data and inversion. In the case of the complete model for data and asymptotic one for inversion, the quality strongly decreases with respect to ω. This phenomenon was explained in the frequency domain in [25] due to the nature of the guided modes in the waveguide Ω, which are completely different depending on the complete or asymptotic model. In the complete model, there are several propagating modes

in the x direction that are oscillating in the z direction, which is nice for the inversion, while in the asymptotic model, there is a single (surface) propagating mode in the x direction that is exponentially decaying in the z direction, which is bad for the inversion. And these two different behaviours are all the more pronounced as the frequency is large. The case of too large frequencies ω cannot be correctly interpreted since our mesh is here not fine enough.

Acknowledgement

The authors thank Eric Lunéville and Nicolas Kielbasiewicz for helping us with the Finite Element code. They also thank Patrick Joly for his unpublished lecture notes proving Theorem 3.2.3 and Marc Bonnet for discussions about the SOR method.

A Proof of Theorem 3.2.3.

We first recall the following result proved in [89] and [9, Proposition 3.3].

We consider two Hilbert spaces $\mathcal{V}$ and $\mathcal{H}$, equipped with their scalar products $(\cdot,\cdot)_{\mathcal{V}}$ and $(\cdot,\cdot)_{\mathcal{H}}$, respectively. We assume that $\mathcal{V}$ is continuously embedded and dense in $\mathcal{H}$. We consider a bilinear form a on $\mathcal{V}$ which is continuous, symmetric and satisfies the following coercivity property: there exist $\alpha > 0$ and $\nu \geq 0$ such that

$$a(u,u) + \nu \|u\|^2_{\mathcal{H}} \geq \alpha \|u\|^2_{\mathcal{V}}, \quad \forall u \in \mathcal{V}.$$

From the above assumptions, we can define an unbounded operator $\mathcal{A}$ in $\mathcal{H}$, the domain $D(\mathcal{A})$ of which is defined by

$$D(\mathcal{A}) = \{\forall u \in \mathcal{V},\, \exists w \in \mathcal{H}, \quad a(u,v) = (w,v)_{\mathcal{H}}, \quad \forall v \in \mathcal{V}\},$$

and $\mathcal{A}$ is defined for $u \in D(\mathcal{A})$ by $\mathcal{A}u = w$, the uniqueness of such w resulting from the density of $\mathcal{V}$ in $\mathcal{H}$. In particular, we have the inclusions $D(\mathcal{A}) \subset \mathcal{V} \subset \mathcal{H}$. We have the following theorem.

Theorem A.1. *For $(u_0, u_1) \in D(\mathcal{A}) \times \mathcal{V}$ and $f \in C^1([0,T];\mathcal{H})$ for $T > 0$, the problem*

$$\left\{ \begin{array}{rcll} \dfrac{d^2u}{dt^2} + \mathcal{A}u & = & f & t \in (0,T), \\ u(0) & = & u_0, & \\ \dfrac{du}{dt}(0) & = & u_1, & \end{array} \right. \tag{3.53}$$

has a unique (strong) solution

$$u \in C^2([0,T];\mathcal{H}) \cap C^1([0,T];\mathcal{V}) \cap C^0([0,T];D(\mathcal{A})).$$

In order to prove Theorem 3.2.3, we set $\mathcal{H} = L^2(\Gamma_0)$, $\mathcal{V} = H^{1/2}(\Gamma_0)$ and we choose the bilinear form a as

$$a(u,v) = g \int_\Omega \nabla(Ru) \cdot \nabla(Rv)\, dX,$$

where the operator $R : H^{1/2}(\Gamma_0) \to H^1(\Omega)$ is defined for $u \in H^{1/2}(\Gamma_0)$ and denoting $U = Ru$, by

$$\left\{ \begin{array}{rcll} \Delta U & = & 0 & \text{in } \Omega, \\ \partial_z U & = & 0 & \text{on } \Gamma_{-H}, \\ U & = & u & \text{on } \Gamma_0. \end{array} \right.$$

From the Poincaré-Wirtinger inequality, we have there exists some constant $C > 0$ such that for all $U \in H^1(\Omega)$,

$$\|U\|^2_{H^1(\Omega)} \leq C \left(\int_\Omega |\nabla U|^2\, dX + \int_{\Gamma_0} U^2\, dx \right).$$

This implies the coercivity of the bilinear form a, since for all $u \in H^{1/2}(\Gamma_0)$, there exists $c > 0$ such that

$$a(u,u) + g\int_{\Gamma_0} u^2\,dx = g\left(\int_\Omega |\nabla(Ru)|^2\,dX + \int_{\Gamma_0} u^2\,dx\right) \geq \frac{g}{C}\|Ru\|^2_{H^1(\Omega)} \geq c\,\|u\|^2_{H^{1/2}(\Gamma_0)},$$

where the last inequality results from the trace inequality. The operator $\mathcal{A}$ is then defined on the domain

$$D(\mathcal{A}) = \{u \in H^{1/2}(\Gamma_0),\ \partial_z(Ru)|_{\Gamma_0} \in L^2(\Gamma_0)\},$$

which happens to coincide, in view of classical regularity results for the Laplace equation, with $H^1(\Gamma_0)$, while for $u \in H^1(\Gamma_0)$,

$$\mathcal{A}u = g\partial_z(Ru)|_{\Gamma_0}.$$

We introduce the operator $R_{-H} : L^2(\Gamma_{-H}) \to H^{3/2}(\Omega)$ defined for $\chi \in L^2(\Gamma_{-H})$, by the solution $U = R_{-H}\chi$ to

$$\left\{\begin{array}{rcll} \Delta U &=& 0 & \text{in } \Omega,\\ -\partial_z U &=& \chi & \text{on } \Gamma_{-H},\\ U &=& 0 & \text{on } \Gamma_0. \end{array}\right.$$

The fact that the function φ satisfies (3.4) is equivalent to the fact that the function $\Phi = \varphi - R_{-H}\chi$ satisfies the system

$$\left\{\begin{array}{rcll} \Delta\Phi &=& 0 & \text{in } \Omega\times(0,T),\\ \partial_t^2\Phi + g\,\partial_z\Phi &=& f & \text{on } \Gamma_0\times(0,T),\\ \partial_z\Phi &=& 0 & \text{on } \Gamma_{-H}\times(0,T),\\ \Phi = \partial_t\Phi &=& 0 & \text{on } \Gamma_0\times\{0\}, \end{array}\right. \tag{3.54}$$

The fact that Φ satisfies (3.54) implies that $u = \Phi|_{\Gamma_0}$ satisfies (3.53). Conversely, that u satisfies (3.53) implies that $\Phi = Ru$ satisfies (3.54). Since $\chi \in C^2([0,T];L^2(\Gamma_{-H}))$, we have that $f = -g\partial_z U \in C^1([0,T];L^2(\Gamma_0))$. Using Theorem A.1 with $u_0 = u_1 = 0$ and $f \in C^1([0,T];L^2(\Gamma_0))$, we obtain that the Problem (3.53) has a unique solution

$$u \in C^2([0,T];L^2(\Gamma_0)) \cap C^1([0,T];H^{1/2}(\Gamma_0)) \cap C^0([0,T];H^1(\Gamma_0)),$$

so that the Problem (3.54) has a unique solution

$$\Phi \in C^2([0,T];H^{1/2}(\Omega)) \cap C^1([0,T];H^1(\Omega)) \cap C^0([0,T];H^{3/2}(\Omega)),$$

and lastly the Problem (3.4) has a unique solution φ in the same space.

Chapter 4

Boundary optimal control and iterative approach

In this chapter, we study the tsunami inverse problem with an optimal control approach. Firstly, we recast the problem in a handful order 1 variational framework by adding a parabolic regularization. This regularisation allows us to obtain a notion of solution for which we can get an inf-sup condition. In Appendix A, we also obtain a notion of variational solutions without any regularization, however the spaces in which the solutions are defined are not well-fitted to get any type of inf-sup conditions. Thanks to this inf-sup condition that we prove, we derive a rigorous and precise numerical analysis of a finite difference scheme to solve the optimal control approach problem. In this sense, this work reconcile the engineering community, which used the 4D-Var approach, with the mathematical community. This work thus presents for a slightly different equation an error estimate that will be verified numerically. In Chapter 5, we will present another framework to handle the case where there is no regularisation, while still being able to obtain a continuous inf-sup condition. This new framework could then be used to get the same type of results as the ones obtained in this chapter. However, the analysis of this framework, especially the discrete inf-sup condition, still represents a challenge we would need to tackle.
The following chapter contains a submitted article, realized in collaboration with Laurent Bourgeois and Philippe Moireau.

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Numerical analysis of an optimal control approach to solve a tsunami inverse problem

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Abstract

This paper concerns the reconstruction of a brutal bottom displacement of the ocean from the measurement of the induced perturbation of the free surface, which is a severely ill-posed inverse problem. This problem is solved by using an optimal control approach, the physics being governed by a time evolution system based on a simple oceanography model. We firstly recast the problem in an abstract framework, secondly propose an implicit Euler scheme for the time discretization combined with a Finite Elements method for the space discretization. The main result is an error estimate between the solution to the discrete optimal control problem and the solution to the continuous optimal problem, which is obtained by considering the discrete and continuous weak mixed formulations that characterize the optimality for these two problems. Some numerical experiments illustrate the efficiency of our approach and the consistency of our error estimate.

4.1 Introduction

In this paper, we study an inverse problem related to oceanography. We consider the case where the ocean bottom is subjected to a sudden, localized disturbance, specifically an underwater landslide, which generates a tsunami. Assuming that we can measure the resulting free surface perturbation, our goal is to identify the bottom displacement from these measurements. This problem is, of course, strongly ill-posed. In [42], however, it was shown that such an inverse problem could be solved with real data obtained in a laboratory by producing an artificial tsunami in a tank. This motivated us to go on working on that subject.

In this contribution, we adopt an optimal control approach as described in [88] to formulate the inverse problem, combining time-domain partial differential equations that model the physical phenomena with both boundary control to be identified – here, the bottom displacement – and boundary observations to assimilate – here, the free surface displacement.

More precisely, we model the propagation of acoustic and surface gravity waves in a stratified free-surface ocean using a linear model derived from the Euler equations [51]. This model consists

of a wave equation in the volume with a specific second-order boundary condition at the top. Such a model is well-suited for tsunami detection, as it accounts for both acoustic and gravity waves.

The strategy is to minimize a cost functional that balances observation error and bottom displacement intensity, subject to the constraint of satisfying the model dynamics. This classical framework is widely used in the data assimilation community and is often referred to as the variational approach, popularized as the 4D-VAR strategy [84, 10]. In practical applications, the gradient of the cost functional is computed by solving a state problem and an adjoint problem at each iteration of a descent algorithm. The state dynamics evolve forward in time, while the adjoint dynamics evolve backward in time. Thus, in this method, the gradient is obtained by solving only time-evolution Cauchy problems, which allows us to address large problems in terms of the total number of degrees of freedom.

By comparison, another approach consists of a mixed formulation of the Tikhonov regularization, as in [23] for the heat equation and [28] for the wave equation. These mixed formulations are non-iterative – a single weak formulation is solved once and for all – but they involve space-time formulations that numerically lead to larger systems to invert. Such an approach is applied in [26] to the same tsunami inverse problem as the one considered in this article. Similarly, space-time formulations of the optimal control problem have been proposed for wave-like equations [41, 39] and heat-like equations [34]. Another alternative is to use a sequential approach by formulating an observer [47], where the model dynamics are corrected in time by assimilating new available data, and the source term is registered simultaneously. These approaches benefit from solving only one time-evolution Cauchy problem. However, the feedback correction can be costly to store for large discretized problems [2]. Moreover, the observer approach is more suited to monitoring wave propagation than reconstructing the source term. Finally, for wave propagation problems, frequency approaches can also be considered, which clearly reduce the problem dimensionality as proposed in [25] for inverse tsunami problems.

Returning to the optimal control approach, the classical numerical methods defined by the data assimilation community for solving a 4D-VAR problem typically follow a discretize-then-optimize strategy. First, the direct problem is discretized in space, along with an appropriate choice of time discretization. The cost functional is then discretized in time, and the adjoint is defined from the discretized problem and functional. This approach computes an exact gradient, which is advantageous when implementing the gradient descent method. However, analyzing the convergence of this method can be challenging, especially when seeking convergence rates; when possible, such analysis rely on the inclusion of an appropriate regularization term in the functional [36]. In contrast, the numerical analysis of space-time domain data assimilation formulations may seem more natural, as they directly discretize the mixed formulation [34]. However, this optimize-then-discretize path often requires inf-sup stabilization terms, necessitating the solution of the full space-time formulation.

In this work, we propose to analyze the computationally attractive discretize-then-optimize approach by demonstrating that it can be interpreted as a specific discretization of the mixed formulation, with a specific time-scheme and quadrature rules. In this context, we establish a discrete inf-sup condition, which, combined with Strang-type techniques [56, 57], enables a complete numerical analysis with error estimates. To obtain our results, we acknowledge two simplifications in the initial model. First, to fit the abstract framework of [89] for time-evolution problems in the form of a first-order system, we introduce a parabolic regularization of the wave equation. This step is essential to obtain the necessary inf-sup condition at both the continuous and discrete levels. Second, to bound the domain laterally, we employ a simple first-order absorbing boundary condition. These two simplifications can be adjusted to remain as perturbations of the initial formulation, so that when they are sufficiently small, we can perform a complete numerical analysis of the optimal control strategy and reconstruct underwater landslides from measurements synthetically generated by the original unbounded wave model.

The paper is organized as follows. We describe the inverse tsunami problem in Section 2. In Section 3, we introduce an abstract optimal control problem, which is applied to the inverse tsunami problem at the continuous level in Section 4. In Section 5 we propose a time-scheme for the abstract optimal control problem, the convergence of which is the subject of Section 6. Section 7 is devoted to the space discretization in the case of the inverse tsunami problem, leading to our final

convergence result with respect to the time step and the space mesh size. Lastly, some numerical experiments are presented in Section 8, in particular to compare the theoretical predictions and the numerical observations.

4.2 Description of the problem

We consider a two-dimensional ocean $\Omega = \mathbb{R} \times (-H, 0)$ bounded from above by the free surface at rest $\Gamma_0 = \mathbb{R} \times \{0\}$ and from below by the straight bottom $\Gamma_{-H} = \mathbb{R} \times \{-H\}$. A generic point of Ω is denoted $X = (x, y)$ while the time is denoted t, t lying in the interval $(0, T)$, where $T > 0$. Let us denote $\varphi(X, t)$ the velocity potential, $s(x, t)$ the upward free surface perturbation and $\chi(x, t) = -\partial_t \zeta(x, t)$, where ζ is the upward displacement of the sea bottom. We assume that those variables satisfy the system

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2\varphi - \Delta\varphi &=& 0 & \text{in } \Omega \times (0,T), \\ \partial_t\varphi + g\, s &=& 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_z\varphi - \partial_t s &=& 0 & \text{on } \Gamma_0 \times (0,T), \\ \partial_\nu\varphi &=& \chi & \text{on } \Gamma_{-H} \times (0,T), \\ \varphi = \partial_t\varphi &=& 0 & \text{on } \Omega \times \{0\}, \\ s &=& 0 & \text{on } \Gamma_0 \times \{0\}, \end{array}\right. \tag{4.1}$$

where ν is the unit outward normal to Ω, g is the gravitational acceleration and c is the celerity of the acoustic waves. The system (4.1), where the surface function χ can be viewed as the origin of the tsunami, is for example considered in [103, 108] and is rigorously derived from the Euler equation in [51]. Such a model is relevant in the context of tsunami detection because it enables us to describe both gravity and acoustic waves. Merging the second and the third lines of (4.1), we obtain

$$\partial_t^2\varphi + g\,\partial_z\varphi = 0 \text{ on } \Gamma_0 \times (0,T),$$

which implies that (4.1) may be written as a system of the sole velocity potential φ. Taking inspiration from [89, 9], we obtained in [26] the following existence and uniqueness result for strong and weak solutions of the system (4.1) – see also [51] for an analogous system in a more complicated situation.

Theorem 4.2.1. *Let us assume that $\chi \in H^2(0,T; H^{-1/2}(\Gamma_{-H}))$ with $\chi(\cdot, 0) = 0$ on Γ_{-H}. There exists a unique strong solution φ to Problem (4.1), that is*

$$\varphi \in C^2([0,T]; L^2(\Omega)) \cap C^1([0,T]; H^1(\Omega)) \cap C^0([0,T]; X),$$

where

$$X = \{\varphi \in H^1(\Omega),\ \Delta\varphi \in L^2(\Omega),\ \partial_z\varphi|_{\Gamma_0} \in L^2(\Gamma_0)\}.$$

Moreover, if we only assume that $\chi \in H^1(0,T; H^{-1/2}(\Gamma_{-H}))$, with $\chi(\cdot, 0) = 0$ on Γ_{-H}, there exists a unique variational solution $\varphi \in \widetilde{W}_0(0,T)$ to Problem (4.1), where

$$\widetilde{W}_0(0,T) = \Big\{\varphi \in L^2(0,T; H^1(\Omega)),\ \partial_t\varphi \in L^2(0,T; L^2(\Omega)), \\ \partial_t(\varphi|_{\Gamma_0}) \in L^2(0,T; L^2(\Gamma_0)),\ \textit{and } \varphi(0) = 0\Big\}, \tag{4.2}$$

in the sense that

$$\partial_t^2\varphi \in L^2(0,T; H^1(\Omega)') \textit{ and } \partial_t^2(\varphi|_{\Gamma_0}) \in L^2(0,T; H^{-1/2}(\Gamma_0)), \\ \textit{while } \partial_t\varphi(0) = 0 \textit{ and } \partial_t(\varphi|_{\Gamma_0})(0) = 0,$$

and for all $\tilde\varphi \in L^2(0,T; H^1(\Omega))$,

$$\int_0^T \frac{1}{c^2}\langle\partial_t^2\varphi, \tilde\varphi\rangle_{H^1(\Omega)', H^1(\Omega)} + \frac{1}{g}\left\langle\partial_t^2(\varphi|_{\Gamma_0}), \tilde\varphi|_{\Gamma_0}\right\rangle_{H^{-1/2}(\Gamma_0), H^{1/2}(\Gamma_0)} \\ + (\nabla\varphi, \nabla\tilde\varphi)_{L^2(\Omega)}\, dt = \int_0^T \langle\chi, \tilde\varphi\rangle_{H^{-1/2}(\Gamma_{-H}), H^{1/2}(\Gamma_{-H})}\, dt. \tag{4.3}$$

Remark 4.2.1. *We understand from Theorem 4.2.1 that the notion of solution is particularly intricate, especially when considering variational solutions where the trace of the function may have a weak time derivative that does not necessarily correspond to the trace of $\partial_t\varphi$, which only belongs to $L^2(\Omega)$ and therefore does not necessarily have a trace on the boundary. Moreover, we see that even if the notion of weak solution is compatible with a source term $\chi \in L^2(0,T;H^{-1/2}(\Gamma_{-H}))$, the existence of a solution in the energy space requires additional regularity in time for χ. Note finally that the upward free surface perturbation can be recovered as*

$$s = -(1/g)\partial_t(\varphi|_{\Gamma_0}) \in L^2(0,T;L^2(\Gamma_0)).$$

In the system (4.1), solving the forward problem consists in computing the free surface perturbation s knowing the velocity of the sea bottom χ. The inverse problem consists in recovering χ from a given measurement of s. From (4.1), such inverse problem can be viewed as an ill-posed problem for the wave equation in the presence of lateral Cauchy data. A first important question is uniqueness: assume that we measure the free surface perturbation s on $\Gamma_0 \times (0,T)$, on which time interval are we able to retrieve the sea bottom velocity χ ? We have the following theorem.

Theorem 4.2.2. *Assume that (φ, s) satisfies (4.1) for some χ. If $s = 0$ on $\Gamma_0 \times (0,T)$, then $\chi = 0$ on $\Gamma_{-H} \times (0, T - H/c)$.*

Proof. The theorem is essentially a consequence of the Holmgren Theorem – see Theorem 5.3.3 in [71]. Assuming that $s = 0$ on $\Gamma_0 \times (0,T)$ implies from the second and third equations of (4.1) that both $\partial_t\varphi$ and $\partial_z\varphi$ vanish on $\Gamma_0 \times (0,T)$. Then the initial condition implies that the pair of lateral Cauchy data $(\varphi, \partial_z\varphi)$ vanish on $\Gamma_0 \times (0,T)$. By using again the initial condition satisfied by φ, we can extend φ by 0 for negative times so that φ satisfies the wave equation in $\Omega \times (-T,T)$ and $(\varphi, \partial_z\varphi) = 0$ on $\Gamma_0 \times (-T,T)$. As a consequence of the Holmgren Theorem for the wave equation – see for example [28] – we obtain that $\varphi = 0$ in

$$Q_0 = \{(X,t) \in \Omega \times (-T,T),\ d(X,\Gamma_0) < c(T - |t|)\}.$$

Since $d(X,\Gamma_0) = -y$, we get that

$$Q_0 = \{(x,y,t) \in \Omega \times (-T,T),\ |t| < T + y/c\}.$$

Using the fourth equation in (4.1), we finally get that $\chi = 0$ on $\Gamma_{-H} \times (0, T - H/c)$. □

Remark 4.2.2. *As a consequence of the uniqueness theorem above, we observe that if the time support of the sea bottom velocity χ to be retrieved is $(0,T_0)$, then we have to measure the free surface velocity in the larger time interval $(0, T_0 + H/c)$.*

We wish now to solve the problem in the uniqueness region by using an optimal control approach. Such approach is first described in an abstract variational framework. The proposed framework will not be directly compatible with the specific notion of variational solution for System (4.1). However, we will see that it is compatible with a parabolic regularization of System (4.1), which allows us to work in a much more natural setup, even for more general source terms $\chi \in L^2(0,T;H^{-1/2}(\Gamma_{-H}))$. The regularized model will also include absorbing boundary conditions to ensure numerical tractability and allow for a complete analysis of discretization convergence.

4.3 An abstract optimal control problem

Let us consider the setting proposed in [89] and reused in [9]. We introduce two Hilbert spaces V and H, equipped with the norms $\|\cdot\|_V$ and $\|\cdot\|_H$, respectively, and the dual of which are denoted V' and H', respectively. We assume in addition that V is dense in H and

$$V \hookrightarrow H = H' \hookrightarrow V',$$

the symbol $\hookrightarrow$ meaning continuous embedding. We consider two additional Hilbert spaces U and Y, which are both identified with their duals. We now introduce a continuous bilinear form a on $V \times V$, which is coercive in the sense that there exist $\lambda, \alpha > 0$ such that

$$a(z,z) + \lambda \|z\|_H^2 \geq \alpha \|z\|_V^2, \quad \forall z \in V, \tag{4.4}$$

and a continuous bilinear form b on $U \times V$. From the bilinear forms a and b, we define two operators $A : V \to V'$ and $B : U \to V'$ such that

$$\langle Az, \tilde{z} \rangle_{V',V} = a(z,\tilde{z}), \quad \langle B\chi, \tilde{z} \rangle_{V',V} = b(\chi,\tilde{z}), \quad \forall z, \tilde{z} \in V, \quad \forall \chi \in U,$$

where $\langle \cdot, \cdot \rangle_{V',V}$ denotes the duality bracket between V' and V. Finally, we introduce a third operator $C : H \to Y$.

In order to introduce an evolution problem, for $T > 0$ we define the Hilbert space $W(0,T)$ such that

$$W(0,T) = \left\{ z \in L^2(0,T;V),\ \frac{dz}{dt} \in L^2(0,T;V') \right\}.$$

It is remarkable from [89] that we have a continuous embedding $W(0,T) \hookrightarrow C^0([0,T];H)$, which in particular gives a meaning to the initial condition $z(0) \in H$ for any function $z \in W(0,T)$. We will also need to introduce the subspaces

$$W_0(0,T) = \{ z \in W(0,T),\ z(0) = 0 \}, \quad W_T(0,T) = \{ p \in W(0,T),\ p(T) = 0 \}.$$

We have the following existence/uniqueness theorem, which is proved in [89].

Theorem 4.3.1. *For $f \in L^2(0,T;V')$, there exists a unique solution $z \in W(0,T)$ to the problem*

$$\begin{cases} \dfrac{dz}{dt} + Az = f \\ z(0) = 0. \end{cases} \tag{4.5}$$

and the operator $f \in L^2(0,T,V') \to z \in W(0,T)$ is continuous.

The optimal control problem we consider, for $y \in L^2(0,T;Y)$ and $\varepsilon > 0$, is

$$\inf_{\chi \in L^2(0,T;U)} J(\chi) = \int_0^T \left(\frac{1}{2} \|y(t) - Cz(t)\|_Y^2 + \frac{\varepsilon}{2} \|\chi(t)\|_U^2 \right) dt, \tag{4.6}$$

where $z \in W(0,T)$ is the solution to the problem

$$\begin{cases} \dfrac{dz}{dt} + Az = B\chi \\ z(0) = 0. \end{cases} \tag{4.7}$$

In the problem above, U and Y can be seen as the control and the observation spaces, respectively, $\chi \in L^2(0,T;U)$ and $y \in L^2(0,T,Y)$ as the control and the observation variables, respectively. As it is classical in optimal control [88], the cost function (4.6) consists of two parts: the first one is the error in the observation while the second one is the regularization. Since the cost function is continuous, coercive and strictly convex, we have the following proposition.

Proposition 4.3.1. *The problem (4.6) (4.7) has a unique solution.*

In the following proposition, we compute the gradient of the cost function J with respect to χ. To this end, we will need the following Green formula – see [45].

Lemma 4.3.1. *For all $z, p \in W(0,T)$, we have*

$$\int_0^T \left\langle \frac{dz}{dt}(t), p(t) \right\rangle_{V',V} dt = -\int_0^T \left\langle \frac{dp}{dt}(t), z(t) \right\rangle_{V',V} dt + (z(T),p(T))_H - (z(0),p(0))_H.$$

We have the following result.

Proposition 4.3.2. *The differential of J at point $\chi \in L^2(0,T;U)$ is given, for $\tilde{\chi} \in L^2(0,T;U)$, by*

$$\langle D_\chi J(\chi), \tilde{\chi} \rangle_{(L^2(0,T;U))',L^2(0,T;U)} = \int_0^T (\varepsilon \chi - B^* p, \tilde{\chi})_U \, dt, \tag{4.8}$$

where p is the solution in $W(0,T)$ to the problem

$$\begin{cases} \dfrac{dp}{dt} - A^* p = C^*(Cz - y), \\ p(T) = 0 \end{cases} \tag{4.9}$$

and z is the solution to the problem (4.7). Here $A^* : V \to V'$, $B^* : V \to U$ and $C^* : Y \to H$ are the adjoint operators of A, B and C, respectively.

Proof. Let us first check that the problem (4.9) is well-posed. The operator $A^* : V \to V'$ satisfies the coercivity property (4.4) and since $z(t) \in V$, we have $Cz(t) - y(t) \in Y$ and then $C^*(Cz(t) - y(t)) \in V'$. This enables us to apply Theorem 4.3.1 (note that the evolution problem (4.9) is backward, which requires a minus sign in front of A^*p), which ensures that the problem (4.9) has a unique solution in $W(0,T)$.

To obtain (4.8) we now introduce the Lagrangian $\mathcal{L}$ defined for $(\tilde{z}, \tilde{p}, \chi) \in W_0(0,T) \times W_T(0,T) \times L^2(0,T;U)$ by

$$\mathcal{L}(\tilde{z}, \tilde{p}, \chi) = \int_0^T \left(\frac{1}{2} \|y(t) - C\tilde{z}(t)\|_Y^2 + \frac{\varepsilon}{2} \|\chi(t)\|_U^2 + \left\langle \frac{d\tilde{z}}{dt}(t) + A\tilde{z}(t) - B\chi(t), \tilde{p}(t) \right\rangle_{V',V} \right) dt.$$

We observe that for $\chi \in L^2(0,T;U)$, we have

$$J(\chi) = \mathcal{L}(z(\chi), \tilde{p}, \chi), \quad \forall \tilde{p} \in W_T(0,T),$$

where $z(\chi)$ is the solution to the problem (4.7). Using the chain rule, we have

$$\begin{aligned} &\langle D_\chi J(\chi), \tilde{\chi} \rangle_{(L^2(0,T;U))',L^2(0,T;U)} \\ &= \langle D_z \mathcal{L}(z(\chi), \tilde{p}, \chi), z(\tilde{\chi}) \rangle_{(W_0(0,T))',W_0(0,T)} + \langle D_\chi \mathcal{L}(z(\chi), \tilde{p}, \chi), \tilde{\chi} \rangle_{(L^2(0,T;U))',L^2(0,T;U)}. \end{aligned}$$

On the one hand, we compute

$$\begin{aligned} &\langle D_z \mathcal{L}(z, \tilde{p}, \chi), \tilde{z}) \rangle_{(W_0(0,T))',W_0(0,T)} \\ &= -\int_0^T (y(t) - Cz(t), C\tilde{z}(t))_Y \, dt + \int_0^T \left\langle \frac{d\tilde{z}}{dt}(t) + A\tilde{z}(t), \tilde{p}(t) \right\rangle_{V',V} dt. \end{aligned}$$

By using Lemma 4.3.1, we get

$$\begin{aligned} &\langle D_z \mathcal{L}(z, \tilde{p}, \chi), \tilde{z}) \rangle_{(W_0(0,T))',W_0(0,T)} \\ &= \int_0^T \langle C^*(Cz(t) - y(t)), \tilde{z}(t) \rangle_{V',V} \, dt - \int_0^T \left\langle \frac{d\tilde{p}}{dt}(t) - A^*\tilde{p}(t), \tilde{z}(t) \right\rangle_{V',V} dt. \end{aligned}$$

On the other hand, we compute

$$\begin{aligned} &\langle D_\chi \mathcal{L}(z, \tilde{p}, \chi), \tilde{\chi} \rangle_{(L^2(0,T;U))',L^2(0,T;U)} \\ &= \int_0^T (\varepsilon(\chi(t), \tilde{\chi}(t))_U - \langle B\tilde{\chi}(t), \tilde{p}(t) \rangle_{V',V}) \, dt \\ &= \int_0^T (\varepsilon\chi(t) - B^*\tilde{p}(t), \tilde{\chi}(t))_U \, dt. \end{aligned}$$

It remains to choose $\tilde{p} = p$, where p is the solution to problem (4.9), which implies that

$$D_z \mathcal{L}(z(\chi), p, \chi) = 0,$$

which yields (4.8). □

As a consequence of Propositions 4.3.1 and 4.3.2, the unique solution χ to the problem (4.6) (4.7) satisfies

$$\varepsilon\chi - B^*p = 0, \tag{4.10}$$

where p is the solution to the problem (4.9), in which z is the solution to the problem (4.7). Indeed, the gradient of the function J vanishes at the minimum argument.

We wish to obtain a characterization of the optimal solution χ with the help of a mixed formulation. Let us introduce the following weak formulation: find $Z = (z, \chi) \in W_0(0,T) \times L^2(0,T;U)$ and $p \in L^2(0,T;V)$ such that for all $\tilde{Z} = (\tilde{z}, \tilde{\chi}) \in W_0(0,T) \times L^2(0,T;U)$ and $\tilde{p} \in L^2(0,T;V)$,

$$\begin{cases} j(Z,\tilde{Z}) + d(\tilde{Z},p) = \ell(\tilde{Z}), \\ d(Z,\tilde{p}) = 0, \end{cases} \tag{4.11}$$

with

$$j(Z,\tilde{Z}) = \int_0^T (\varepsilon(\chi,\tilde{\chi})_U + (Cz, C\tilde{z})_Y)\, dt, \tag{4.12}$$

$$d(Z,p) = \int_0^T \left(\left\langle \frac{dz}{dt}, p \right\rangle_{V',V} + a(z,p) - b(\chi,p) \right) dt \tag{4.13}$$

and

$$\ell(Z) = \int_0^T (y, Cz)_Y\, dt. \tag{4.14}$$

We aim to benefit from the weak formulation (4.11) to study our inverse problem, which is defined as an optimal control problem. In particular, we anticipate that a convergent discretization of the optimal control problem can be derived as a convergent discretization of the mixed formulation (4.11), with quantitative convergence estimates. First, let us prove the well-posedness of the weak formulation (4.11).

Theorem 4.3.2. *The weak formulation (4.11) has a unique variational solution in* $W_0(0,T) \times L^2(0,T;U) \times L^2(0,T;V)$.

Proof. It suffices to prove that the bilinear form j is coercive on $\operatorname{Ker} d$ and that the bilinear form d satisfies the inf-sup condition (see for example [58]). For the first point, we have for all $Z \in W_0(0,T) \times L^2(0,T;U)$,

$$j(Z,Z) \geq \varepsilon \int_0^T \|\chi\|_U^2\, dt = \frac{\varepsilon}{2} \int_0^T \|\chi\|_U^2\, dt + \frac{\varepsilon}{2} \int_0^T \|\chi\|_U^2\, dt,$$

Since $Z \in \operatorname{Ker} d$, z is the solution to problem (4.7). By Theorem 4.3.1, there exists a constant $C_{\mathrm{st}} > 0$ such that for all $\chi \in L^2(0,T;U)$,

$$\|z\|_{W(0,T)}^2 \leq C_{\mathrm{st}}^2 \int_0^T \|B\chi\|_{V'}^2\, dt \leq C_{\mathrm{st}}^2 \|B\|^2 \int_0^T \|\chi\|_U^2\, dt,$$

hence

$$j(Z,Z) \geq \min\left(\frac{\varepsilon}{2\, C_{\mathrm{st}}^2 \|B\|^2}, \frac{\varepsilon}{2} \right) (\|z\|_{W(0,T)}^2 + \|\chi\|_{L^2(0,T;U)}^2),$$

which is the desired result.

Now consider the second point, which is equivalent to proving that there exists some $\beta > 0$ such that for all $p \in L^2(0,T;V)$,

$$\sup_{Z \in W_0(0,T) \times L^2(0,T;U)} \frac{d(Z,p)}{\|Z\|_{W_0(0,T) \times L^2(0,T;U)}} \geq \beta \|p\|_{L^2(0,T;V)}.$$

Let $z \in W(0,T)$ be the solution to the problem (4.5) for $f = Ap + \lambda p \in L^2(0,T;V')$, where λ is the constant of (4.4), and let us set $Z = (z,0) \in W_0(0,T) \times L^2(0,T;U)$. By (4.4), we have on the one hand

$$d(Z,p) = \int_0^T (\langle Ap, p\rangle_{V',V} + \lambda \langle p, p\rangle_{V',V})\, dt = \int_0^T \left(a(p,p) + \lambda \|p\|_H^2 \right) dt \geq \alpha \|p\|_{L^2(0,T;V)}^2.$$

Using Theorem 4.3.1 again, we have on the other hand

$$\|Z\|_{W_0(0,T)\times L^2(0,T;U)} = \|z\|_{W(0,T)} \leq C_{\mathrm{st}} \|Ap + \lambda p\|_{L^2(0,T;V')} \leq C_{\mathrm{st}} (\|A\| + \lambda)\|p\|_{L^2(0,T;V)}.$$

Combining the two estimates above, we conclude that for all $p \in L^2(0,T;V)$,

$$\frac{d(Z,p)}{\|Z\|_{W_0(0,T)\times L^2(0,T;U)}} \geq \frac{\alpha}{C_{\mathrm{st}} (\|A\| + \lambda)} \|p\|_{L^2(0,T;V)},$$

which is the result, with $\beta = \alpha/(C_{\mathrm{st}}(\|A\| + \lambda))$. □

Then, the following proposition establishes a link between the optimal control problem (4.6) (4.7) and the weak formulation (4.11).

Proposition 4.3.3. *Let us consider the solution $\chi \in L^2(0,T;U)$ to the problem (4.6) (4.7) as well as the corresponding solutions $z \in W(0,T)$ to problem (4.7) and $p \in W(0,T)$ to problem (4.9). Setting $Z = (z,\chi)$, the pair (Z,p) is the unique solution to problem (4.11).*

Proof. The function $z \in W(0,T)$ satisfies (4.7), that is $z \in W_0(0,T)$ and for all $\tilde{p} \in L^2(0,T;V)$,

$$\int_0^T \left\langle \frac{dz}{dt} + Az - B\chi, \tilde{p} \right\rangle_{V',V} dt = 0,$$

that is

$$\int_0^T \left(\left\langle \frac{dz}{dt}, \tilde{p} \right\rangle_{V',V} + a(z,\tilde{p}) - b(\chi,\tilde{p}) \right) dt = 0,$$

and finally $d(Z,\tilde{p}) = 0$, which is the second equation of (4.11).

Let us prove the first one. For all $\tilde{Z} = (\tilde{z},\tilde{\chi}) \in W_0(0,T) \times L^2(0,T;U)$, we have

$$\begin{aligned} & j(Z,\tilde{Z}) + d(\tilde{Z},p) - \ell(\tilde{Z}) \\ &= \int_0^T (\varepsilon(\chi,\tilde{\chi})_U + (Cz, C\tilde{z})_Y)\, dt \\ &+ \int_0^T \left(\left\langle \frac{d\tilde{z}}{dt}, p \right\rangle_{V',V} + a(\tilde{z},p) - b(\tilde{\chi},p) \right) dt - \int_0^T (y, C\tilde{z})_Y\, dt, \end{aligned}$$

that is, by using Lemma 4.3.1 again,

$$\begin{aligned} & j(Z,\tilde{Z}) + d(\tilde{Z},p) - \ell(\tilde{Z}) \\ &= \int_0^T (\varepsilon(\chi,\tilde{\chi})_U - (B^*p,\tilde{\chi})_U)\, dt \\ &- \int_0^T \left(\left\langle \frac{dp}{dt}, \tilde{z} \right\rangle_{V',V} - \langle A^*p, \tilde{z}\rangle_{V',V} - \langle C^*(Cz - y), \tilde{z}\rangle_{V',V} \right) dt. \end{aligned}$$

By using (4.9) and (4.10), we conclude that

$$j(Z,\tilde{Z}) + d(\tilde{Z},p) - \ell(\tilde{Z}) = 0,$$

which is the first equation of (4.11). □

We hence observe from Proposition 4.3.3 and Theorem 4.3.2 that if χ is the unique solution to the optimal control problem (4.6) (4.7), z and p the corresponding solutions to the problems (4.7) and (4.9), respectively, then $(Z,p) = (z,\chi,p)$ is the unique solution to the weak formulation (4.11). In other words, at the continuous level, solving the optimal control problem amounts to solve the weak formulation.

Remark 4.3.1 (Additional regularity of the optimal solution)**.** *We observe that the solution $p \in L^2(0,T;V)$ to the problem (4.11) happens to be more regular in the sense that $p \in W_T(0,T)$.*

4.4 Application to the tsunami problem

In this section, we apply the previous abstract framework to the inverse tsunami problem. Since we have the numerical resolution of the problem in mind, we have first to approximate the initial problem (4.1) in the bounded domain $\Omega^R = \{X = (x, y) \in \Omega,\ -R < x < R\}$ by adding the artificial first order absorbing boundary conditions on $\Sigma_{\pm R} = \{\pm R\} \times (-H, 0)$ given by

$$\pm\partial_x\varphi + \frac{1}{c}\,\partial_t\varphi = 0 \quad \text{on} \quad \Sigma_{\pm R} \times (0, T),$$

in the spirit of [54] (see also [78] in the pure gravity case, that is $c = +\infty$). We also denote Γ_0^R and Γ_{-H}^R the subparts of Γ_0 and Γ_{-H} delimited by Σ_{-R} and Σ_R, respectively.

Hence the problem (4.1) in the unbounded domain Ω becomes in the bounded domain Ω^R:

$$\left\{\begin{array}{rcll} (1/c^2)\partial_t^2\varphi - \Delta\varphi &=& 0 & \text{in } \Omega^R \times (0,T), \\ \partial_t\varphi + g\,s &=& 0 & \text{on } \Gamma_0^R \times (0,T), \\ \partial_z\varphi - \partial_t s &=& 0 & \text{on } \Gamma_0^R \times (0,T), \\ \partial_\nu\varphi &=& \chi & \text{on } \Gamma_{-H}^R \times (0,T), \\ \pm\partial_x\varphi + (1/c)\,\partial_t\varphi &=& 0 & \text{on } \Sigma_{\pm R} \times (0,T), \\ \varphi = \partial_t\varphi &=& 0 & \text{on } \Omega^R \times \{0\}, \\ s &=& 0 & \text{on } \Gamma_0^R \times \{0\}. \end{array}\right. \tag{4.15}$$

We have now to specify the spaces and bilinear forms of the abstract framework. We choose

$$H = H^1(\Omega^R) \times L^2(\Omega^R) \times L^2(\Gamma_0^R),$$

and the norm of H is defined, for $z = (\varphi, \psi, s) \in H$, by

$$\|z\|_H^2 = \|\varphi\|_{H^1(\Omega^R)}^2 + \frac{1}{c^2}\|\psi\|_{L^2(\Omega^R)}^2 + g\,\|s\|_{L^2(\Gamma_0^R)}^2.$$

We also choose

$$V = \{z = (\varphi, \psi, s) \in H^1(\Omega^R) \times H^1(\Omega^R) \times L^2(\Gamma_0^R),\ \psi|_{\Gamma_0^R} + g\,s = 0\},$$

equipped with the norm given, for $z = (\varphi, \psi, s) \in V$, by

$$\|z\|_V^2 = \|\varphi\|_{H^1(\Omega^R)}^2 + \|\psi\|_{H^1(\Omega^R)}^2 + \|s\|_{L^2(\Gamma_0^R)}^2.$$

It is clear that V is a Hilbert space. Let us check that V is dense in H.

Lemma 4.4.1. *The space V is dense in H.*

Proof. Assume that $z = (\varphi, \psi, s) \in V^\perp$, where $V^\perp$ is the orthogonal of V in H. We have that

$$(\varphi, \tilde\varphi)_{H^1(\Omega^R)} + \frac{1}{c^2}(\psi, \tilde\psi)_{L^2(\Omega^R)} + g\,(s, \tilde s)_{L^2(\Gamma_0^R)} = 0, \quad \forall \tilde z = (\tilde\varphi, \tilde\psi, \tilde s) \in V.$$

Choosing $\tilde\psi = 0$ and $\tilde s = 0$, we first obtain that $\varphi = 0$. Then we choose $\tilde\psi \in C_0^\infty(\Omega_R)$ and $\tilde s = 0$ (which satisfy $\tilde\psi|_{\Gamma_0^R} + g\,\tilde s = 0$), this yields $\psi = 0$. We conclude that $s = 0$, which completes the proof. □

We also set $U = L^2(\Gamma_{-H}^R)$ and $Y = L^2(\Gamma_0^R)$, both equipped with their usual norms. For $\eta > 0$, let us now define the bilinear form $a = a_\eta$ on V as: for all $z = (\varphi, \psi, s) \in V$, for all $\tilde z = (\tilde\varphi, \tilde\psi, \tilde s) \in V$,

$$a_\eta(z, \tilde z) = -(\psi, \tilde\varphi)_{H^1(\Omega^R)} + (\nabla\varphi, \nabla\tilde\psi)_{L^2(\Omega^R)} + \eta\,(\nabla\psi, \nabla\tilde\psi)_{L^2(\Omega^R)} + \frac{1}{c}\sum_{\pm}(\psi, \tilde\psi)_{L^2(\Sigma_{\pm R})},$$

where we have used the notation

$$(\nabla\varphi, \nabla\tilde\psi)_{L^2(\Omega^R)} = \int_{\Omega^R} \nabla\varphi \cdot \nabla\tilde\psi\, dX.$$

The bilinear form a_η is clearly continuous on V. Here, $\eta > 0$ is a small regularization parameter which has the effect to provide the coercivity of a_η.

Lemma 4.4.2. *The bilinear form a_η is coercive in the sense of (4.4) as soon as $\eta > 0$.*

Proof. For all $z = (\varphi, \psi, s) \in V$, we have

$$\begin{aligned} a_\eta(z,z) &= -(\varphi,\psi)_{H^1(\Omega^R)} + (\nabla\varphi, \nabla\psi)_{L^2(\Omega^R)} + \eta \|\nabla\psi\|^2_{L^2(\Omega^R)} + \frac{1}{c}\sum_{\pm} \|\psi\|^2_{L^2(\Sigma_{\pm R})} \\ &\geq -(\varphi,\psi)_{L^2(\Omega^R)} + \eta\|\nabla\psi\|^2_{L^2(\Omega^R)}, \end{aligned}$$

so that for all $\lambda > 0$,

$$\begin{aligned} a_\eta(z,z) + \lambda\|z\|^2_H \geq -\frac{1}{2}\|\varphi\|^2_{L^2(\Omega^R)} - \frac{1}{2}\|\psi\|^2_{L^2(\Omega^R)} \\ + \eta\|\nabla\psi\|^2_{L^2(\Omega^R)} + \lambda\|\varphi\|^2_{H^1(\Omega^R)} + \frac{\lambda}{c^2}\|\psi\|^2_{L^2(\Omega^R)} + \lambda\, g\, \|s\|^2_{L^2(\Gamma_0^R)}. \end{aligned}$$

Choosing $\lambda > \max(1/2, c^2/2)$ with $g > 1$, we obtain that there exists some $\alpha > 0$ such that

$$a_\eta(z,z) + \lambda\|z\|^2_H \geq \alpha \left(\|\varphi\|^2_{H^1(\Omega^R)} + \|\psi\|^2_{H^1(\Omega^R)} + \|s\|^2_{L^2(\Gamma_0^R)} \right) = \alpha\, \|z\|^2_V,$$

which is the result. □

Remark 4.4.1. *We observe that a_0 is not coercive.*

The bilinear form b is defined on $L^2(\Gamma^R_{-H}) \times V$ as: for all $\chi \in L^2(\Gamma^R_{-H})$, for all $z = (\varphi, \psi, s) \in V$,

$$b(\chi, z) = (\chi, \psi|_{\Gamma^R_{-H}})_{L^2(\Gamma^R_{-H})},$$

which is clearly continuous. Lastly, the operator $C : H \to L^2(\Gamma_0^R)$ is defined, for all $z = (\varphi, \psi, s) \in H$, by $Cz = s$. In the previous section, we have seen that we can define two operators $A_\eta : V \to V'$ and $B : L^2(\Gamma^R_{-H}) \to V'$ from the bilinear forms a_η and b, respectively. Now the following proposition establishes a relationship between the abstract problem (4.7) for $A = A_\eta$ and the concrete problem (4.15).

Proposition 4.4.1. *The vector of functions $z = (\varphi, \psi, s) \in W(0,T)$ satisfies the problem (4.7) if and only if it satisfies the strong problem:*

$$\left\{ \begin{array}{rcll} (1/c^2)\partial_t^2\varphi - \Delta\varphi - \eta\,\Delta\partial_t\varphi &=& 0 & \text{in } \Omega^R \times (0,T), \\ \partial_t\varphi + g\, s &=& 0 & \text{on } \Gamma_0^R \times (0,T), \\ \partial_z\varphi + \eta\,\partial_z\partial_t\varphi - \partial_t s &=& 0 & \text{on } \Gamma_0^R \times (0,T), \\ \partial_\nu\varphi + \eta\,\partial_\nu\partial_t\varphi &=& \chi & \text{on } \Gamma^R_{-H} \times (0,T), \\ \pm(\partial_x\varphi + \eta\,\partial_x\partial_t\varphi) + (1/c)\,\partial_t\varphi &=& 0 & \text{on } \Sigma_{\pm R} \times (0,T), \\ \varphi = \partial_t\varphi &=& 0 & \text{on } \Omega^R \times \{0\}, \\ s &=& 0 & \text{on } \Gamma_0^R \times \{0\}. \end{array} \right. \tag{4.16}$$

Proof. Let us assume that $z = (\varphi, \psi, s) \in W(0,T)$ satisfies the problem (4.7), that is $z(0) = 0$ and for all $\tilde{z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s}) \in V$,

$$\left\langle \frac{dz}{dt}, \tilde{z} \right\rangle_{V',V} + a_\eta(z, \tilde{z}) = b(\chi, \tilde{z}),$$

in other words

$$\begin{aligned} \langle \partial_t\varphi, \tilde{\varphi}\rangle_{(H^1(\Omega^R))',H^1(\Omega^R)} + \frac{1}{c^2}\langle \partial_t\psi, \tilde{\psi}\rangle_{(H^1(\Omega^R))',H^1(\Omega^R)} + g\ (\partial_t s, \tilde{s})_{L^2(\Gamma_0^R)} \\ - (\psi, \tilde{\varphi})_{H^1(\Omega^R)} + (\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega^R)} + \eta\,(\nabla\psi, \nabla\tilde{\psi})_{L^2(\Omega^R)} + \frac{1}{c}\sum_{\pm}(\psi, \tilde{\psi})_{L^2(\Sigma_{\pm R})} \\ = (\chi, \tilde{\psi})_{L^2(\Gamma^R_{-H})}. \end{aligned}$$

First choosing $\tilde{\psi} = 0$ and $\tilde{s} = 0$, we get that $\psi = \partial_t \varphi \in H^1(\Omega^R)$. The initial condition $z(0) = 0$ then immediately implies the initial conditions satisfied by φ and s in (4.16). The fact that $\psi|_{\Gamma_0^R} + g\, s = 0$ also implies the second equation in (4.16). We finally obtain that for all $\tilde{z} = (0, \tilde{\psi}, \tilde{s}) \in V$, then

$$\frac{1}{c^2}\langle \partial_t^2 \varphi, \tilde{\psi}\rangle_{(H^1(\Omega^R))',H^1(\Omega^R)} + g\ (\partial_t s, \tilde{s})_{L^2(\Gamma_0^R)} + (\nabla(\varphi + \eta\partial_t\varphi), \nabla\tilde{\psi})_{L^2(\Omega^R)} + \frac{1}{c}\sum_{\pm}(\partial_t\varphi, \tilde{\psi})_{L^2(\Sigma_{\pm R})} = (\chi, \tilde{\psi})_{L^2(\Gamma_{-H}^R)}.$$

Now we choose $\tilde{\psi} \in C_0^\infty(\Omega^R)$ and $\tilde{s} = 0$, which satisfy $\tilde{\psi}|_{\Gamma_0^R} + g\,\tilde{s} = 0$. We obtain

$$(1/c^2)\partial_t^2\varphi - \Delta(\varphi + \eta\,\partial_t\varphi) = 0 \tag{4.17}$$

in $\Omega^R \times (0,T)$, that is the first equation of (4.16). Let us establish the other boundary conditions. We assume that $\tilde{\psi} \in H^1(\Omega^R)$, which implies the Green formula

$$(\nabla(\varphi + \eta\partial_t\varphi), \nabla\tilde{\psi})_{L^2(\Omega^R)} = -\int_{\Omega^R} \Delta(\varphi + \eta\partial_t\varphi)\tilde{\psi}\, dX + \langle \partial_\nu(\varphi + \eta\partial_t\varphi), \tilde{\psi}\rangle_{H^{-1/2}(\partial\Omega^R),H^{1/2}(\partial\Omega^R)}.$$

Using (4.17) we obtain that for all $\tilde{\psi} \in H^1(\Omega^R)$ and $\tilde{s} = -(1/g)\tilde{\psi}|_{\Gamma_0^R}$,

$$-(\partial_t s, \tilde{\psi})_{L^2(\Gamma_0^R)} + \langle \partial_\nu(\varphi + \eta\partial_t\varphi), \tilde{\psi}\rangle_{H^{-1/2}(\partial\Omega^R),H^{1/2}(\partial\Omega^R)} + \frac{1}{c}\sum_{\pm}(\partial_t\varphi, \tilde{\psi})_{L^2(\Sigma_{\pm R})} = (\chi, \tilde{\psi})_{L^2(\Gamma_{-H}^R)}.$$

Let us impose in addition that the function $\tilde{\psi}$ vanishes in a volumic vicinity of $\partial\Omega^R \setminus \Gamma_0^R$, we obtain that

$$\partial_\nu(\varphi + \eta\partial_t\varphi) - \partial_t s = 0$$

on Γ_0^R, that is the third equation of (4.16). The boundary conditions on Γ_{-H}^R and on $\Sigma_{\pm R}$ are obtained similarly. We have hence proved that z satisfies the strong problem (4.16). The converse statement follows the same lines. □

We observe that for $\eta = 0$, the problem (4.16) coincides with the problem (4.15) we are interested in.

Let us now return to our initial objective: reconstructing the sea bottom displacement from measurements of the free surface. From this point forward, we will base our inversion strategy on the regularized problem (4.16) with a sufficiently small regularization parameter $\eta > 0$, which must be strictly positive because the bilinear form a_η loses coercivity when $\eta = 0$. We assume that the regularization introduces a small model error when recovering χ from measurements of the free surface perturbation s, which was initially modeled by (4.1). This model error will be reasonable compared to the measurement error that will naturally affect s. As presented in the abstract framework, denoting by $y \in L^2(0,T;L^2(\Gamma_0))$ the available measurements, we choose to rely on minimizing the least squares criterion

$$\inf_{\chi \in L^2(0,T;L^2(\Gamma_{-H}^R))} J(\chi) = \int_0^T \left(\frac{1}{2}\|y(t) - s(t)\|^2_{L^2(\Gamma_0)} + \frac{\varepsilon}{2}\|\chi(t)\|^2_{L^2(\Gamma_{-H}^R)}\right) dt, \tag{4.18}$$

where $(\varphi, \psi, s) \in W(0,T)$ is the solution to (4.7). This criterion is the direct transcription of (4.6), with $z = (\varphi, \psi, s) \in W(0,T)$, $Y = L^2(\Gamma_0^R)$, $U = L^2(\Gamma_{-H}^R)$, and $C = \begin{pmatrix} 0 & 0 & Id \end{pmatrix}$.

Remark 4.4.2 (Source regularity for the inverse problem)**.** *Note that taking $\chi \in L^2(0,T;L^2(\Gamma_{-H}^R))$ simplifies the implementation of the least-squares minimization. However, we could also consider a more regular source space $\chi \in H^1(0,T;L^2(\Gamma_{-H}^R))$, since such time regularity is necessary for the initial conservative System (4.1) to be well posed in the energy space.*

4.5 A time-scheme for the abstract problem

In this section, we wish to consider a discretized version of the optimal control problem (4.6)– or equivalently (4.18) – under the dynamics (4.7) in a view to solving the inverse tsunami problem numerically, such discretization being based on a simple implicit Euler scheme. On the one hand, the time discretization is based on a classical finite difference scheme, that is the time interval $(0,T)$ is divided into N sub-intervals $(n\tau,(n+1)\tau)$, with $\tau = T/N$ and $n = 0,\cdots,N-1$. On the other hand, we introduce some finite dimensional spaces $U_h \subset U$, $V_h \subset V$ and $Y_h \subset Y$, such that $CV_h \subset Y_h$. Those finite dimensional spaces are equipped with the induced norms of U, V and Y, respectively. We also denote $A_h : V_h \to V_h'$, $B_h : U_h \to V_h'$ and $C_h : V_h \to Y_h$ the restrictions of the operators A, B and C to V_h, U_h and V_h, respectively. Based on such discretization, instead of χ, z and y we consider $\chi_h^\tau = (\chi_h^n)_{n=1,\cdots,N} \in U_h^N$, $z_h^\tau = (z_h^n)_{n=0,\cdots,N} \in V_h^{N+1}$ and $y_h^\tau = (y_h^n)_{n=1,\cdots,N} \in Y_h^N$.

The discretized optimal control problem we consider, for $y_h^\tau \in Y_h^N$ and $\varepsilon > 0$, is

$$\inf_{\chi_h^\tau \in U_h^N} J_h^\tau(\chi_h^\tau) = \sum_{n=1}^{N} \left(\frac{\tau}{2} \|y_h^n - C_h z_h^n\|_{Y_h}^2 + \frac{\tau\varepsilon}{2} \|\chi_h^n\|_{U_h}^2 \right), \tag{4.19}$$

where z_h^τ is the solution in V_h^{N+1} to the problem

$$\begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = B_h \chi_h^n, \quad n = 1,\cdots,N \\ z_h^0 = 0. \end{cases} \tag{4.20}$$

Clearly, we have the following proposition.

Proposition 4.5.1. *The problem (4.19) (4.20) has a unique solution.*

As in the continuous case, the gradient of the cost function J_h^τ with respect to χ_h^τ is given by the following proposition.

Proposition 4.5.2. *The differential of J_h^τ at point $\chi_h^\tau \in U_h^N$ is given, for $\tilde{\chi}_h^\tau \in U_h^N$, by*

$$(D_{\chi_h^\tau} J_h^\tau(\chi_h^\tau), \tilde{\chi}_h^\tau)_{U_h^N} = \sum_{n=1}^{N} (\tau\varepsilon \chi_h^n - \tau B_h^* p_h^n, \tilde{\chi}_h^n)_{U_h}, \tag{4.21}$$

where $p_h^\tau = (p_h^n)_{n=1,\cdots,N+1}$ is the solution in V_h^{N+1} to the problem

$$\begin{cases} \dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^* p_h^n = C_h^*(y_h^n - C_h z_h^n), \quad n = 1,\cdots,N, \\ p_h^{N+1} = 0. \end{cases} \tag{4.22}$$

and z_h^τ is the solution to the problem (4.20).

Proof. To obtain (4.21) we now introduce the discrete Lagrangian $\mathcal{L}_h^\tau$ defined for $(\tilde{z}_h^\tau, \tilde{p}_h^\tau, \chi_h^\tau) \in V_h^{N+1} \times V_h^{N+1} \times U_h^N$ by

$$\mathcal{L}_h^\tau(\tilde{z}_h^\tau, \tilde{p}_h^\tau, \chi_h^\tau) = \sum_{n=1}^{N} \left(\frac{\tau}{2} \|y_h^n - C_h \tilde{z}_h^n\|_{Y_h}^2 + \frac{\tau\varepsilon}{2} \|\chi_h^n\|_{U_h}^2 \right) + \sum_{n=1}^{N} \tau \left\langle \frac{\tilde{z}_h^n - \tilde{z}_h^{n-1}}{\tau} + A_h \tilde{z}_h^n - B_h \chi_h^n, \tilde{p}_h^n \right\rangle_{V_h', V_h}.$$

We observe that for $\chi_h^\tau \in U_h^N$, we have

$$J_h^\tau(\chi_h^\tau) = \mathcal{L}_h^\tau(z_h^\tau(\chi_h^\tau), \tilde{p}_h^\tau, \chi_h^\tau), \quad \forall \tilde{p}_h^\tau \in V_h^{N+1},$$

where $z_h^\tau(\chi_h^\tau)$ is the solution to the problem (4.20). Using the chain rule, we have

$$(D_{\chi_h^\tau} J_h^\tau(\chi_h^\tau), \tilde{\chi}_h^\tau)_{U_h^N} = \left\langle D_{z_h^\tau} \mathcal{L}_h^\tau(z_h^\tau(\chi_h^\tau), \tilde{p}_h^\tau, \chi), z_h^\tau(\tilde{\chi}_h^\tau) \right\rangle_{(V_h^{N+1})', V_h^{N+1}} + \left(D_{\chi_h^\tau} \mathcal{L}_h^\tau(z_h^\tau(\chi_h^\tau), \tilde{p}_h^\tau, \chi_h^\tau), \tilde{\chi}_h^\tau \right)_{U_h^N}.$$

On the one hand, we compute, for $n = 1, \cdots, N-1$,

$$\begin{aligned} &\left\langle D_{z_h^n} \mathcal{L}_h^\tau(z_h^\tau, \tilde{p}_h^\tau, \chi_h^\tau), \tilde{z}_h^n \right\rangle_{V_h', V_h} \\ &\quad = -\tau(y_h^n - C_h z_h^n, C_h \tilde{z}_h^n)_{Y_h} + \langle \tilde{p}_h^n - \tilde{p}_h^{n+1}, \tilde{z}_h^n \rangle_{V_h', V_h} + \tau \langle A_h \tilde{z}_h^n, \tilde{p}_h^n \rangle_{V_h', V_h} \\ &\quad = \tau \langle C_h^*(C_h z_h^n - y_h^n), \tilde{z}_h^\tau \rangle_{V_h', V_h} + \tau \left\langle \frac{\tilde{p}_h^n - \tilde{p}_h^{n+1}}{\tau} + A_h^* \tilde{p}_h^n, \tilde{z}_h^n \right\rangle_{V_h', V_h}. \end{aligned}$$

Note that the computation above is also valid for $n = N$ provided we assume that $\tilde{p}_h^{N+1} = 0$.

On the other hand, we compute, for $n = 1, \cdots, N$,

$$\left(D_{\chi_h^n} \mathcal{L}_h^\tau(z_h^\tau, \tilde{p}_h^\tau, \chi_h^\tau), \tilde{\chi}_h^n \right)_{U_h} = \tau \varepsilon (\chi_h^n, \tilde{\chi}_h^n)_{U_h} - \tau \langle B_h \tilde{\chi}_h^n, \tilde{p}_h^n \rangle_{V_h', V_h} = (\tau \varepsilon \chi_h^n - \tau B_h^* p_h^n, \tilde{\chi}_h^n)_{U_h}.$$

It remains to choose $\tilde{p}_h^\tau = p_h^\tau$, where p_h^τ is the solution to problem (4.22), which implies that

$$D_{z_h^\tau} \mathcal{L}_h^\tau(z_h^\tau(\chi_h^\tau), p_h^\tau, \chi_h^\tau) = 0.$$

The result is proved. □

In view of Proposition 4.5.2, the unique solution to the problem (4.19) (4.20) is characterized by

$$\varepsilon \chi_h^n - B_h^* p_h^n = 0, \quad n = 1, \cdots, N, \tag{4.23}$$

where p_h^τ is the solution to the problem (4.22), in which z_h^τ is the solution to the problem (4.20).

In the particular case of the inverse tsunami problem, solving the optimal control problem numerically amounts to solve the problem (4.19) (4.20) with the help of a gradient method based on the explicit form of such gradient given by Proposition 4.5.2. Our objective is to estimate the error between the solution χ to the continuous optimal control problem (4.6) (4.7) and the solution χ_h^τ to the discrete optimal control problem (4.19) (4.20) obtained after a time and space discretization. Such an estimate is quite difficult to obtain directly. However, we have proved that solving the optimal control problem (4.6) (4.7) is equivalent to solving the mixed weak formulation (4.11). This is why we now introduce a well-chosen discrete mixed weak formulation which will happen to be equivalent to the discrete optimal control problem (4.19) (4.20). So that the problem will boil down to estimating the error between the solutions to the continuous and the discrete mixed weak formulations.

Let us introduce the discrete spaces

$$W_h^\tau = \{z_h^\tau \in \mathbb{P}_1(0,T) \otimes V_h,\ z_h^\tau(0) = 0\}, \quad U_h^\tau = \mathbb{P}_0(0,T) \otimes U_h,$$
$$V_h^\tau = \mathbb{P}_0(0,T) \otimes V_h, \quad Y_h^\tau = \mathbb{P}_0(0,T) \otimes Y_h,$$

where $\mathbb{P}_0(0,T)$ denotes the subspace of $L^2(0,T)$ formed by the functions that are constant on each sub-interval $(n\tau, (n+1)\tau)$, for $n = 0, \cdots, N-1$, while $\mathbb{P}_1(0,T)$ denotes the subspace of $H^1(0,T)$ formed by continuous functions which are polynomials of degree one on each sub-interval $(n\tau, (n+1)\tau)$, for $n = 0, \cdots, N-1$. We observe that $W_h^\tau \subset W_0(0,T)$, $U_h^\tau \subset L^2(0,T;U)$, $V_h^\tau \subset L^2(0,T;V)$ and $Y_h^\tau \subset L^2(0,T;Y)$. Let us denote ω_0^n the basis functions of $\mathbb{P}_0(0,T)$, for $n = 1, \cdots, N$, such that $\omega_0^n = 1$ on the segment $[(n-1)\tau, n\tau]$ and 0 elsewhere. Let us denote ω_1^n the basis functions of $\mathbb{P}_1(0,T)$, for $n = 0, \cdots, N$, such that $\omega_n(m\tau) = \delta_{mn}$. We finally denote, for all $z_h^\tau \in W_h^\tau$, $\chi_h^\tau \in U_h^\tau$, $p_h^\tau \in V_h^\tau$ and $y_h^\tau \in Y_h^\tau$,

$$\begin{aligned} z_h^\tau(t) &= \sum_{n=1}^N z_h^n\, \omega_1^n(t), \quad \chi_h^\tau(t) = \sum_{n=1}^N \chi_h^n\, \omega_0^n(t), \\ p_h^\tau(t) &= \sum_{n=1}^N p_h^n\, \omega_0^n(t), \quad y_h^\tau(t) = \sum_{n=1}^N y_h^n\, \omega_0^n(t), \end{aligned} \tag{4.24}$$

where $(z_h^n, \chi_h^n, p_h^n, y_h^n) \in V_h \times U_h \times V_h \times Y_h$ for $n = 1, \cdots, N$. Hence W_h^τ can be identified with V_h^N, U_h^τ with U_h^N, V_h^τ with V_h^N and Y_h^τ with Y_h^N.

Let us introduce the following weak formulation: find $Z_h^\tau = (z_h^\tau, \chi_h^\tau) \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$ such that for all $\tilde{Z}_h^\tau = (\tilde{z}_h^\tau, \tilde{\chi}_h^\tau) \in W_h^\tau \times U_h^\tau$ and $\tilde{p}_h^\tau \in V_h^\tau$,

$$\left\{ \begin{array}{l} j^\tau(Z_h^\tau, \tilde{Z}_h^\tau) + d^\tau(\tilde{Z}_h^\tau, p_h^\tau) = \ell^\tau(\tilde{Z}_h^\tau), \\ d^\tau(Z_h^\tau, \tilde{p}_h^\tau) = 0, \end{array} \right. \tag{4.25}$$

with

$$j^\tau(Z_h^\tau, \tilde{Z}_h^\tau) = \int_0^T \varepsilon(\chi_h^\tau, \tilde{\chi}_h^\tau)_U \, dt + \fint_0^T (Cz_h^\tau, C\tilde{z}_h^\tau)_Y \, dt, \tag{4.26}$$

$$d^\tau(Z_h^\tau, \tilde{p}_h^\tau) = \int_0^T \left\langle \frac{dz_h^\tau}{dt}, \tilde{p}_h^\tau \right\rangle_{V',V} dt + \fint_0^T a(z_h^\tau, \tilde{p}_h^\tau) \, dt - \int_0^T b(\chi_h^\tau, \tilde{p}_h^\tau) \, dt, \tag{4.27}$$

and

$$\ell^\tau(Z_h^\tau) = \fint_0^T (y_h^\tau, C\tilde{z}_h^\tau)_{Y_h} \, dt. \tag{4.28}$$

Here, the symbol $\fint_0^T$ denotes, for a function f defined on $[0,T]$ which is piecewise continuous on each interval $[n\tau, (n+1)\tau]$, $n = 0, \cdots, N-1$, the Riemann sum

$$\fint_0^T f(t) \, dt = \tau \sum_{n=1}^N f(n\tau_-),$$

where $f(n\tau_-)$ is the left value of f at point $n\tau$. The definitions of the bilinear form j^τ, d^τ and the linear form ℓ^τ have to be compared to those of j, d and ℓ given by (4.12), (4.13) and (4.14), respectively. The introduction of those new bilinear forms is justified by the following proposition, which establishes the link between the discrete optimal problem (4.19) (4.20) and the discrete weak formulation (4.25).

Proposition 4.5.3. *The element $\chi_h^\tau \in U_h^N$ is the solution to the discrete optimal control (4.19) (4.20) if and only if the pair $Z_h^\tau = (z_h^\tau, \chi_h^\tau) \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$, where z_h^τ is the solution to problem (4.20) and p_h^τ is the solution to problem (4.22), satisfy the discrete weak formulation (4.25).*

Proof. Let us assume that $Z_h^\tau = (z_h^\tau, \chi_h^\tau) \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$ are solutions to the weak formulation (4.25). The second equation of (4.25) implies that for all $\tilde{p}_h^\tau \in V_h^\tau$,

$$\int_0^T \left\langle \frac{dz_h^\tau}{dt}, \tilde{p}_h^\tau \right\rangle_{V',V} dt + \fint_0^T a(z_h^\tau, \tilde{p}_h^\tau) \, dt - \int_0^T b(\chi_h^\tau, \tilde{p}_h^\tau) \, dt = 0.$$

By using the decomposition (4.24) for z_h^τ and χ_h^τ and choosing $\tilde{p}_h^\tau = \tilde{p}_h \, \omega_0^n(t)$ for any $n = 1, \cdots, N$ and any $\tilde{p}_h \in V_h$, we obtain that for any $n = 1, \cdots, N$,

$$\sum_{m=1}^N \left(\int_0^T \frac{d\omega_1^m}{dt} \omega_0^n \, dt \right) z_h^m + \tau \left(\sum_{m=1}^N \omega_1^m(m\tau_-) \, \omega_0^n(m\tau_-) \, dt \right) A_h z_h^m \\ - \sum_{m=1}^N \left(\int_0^T \omega_0^m \, \omega_0^n \, dt \right) B_h \chi_h^m = 0.$$

A simple computation shows that for $m, n = 1, \cdots, N$, we have

$$\int_0^T \frac{d\omega_1^m}{dt} \omega_0^n \, dt = \left\{ \begin{array}{lll} -1 & \text{if} & m = n-1 \\ 1 & \text{if} & m = n \\ 0 & \text{otherwise} & \end{array} \right.$$

and

$$\int_0^T \omega_0^m \omega_0^n \, dt = \tau \delta_{mn},$$

while $\omega_1^m(m\tau_-) = 1$ and $\omega_0^n(m\tau_-) = \delta_{mn}$. Hence we have that z_h^τ satisfies the problem (4.20).

Let us consider the first equation of (4.25) with $\tilde{\chi}_h^\tau = 0$. We obtain that for all $\tilde{z}_h^\tau \in W_h^\tau$,

$$\int_0^T \left\langle \frac{d\tilde{z}_h^\tau}{dt}, p_h^\tau \right\rangle_{V',V} dt + \fint_0^T a(\tilde{z}_h^\tau, p_h^\tau)\, dt = \fint_0^T (y_h^\tau - C_h z_h^\tau, C_h \tilde{z}_h^\tau)_Y\, dt.$$

By using the decomposition (4.24) for p_h^τ and y_h^τ and choosing $\tilde{z}_h^\tau = \tilde{z}_h\, \omega_1^n(t)$ for any $n = 1, \cdots, N$ and any $\tilde{z}_h \in V_h$, we obtain that for any $n = 1, \cdots, N$, we obtain

$$\sum_{m=1}^N \left(\int_0^T \frac{d\omega_1^n}{dt} \omega_0^m\, dt \right) p_h^m + \tau \sum_{m=1}^N \omega_1^n(m\tau_-)\, \omega_0^m(m\tau_-) A_h^* p_h^m = \tau \sum_{m=1}^N C_h^* \left(\omega_0^m(n\tau_-) y_h^m - \omega_1^m(n\tau_-) C_h z_h^m \right). \quad (4.29)$$

We hence obtain (4.22).

Lastly we consider the first equation of (4.25) with $\tilde{z}_h^\tau = 0$. We obtain that for all $\tilde{\chi}_h^\tau \in U_h^\tau$,

$$\int_0^T \varepsilon(\chi_h^\tau, \tilde{\chi}_h^\tau)_U\, dt - \int_0^T b(\tilde{\chi}_h^\tau, p_h^\tau)\, dt = 0.$$

By using the decomposition (4.24) for χ_h^τ and choosing $\tilde{\chi}_h^\tau = \tilde{\chi}_h\, \omega_0^n(t)$ for any $n = 1, \cdots, N$ and any $\tilde{\chi}_h \in U_h$, we obtain that χ_h^τ satisfies (4.23). Given that $(\chi_h^\tau, z_h^\tau, p_h^\tau)$ satisfies the system (4.23), (4.20) and (4.22), we conclude that χ_h^τ is the solution to the discrete optimal control (4.19) (4.20). The converse statement is obtained similarly. □

Remark 4.5.1. *The reader may wonder why we have introduced the bilinear forms j^τ, d^τ and the linear form ℓ^τ in the weak formulation (4.25) instead of j, d and ℓ given by (4.12), (4.13) and (4.14), respectively. Let us introduce the following "more natural" weak formulation: find $Z_h^\tau = (z_h^\tau, \chi_h^\tau) \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$ such that for all $\tilde{Z}_h^\tau = (\tilde{z}_h^\tau, \tilde{\chi}_h^\tau) \in W_h^\tau \times U_h^\tau$ and $\tilde{p}_h^\tau \in V_h^\tau$,*

$$\begin{cases} j(Z_h^\tau, \tilde{Z}_h^\tau) + d(\tilde{Z}_h^\tau, p_h^\tau) = \ell(\tilde{Z}_h^\tau), \\ d(Z_h^\tau, \tilde{p}_h^\tau) = 0. \end{cases} \quad (4.30)$$

A natural question is: what is the discretized optimal control problem which is equivalent to (4.30) ?

We consider the discretized optimal control problem: for $y_h^\tau \in Y_h^N$ and $\varepsilon > 0$,

$$\inf_{\chi_h^\tau \in U_h^N} J_h^\tau(\chi_h^\tau) = \sum_{n=1}^N \left(\frac{\tau}{2} \left\| \frac{y_h^{n-1}}{2} - C_h \left(\frac{1}{3-\sqrt{3}} z_h^{n-1} + \frac{3-\sqrt{3}}{6} z_h^n \right) \right\|_{Y_h}^2 + \frac{\tau\varepsilon}{2} \|\chi_h^n\|_{U_h}^2 \right) + \frac{\tau}{2} \frac{\sqrt{3}}{6} \left\| \sqrt{3} y_h^N + \sqrt{3} \frac{3+\sqrt{3}}{6} y_h^{N-1} - C_h z_h^N \right\|_{Y_h}^2, \quad (4.31)$$

where z_h^τ is the solution in V_h^{N+1} to the problem

$$\begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h \dfrac{z_h^n + z_h^{n-1}}{2} = B_h \chi_h^n, \quad n = 1, \cdots, N \\ z_h^0 = 0. \end{cases} \quad (4.32)$$

We also consider the new following adjoint system

$$\begin{cases} \dfrac{p_h^n - p_h^{n+1}}{\tau} + A_h^* \dfrac{p_h^n + p_h^{n+1}}{2} \\ = C_h^* \left(\frac{y_h^{n-1} + y_h^n}{2} \right) - C_h^* C_h \left(\frac{z_h^{n-1} + 4z_h^n + z_h^{n+1}}{6} \right), \quad n = 1, \cdots, N-1, \\ \dfrac{p_h^N - p_h^{N+1}}{\tau} + A_h^* \dfrac{p_h^N + p_h^{N+1}}{2} \\ = C_h^* \left(\frac{y_h^{N-1} + y_h^N}{2} \right) - C_h^* C_h \left(\frac{z_h^{N-1} + 2z_h^N}{6} \right), \\ p_h^{N+1} = 0. \end{cases} \quad (4.33)$$

It can be checked that the link between the problem (4.31) (4.32) and the problem (4.30) is the following. The element $\chi_h^\tau \in U_h^N$ is the solution to the discrete optimal control (4.31) (4.32) if and only if the pair $Z_h^\tau = (z_h^\tau, \chi_h^\tau) \in W_h^\tau \times U_h^\tau$ and $p_h^\tau \in V_h^\tau$, where z_h^τ is the solution to problem (4.32) and p_h^τ is the solution to problem (4.33), satisfy the discrete weak formulation (4.30). On the one hand, from the practical point of view, we wish to solve the simplest discretized optimal control problem, that is we will minimize the cost function (4.19) rather than (4.31). On the other hand, from the numerical analysis point of view, we did not manage to obtain the energy identity of Lemma 4.6.1 using the Crank-Nicolson scheme (4.32) instead of the implicit Euler scheme (4.20).

4.6 Analysis of convergence

The objective is now to obtain an error estimate between the solution $(z, \chi, p) \in W_0(0,T) \times L^2(0,T;U) \times L^2(0,T;V)$ to the continuous mixed formulation (4.11) and the solution $(z_h^\tau, \chi_h^\tau, p_h^\tau) \in W_h^\tau \times U_h^\tau \times V_h^\tau$ to the discrete mixed formulation (4.25). Let us introduce a new norm of V_h given by

$$\|\varphi_h\|_{V_h'} := \sup_{v_h \in V_h} \frac{\langle \varphi_h, v_h \rangle_{V',V}}{\|v_h\|_V}, \quad \forall \varphi_h \in V_h. \tag{4.34}$$

We will need the following lemmas.

Lemma 4.6.1. *For $f_h^\tau \in (V_h^\tau)'$, let us consider the unique solution $z_h^\tau \in W_h^\tau$ to the problem*

$$d^\tau((z_h^\tau, 0), \tilde{p}_h^\tau) = \int_0^T \langle f_h^\tau, \tilde{p}_h^\tau \rangle_{V',V} \, dt, \quad \forall \tilde{p}_h^\tau \in V_h^\tau, \tag{4.35}$$

where d^τ is defined by (4.27). Then there exists some constant $C_{\rm st} > 0$, which depends neither on h nor on τ, such that

$$\|z_h^\tau\|_{W_h^\tau} := \sqrt{\|z_h^\tau\|^2_{L^2(0,T;V_h)} + \left\| \frac{dz_h^\tau}{dt} \right\|^2_{L^2(0,T;V_h')}} \leq C_{\rm st} \, \|f_h^\tau\|_{L^2(0,T;V')}. \tag{4.36}$$

Proof. We first observe that $z_h^\tau \in W_h^\tau$ is the solution to the well-posed problem:

$$\begin{cases} \dfrac{z_h^n - z_h^{n-1}}{\tau} + A_h z_h^n = f_h^n, \quad n = 1, \cdots, N \\ z_h^0 = 0, \end{cases} \tag{4.37}$$

where

$$z_h^\tau(t) = \sum_{n=1}^N z_h^n \, \omega_1^n(t), \quad f_h^\tau(t) = \sum_{n=1}^N f_h^n \, \omega_0^n(t).$$

Let us first prove that there exists two constants $c_1, c_2 > 0$, which depends neither on h nor on τ, such that

$$c_1 \sum_{n=1}^N \tau \|z_h^n\|_V^2 \leq \|z_h^\tau\|^2_{L^2(0,T;V)} \leq c_2 \sum_{n=1}^N \tau \|z_h^n\|_V^2.$$

We have

$$\begin{aligned} \|z_h^\tau\|^2_{L^2(0,T;V)} &= \int_0^T \|z_h^\tau(t)\|_V^2 \, dt \\ &= \sum_{n=0}^{N-1} \int_{n\tau}^{(n+1)\tau} \|z_h^\tau(t)\|_V^2 \, dt \\ &= \sum_{n=0}^{N-1} \tau \int_0^1 \|(1-s) z_h^n + s z_h^{n+1}\|_V^2 \, ds \\ &= \sum_{n=0}^{N-1} \frac{\tau}{3} \left(\|z_h^n\|_V^2 + \|z_h^{n+1}\|_V^2 + (z_h^n, z_h^{n+1})_V \right) \end{aligned}$$

Using that

$$-\frac{1}{2}\left(\|z_h^n\|_V^2+\|z_h^{n+1}\|_V^2\right)\le(z_h^n,z_n^{n+1})_V\le\frac{1}{2}\left(\|z_h^n\|_V^2+\|z_h^{n+1}\|_V^2\right),$$

we obtain

$$\frac{1}{6}\sum_{n=1}^{N}\tau\|z_h^n\|_V^2\le\|z_h^\tau\|_{L^2(0,T;V)}^2\le\sum_{n=1}^{N}\tau\|z_h^n\|_V^2, \tag{4.38}$$

which is the expected result. Now, in view of (4.37) we have for all $n=1,\cdots,N$,

$$\left(\frac{z_h^n-z_h^{n-1}}{\tau},z_h^n\right)_H+a(z_h^n,z_h^n)=\langle f_h^n,z_h^n\rangle_{V',V}\,.$$

Since

$$\left(\frac{z_h^n-z_h^{n-1}}{\tau},z_h^n\right)_H=\frac{\|z_h^n\|_H^2}{\tau}-\frac{(z_h^n,z_h^{n-1})_H}{\tau}\ge\frac{\|z_h^n\|_H^2-\|z_h^{n-1}\|_H^2}{2\tau},$$

we get that for all $n=1,\cdots,N$,

$$\frac{\|z_h^n\|_H^2-\|z_h^{n-1}\|_H^2}{2\tau}+a(z_h^n,z_h^n)\le\langle f_h^n,z_h^n\rangle_{V',V}\,.$$

Adding the above inequalities, we obtain that for all $n=1,\cdots,N$,

$$\frac{\|z_h^n\|_H^2}{2\tau}+\sum_{m=1}^{n}a(z_h^m,z_h^m)\le\sum_{m=1}^{n}\langle f_h^m,z_h^m\rangle_{V',V}\,.$$

On the one hand, since we easily check that

$$\|f_h^\tau\|_{L^2(0,T;V')}^2=\sum_{n=1}^{N}\tau\|f_h^n\|_{V'}^2,$$

we have

$$\begin{aligned}\frac{\|z_h^n\|_H^2}{2}&\le\sum_{m=1}^{n}\tau\,\langle f_h^m,z_h^m\rangle_{V',V}\\&\le\sqrt{\sum_{m=1}^{N}\tau\|f_h^m\|_{V'}^2}\sqrt{\sum_{m=1}^{N}\tau\|z_h^m\|_V^2}\\&\le\sqrt{6}\,\|f_h^\tau\|_{L^2(0,T;V')}\|z_h^\tau\|_{L^2(0,T;V)},\end{aligned}$$

where we have used the Cauchy-Schwarz inequality in the second inequality and the left inequality of (4.38) to obtain the third inequality. Hence, observing that $N\tau=T$,

$$\sum_{n=1}^{N}\tau\|z_h^n\|_H^2\le2\sqrt{6}T\,\|f_h^\tau\|_{L^2(0,T;V')}\|z_h^\tau\|_{L^2(0,T;V)}.$$

On the other hand, we have

$$\sum_{n=1}^{N}\tau a(z_h^n,z_h^n)\le\sqrt{6}\,\|f_h^\tau\|_{L^2(0,T;V')}\|z_h^\tau\|_{L^2(0,T;V)},$$

so that using the coercivity property (4.4), we get

$$\alpha\sum_{n=0}^{N}\tau\|z_h^n\|_V^2\le(2\sqrt{6}\lambda T+\sqrt{6})\,\|f_h^\tau\|_{L^2(0,T;V')}\|z_h^\tau\|_{L^2(0,T;V)},$$

which implies in view of the right inequality of (4.38) that

$$\|z_h^\tau\|_{L^2(0,T;V)}\le\frac{2\sqrt{6}\lambda T+\sqrt{6}}{\alpha}\|f_h^\tau\|_{L^2(0,T;V')}. \tag{4.39}$$

Reusing again (4.37), we have for $n = 1, \cdots, N$, for all $q_h \in V_h$,

$$\left\langle \frac{z_h^n - z_h^{n-1}}{\tau}, q_h \right\rangle_{V',V} = -\langle A_h z_h^n, q_h\rangle_{V',V} + \langle f_h^n, q_h\rangle_{V',V},$$

which implies from the definition of $\|\cdot\|_{V_h'}$ that

$$\left\| \frac{z_h^n - z_h^{n-1}}{\tau} \right\|_{V_h'}^2 \leq 2\left(\|A\|^2 \|z_h^n\|_V^2 + \|f_h^n\|_{V'}^2\right),$$

hence

$$\sum_{n=1}^{N} \tau \left\| \frac{z_h^n - z_h^{n-1}}{\tau} \right\|_{V_h'}^2 \leq 2\left(6\|A\|^2 \|z_h^\tau\|_{L^2(0,T;V)}^2 + \|f_h^\tau\|_{L^2(0,T;V')}^2\right).$$

We observe that

$$\begin{aligned}
\int_0^T \left\| \frac{dz_h^\tau}{dt} \right\|_{V_h'}^2 dt &= \sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} \left\| \frac{dz_h^\tau}{dt} \right\|_{V_h'}^2 dt \\
&= \sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} \left\| \frac{z_h^n - z_h^{n-1}}{\tau} \right\|_{V_h'}^2 dt \\
&= \sum_{n=1}^{N} \tau \left\| \frac{z_h^n - z_h^{n-1}}{\tau} \right\|_{V_h'}^2.
\end{aligned}$$

We finally obtain, by using (4.39), that exists a constant $C_{\rm st} > 0$ which depends neither on h nor on τ such that

$$\left\| \frac{dz_h^\tau}{dt} \right\|_{L^2(0,T;V_h')} \leq C_{\rm st} \|f_h^\tau\|_{L^2(0,T;V')}. \tag{4.40}$$

We complete the proof gathering the estimates (4.39) and (4.40). □

Let us first prove the continuity of the discrete bilinear forms j^τ and d^τ, that we address in two propositions below.

Proposition 4.6.1. *The bilinear form j^τ satisfies, for some constant $C_{\rm st}$ which depends neither on τ nor on h, for all $Z, \tilde{Z} \in W_0(0,T) \times L^2(0,T;U)$,*

$$|j^\tau(Z, \tilde{Z})| \leq C_{\rm st} \|Z\|_{W_0(0,T)\times L^2(0,T;U)} \|\tilde{Z}\|_{W_0(0,T)\times L^2(0,T;U)}.$$

In particular, the inequality holds for $Z_h^\tau, \tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau$.

Proof. In view of (4.26), we have

$$|j^\tau((z,\chi),(\tilde{z},\tilde{\chi}))| \leq \left| \int_0^T \varepsilon(\chi,\tilde{\chi})_U \, dt \right| + \left| \fint_0^T (Cz, C\tilde{z})_Y \, dt \right|.$$

On the one hand,

$$\left| \int_0^T \varepsilon(\chi,\tilde{\chi})_U \, dt \right| \leq \varepsilon \|\chi\|_{L^2(0,T;U)} \|\tilde{\chi}\|_{L^2(0,T;U)}.$$

On the other hand,

$$\begin{aligned}
\left| \fint_0^T (Cz, C\tilde{z})_Y \, dt \right| = \left| \tau \sum_{n=1}^{N} (Cz, C\tilde{z})_Y (n\tau_-) \right| &\leq \|C\|^2 \sum_{n=1}^{N} \tau \|z(n\tau)\|_H \|\tilde{z}(n\tau)\|_H \\
&\leq \|C\|^2 \sum_{n=1}^{N} \tau \|z\|_{C^0([0,T];H)} \|\tilde{z}\|_{C^0([0,T];H)} \\
&\leq C_{\rm st} T \|C\|^2 \|z\|_{W_0(0,T)} \|\tilde{z}\|_{W_0(0,T)},
\end{aligned}$$

where we have used the continuous embedding $W(0,T) \hookrightarrow C^0([0,T];H)$, which completes the proof. □

Let us define $C_0(0,T) = C^0([0,T];V) \cap W_0(0,T)$, equipped with the norm given for $z \in C_0(0,T)$ by

$$\|z\|^2_{C_0(0,T)} := \|z\|^2_{L^\infty(0,T;V)} + \left\| \frac{dz}{dt} \right\|^2_{L^2(0,T;V')}.$$

Proposition 4.6.2. *The bilinear form d^τ satisfies, for some constant $C_{\rm st}$ which depends neither on τ nor on h, for all $Z_h^\tau \in W_h^\tau \times U_h^\tau$, for all $p_h^\tau \in V_h^\tau$,*

$$|d^\tau(Z_h^\tau, p_h^\tau)| \le C_{\rm st} \, \|Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \, \|p_h^\tau\|_{L^2(0,T;V)}. \tag{4.41}$$

In addition, for all $p \in C^0([0,T];V)$ and all $Z \in C_0(0,T) \times L^2(0,T;U)$,

$$|d^\tau(Z_h^\tau, p - p_h^\tau)| \le C_{\rm st} \, \|Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \, \|p - p_h^\tau\|_{L^\infty(0,T;V)} \tag{4.42}$$

and

$$|d^\tau(Z - Z_h^\tau, p_h^\tau)| \le C_{\rm st} \, \|Z - Z_h^\tau\|_{C_0(0,T)\times L^2(0,T;U)} \, \|p_h^\tau\|_{L^2(0,T;V)}. \tag{4.43}$$

Proof. Let us prove (4.41). In view of (4.27), we have

$$|d^\tau((z_h^\tau, \chi_h^\tau), p_h^\tau)| \le \left| \int_0^T \left\langle \frac{dz_h^\tau}{dt}, p_h^\tau \right\rangle_{V',V} dt \right| + \left| \fint_0^T a(z_h^\tau, p_h^\tau)\, dt \right| + \left| \int_0^T b(\chi_h^\tau, p_h^\tau)\, dt \right|.$$

Firstly we observe that

$$\left| \int_0^T \left\langle \frac{dz_h^\tau}{dt}, p_h^\tau \right\rangle_{V',V} dt \right| \le \left\| \frac{dz_h^\tau}{dt} \right\|_{L^2(0,T;V')} \|p_h^\tau\|_{L^2(0,T;V)} \le \|z_h^\tau\|_{W_0(0,T)} \, \|p_h^\tau\|_{L^2(0,T;V)},$$

and

$$\left| \int_0^T b(\chi_h^\tau, p_h^\tau)\, dt \right| \le \|B\| \|\chi_h^\tau\|_{L^2(0,T;U)} \, \|p_h^\tau\|_{L^2(0,T;V)}.$$

Secondly we have

$$\begin{aligned} \left| \fint_0^T a(z_h^\tau, p_h^\tau)\, dt \right| &= \left| \tau \sum_{n=1}^N a(z_h^\tau, p_h^\tau)(n\tau_-) \right| \le \|A\| \sum_{n=1}^N \tau \|z_h^\tau(n\tau_-)\|_V \|p_h^\tau(n\tau_-)\|_V \\ &= \|A\| \sum_{n=1}^N \tau \|z_h^n\|_V \|p_h^n\|_V \le 6\|A\| \|z_h^\tau\|_{W_0(0,T)} \|p_h^\tau\|_{L^2(0,T;V)}, \end{aligned}$$

which completes the proof of (4.41). Concerning the proof of (4.42), we only have to adapt the estimate of $\fint_0^T a(z_h^\tau, p - p_h^\tau)\, dt$.

$$\begin{aligned} \left| \fint_0^T a(z_h^\tau, p - p_h^\tau)\, dt \right| &= \left| \tau \sum_{n=1}^N a(z_h^\tau, p - p_h^\tau)(n\tau_-) \right| \le \|A\| \sum_{n=1}^N \tau \|z_h^\tau(n\tau_-)\|_V \|(p - p_h^\tau)(n\tau_-)\|_V \\ &\le \|A\| \left(\sum_{n=1}^N \tau \|z_h^n\|_V \right) \|p - p_h^\tau\|_{L^\infty(0,T;V)} \\ &\le \|A\| \sqrt{\sum_{n=1}^N \tau} \sqrt{\sum_{n=1}^N \tau \|z_h^n\|_V^2} \, \|p - p_h^\tau\|_{L^\infty(0,T;V)} \\ &\le \sqrt{6} \|A\| \sqrt{T} \|z_h^\tau\|_{W_0(0,T)} \|p - p_h^\tau\|_{L^\infty(0,T;V)}. \end{aligned}$$

The inequality (4.43) is proved the same way. □

Now we wish to prove that the bilinear form d^τ satisfies a discrete inf-sup condition and that the bilinear form j^τ satisfies a discrete coercivity property on the kernel of d^τ. For we need to introduce the following quantity r_h related to the space discretization, which in general depends on h.

$$r_h := \inf_{z_h \in V_h} \frac{\|z_h\|_{V_h'}}{\|z_h\|_{V'}} = \frac{1}{\|P_h\|_V} \leq 1, \tag{4.44}$$

where the norm $\|\cdot\|_{V_h'}$ is given by (4.34) and $\|P_h\|_V$ is the norm of the operator $P_h : V \to V$ defined as follows: for $z \in V$, $P_h z$ denotes the H-orthogonal projection of z onto $V_h \subset V$. The second equality of (4.44) is proven in [56] (see Example 26.22) and recalled in [57] (see Lemma 71.8).

Proposition 4.6.3. *The bilinear form d^τ satisfies the discrete inf-sup condition: for sufficiently small h, there exists some constant $\beta > 0$ which does not depend on h, τ such that*

$$\inf_{p_h^\tau \in V_h^\tau} \sup_{Z_h^\tau \in W_h^\tau \times U_h^\tau} \frac{d^\tau(Z_h^\tau, p_h^\tau)}{\|Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \|p_h^\tau\|_{L^2(0,T;V)}} \geq \beta\, r_h.$$

Proof. For a given $p_h^\tau \in V_h^\tau$, let us consider $Z_h^\tau = (z_h^\tau, 0) \in W_h^\tau$ such that

$$d^\tau(Z_h^\tau, \tilde{p}_h^\tau) = \int_0^T a(p_h^\tau, \tilde{p}_h^\tau) + \lambda(p_h^\tau, \tilde{p}_h^\tau)_H \, dt, \quad \forall \tilde{p}_h^\tau \in V_h^\tau,$$

that is $z_h^\tau \in W_h^\tau$ is the unique solution to the problem (4.37) with $f_h^\tau = A p_h^\tau + \lambda p_h^\tau \in \mathbb{P}_0(0,T) \otimes V'$. Note that we have the estimate

$$\|f_h^\tau\|_{L^2(0,T;V')} \leq (\|A\| + \lambda) \|p_h^\tau\|_{L^2(0,T;V)}. \tag{4.45}$$

Then

$$\begin{aligned} \frac{d^\tau(Z_h^\tau, p_h^\tau)}{\|Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \|p_h^\tau\|_{L^2(0,T;V)}} &\geq \frac{\alpha \|p_h^\tau\|^2_{L^2(0,T;V)}}{\|z_h^\tau\|_{W_0(0,T)} \|p_h^\tau\|_{L^2(0,T;V)}} \\ &= \alpha \frac{\|p_h^\tau\|_{L^2(0,T;V)}}{\sqrt{\|z_h^\tau\|^2_{L^2(0,T;V)} + \left\|\frac{dz_h^\tau}{dt}\right\|^2_{L^2(0,T;V')}}}. \end{aligned}$$

In view of Lemma 4.6.1 and (4.45), we get that

$$\frac{d^\tau(Z_h^\tau, p_h^\tau)}{\|Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \|p_h^\tau\|_{L^2(0,T;V)}} \geq \frac{\alpha}{C_{\mathrm{st}}(\|A\| + \lambda)} \frac{\sqrt{\|z_h^\tau\|^2_{L^2(0,T;V_h)} + \left\|\frac{dz_h^\tau}{dt}\right\|^2_{L^2(0,T;V_h')}}}{\sqrt{\|z_h^\tau\|^2_{L^2(0,T;V)} + \left\|\frac{dz_h^\tau}{dt}\right\|^2_{L^2(0,T;V')}}}.$$

Having in mind that $\|\cdot\|_{V_h} = \|\cdot\|_V$ and using the definition of r_h given by (4.44), we finally obtain the result with $\beta = \alpha/(C_{\mathrm{st}}(\|A\| + \lambda))$. □

Proposition 4.6.4. *The bilinear form j^τ satisfies the discrete coercivity property on the kernel of d^τ: for sufficiently small h, there exists some constant $\gamma > 0$ which does not depend on h, τ such that for all $Z_h^\tau \in W_h^\tau \times V_h^\tau$ satisfying*

$$d^\tau(Z_h^\tau, p_h^\tau) = 0, \quad \forall p_h^\tau \in V_h^\tau, \tag{4.46}$$

we have

$$j^\tau(Z_h^\tau, Z_h^\tau) \geq \gamma\, r_h^2 \, \|Z_h^\tau\|^2_{W_0(0,T)\times L^2(0,T;U)}.$$

Proof. Let us consider $Z_h^\tau = (z_h^\tau, \chi_h^\tau) \in W_h^\tau \times V_h^\tau$ satisfying (4.46), which is equivalent to the fact that (z_h^τ, χ_h^τ) satisfies (4.20). By using the estimate (4.36) for $f_h^\tau = B\chi_h^\tau$, we see that

$$\|z_h^\tau\|_{W_h^\tau} \leq C_{\mathrm{st}} \, \|\chi_h^\tau\|_{L^2(0,T;U)}.$$

Hence, we get

$$j^\tau(Z_h^\tau, Z_h^\tau) \geq \varepsilon \int_0^T \|\chi_h^\tau\|_U^2 \, dt = \frac{\varepsilon}{2}\|\chi_h^\tau\|^2_{L^2(0,T;U)} + \frac{\varepsilon}{2}\|\chi_h^\tau\|^2_{L^2(0,T;U)} \geq \frac{\varepsilon}{2C_{\mathrm{st}}^2}\|z_h^\tau\|^2_{W_h^\tau} + \frac{\varepsilon}{2}\|\chi_h^\tau\|^2_{L^2(0,T;U)}.$$

By using again the definition of r_h given by (4.44) and reasoning as in the proof of Proposition 4.6.3, we obtain

$$j^\tau(Z_h^\tau, Z_h^\tau) \geq \frac{\varepsilon}{2C_{\mathrm{st}}^2} r_h^2(\|z_h^\tau\|^2_{W_0(0,T)} + \|\chi_h^\tau\|^2_{L^2(0,T;U)}),$$

which is the result. □

We are now in a position to obtain a Strang-type estimate. If the constants in Proposition 4.6.3 and 4.6.4 were independent of h, this would correspond to Proposition 2.16 in [33]. But since they depend on h through r_h, we have to adapt the proof of Proposition 2.16 in [33] accordingly. It is important to note that, because of our quadrature formula in (4.25), the error estimate require some additional regularity for the exact solution (Z, p).

Proposition 4.6.5. *Let $(Z,p) \in (W_0(0,T) \times L^2(0,T;U)) \times L^2(0,T;V)$ be the solution to problem (4.11) for $y \in L^2(0,T;Y)$. For $y_h^\tau \in Y_h^\tau$, the problem (4.25) has a unique solution $(Z_h^\tau, p_h^\tau) \in (W_h^\tau \times U_h^\tau) \times V_h^\tau$.*

If moreover we assume that $(Z,p) \in (C_0(0,T) \times L^2(0,T;U)) \times C^0([0,T];V)$, we have the error estimates

$$\|Z_h^\tau - Z\|_{W_0(0,T)\times L^2(0,T;U)} \leq \frac{C_{\mathrm{st}}}{r_h^3}(I_h^\tau + C_h^\tau)$$

and

$$\|p_h^\tau - p\|_{L^2(0,T;V)} \leq \frac{C_{\mathrm{st}}}{r_h^4}(I_h^\tau + C_h^\tau),$$

where $C_{\mathrm{st}} > 0$ is a constant which depends neither on h nor on τ, while I_h^τ and C_h^τ are the "interpolation" and "consistency" terms, respectively, which are given by

$$I_h^\tau := \inf_{Z_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - Z_h^\tau\|_{C_0(0,T)\times L^2(0,T;U)} + \inf_{p_h^\tau \in V_h^\tau} \|p - p_h^\tau\|_{L^\infty(0,T;V)} \tag{4.47}$$

and

$$\begin{aligned} C_h^\tau := &\sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|d(\tilde{Z}_h^\tau, p) - d^\tau(\tilde{Z}_h^\tau, p)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}} + \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|j(Z, \tilde{Z}_h^\tau) - j^\tau(Z, \tilde{Z}_h^\tau)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}} \\ &+ \sup_{\tilde{p}_h^\tau \in V_h^\tau} \frac{|d(Z, \tilde{p}_h^\tau) - d^\tau(Z, \tilde{p}_h^\tau)|}{\|\tilde{p}_h^\tau\|_{L^2(0,T;V)}} + \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{|\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)|}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}}. \end{aligned} \tag{4.48}$$

Proof. The well-posedness of problem (4.25) is a consequence of Proposition 2.15 in [33]. We first estimate $\|Z_h^\tau - Z\|_{W_0(0,T)\times L^2(0,T;U)}$. In view of (4.11) and (4.25), we have for all $\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau$,

$$\begin{aligned} &j^\tau(Z - Z_h^\tau, \tilde{Z}_h^\tau) + d^\tau(\tilde{Z}_h^\tau, p - p_h^\tau) \\ &= \left(j^\tau(Z, \tilde{Z}_h^\tau) + d^\tau(\tilde{Z}_h^\tau, p) - \ell(\tilde{Z}_h^\tau)\right) + \left(\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)\right) \\ &= \left(j^\tau(Z, \tilde{Z}_h^\tau) - j(Z, \tilde{Z}_h^\tau)\right) + \left(d^\tau(\tilde{Z}_h^\tau, p) - d(\tilde{Z}_h^\tau, p)\right) + \left(\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)\right). \end{aligned} \tag{4.49}$$

Let us define the operator $D : W_0(0,T) \times L^2(0,T;U) \to L^2(0,T;V)$ such that for all $\tilde{Z} \in W_0(0,T) \times L^2(0,T;U)$, for all $\tilde{p} \in L^2(0,T;V)$,

$$d(\tilde{Z}, \tilde{p}) = (D\tilde{Z}, \tilde{p})_{L^2(0,T;V)}$$

and the operator $D^\tau : W_h^\tau \times U_h^\tau \to V_h^\tau$ such that for all $\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau$, for all $\tilde{p}_h^\tau \in V_h^\tau$,

$$d^\tau(\tilde{Z}_h^\tau, \tilde{p}_h^\tau) = (D^\tau \tilde{Z}_h^\tau, \tilde{p}_h^\tau)_{V_h^\tau}.$$

We now choose $\tilde{Z}_h^\tau = Z_h^\tau - \hat{Z}_h^\tau$, where $\hat{Z}_h^\tau$ is any element of $\mathrm{Ker}\, D^\tau$, so that (4.49) may be rewritten

$$\begin{aligned} j^\tau(\tilde{Z}_h^\tau, \tilde{Z}_h^\tau) &= j^\tau(Z - \hat{Z}_h^\tau, \tilde{Z}_h^\tau) + d^\tau(\tilde{Z}_h^\tau, p - p_h^\tau) \\ &- \big(j^\tau(Z, \tilde{Z}_h^\tau) - j(Z, \tilde{Z}_h^\tau)\big) - \big(d^\tau(\tilde{Z}_h^\tau, p) - d(\tilde{Z}_h^\tau, p)\big) - \big(\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)\big), \end{aligned} \tag{4.50}$$

in which we can replace p_h^τ by any $\hat{p}_h^\tau \in V_h^\tau$ since $d^\tau(\tilde{Z}_h^\tau, p_h^\tau - \hat{p}_h^\tau) = 0$. By using the coercivity of j^τ given by Proposition 4.6.4, as well as the continuity properties given by Propositions 4.6.1 and 4.6.2 (inequality 4.42), we get, for all $\hat{Z}_h^\tau \in W_h^\tau \times U_h^\tau$ and for all $\hat{p}_h^\tau \in V_h^\tau$,

$$\gamma\, r_h^2 \|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \le C_{\mathrm{st}} \|Z - \hat{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} + C_{\mathrm{st}} \|p - \hat{p}_h^\tau\|_{L^\infty(0,T;V)} + C_h^\tau.$$

Since $Z - Z_h^\tau = (Z - \hat{Z}_h^\tau) - \tilde{Z}_h^\tau$, we obtain that

$$\begin{aligned} \|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \le \left(1 + \frac{C_{\mathrm{st}}}{\gamma r_h^2}\right) \|Z - \hat{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \\ + \frac{C_{\mathrm{st}}}{\gamma r_h^2} \|p - \hat{p}_h^\tau\|_{L^\infty(0,T;V)} + \frac{C_h^\tau}{\gamma r_h^2}, \end{aligned}$$

hence

$$\begin{aligned} \|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \le \left(1 + \frac{C_{\mathrm{st}}}{\gamma r_h^2}\right) \inf_{\hat{Z}_h^\tau \in \mathrm{Ker}\, D^\tau} \|Z - \hat{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \\ + \frac{C_{\mathrm{st}}}{\gamma r_h^2} \inf_{\hat{p}_h^\tau \in V_h^\tau} \|p - \hat{p}_h^\tau\|_{L^\infty(0,T;V)} + \frac{C_h^\tau}{\gamma r_h^2}, \end{aligned} \tag{4.51}$$

Now, let us check that there exists a constant C'_{st} such that

$$\begin{aligned} &\inf_{\hat{Z}_h^\tau \in \mathrm{Ker}\, D^\tau} \|Z - \hat{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \\ &\le \left(C'_{\mathrm{st}} + \frac{C_{\mathrm{st}}}{\beta r_h}\right) \inf_{\hat{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - \hat{Z}_h^\tau\|_{C_0(0,T)\times L^2(0,T;U)} + \frac{C_h^\tau}{\beta r_h}. \end{aligned} \tag{4.52}$$

From Lemma 4.1 in [64] (see also Proposition 1.2 in [33]), the inf-sup condition given by Proposition 4.6.3 implies the inverse inf-sup condition

$$\inf_{Z_h^\tau \in (\mathrm{Ker} D^\tau)^\perp} \sup_{p_h^\tau \in V_h^\tau} \frac{d^\tau(Z_h^\tau, p_h^\tau)}{\|Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}\, \|p_h^\tau\|_{L^2(0,T;V)}} \ge \beta\, r_h,$$

which itself implies that for all $\hat{Z}_h^\tau \in (\mathrm{Ker} D^\tau)^\perp$, there exists $\tilde{p}_h^\tau \in V_h^\tau$ such that

$$\beta\, r_h \|\hat{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \|\tilde{p}_h^\tau\|_{L^2(0,T;V)} \le d^\tau(\hat{Z}_h^\tau, \tilde{p}_h^\tau). \tag{4.53}$$

Now for all $\hat{Z}_h^\tau \in W_h^\tau \times U_h^\tau$, let us use the decomposition $\hat{Z}_h = \hat{Z}_h^{\tau,0} + \hat{Z}_h^{\tau,1}$, where $\hat{Z}_h^{\tau,0} \in \mathrm{Ker} D^\tau$ and $\hat{Z}_h^{\tau,1} \in (\mathrm{Ker} D^\tau)^\perp$. Applying (4.53) to $\hat{Z}_h^{\tau,1}$ yields there exists $\tilde{p}_h^\tau \in V_h^\tau$ such that

$$\|\hat{Z}_h^{\tau,1}\|_{W_0(0,T)\times L^2(0,T;U)} \le \frac{1}{\beta r_h} \frac{d^\tau(\hat{Z}_h^{\tau,1}, \tilde{p}_h^\tau)}{\|\tilde{p}_h^\tau\|_{L^2(0,T;V)}} = \frac{1}{\beta r_h} \frac{d^\tau(\hat{Z}_h^\tau, \tilde{p}_h^\tau)}{\|\tilde{p}_h^\tau\|_{L^2(0,T;V)}}.$$

Since

$$d^\tau(\hat{Z}_h^\tau, \tilde{p}_h^\tau) = d^\tau(\hat{Z}_h^\tau - Z, \tilde{p}_h^\tau) + d^\tau(Z, \tilde{p}_h^\tau) - d(Z, \tilde{p}_h^\tau),$$

we get, using Proposition 4.6.2 (inequality (4.43)) that

$$\begin{aligned} &\|\hat{Z}_h^{\tau,1}\|_{W_0(0,T)\times L^2(0,T;U)} \\ &\le \frac{C_{\mathrm{st}}}{\beta r_h} \|Z - \hat{Z}_h^\tau\|_{C_0(0,T)\times L^2(0,T;U)} + \frac{1}{\beta r_h} \sup_{\tilde{p}_h^\tau \in V_h^\tau} \frac{|d(Z, \tilde{p}_h^\tau) - d^\tau(Z, \tilde{p}_h^\tau)|}{\|\tilde{p}_h^\tau\|_{L^2(0,T;V)}} \\ &\le \frac{C_{\mathrm{st}}}{\beta r_h} \|Z - \hat{Z}_h^\tau\|_{C_0(0,T)\times L^2(0,T;U)} + \frac{C_h^\tau}{\beta r_h}. \end{aligned}$$

Then then exists a constant $C'_{\rm st}$ such that

$$\begin{aligned}
&\|Z - \hat{Z}_h^{\tau,0}\|_{W_0(0,T)\times L^2(0,T;U)} \\
&\leq \|Z - \hat{Z}_h^{\tau}\|_{W_0(0,T)\times L^2(0,T;U)} + \|\hat{Z}_h^{\tau,1}\|_{W_0(0,T)\times L^2(0,T;U)} \\
&\leq \left(C'_{\rm st} + \frac{C_{\rm st}}{\beta r_h}\right) \|Z - \hat{Z}_h^{\tau}\|_{C_0(0,T)\times L^2(0,T;U)} + \frac{C_h^\tau}{\beta r_h},
\end{aligned}$$

which implies the estimate (4.52). Gathering (4.51) and (4.52) and optimizing with respect to $\hat{Z}_h^\tau \in W_h^\tau \times U_h^\tau$ and $\hat{p}_h^\tau \in V_h^\tau$, we get

$$\begin{aligned}
\|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \leq \left(1 + \frac{C_{\rm st}}{\gamma r_h^2}\right)\left(C'_{\rm st} + \frac{C_{\rm st}}{\beta r_h}\right) \inf_{\hat{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \|Z - \hat{Z}_h^\tau\|_{C_0(0,T)\times L^2(0,T;U)} \\
+ \frac{C_{\rm st}}{\gamma r_h^2} \inf_{\hat{p}_h^\tau \in V_h^\tau} \|p - \hat{p}_h^\tau\|_{L^\infty(0,T;V)} \\
+ C_h^\tau \left(\left(1 + \frac{C_{\rm st}}{\gamma r_h^2}\right)\frac{1}{\beta r_h} + \frac{1}{\gamma r_h^2}\right),
\end{aligned} \tag{4.54}$$

which is the first estimate of the proposition.

We have now to estimate $\|p_h^\tau - p\|_{L^2(0,T;V)}$. From the inf-sup condition given by Proposition 4.6.3, we get that for all $\hat{p}_h^\tau \in V_h^\tau$,

$$\beta\, r_h \, \|p_h^\tau - \hat{p}_h^\tau\|_{L^2(0,T;V)} \leq \sup_{\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau} \frac{d^\tau(\tilde{Z}_h^\tau, p_h^\tau - \hat{p}_h^\tau)}{\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}}.$$

From (4.49), we get that for all $\hat{p}_h^\tau \in V_h^\tau$,

$$\begin{aligned}
&d^\tau(\tilde{Z}_h^\tau, p_h^\tau - \hat{p}_h^\tau) = j^\tau(Z - Z_h^\tau, \tilde{Z}_h^\tau) + d^\tau(\tilde{Z}_h^\tau, p - \hat{p}_h^\tau) \\
&- \left(j^\tau(Z, \tilde{Z}_h^\tau) - j(Z, \tilde{Z}_h^\tau)\right) - \left(d^\tau(\tilde{Z}_h^\tau, p) - d(\tilde{Z}_h^\tau, p)\right) - \left(\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)\right).
\end{aligned}$$

By using again Propositions 4.6.1 and 4.6.2, we obtain that for all $\hat{p}_h^\tau \in V_h^\tau$,

$$\beta\, r_h \, \|p_h^\tau - \hat{p}_h^\tau\|_{L^2(0,T;V)} \leq C_{\rm st}\, \|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} + C_{\rm st}\, \|p - \hat{p}_h^\tau\|_{L^\infty(0,T;V)} + C_h^\tau,$$

hence there exists a constant $C'_{\rm st}$ such that

$$\begin{aligned}
\|p - p_h^\tau\|_{L^2(0,T;V)} \leq \frac{C_{\rm st}}{\beta r_h}\, \|Z - Z_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)} \\
+ \left(C'_{\rm st} + \frac{C_{\rm st}}{\beta r_h}\right) \|p - \hat{p}_h^\tau\|_{L^\infty(0,T;V)} + \frac{C_h^\tau}{\beta r_h}.
\end{aligned} \tag{4.55}$$

By plugging the estimate (4.54) in (4.55), we obtain the result. □

In the following proposition, we estimate the"consistency" error C_h^τ with the help of additional regularity assumptions.

Proposition 4.6.6. *Assuming that $z, p \in W^{1,\infty}(0,T;V)$, and $y \in W^{1,\infty}(0,T;Y)$ satisfies the estimate*

$$\left\| \frac{y((n-1)\tau) + y(n\tau)}{2} - y_h^n \right\|_Y \leq C_{\rm st}\, h^k, \quad \forall n = 1, \cdots, N \tag{4.56}$$

for some constant $C_{\rm st}$ which depends neither on h nor on τ and some $k \in \mathbb{N} \setminus \{0\}$, then there exists another such constant $C_{\rm st}$ such that

$$\|C_h^\tau\| \leq C_{\rm st}(\tau + h^k).$$

Proof. From the regularity assumptions on z and p, we have

$$\|z(t) - z(t')\|_V \leq \|z\|_{W^{1,\infty}(0,T;V)}|t - t'|, \quad \forall t, t' \in [0,T] \tag{4.57}$$

and

$$\|p(t) - p(t')\|_V \leq \|p\|_{W^{1,\infty}(0,T;V)}|t - t'|, \quad \forall t, t' \in [0,T]. \tag{4.58}$$

In order to estimate the first term in (4.48), for all $\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau$, in view of (4.13) and (4.27),

$$\begin{aligned} d(\tilde{Z}_h^\tau, p) - d^\tau(\tilde{Z}_h^\tau, p) &= \int_0^T a(\tilde{z}_h^\tau, p)\,dt - \boxed{\int_0^T} a(\tilde{z}_h^\tau, p)\,dt \\ &= \int_0^T \langle A^* p, \tilde{z}_h^\tau \rangle_{V',V}\,dt - \tau \sum_{n=1}^N \langle A^* p(n\tau), \tilde{z}_h^n \rangle_{V',V}. \end{aligned}$$

Working on the first term above, we have

$$\begin{aligned} \int_0^T \langle A^* p, \tilde{z}_h^\tau \rangle_{V',V}\,dt &= \sum_{n=1}^N \int_{(n-1)\tau}^{n\tau} \langle A^* p, \tilde{z}_h^\tau \rangle_{V',V}\,dt \\ &= \sum_{n=1}^N \tau \int_0^1 \langle A^* p(s\tau + (n-1)\tau), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1} \rangle_{V',V}\,ds \\ &= \sum_{n=1}^N \tau \int_0^1 \langle A^*(p(s\tau + (n-1)\tau) - p(n\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1} \rangle_{V',V}\,ds \\ &+ \sum_{n=1}^N \tau \int_0^1 \langle A^* p(n\tau), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1} \rangle_{V',V}\,ds, \end{aligned}$$

so that

$$\begin{aligned} d(\tilde{Z}_h^\tau, p) - d^\tau(\tilde{Z}_h^\tau, p) &= \sum_{n=1}^N \tau \int_0^1 \langle A^*(p(s\tau + (n-1)\tau) - p(n\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1} \rangle_{V',V}\,ds \\ &- \sum_{n=1}^N \tau \int_0^1 \langle A^* p(n\tau), (1-s)(\tilde{z}_h^n - \tilde{z}_h^{n-1}) \rangle_{V',V}\,ds. \end{aligned}$$

We observe that

$$\begin{aligned} \sum_{n=1}^N \tau \int_0^1 \langle A^* p(n\tau), (1-s)(\tilde{z}_h^n - \tilde{z}_h^{n-1}) \rangle_{V',V}\,ds &= \sum_{n=1}^N \tau \left\langle A^* p(n\tau), \frac{\tilde{z}_h^n - \tilde{z}_h^{n-1}}{2} \right\rangle_{V',V} \\ &= \sum_{n=1}^{N-1} \tau \left\langle A^* \frac{p(n\tau) - p((n+1)\tau)}{2}, \tilde{z}_h^n \right\rangle_{V',V}, \end{aligned}$$

hence

$$\begin{aligned} d(\tilde{Z}_h^\tau, p) - d^\tau(\tilde{Z}_h^\tau, p) &= \sum_{n=1}^N \tau \int_0^1 \langle A^*(p(s\tau + (n-1)\tau) - p(n\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1} \rangle_{V',V}\,ds \\ &- \sum_{n=1}^{N-1} \tau \left\langle A^* \frac{p(n\tau) - p((n+1)\tau)}{2}, \tilde{z}_h^n \right\rangle_{V',V}. \end{aligned}$$

Using (4.58), the above identity yields

$$
\begin{aligned}
|d(\tilde{Z}_h^\tau, p) - d^\tau(\tilde{Z}_h^\tau, p)| &\le \sum_{n=1}^{N} \frac{1}{2}\tau^2 \|A^*\| \|p\|_{W^{1,\infty}(0,T;V)} (\|\tilde{z}_h^n\|_V + \|\tilde{z}_h^{n-1}\|_V) \\
&+ \sum_{n=1}^{N-1} \frac{1}{2}\tau^2 \|A^*\| \|p\|_{W^{1,\infty}(0,T;V)} \|\tilde{z}_h^n\|_V \\
&\le \sum_{n=1}^{N} \frac{3}{2}\tau^2 \|A^*\| \|p\|_{W^{1,\infty}(0,T;V)} \|\tilde{z}_h^n\|_V \\
&\le \frac{3}{2}\tau \|A^*\| \|p\|_{W^{1,\infty}(0,T;V)} \sqrt{T} \sqrt{\sum_{n=1}^{N} \tau \|\tilde{z}_h^n\|_V^2} \\
&\le C_{\mathrm{st}} \tau \, \|\tilde{z}_h^\tau\|_{L^2(0,T;V)} \le C_{\mathrm{st}} \tau \, \|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)},
\end{aligned}
\tag{4.59}
$$

where $C_{\mathrm{st}} = (3\sqrt{6}/2)\|A^*\|\|p\|_{W^{1,\infty}(0,T;V)}\sqrt{T}$.

Now let us estimate the second term in (4.48). In view of (4.12) and (4.26), for all $\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau$,

$$
\begin{aligned}
j(Z, \tilde{Z}_h^\tau) - j^\tau(Z, \tilde{Z}_h^\tau) &= \int_0^T (Cz, C\tilde{z}_h^\tau)_Y \, dt - \int_0^T (Cz, C\tilde{z}_h^\tau)_Y \, dt \\
&= \int_0^T (C^*Cz, \tilde{z}_h^\tau)_H \, dt - \tau \sum_{n=1}^{N} (C^*Cz(n\tau), \tilde{z}_h^n)_H.
\end{aligned}
$$

The first term above is rewritten

$$
\begin{aligned}
\int_0^T (C^*Cz, \tilde{z}_h^\tau)_H \, dt &= \sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} (C^*Cz, \tilde{z}_h^\tau)_H \, dt \\
&= \sum_{n=1}^{N} \tau \int_0^1 \left(C^*Cz(s\tau + (n-1)\tau), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds \\
&= \sum_{n=1}^{N} \tau \int_0^1 \left(C^*Cz(s\tau + (n-1)\tau) - z((n-1)\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds \\
&+ \sum_{n=1}^{N} \tau \int_0^1 \left(C^*Cz((n-1)\tau), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds.
\end{aligned}
$$

We observe that

$$
\begin{aligned}
&\sum_{n=1}^{N} \tau \int_0^1 \left(C^*Cz((n-1)\tau), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds \\
&= \sum_{n=1}^{N} \tau \left(C^*Cz((n-1)\tau), \frac{\tilde{z}_h^n + \tilde{z}_h^{n-1}}{2}\right)_H \\
&= \sum_{n=1}^{N-1} \tau \left(C^*C\frac{z((n-1)\tau) + z(n\tau)}{2}, \tilde{z}_h^n\right)_H + \tau \left(C^*C\frac{z((N-1)\tau)}{2}, \tilde{z}_h^N\right)_H.
\end{aligned}
$$

Then

$$\begin{aligned} j(Z,\tilde{Z}_h^\tau) - j^\tau(Z,\tilde{Z}_h^\tau) &= \sum_{n=1}^{N} \tau \int_0^1 \left(C^*C(z(s\tau+(n-1)\tau) - z((n-1)\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds \\ &+ \sum_{n=1}^{N-1} \tau \left(C^*C\frac{z((n-1)\tau)+z(n\tau)}{2} - C^*Cz(n\tau), \tilde{z}_h^n\right)_H \\ &+ \tau \left(C^*C\frac{z((N-1)\tau)}{2} - C^*Cz(N\tau), \tilde{z}_h^N\right)_H \\ &= \sum_{n=1}^{N} \tau \int_0^1 \left(C^*C(z(s\tau+(n-1)\tau) - z((n-1)\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds \\ &+ \sum_{n=1}^{N} \tau \left(C^*C\frac{z((n-1)\tau)-z(n\tau)}{2}, \tilde{z}_h^n\right)_H - \tau \left(C^*C\frac{z(N\tau)}{2}, \tilde{z}_h^N\right)_H. \end{aligned}$$

Using (4.57), we obtain

$$\begin{aligned} |j(Z,\tilde{Z}_h^\tau) - j^\tau(Z,\tilde{Z}_h^\tau)| &\le \tau^2\|C\|^2\|z\|_{W^{1,\infty}(0,T;V)} \sum_{n=1}^{N} \frac{\|\tilde{z}_h^n\|_H + \|\tilde{z}_h^{n-1}\|_H}{2} \\ &+ \frac{1}{2}\tau^2\|C\|^2\|z\|_{W^{1,\infty}(0,T;V)} \sum_{n=1}^{N} \|\tilde{z}_h^n\|_H + \frac{1}{2}\tau\|C\|^2\|z\|_{W^{1,\infty}(0,T;V)}\|\tilde{z}_h^\tau(T)\|_H. \end{aligned}$$

Completing the computation as in (4.59) and having in mind the continuous embeddings $V \hookrightarrow H$ and $W(0,T) \hookrightarrow C^0([0,T];H)$, we obtain that there exists a constant $C_{\rm st} > 0$, which is independent of h and τ, such that

$$|j(Z,\tilde{Z}_h^\tau) - j^\tau(Z,\tilde{Z}_h^\tau)| \le C_{\rm st}\tau\,\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)}.$$

Let us estimate the third term in (4.48). For all $\tilde{p}_h^\tau \in V_h^\tau$,

$$\begin{aligned} d(Z,\tilde{p}_h^\tau) - d^\tau(Z,\tilde{p}_h^\tau) &= \int_0^T a(z,\tilde{p}_h^\tau)\,dt - \tau\sum_{n=1}^{N} a(z,\tilde{p}_h^\tau)(n\tau_-) \\ &= \sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} a(z,\tilde{p}_h^\tau)(t) - a(z,\tilde{p}_h^\tau)(n\tau_-)\,dt \\ &= -\sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} \left(\int_t^{n\tau} \frac{d}{dt}a(z,\tilde{p}_h^\tau)\,ds\right) dt \\ &= -\sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} \left(\int_t^{n\tau} a\left(\frac{dz}{dt},\tilde{p}_h^\tau\right) ds\right) dt, \end{aligned}$$

where we have used the fact that $\tilde{p}_h^\tau$ is a constant on each interval $((n-1)\tau, n\tau)$. Hence

$$\begin{aligned} |d(Z,\tilde{p}_h^\tau) - d^\tau(Z,\tilde{p}_h^\tau)| &\le \sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} \left(\int_{(n-1)\tau}^{n\tau} \|A\| \left\|\frac{dz}{dt}\right\|_V \|\tilde{p}_h^\tau\|_V\,ds\right) dt \\ &\le \tau\|A\| \sum_{n=1}^{N} \int_{(n-1)\tau}^{n\tau} \left\|\frac{dz}{dt}\right\|_V \|\tilde{p}_h^\tau\|_V\,ds \le \tau\|A\| \left\|\frac{dz}{dt}\right\|_{L^2(0,T;V)} \|\tilde{p}_h^\tau\|_{L^2(0,T;V)}. \end{aligned}$$

Using that

$$\left\|\frac{dz}{dt}\right\|_{L^2(0,T;V)} \le \sqrt{T}\|z\|_{W^{1,\infty}(0,T;V)},$$

we finally get

$$|d(Z,\tilde{p}_h^\tau) - d^\tau(Z,\tilde{p}_h^\tau)| \le C_{\rm st}\tau\,\|\tilde{p}_h^\tau\|_{L^2(0,T;V)},$$

with $C_{\rm st} = \|A\|\sqrt{T}\|z\|_{W^{1,\infty}(0,T;V)}$.

It remains to estimate the fourth term in (4.48). Given (4.14) and (4.28), we have for all $\tilde{Z}_h^\tau \in W_h^\tau \times U_h^\tau$,

$$\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau) = \int_0^T (y, C\tilde{z}_h^\tau)_Y\,dt - \tau\sum_{n=1}^N (y_h^\tau, C\tilde{z}_h^\tau)_{Y_h}(n\tau_-).$$

By reproducing the computations done above to estimate the second term in (4.48), we get

$$\begin{aligned}\ell(\tilde{Z}_h^\tau) = &\sum_{n=1}^N \tau\int_0^1 \left(C^*(y(s\tau+(n-1)\tau) - y((n-1)\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds\\ &+ \sum_{n=1}^{N-1}\tau\left(C^*\frac{y((n-1)\tau)+y(n\tau)}{2}, \tilde{z}_h^n\right)_H + \tau\left(C^*\frac{y((N-1)\tau)}{2}, \tilde{z}_h^N\right)_H.\end{aligned}$$

Hence

$$\begin{aligned}\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau) = &\sum_{n=1}^N \tau\int_0^1 \left(C^*(y(s\tau+(n-1)\tau) - y((n-1)\tau)), s\tilde{z}_h^n + (1-s)\tilde{z}_h^{n-1}\right)_H ds\\ &+ \sum_{n=1}^N \tau\left(C^*\left(\frac{y((n-1)\tau)+y(n\tau)}{2} - y_h^n\right), \tilde{z}_h^n\right)_H - \tau\left(C^*\frac{y(N\tau)}{2}, \tilde{z}_h^N\right)_H.\end{aligned}$$

We hence obtain, using the assumption (4.56),

$$\begin{aligned}|\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)| \le\ &\tau^2\|C\|\|y\|_{W^{1,\infty}(0,T;Y)}\sum_{n=1}^N \frac{\|\tilde{z}_h^n\|_H + \|\tilde{z}_h^{n-1}\|_H}{2}\\ &+ C_{\rm st}\tau\,h^k\|C\|\sum_{n=1}^N\|\tilde{z}_h^n\|_H + \frac{1}{2}\tau\|C\|\|y\|_{W^{1,\infty}(0,T;Y)}\|\tilde{z}_h^\tau\|_{W_0(0,T)}.\end{aligned}$$

We conclude that

$$|\ell(\tilde{Z}_h^\tau) - \ell^\tau(\tilde{Z}_h^\tau)| \le C_{\rm st}\,(\tau + h^k)\|\tilde{Z}_h^\tau\|_{W_0(0,T)\times L^2(0,T;U)},$$

for some constant $C_{\rm st} > 0$ which depends neither on h nor on τ. This completes the proof. □

Gathering Propositions 4.6.5 and 4.6.6 yields the following error estimate between the continuous and the discretized solution.

Proposition 4.6.7. *Assume that $z, p \in W^{1,\infty}(0,T;V)$, and $y \in W^{1,\infty}(0,T;Y)$ satisfies the estimate (4.56). Assume in addition that there exists a constant $C_{\rm st}$ which depends neither on h nor on τ and some $k \in \mathbb{N}\setminus\{0\}$ such that*

$$|I_h^\tau| \le C_{\rm st}(\tau + h^k), \tag{4.60}$$

where I_h^τ is defined by (4.47). Then there exists some constant $C_{\rm st} > 0$ such that

$$\|Z_h^\tau - Z\|_{W_0(0,T)\times L^2(0,T;U)} \le \frac{C_{\rm st}}{r_h^3}(\tau + h^k),$$

and

$$\|p_h^\tau - p\|_{L^2(0,T;V)} \le \frac{C_{\rm st}}{r_h^4}(\tau + h^k).$$

where r_h is the quantity defined by (4.44).

In order to specify the above estimates in the case of the tsunami problem, in particular to quantity r_h, we have to describe the space discretization, which is the subject of the section hereafter.

4.7 Space discretization of the tsunami problem

In this section, we specify the space discretization of the continuous tsunami problem presented in Section 4.4, that is we have to specify the discretized spaces U_h, V_h and Y_h. Let us consider a family of triangular meshes $\mathcal{T}_h$ of the domain Ω_R of size h which is regular in the sense of [40]. Let us consider $k \in \mathbb{N} \setminus \{0\}$. We choose $U_h = \mathbb{P}_k(\Gamma^R_{-H})$, where $\mathbb{P}_k(\Gamma^R_{-H})$ is the subspace of continuous functions on the closure of Γ^R_{-H}, the restriction of which on each interval of the subdivision induced on Γ^R_{-H} by the mesh is a polynomial of degree k. Similarly, we set $Y_h = \mathbb{P}_k(\Gamma^R_0)$. We choose $V_h = V \cap (\mathbb{P}_k(\Omega^R) \times \mathbb{P}_k(\Omega^R) \times \mathbb{P}_k(\Gamma^R_0))$, where $\mathbb{P}_k(\Omega^R)$ is the subspace of continuous functions on $\overline{\Omega_R}$ which are polynomials of degree k on each triangle of the mesh. Obviously, we have $U_h \subset U$, $V_h \subset V$, $Y_h \subset Y$ and $CV_h \subset Y_h$.

We now introduce the following quasi-uniformity assumption of the mesh.

Assumption 4.7.1. There exists a constant $c_{\rm st} > 0$ such that for each triangle $K \in \mathcal{T}_h$, the diameter h_K of K satisfies

$$h_K \geq c_{\rm st}\, h.$$

We recall that from the definition of the size h of $\mathcal{T}_h$, we have $h_K \leq h$, for all $K \in \mathcal{T}_h$.

Let us prove the following important result.

Proposition 4.7.1. *With Assumption 4.7.1, there exists a constant $c_{\rm st} > 0$ such that*

$$r_h \geq c_{\rm st}\, \sqrt{h},$$

where r_h is given by (4.44).

Note that the statement of Proposition 4.7.1 is equivalent to the existence of a constant $C_{\rm st} > 0$ such that $\|P_h\|_V \leq C_{\rm st}/\sqrt{h}$. The main difficulty here relies on the fact that the space V is characterized by a constraint relating the second and third components of an element z in V. For $z = (\varphi, \psi, s) \in V$, let us denote $(P^\varphi_h z, P^\psi_h z, P^s_h z) \in \mathbb{P}_k(\Omega^R) \times \mathbb{P}_k(\Omega^R) \times \mathbb{P}_k(\Gamma^R_0)$ the three components of $P_h z \in V_h$. In order to prove Proposition 4.7.1 we need the following lemmas.

Lemma 4.7.1. *For all $z = (\varphi, \psi, s) \in V$, we have*

$$\|P^\varphi_h z\|_{H^1(\Omega^R)} \leq \|\varphi\|_{H^1(\Omega^R)},$$

$$\|P^\psi_h z\|_{L^2(\Omega^R)} \leq \|\psi\|_{L^2(\Omega^R)} + \frac{c}{\sqrt{g}} \|\psi\|_{L^2(\Gamma^R_0)}.$$

Proof. From the definition of the operator P_h, for $z \in V$, $P_h z$ is defined by $P_h z \in V_h$ and

$$(P_h z, \tilde{z}_h)_H = (z, \tilde{z}_h)_H, \quad \forall \tilde{z}_h \in V_h. \tag{4.61}$$

Choosing firstly $\tilde{z_h} = (\tilde{\varphi}_h, 0, 0)$ with $\tilde{\varphi}_h \in \mathbb{P}_k(\Omega^R)$, we obtain that $P^\varphi_h z \in \mathbb{P}_k(\Omega^R)$ and

$$(P^\varphi_h z, \tilde{\varphi}_h)_{H^1(\Omega^R)} = (\varphi, \tilde{\varphi}_h)_{H^1(\Omega^R)}, \quad \forall \tilde{\varphi}_h \in \mathbb{P}_k(\Omega^R).$$

In other words, $P^\varphi_h z$ is the orthogonal projection of $\varphi \in H^1(\Omega^R)$ onto $\mathbb{P}_k(\Omega^R)$, which implies the first inequality.

Choosing secondly $\tilde{z_h} = (0, \tilde{\psi}_h, -\tilde{\psi}_h|_{\Gamma^R_0}/g)$ with $\tilde{\psi}_h \in \mathbb{P}_k(\Omega^R)$, we obtain from (4.61) that for all $\tilde{\psi}_h \in \mathbb{P}_k(\Omega^R)$,

$$\frac{1}{c^2}(P^\psi_h z, \tilde{\psi}_h)_{L^2(\Omega_R)} + \frac{1}{g}(P^\psi_h z|_{\Gamma^H_0}, \tilde{\psi}_h|_{\Gamma^H_0})_{L^2(\Gamma^R_0)} = \frac{1}{c^2}(\psi, \tilde{\psi}_h)_{L^2(\Omega_R)} + \frac{1}{g}(\psi|_{\Gamma^H_0}, \tilde{\psi}_h|_{\Gamma^H_0})_{L^2(\Gamma^R_0)}.$$

Choosing now $\tilde{\psi}_h = P^\psi_h z$ and using the Cauchy-Schwarz inequality, yields

$$\frac{1}{c^2}\|P^\psi_h z\|^2_{L^2(\Omega_R)} + \frac{1}{g}\|P^\psi_h z|_{\Gamma^H_0}\|^2_{L^2(\Gamma^R_0)} \leq \frac{1}{c^2}\|\psi\|^2_{L^2(\Omega_R)} + \frac{1}{g}\|\psi|_{\Gamma^H_0}\|^2_{L^2(\Gamma^R_0)},$$

which implies the second inequality. □

The following lemma is proved in [55].

Lemma 4.7.2. *There exists a constant $C_{\rm st}$ such that for any triangle $K \in \mathcal{T}_h$ and any face $F \subset \partial K$, for all function $\psi \in H^1(K)$,*

$$\|\psi\|^2_{L^2(F)} \leq C_{\rm st}\left(h_K^{-1}\|\psi\|^2_{L^2(K)} + h_K\|\nabla\psi\|^2_{L^2(K)}\right).$$

We are now in a position to prove Proposition 4.7.1.

Proof of Proposition 4.7.1. We have to estimate

$$\sup_{z\in V}\frac{\|P_h z\|_V}{\|z\|_V}, \quad \text{with} \quad \|P_h z\|^2_V = \|P_h^\varphi z\|^2_{H^1(\Omega^R)} + \|P_h^\psi z\|^2_{H^1(\Omega^R)} + \|P_h^s z\|^2_{L^2(\Gamma_0^R)},$$

as a function of h. From the first inequality of Lemma 4.7.1, we have

$$\|P_h^\varphi z\|_{H^1(\Omega^R)} \leq \|z\|_V.$$

In order to estimate $\|P_h^\psi z\|_{H^1(\Omega^R)}$, we introduce the interpolate $I_h z$ of z in V_h, and $I_h^\psi z$ its component along ψ. We have

$$\|P_h^\psi z\|_{H^1(\Omega^R)} \leq \|P_h^\psi z - I_h^\psi z\|_{H^1(\Omega^R)} + \|I_h^\psi z\|_{H^1(\Omega^R)} = \|P_h^\psi(z - I_h^\psi z)\|_{H^1(\Omega^R)} + \|I_h^\psi z\|_{H^1(\Omega^R)}.$$

By using the inverse inequality of Lemma 1.138 in [58] (in particular the Example 1.139) in view of Assumption 4.7.1, there exists a constant $C_{\rm st} > 0$ such that

$$\|P_h^\psi(z - I_h^\psi z)\|_{H^1(\Omega^R)} \leq \frac{C_{\rm st}}{h}\|P_h^\psi(z - I_h^\psi z)\|_{L^2(\Omega^R)}.$$

From the second inequality of Lemma 4.7.1, we have

$$\|P_h^\psi(z - I_h^\psi z)\|_{H^1(\Omega^R)} \leq \frac{C_{\rm st}}{h}\left(\|\psi - I_h^\psi z\|_{L^2(\Omega^R)} + \frac{c}{\sqrt{g}}\|\psi - I_h^\psi z\|_{L^2(\Gamma_0^R)}\right)$$

From the estimates of local interpolation given by Theorem 1.103 in [58], there exists a constant $C_{\rm st} > 0$ such that

$$\|\psi - I_h^\psi z\|_{L^2(\Omega^R)} \leq C_{\rm st}\, h\|\psi\|_{H^1(\Omega^R)}, \quad \|\psi - I_h^\psi z\|_{H^1(\Omega^R)} \leq C_{\rm st}\,\|\psi\|_{H^1(\Omega^R)}.$$

From Lemma 4.7.2 and using Assumption 4.7.1, we get

$$\|\psi - I_h^\psi z\|_{L^2(\Gamma_0^R)} \leq C_{\rm st}\left(h^{-1/2}\|\psi - I_h^\psi z\|_{L^2(\Omega^R)} + h^{1/2}\|\psi - I_h^\psi z\|_{H^1(\Omega^R)}\right),$$

hence

$$\|\psi - I_h^\psi z\|_{L^2(\Gamma_0^R)} \leq C_{\rm st}\,\sqrt{h}\|\psi\|_{H^1(\Omega^R)}.$$

Gathering the above estimates yields

$$\|P_h^\psi z\|_{H^1(\Omega^R)} \leq \frac{C_{\rm st}}{\sqrt{h}}\|z\|_V.$$

Finally, in view of $P_h^s z = -P_h^\psi z|_{\Gamma_0^R}/g$, we have from the trace inequality,

$$\|P_h^s z\|_{L^2(\Gamma_0^R)} \leq \frac{C_{\rm st}}{\sqrt{h}}\|z\|_V.$$

Gathering the above estimates of the three components $P_h^\varphi z$, $P_h^\psi z$ and $P_h^s z$, we conclude that $\|P_h\|_V \leq C_{\rm st}/\sqrt{h}$, which is the result. □

It remains to check that the assumptions (4.56) and (4.60) are satisfied with the help of appropriate regularity assumptions.

Lemma 4.7.3. *Assume that $y \in W^{1,\infty}(0,T;H^{k+1}(\Gamma_0^R))$ for $k \in \mathbb{N}\setminus\{0\}$ and let us define y_h^n, for $n = 1, \cdots, N$, as the interpolate of the function $(y((n-1)\tau)+y(n\tau))/2$ in $\mathbb{P}_k(\Gamma_0^R)$. Then (4.56) is satisfied.*

Proof. Since y_h^n is the interpolate in $\mathbb{P}_k(\Gamma_0^R)$ of the function $(y((n-1)\tau)+y(n\tau))/2$, by using Theorem 1.103 in [58], there exists a constant $C_{\rm st}>0$ such that for all $n=1,\cdots,N$ and all τ,

$$\left\| \frac{y((n-1)\tau)+y(n\tau)}{2} - y_h^n \right\|_{L^2(\Gamma_0^R)} \leq C_{\rm st}\, h^{k+1} \left\| \frac{y((n-1)\tau)+y(n\tau)}{2} \right\|_{H^{k+1}(\Gamma_0^R)}.$$

Since

$$\left\| \frac{y((n-1)\tau)+y(n\tau)}{2} \right\|_{H^{k+1}(\Gamma_0^R)} \leq \|y\|_{C^0([0,T];H^{k+1}(\Gamma_0^R))} \leq C_{\rm st}\, \|y\|_{W^{1,\infty}(0,T;H^{k+1}(\Gamma_0^R))},$$

the inequality (4.56) is satisfied. □

The following lemma is a standard result related to the interpolation error in the tensor products W_h^τ, U_h^τ and V_h^τ (see for example [23]) and is based on the local interpolation inequalities of [58] (see Theorem 1.103, with both $p=2$ and $p=\infty$).

Lemma 4.7.4. *Assume that $z \in H^2(0,T;H^{k+1}(\Omega^R)\times H^{k+1}(\Omega^R)\times H^{k+1}(\Gamma_0^R))$, $p \in W^{1,\infty}(0,T;H^{k+1}(\Omega^R)\times H^{k+1}(\Omega^R)\times H^{k+1}(\Gamma_0^R))$ and $\chi \in H^1(0,T;H^{k+1}(\Gamma_{-H}^R))$ for $k\in\mathbb{N}\setminus\{0\}$, then (4.60) is satisfied.*

In view of Propositions 4.6.7 and 4.7.1, we conclude by the main theorem of this section.

Theorem 4.7.2. *Let us assume that in the case of the tsunami problem, χ is the unique solution to the optimal control problem (4.6), where z is the solution to the problem (4.7), p the corresponding solution to the problem (4.9). Let us assume that χ_h^τ is the unique solution to the problem (4.19), where z_h^τ is the solution to the problem (4.20), p_h^τ the corresponding solution to the problem (4.22). We suppose that Assumption 4.7.1 is true.*

For $k\in\mathbb{N}\setminus\{0\}$ and the regularity assumptions on y,z,p,χ of Lemmas 4.7.3 and 4.7.4 which depends on k, we have the estimates

$$\|z_h^\tau - z\|_{W_0(0,T)} + \|\chi_h^\tau - \chi\|_{L^2(0,T;U)} \leq \frac{C_{\rm st}}{h^{3/2}}(\tau + h^k),$$

and

$$\|p_h^\tau - p\|_{L^2(0,T;V)} \leq \frac{C_{\rm st}}{h^2}(\tau + h^k).$$

4.8 Numerical experiments

All our numerical results were obtained by using the Library XLIFE++ [80]. In our numerical experiments, we first want to check how successful is our identification method. In this view we choose the physical parameters $H=0.15$, $R=0.9$, $T=3$, $g=10$ and $c=1500$. Note that those parameters are consistent with the real experiment described in [42]. This choice implies that H/c is negligible with respect to T, which means in view of Remark 4.2.2 that from the measurements of s on $(0,T)$ we are able to identify χ almost on $(0,T)$. The discretization parameters are $h=0.1$, $k=2$ and $\tau=h^k$, while the parabolic parameter is $\eta=10^{-6}$. In order to obtain artificial data, we solve the forward problem (4.15) to obtain s with bottom velocity

$$\chi_0(x,t) = 0.2\left(x+\frac{7}{8}R\right)\left(x+\frac{3}{8}R\right)\left(x+\frac{1}{2}R\right)1_{[-7R/8,3R/8]}(x)\frac{d\cdot}{dt}\left(\sin(t)e^{-10(t/t_0-1)^2}\right),$$

where 1_I is the indicator function of the segment I and $t_0=(T-H/c)/2$. Such computation is based on the scheme (4.20) with $\eta=0$. The computed free surface perturbation s_h^τ is plotted at

different times in Figure 4.1. From such free surface perturbation s_h^τ, the artificial noisy data $s_h^{\tau,\delta}$ on $\Gamma_0^R \times (0,T)$ is given by

$$s_h^{\tau,\delta} = s_h^\tau + \rho\, b_h^\tau.$$

Here b_h^τ is a function defined from its values at each discretization point of $\Gamma_0^R \times (0,T)$, which are random numbers following a uniform distribution $\mathcal{U}(-1,1)$. We then calibrate the number $\rho > 0$ so that

$$\frac{\|s_h^{\tau,\delta} - s\|_{L^2(\Gamma_0^R \times (0,T))}}{\|s_h^\tau\|_{L^2(\Gamma_0^R \times (0,T))}} = \sigma,$$

where σ is the relative noise amplitude that we impose. From noisy data $s_h^{\tau,\delta}$ we form the data y_h^τ of Problem (4.19)-(4.20), that we solve with a Limited-memory BFGS descent method [12, 91] based on the gradient given by (4.21) and the adjoint state (4.22). More precisely, we use the library NLopt [77]. In the presence of noise ($\sigma \neq 0$), the regularization parameter ε is chosen as a function of δ by using a $L-$curve strategy (see for example [67]), while $\varepsilon = 10^{-6}$ with uncontaminated data ($\sigma = 0$). A comparison between the actual bottom velocity χ and the reconstructed velocity using exact data is shown in Figure 4.2. Figures 4.3 and 4.4 show the same comparison with noisy data, specifically for $\sigma = 1\%$ and $\sigma = 5\%$, respectively. Note that in our computations, all distances are divided by H and are therefore dimensionless. The $L-$curve which enabled us to compute ε from δ corresponding to $\sigma = 0.05$ is given on Figure 4.5.

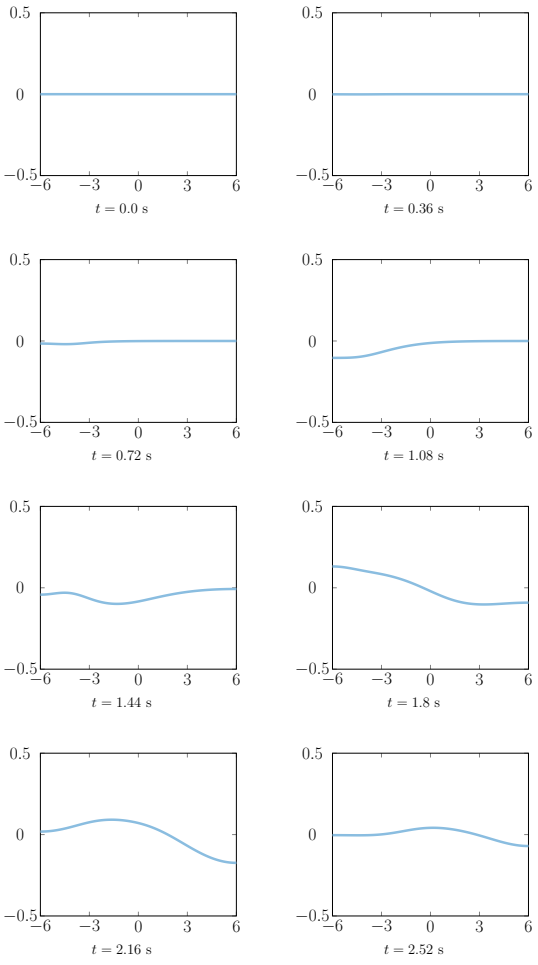


Figure 4.1: Computed free surface s at different times t.

Now we wish to check how consistent is the theoretical error predicted by Theorem 4.7.2. In this view we keep the same parameters as before except the final time $T = 1$ and the simpler bottom

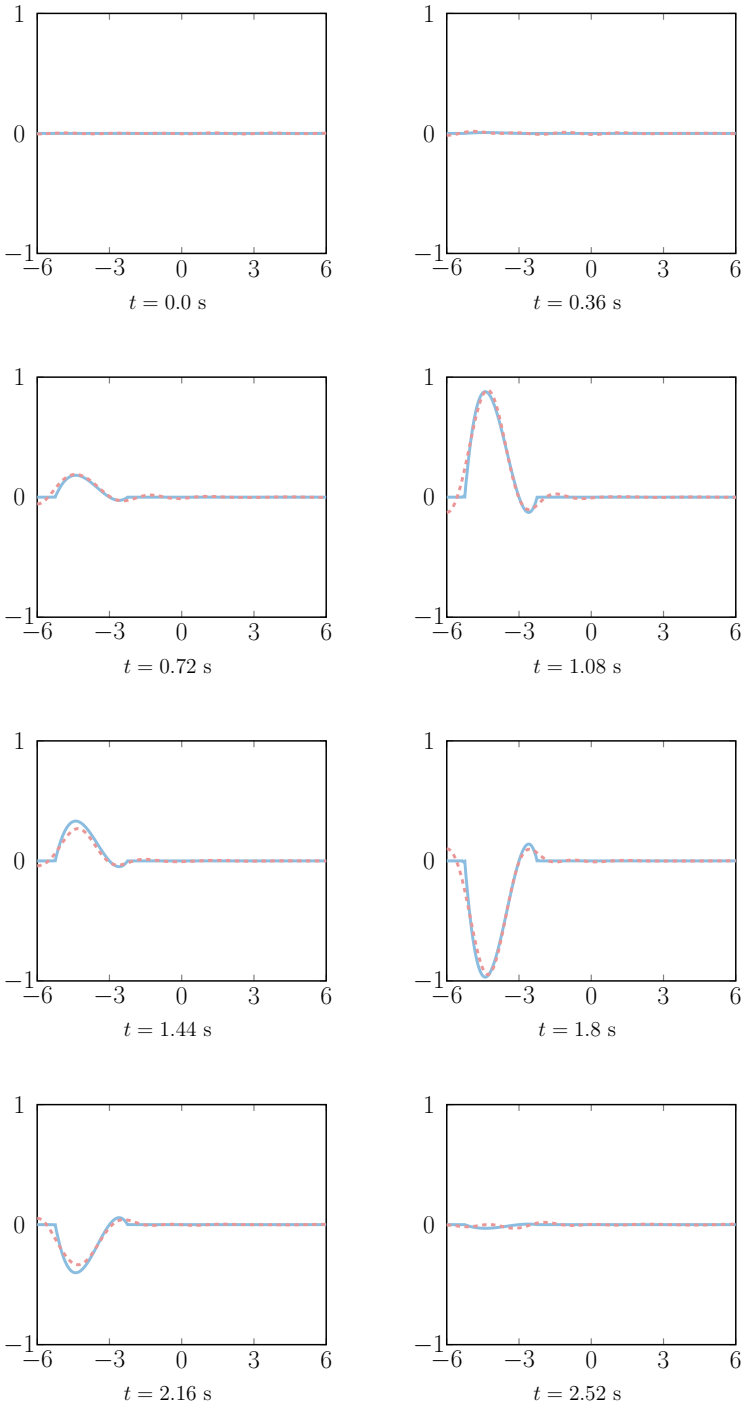


Figure 4.2: Reconstructed χ at different times t for $\sigma = 0$. Continuous line: true χ_0. Dashed line: reconstructed χ.

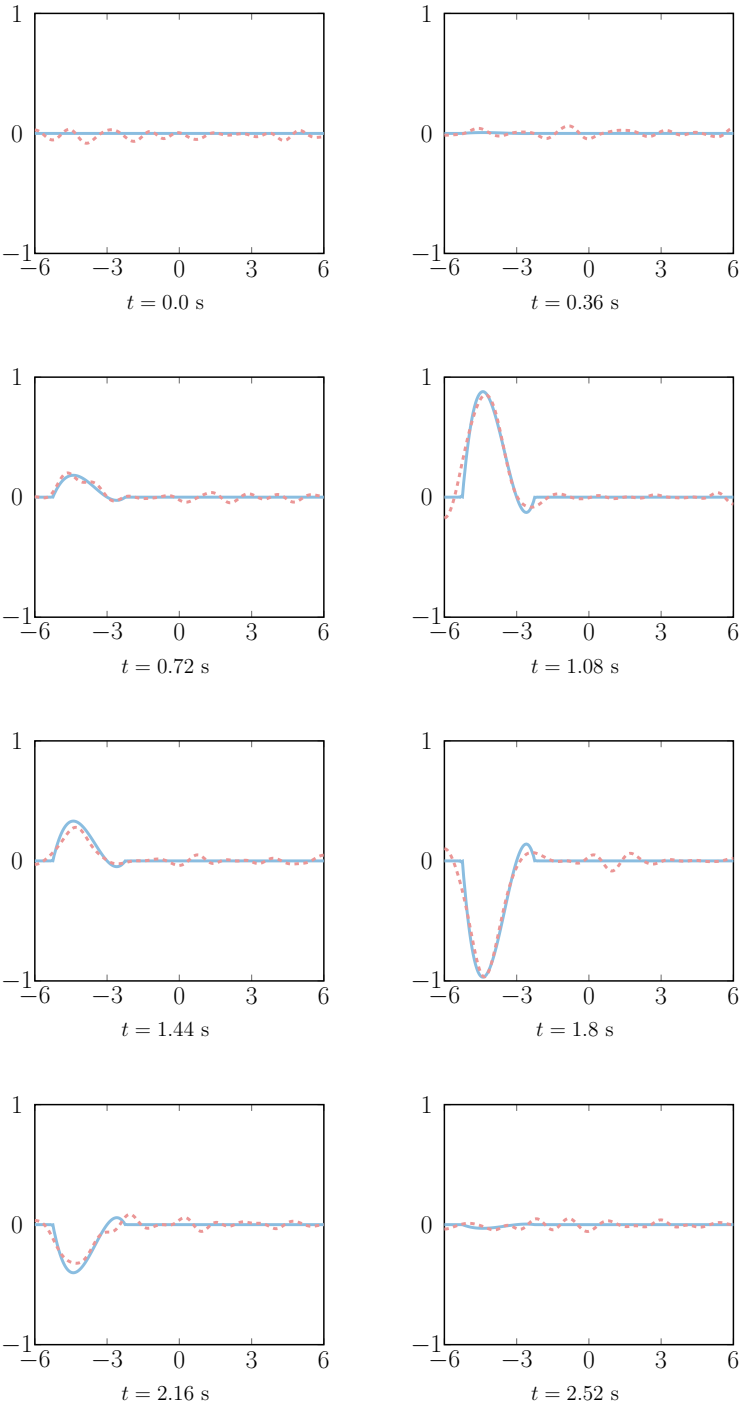


Figure 4.3: Reconstructed χ at different times t for $\sigma = 1\%$. Continuous line: true χ_0. Dashed line: reconstructed χ.

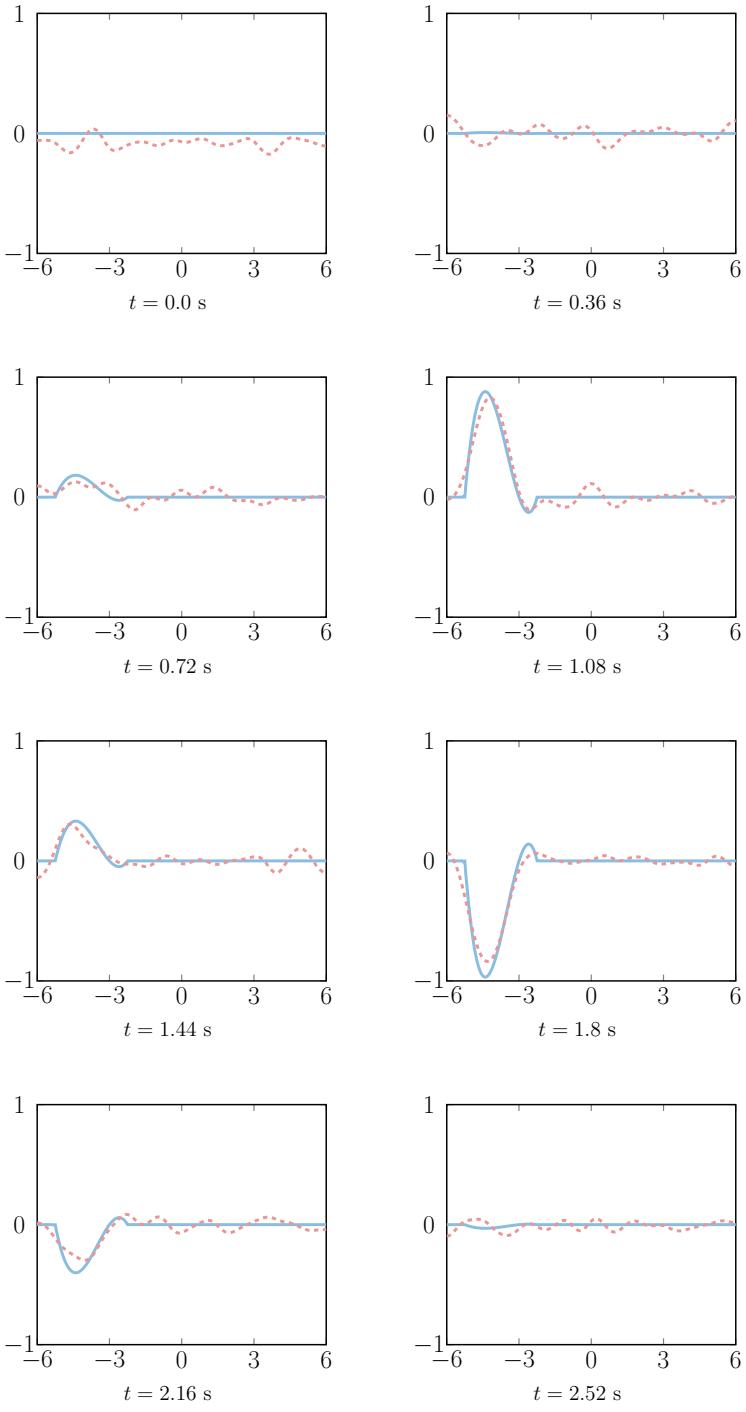


Figure 4.4: Reconstructed χ at different times t for $\sigma = 5\%$. Continuous line: true χ_0. Dashed line: reconstructed χ.

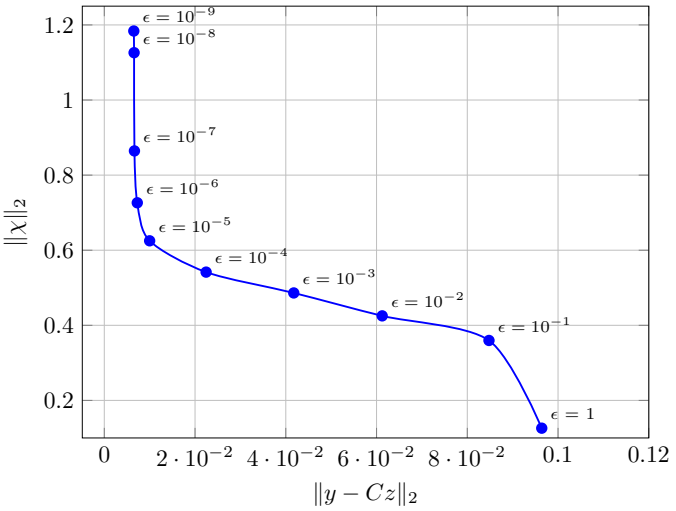


Figure 4.5: The $L-$ curve for $\sigma = 5\%$ noise.

velocity

$$\chi_0(x,t) = 0.25\cos^2\left(\pi\frac{x}{R}\right)1_{[-R/2,R/2]}\frac{d\cdot}{dt}\left(\sin(t)e^{-10(t/t_0-1)^2}\right).$$

In order to compare our numerical results with the error estimate of Theorem 4.7.2, which in particular states that

$$\|\chi_h^\tau - \chi\|_{L^2((0,T)\times\Gamma^R_{-H})} \leq \frac{C_{\text{st}}}{h^{3/2}}(\tau + h^k), \tag{4.62}$$

we naturally work with uncontaminated data. We recall that in the estimate (4.62), χ is the solution of the continuous optimal control problem while χ_h^τ is the solution of the discrete optimal control problem. Such estimate imposes the CFL condition $\tau = h^k$, where τ is the time step and h is the mesh size, which provides a $h^{1/2}$ error when $k = 2$ and a $h^{3/2}$ error when $k = 3$. In Figure 4.6, we have plotted the error $\|\chi_h^\tau - \chi\|_{L^2((0,T)\times\Gamma^R_{-H})}$ for $\tau = h^k$ for $k = 1$, $k = 2$ and $k = 3$, where χ is approximated by χ_h^τ for $\tau = h^k$ and very small $h = 0.07$. Comparing this experimental error and the theoretical error (4.62), we observe that the experimental error is slightly better, with of factor h^0 instead of $h^{-1/2}$ for $k = 1$, h^1 instead of $h^{1/2}$ for $k = 2$ and h^2 instead of $h^{3/2}$ for $k = 3$.

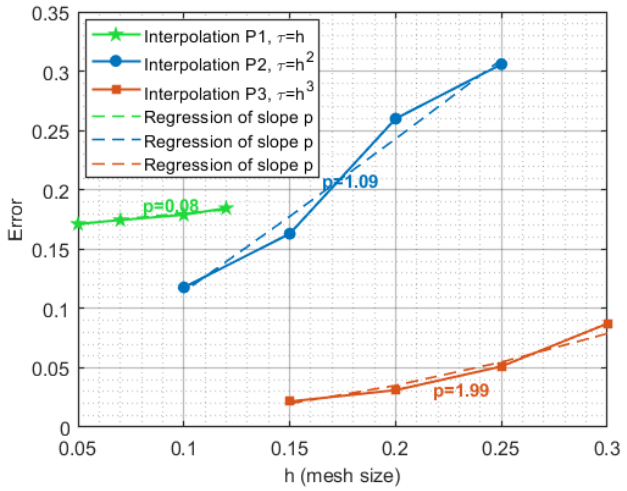


Figure 4.6: Error $\|\chi_h^\tau - \chi\|_{L^2((0,T)\times\Gamma^R_{-H})}$ for $\tau = h^k$ as a function of h in log-scale, for $k = 1$, $k = 2$ and $k = 3$..

Acknowledgement

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Chapter 5

Conclusion and Perspectives

In this final chapter, we present some final results and future axis of work. Firstly, we show in two easier setup the link between the two approaches developed in Chapter 4 and Chapter 2 and 3. Even though, these two approaches seem to differ in their approach and their philosophy, it follows that the two approaches are quite similar. Indeed, up to a penalisation term and a right choice for the optimal control criterion, the two approaches amount to solve a mixed formulation, the analysis is thus quite similar for both of these methods. We will present these similarities in the case of the Poisson and the heat equation, this link being also true for the tsunami we are interested in. Secondly, we will generalize the results obtained in Chapter 4 in the absence of parabolic regularisation. As presented in Chapter 4, the numerical analysis we made relies on the link between the cost functional and a mixed formulation. Thus, this link relies on the existence of an inf-sup condition for the forward problem. If the numerical analysis will not be detailed in this chapter, we will describe the workable framework we should choose to obtain the same results as before.

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5.1 Summary of contributions

Our objective was to develop efficient and rigorous methods to solve the following inverse problem: how can we retrieve the deformations of the ocean bottom from surface measurements? To this end, we first introduced a one-shot approach based on a Tikhonov regularisation of a mixed formulation. This approach allowed us to choose the regularisation parameter rigorously with respect to the amplitude of the noise. In this sense, it has proved robust as the noise increases, both in a harmonic and in a space–time framework. Since the discretisation of the method relies on the Lax-Milgram theorem, the numerical analysis of the approach is also direct and easily generalizable. However, as solving the mixed formulation is based on the inversion of a single linear system after discretisation, this method is hardly scalable with the size of the domain or the refinement of the mesh. Indeed, the FEM matrices we need to invert rapidly become too large for us to consider more realistic setups with better accuracy.

To prevent this numerical inefficiency, we have developed another approach in which the bottom deformation is obtained as the solution of an optimal control problem. To solve this problem, we have applied a descent algorithm such as L-BFGS so that, at each iteration, the matrices to invert are much smaller. At the cost of additional iterations, we have therefore managed to reduce the size of the matrices and thus increase the scalability of the approach. This has allowed us to tackle larger problems with more refined meshes. However, this new approach has only been rigorously justified by adding a parabolic regularisation to the model. Therefore, we did not solve the problem we initially chose but a slight perturbation of it. Furthermore, the regularisation parameters are no longer chosen in a way that is rigorously consistent with the noise.
Let us now present an interesting link between these two approaches that allows to recast each approach into a common framework. Finally, let us present some final perspectives.

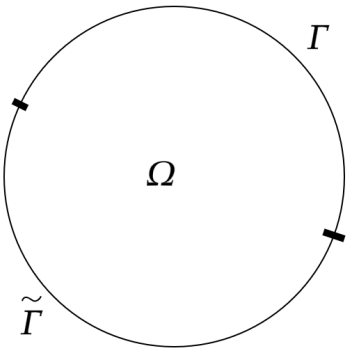
5.2 A link between an optimal control approach and a one shot approach

In this section, we seek to find a link between the two approaches developed in 4 and Chapter 2 and 3 to solve an inverse problem. The first one is a one shot method, where a mixed formulation is solved in one iteration while the other relies on a gradient descent to solve the problem. The goal here is to propose a framework in which the two approaches coincide up to a penalisation. This section unifies in a single framework the two approaches developed in this thesis, so that it is conceivable to benefit from the strength of the two methods in a single approach. This link has been studied in two simpler frameworks, the Poisson and the heat equation, to simplify the redaction. The reasoning made in this section would be equivalent for our problem to which we would have added parabolic regularisation. A future work on extending this link to the problem we studied in this thesis would be interesting as it unifies the two classes of approach we presented in this manuscript. Let us consider the following bounded domain :

The inverse problem we will focus on is now the following: *For f, find u solution of:*

$$\begin{cases} -\Delta u + u = 0 & \text{in } \Omega, \\ \partial_\nu u = 0 & \text{on } \Gamma, \\ u = f & \text{on } \Gamma. \end{cases} \tag{5.1}$$

where ν is the outward unit normal to the domain Ω. In other words, from a surface measurement of u on Γ we call f, how to reconstruct a volumic solution u on Ω. In the following, we will focus on retrieving a surfacic source $g = \partial_\nu u$ on $\tilde{\Gamma}$, where $\tilde{\Gamma}$ corresponds to the complementary of Γ on $\partial\Omega$, but since we did not mention any regularity for the solution u and f, we will introduce g later. This inverse problem can be viewed as an ill-posed laplacian problem with lateral Cauchy condition. A first question is uniqueness: assume that we have a measurement f on Γ, on which subdomain of Ω are we able to retrieve u. We have the following proposition:



Proposition 5.2.1. *Assume that u satisfies (5.1) with $f = 0$ on Γ, then $u = 0$ on Ω. In other words, if the inverse problem has a solution u on Ω, this solution is unique on Ω.*

Proof. The proof is a consequence of the Holmgren Theorem. □

In the two following sections, we present the two approaches developed to solve such an inverse problem.

5.2.1 The one shot approach for the Poisson equation

Let us introduce the following Hilbert space

$$M = \left\{ q \in H^1(\Omega), q = 0 \text{ on } \tilde{\Gamma} \right\},$$

endowed with the H^1 seminorm and the following space

$$\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}) = \left\{ g \in H^{-\frac{1}{2}}(\tilde{\Gamma}), \tilde{g} \in H^{-\frac{1}{2}}(\partial\Omega) \right\},$$

where $\tilde{g}$ is the extension by 0 to the whole surface $\partial\Omega$. Then u is a solution of (5.1) if and only if u is a solution of:
Find $u \in H^1(\Omega)$ such that for all $q \in M$:

$$\left| \begin{array}{l} b(u,q) = 0, \\ u = f \text{ on } \Gamma. \end{array} \right. \tag{5.2}$$

with $b(u,q) = (u,q)_{H^1(\Omega)}$. Let us now introduce two new operators that are going to be useful to recast the previous problem in a more workable framework. The first one is $B : V \to M$ the Riesz lifting of the bilinear form b defined as

$$\forall u \in H^1(\Omega), \forall q \in M, (Bu, q)_M = b(u,q).$$

We then introduce $C : H^1(\Omega) \to L^2(\Gamma)$ the trace operator on Γ. The next proposition introduces the framework in which we are going to study the inverse problem (5.1).

Proposition 5.2.2. *Introducing $F = (0, f) \in H := M \times L^2(\Gamma)$ and $A : u \in H^1(\Omega) \mapsto (Bu, Cu) \in H$, one can show that:*

$$u \text{ solution of (5.2)} \iff Au = F.$$

From now on, the inverse problem (5.1) only amounts to inverse the operator A. Because of the ill-posedness of the problem, we get that A is non invertible but injective thanks to Proposition 5.2.1. The problem being ill-posed, we introduce a regularized problem, that is the following Tikhonov regularisation for a parameter $\epsilon > 0$, the regularization parameter: *Find* $u_\epsilon \in H^1(\Omega)$ *such that:*

$$A^*Au_\epsilon + \epsilon u_\epsilon = A^*F. \tag{5.3}$$

Proposition 5.2.3. *The problem (5.3) is well-posed.*

The problem (5.3) gives us a way to compute a solution u_ϵ that will eventually converges toward a solution of the inverse problem (see Proposition 5.2.5). However, since the operator A is composite, discretizing problem (5.3) is quite tricky. The next proposition of this section gives a system satisfied by u_ϵ that is easy to solve numerically.

Proposition 5.2.4. *The solution u_ϵ of (5.3) is the solution of : Find $(u_\epsilon, p_\epsilon) \in H^1(\Omega) \times M$ such that for all $(v, q) \in H^1(\Omega) \times M$:*

$$\begin{cases} b(v, p_\epsilon) + \epsilon(u_\epsilon, v)_{H^1(\Omega)} + (u_\epsilon, v)_{L^2(\Gamma)} = (f, v)_{L^2(\Gamma)}, \\ b(u_\epsilon, q) - (p_\epsilon, q)_M = 0. \end{cases} \tag{5.4}$$

Finally, we present a result of convergence of the solution (u_ϵ, p_ϵ) of (5.4). This result will allow us to consider u_ϵ as an approximation of u a solution of (5.2) as ϵ is small.

Proposition 5.2.5. *If there exists $u \in H^1(\Omega)$ such that $Au = F$, then $u_\epsilon \to u$ in $H^1(\Omega)$ and $p_\epsilon \to 0$ in $H^1(\Omega)$.*

Proof. Let $u \in H^1(\Omega)$ solution of $Au = F$, then u satisfies $u = f$ on Γ and $b(u, q) = 0, \forall \mu \in M$. From (5.4), we thus get for all $(v, q) \in H^1(\Omega) \times M$ that:

$$\begin{cases} b(v, p_\epsilon) + \epsilon(u_\epsilon, v)_{H^1(\Omega)} + (u_\epsilon - u, v)_{L^2(\Gamma)} = 0, \\ b(u_\epsilon - u, q) - (p_\epsilon, q)_M = 0. \end{cases}$$

Taking $\mu = p_\epsilon$ and $v = u_\epsilon - u$, we get by substracting both lines that:

$$\epsilon(u_\epsilon, u_\epsilon - u)_{H^1(\Omega)} + \|u_\epsilon - u\|^2_{L^2(\Gamma)} + \|p_\epsilon\|^2_M = 0.$$

From this equation, it is easy to show that:

$$\|u_\epsilon\|_{H^1(\Omega)} \leq \|u\|_{H^1(\Omega)}, \quad \|p_\epsilon\|_M \leq \sqrt{\epsilon}\|u\|_{H^1(\Omega)}, \quad \|u_\epsilon - f\|_{L^2(\Gamma)} \leq \sqrt{\epsilon}\|u\|_{H^1(\Omega)}.$$

so that $p_\epsilon \to 0$ in M, $u_\epsilon \to f$ on $L^2(\Gamma)$ and u_ϵ is bounded in $H^1(\Omega)$. This last point means that up to a subsequence there exists $w \in H^1(\Omega)$ such that $u_\epsilon \rightharpoonup w$ in $H^1(\Omega)$. Hence, passing to the limit in the second line of (5.4) gives us :

$$b(w, q) = 0, \forall q \in M.$$

and $w = f$ on Γ by the continuity of the trace operator. Using the injectivity of A, one observes that $w = u$, so that $u_\epsilon \rightharpoonup u$ in $H^1(\Omega)$. Finally, since:

$$\|u_\epsilon - u\|^2_{H^1(\Omega)} \leq -(u, u_\epsilon - u)_{H^1(\Omega)},$$

the weak convergence of u_ϵ implies the strong convergence of u_ϵ towards u in $H^1(\Omega)$. □

This proposition allows us to propose a method to solve the inverse problem (5.2). For ϵ small enough, we solve the problem (5.4). Since the solution obtained u_ϵ is closed to the exact one, the source g satisfies $g \approx \partial_\nu u_\epsilon$ on $\tilde{\Gamma}$, more precisely $\partial_\nu u_\epsilon \to g$ in $H^{-\frac{1}{2}}(\tilde{\Gamma})$ when $\epsilon \to 0$. The problem (5.4) is set to be solved numerically in one iteration through a linear system using finite elements. This approach has the practical advantage of requiring only a single iteration, thereby avoiding the difficulties associated with selecting optimization parameters. However, because the formulation involves two unknowns, the numerical resolution can become very costly as the meshes are refined. Let us now present in the following section an optimal control approach to solve problem (5.1).

5.2.2 The optimal control approach for the Poisson equation

Another idea to solve (5.1) is to retrieve g as the minimizer of a least-square functional, in other words to solve an optimal control problem. The idea is then to compute a gradient of such a functional and compute g thanks to a descent algorithm such as a conjugate gradient or a BFGS algorithm. Classically, the natural least-square functional introduced to solve the problem (5.1) is the following for $\epsilon > 0$:

$$\inf_{g \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})} \mathcal{J}(g) = \frac{1}{2}\|u_g - f\|^2_{L^2(\Gamma)} + \frac{\epsilon}{2}\|g\|^2_{H^{-\frac{1}{2}}(\tilde{\Gamma})} \tag{5.5}$$

with u_g satisfying

$$\begin{cases} -\Delta u_g + u_g = 0 & \text{in } \Omega, \\ \partial_\nu u_g = 0 & \text{on } \Gamma, \\ \partial_\nu u_g = g & \text{on } \tilde{\Gamma}. \end{cases} \tag{5.6}$$

We now propose a different regularisation term such that the mixed formulation (5.4) matches the optimality conditions of the functional. In the following, we will thus study the following least-square functional:

$$\inf_{g \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})} \tilde{\mathcal{J}}(g) = \frac{1}{2}\|u_g - f\|^2_{L^2(\Gamma)} + \frac{\epsilon}{2}\|u_g\|^2_{H^1(\Omega)}. \tag{5.7}$$

where u_g satisfies (5.6) which is equivalent to

$$\forall v \in H^1(\Omega),\ b(u_g, v) = \langle g, v \rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}), H^{\frac{1}{2}}(\tilde{\Gamma})}.$$

Here, to match the mixed formulation (5.4) we have chosen to regularise the functional with the norm of u_g and not g anymore. The functional $\mathcal{J}$ is obviously continuous, strictly convex and coercive, so that the problem (5.5) admits a unique solution. This is less clear for the functional $\tilde{\mathcal{J}}$, so that the problem (5.7) has to be studied.

Proposition 5.2.6. *The problem (5.7) admits a unique solution $g_\epsilon \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$.*

Proof. The functional $\tilde{\mathcal{J}}$ is clearly continuous and strictly convex. It remains to show that $\tilde{\mathcal{J}}$ is coercive on $\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$. We have:

$$\tilde{\mathcal{J}}(g) \geq \frac{\epsilon}{2}\|u_g\|^2_{H^1(\Omega)}.$$

By splitting the norm in two and using the equation (5.6), it follows that:

$$\begin{aligned} \tilde{\mathcal{J}}(g) &\geq \frac{\epsilon}{4}\|u_g\|^2_{H^1(\Omega)} + \frac{\epsilon}{4}\|\Delta u_g\|^2_{L^2(\Omega)} \\ &= \frac{\epsilon}{4}\|u_g\|^2_{H^1(\Omega,\Delta)}, \end{aligned}$$

where $H^1(\Omega, \Delta) = \left\{u \in H^1(\Omega), \Delta u \in L^2(\Omega)\right\}$. Hence, using the continuity of the normal trace from $H^1(\Omega, \Delta)$ to $\tilde{H}^{-\frac{1}{2}}(\partial\Omega)$, we finally get that there exists $C > 0$ such that:

$$\tilde{\mathcal{J}}(g) \geq C\frac{\epsilon}{4}\|\partial_\nu u_g\|^2_{\tilde{H}^{-\frac{1}{2}}(\partial\Omega)},$$

which yields the following inequation since $\partial_\nu u_g = 0$ on Γ

$$\tilde{\mathcal{J}}(g) \geq C\frac{\epsilon}{4}\|g\|^2_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})}.$$

Which concludes the proof. □

Let us now introduce the following mixed formulation, inspired by the one obtained in (5.4). The main difference here being that the regularisation term $(p_\epsilon, q)_M$ has been removed. Find $(u_\epsilon, p_\epsilon) \in H^1(\Omega) \times M$ such that for all $(v, q) \in H^1(\Omega) \times M$,

$$\begin{cases} b(v, p_\epsilon) + \epsilon(u_\epsilon, v)_{H^1(\Omega)} + (u_\epsilon, v)_{L^2(\Gamma)} = (f, v)_{L^2(\Gamma)}, \\ b(u_\epsilon, q) = 0. \end{cases} \tag{5.8}$$

Let us now prove that this mixed formulation is well-posed.

Proposition 5.2.7. *The mixed formulation (5.8) has a unique solution (u_ϵ, p_ϵ) in $H^1(\Omega) \times M$.*

Proof. It suffices to prove that the bilinear form is $j(u, v) = \epsilon(u, v)_{H^1(\Omega)} + (u, v)_{L^2(\Gamma)}$ is coercive on $Ker\ b$ and that the bilinear form b satisfies an inf-sup condition (see [58, Theorem 2.34, Page 100] for example). Concerning the first point, the proof is trivial. We indeed find that for $u \in H^1(\Omega) \supset Ker\ b$ that:

$$j(u, u) \geq \epsilon \|u\|^2_{H^1(\Omega)}.$$

Let us now consider the second point, which is equivalent to prove that there exists some $\beta > 0$ such that for all $p \in M$,

$$\sup_{u \in H^1(\Omega)} \frac{b(u, p)}{\|u\|_{H^1(\Omega)}} \geq \beta \|p\|_M.$$

Since $M \subset H^1(\Omega)$, we can choose $u = p$. The previous inequality is then satisfied with $\beta = 1$. □

The main goal is now to highlight the link between the mixed formulation (5.8) and the optimal control problem (5.7). The next Propositions will set the equivalence between these two problems so that finding the solution g_ϵ to (5.7) amounts to find $g_\epsilon = \partial_\nu u_\epsilon$ where u_ϵ is the solution of (5.8). First, let us compute the gradient of the functional $\tilde{\mathcal{J}}$.

Proposition 5.2.8. *The differential of $\tilde{\mathcal{J}}$ at point $g \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$ is given, for $h \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$ by*

$$\left\langle D_g \tilde{\mathcal{J}}(g), h \right\rangle_{H^{\frac{1}{2}}(\tilde{\Gamma}), \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})} = - \left\langle p_g, h \right\rangle_{H^{\frac{1}{2}}(\tilde{\Gamma}), \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})}. \tag{5.9}$$

where p_g is the solution in $H^1(\Omega)$ to the problem : find $p_g \in H^1(\Omega)$ such that for all $v \in H^1(\Omega)$

$$b(v, p_g) = -\epsilon(u_g, v)_{H^1(\Omega)} - (u_g, v)_{L^2(\Gamma)} + (f, v)_{L^2(\Gamma)} \tag{5.10}$$

and u_g is the solution to the problem (5.6).

Proof. First, using Lax-Milgram theorem, one can show that problem (5.10) is well-posed. To compute, the differential of $\tilde{\mathcal{J}}$, let us introduce the following lagrangian $\tilde{\mathcal{L}}$ defined for $(u, p, g) \in H^1(\Omega) \times H^1(\Omega) \times \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$ by:

$$\tilde{\mathcal{L}}(u, p, g) = \frac{1}{2}\|u|_\Gamma - f\|^2_{L^2(\Gamma)} + \frac{\epsilon}{2}\|u\|^2_{H^1(\Omega)} + b(u, p) - \left\langle g, p \right\rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}), H^{\frac{1}{2}}(\tilde{\Gamma})}.$$

We observe that for $g \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$:

$$\tilde{\mathcal{J}}(g) = \tilde{\mathcal{L}}(u_g, p, g), \quad \forall p \in H^1(\Omega),$$

where u_g is the solution of (5.6). Using the chain rule, it follows that for $h \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$

$$\left\langle D_g \tilde{\mathcal{J}}(g), h \right\rangle_{H^{\frac{1}{2}}(\tilde{\Gamma}), \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})} = \left\langle D_u \tilde{\mathcal{L}}(u_g, p, g), u_h \right\rangle_{(H^1(\Omega))', H^1(\Omega)} + \left\langle D_g \tilde{\mathcal{L}}(u_g, p, g), h \right\rangle_{H^{\frac{1}{2}}(\tilde{\Gamma}), \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})}.$$

On the one hand, we compute

$$\left\langle D_u \tilde{\mathcal{L}}(u_g, p, g), u_h \right\rangle_{(H^1(\Omega))', H^1(\Omega)} = b(u_h, p) + \epsilon(u_g, u_h)_{H^1(\Omega)} + (u_g - f, u_h)_{L^2(\Gamma)}.$$

On the other, we easily get that

$$\left\langle D_g \tilde{\mathcal{L}}(u_g, p, g), h \right\rangle_{H^{\frac{1}{2}}(\tilde{\Gamma}), \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})} = - \left\langle p, h \right\rangle_{H^{\frac{1}{2}}(\tilde{\Gamma}), \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})}.$$

It remains to choose $p = p_g$ where p_g is solution to problem (5.10) which implies that

$$\left\langle D_u \tilde{\mathcal{L}}(u_g, p, g), u_h \right\rangle_{(H^1(\Omega))', H^1(\Omega)} = 0,$$

which finally gives (5.9). □

As a consequence of Proposition 5.2.6 and 5.2.8, the unique solution g_ϵ to the problem (5.7) satisfies :

$$p_{g_\epsilon} = 0 \text{ on } \tilde{\Gamma} \tag{5.11}$$

where p_{g_ϵ} is the solution to the problem (5.10), in which u_{g_ϵ} is the solution to the problem (5.6) associated with g_ϵ. The following Proposition establishes a link between the optimal control problem (5.7) and the mixed formulation (5.8).

Proposition 5.2.9. *Let us consider the solution $g_\epsilon \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$ to the problem (5.7) as well as the corresponding solutions $u_{g_\epsilon} \in H^1(\Omega)$ to problem (5.6) and $p_{g_\epsilon} \in H^1(\Omega)$ to problem (5.10). Then the pair $(u_{g_\epsilon}, p_{g_\epsilon})$ is a solution to problem (5.8).*

Proof. The solution $u_{g_\epsilon} \in H^1(\Omega)$ to problem (5.6) is solution of

$$b(u_{g_\epsilon}, v) = \langle g_\epsilon, v \rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}), H^{\frac{1}{2}}(\tilde{\Gamma})}, \ \forall \mu \in H^1(\Omega),$$

and the solution $p_{g_\epsilon} \in H^1(\Omega)$ to problem (5.10) satisfies

$$b(v, p_{g_\epsilon}) = -\epsilon(u_{g_\epsilon}, v)_{H^1(\Omega)} - (u_{g_\epsilon}, v)_{L^2(\Gamma)} + (f, v)_{L^2(\Gamma)}, \ \forall v \in H^1(\Omega).$$

Finally, the pair $(u_{g_\epsilon}, p_{g_\epsilon})$ is solution of : find $(u_\epsilon, p_\epsilon) \in H^1(\Omega) \times H^1(\Omega)$ such that for all $(v, q) \in H^1(\Omega) \times H^1(\Omega)$:

$$\begin{cases} b(v, p_\epsilon) + \epsilon(u_\epsilon, v)_{H^1(\Omega)} + (u_\epsilon, v)_{L^2(\Gamma)} = (f, v)_{L^2(\Gamma)}, \\ b(u_\epsilon, q) = \langle g_\epsilon, q \rangle_{H^{-\frac{1}{2}}(\tilde{\Gamma}), H^{\frac{1}{2}}(\tilde{\Gamma})}. \end{cases}$$

Furthermore, using equation (5.11), we get that $p_{g_\epsilon} \in M$. Finally since $M \subset H^1(\Omega)$, we get by choosing $q \in M$ that the pair $(u_{g_\epsilon}, p_{g_\epsilon})$ is a solution of (5.8). □

As a conclusion, because the mixed formulation (5.8) admits a unique solution $(u_\epsilon, p_\epsilon) \in H^1(\Omega) \times M$, the unique solution $g_\epsilon \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$ of (5.7) is given by $g_\epsilon = \partial_\nu u_\epsilon$ associated to the unique solution u_ϵ of (5.8). In other words, solving the optimal control problem amounts to solve the mixed formulation. Finally, let us study the convergence as ϵ goes to 0 of the solution (u_ϵ, p_ϵ) of (5.8).

Proposition 5.2.10. *If there exists $u \in H^1(\Omega)$ such that u is solution of (5.1), then $u_\epsilon \to u$ in $H^1(\Omega)$ and $p_\epsilon \to 0$ in M.*

Proof. Let $u \in H^1(\Omega)$ be a solution of (5.1), then u satisfies $u = f$ on Γ and $b(u, \mu) = 0, \forall \mu \in M$. From (5.8), we thus get for all $(v, \mu) \in H^1(\Omega) \times M$ that:

$$\begin{cases} b(v, p_\epsilon) + \epsilon(u_\epsilon, v)_{H^1(\Omega)} + (u_\epsilon - u, v)_{L^2(\Gamma)} = 0, \\ b(u_\epsilon - u, q) = 0. \end{cases}$$

Taking $q = p_\epsilon$ and $v = u_\epsilon - u$, we get by subtracting both lines that:

$$\epsilon(u_\epsilon, u_\epsilon - u)_{H^1(\Omega)} + \|u_\epsilon - u\|^2_{L^2(\Gamma)} = 0.$$

From now on, it is easy to show that:

$$\|u_\epsilon\|_{H^1(\Omega)} \leq \|u\|_{H^1(\Omega)}, \quad \|u_\epsilon - f\|_{L^2(\Gamma)} \leq \sqrt{\epsilon}\|u\|_{H^1(\Omega)}.$$

so that $u_\epsilon \to f$ on $L^2(\Gamma)$ and u_ϵ is bounded in $H^1(\Omega)$. This last point means that up to a subsequence there exists $w \in H^1(\Omega)$ such that $u_\epsilon \rightharpoonup w$ in $H^1(\Omega)$. Hence, passing to the limit in the second line of (5.8) gives us :

$$b(w, q) = 0, \forall q \in M.$$

and $w = f$ on Γ by the continuity of the trace operator. Using the injectivity of A, one can deduce that $w = u$, so that $u_\epsilon \rightharpoonup u$ in $H^1(\Omega)$. Finally, since:

$$\|u_\epsilon - u\|^2_{H^1(\Omega)} \leq -(u, u_\epsilon - u)_{H^1(\Omega)},$$

the weak convergence of u_ϵ impies the strong convergence of u_ϵ towards u in $H^1(\Omega)$. Now using the first line of (5.8) and using the trace inequality, we get that for all $v \in H^1(\Omega)$, there exists $C > 0$ such that:

$$b(v, p_\epsilon) \leq \epsilon \|u_\epsilon\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} + C\|u_\epsilon - u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}$$

It follows that for $v \neq 0$:

$$\frac{b(v, p_\epsilon)}{\|v\|_{H^1(\Omega)}} \leq \epsilon \|u\|_{H^1(\Omega)} + \|u_\epsilon - u\|_{H^1(\Omega)}.$$

Finally taking the sup over v gives us, using the inf sup condition satisfied by b:

$$\beta \|p_\epsilon\|_M \leq \epsilon \|u\|_{H^1(\Omega)} + C\|u_\epsilon - u\|_{H^1(\Omega)},$$

so that the strong convergence of u_ϵ implies the convergence of p_ϵ to 0. □

We just presented the two approaches used to solve the inverse problem (5.1). Although the two methods under consideration originate from markedly different philosophical standpoints, a quick analysis reveals that they are, in essence, closely related. Despite these apparent conceptual differences, the resulting formulations differ only through the introduction of a penalization term for the adjoint state. Consequently, when viewed through the lens of penalization, the two approaches become nearly equivalent, leading to solutions that are mathematically comparable. In this sense, the distinction between the methods is largely philosophical rather than practical, as both ultimately yield almost identical results. Furthermore, the solutions of the two mixed formulations behave the same way as ϵ goes to 0 which solidifies the link between the two approaches.

5.2.3 The heat equation case

Let us now present the same link in the case of a parabolic equation in time. For simplicity we will consider the simple heat equation on the same domain Ω as before but the arguments we will use here would be the same for any parabolic equation. The inverse problem we will now focus on is the following: *For f, find u solution of:*

$$\begin{cases} \partial_t u - \Delta u = 0 & \text{in } \Omega \times (0,T), \\ \partial_\nu u = 0 & \text{on } \Gamma \times (0,T), \\ u = f & \text{on } \Gamma \times (0,T), \\ u = 0 & \text{in } \Omega \times \{0\}. \end{cases} \tag{5.12}$$

Remark 5.2.1. *We did not precise the regularity of the solutions u since in the following the inverse problem considered will not be problem (5.12) but a more handful problem. We will then precise the regularity of the solution u.*

We will denote in the following by V the space $H^1(\Omega)$, H the space $L^2(\Omega)$ and we will consider the following Gelfand triplet:

$$V \hookrightarrow H \equiv H' \hookrightarrow V',$$

where $\hookrightarrow$ means continuous injection. We need to consider solutions of the system (5.12) that will match the optimal control approach, to do so we will consider variational solutions in the Hilbert space

$$W_0(0,T) = \left\{ u \in L^2(0,T;V), \frac{du}{dt} \in L^2(0,T;V'), u(0) = 0 \right\}.$$

Let us introduce the space $M = \left\{ q \in L^2(0,T;V), q = 0 \text{ on } \tilde{\Gamma} \times (0,T) \right\}$ endowed with the norm

$$\|q\|_M^2 = \int_0^T \|\nabla q\|_{L^2(\Omega)}^2 \, dt.$$

Let us introduce the following inverse problem : *For $f \in L^2(0,T;L^2(\Gamma))$, Find $u \in W_0(0,T)$ solution of:*

$$\dot{u} + Au = 0 \text{ in } M' \quad \text{and} \quad u = f \text{ on } \Gamma \times (0,T). \tag{5.13}$$

with $A : V \to V'$ such that for all $u, v \in V$

$$\langle Au, v\rangle_{V',V} := a(u,v) = \int_\Omega \nabla u.\nabla v \; dx,$$

which comes down to solve $u = f$ on $\Gamma \times (0,T)$ and for all $q \in M$:

$$\int_0^T \langle \dot{u}, q\rangle_{V',V} + a(u,q) \; dt = 0$$

This inverse problem is obviously ill-posed for the same reasons as in the previous section. Let us now prove that, if u solution of (5.13) has enough regularity, then u is a solution of (5.12). In this sense, the solution u of (5.13) (if it exists) is a weak solution of (5.12). In the following, we will only consider those weak solutions in $W_0(0,T)$ as they are handful when it comes to optimal control.

Proposition 5.2.11. *Suppose (5.13) has a solution $u \in H^1(0,T;L^2(\Omega)) \cap L^2(0,T;H^2(\Omega))$, then u satisfies (5.12).*

Proof. If u satisfies (5.13) then we trivially get that $u = f$ on $\Gamma \times (0,T)$ and $u = 0$ in $\Omega \times \{0\}$. Furthermore, we get that for all $q \in M$

$$\int_0^T (\dot{u}, q)_H + a(u,q) \; dt = 0$$

First choosing $\mu \in \mathcal{D}((0,T) \times \Omega)$, we get

$$\partial_t u - \Delta u = 0 \text{ in } \Omega \times (0,T)$$

Finally, choosing $q \in M$, we get by integration by parts

$$\int_0^T (\dot{u}, q)_{L^2(\Omega)} - (\Delta u, q)_{L^2(\Omega)} + \langle \partial_\nu u, q\rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}), H^{\frac{1}{2}}(\tilde{\Gamma})} \, dt = 0$$

Which concludes the proof. □

Remark 5.2.2. *The required regularity in Proposition 5.2.11 is probably not optimal.*

As in the previous section, the goal is to retrieve the source g through a measurement f of u on $\Gamma \times (0,T)$. We now develop the two approaches in this framework. An equivalent argumentation to Proposition 5.2.1 can be lead to show that the solution u, if it exists, is unique.

The one shot approach

By the same argument as in Section 1, u is solution of the inverse problem (5.13) if and only if u is a solution of:
Find $u \in W_0(0,T)$ such that for all $q \in M$:

$$\left| \begin{array}{l} d(u,q) = 0, \\ u = f \text{ on } \Gamma \times (0,T). \end{array} \right. \tag{5.14}$$

with $d(u,q) = \displaystyle\int_0^T \langle \partial_t u, q\rangle_{V',V} + a(u,q) \; dt$. Using the same approach as in Section 1, the problem we're meant to solve is the following, for $\epsilon > 0$: *Find $(u_\epsilon, p_\epsilon) \in W_0(0,T) \times M$ such that for all $(v,q) \in W_0(0,T) \times M$:*

$$\begin{cases} d(v,p_\epsilon) + \epsilon(u_\epsilon, v)_{W(0,T)} + (u_\epsilon, v)_{L^2(0,T;L^2(\Gamma))} = (f,v)_{L^2(0,T;L^2(\Gamma))}, \\ d(u_\epsilon, q) - (p_\epsilon, q)_M = 0. \end{cases} \tag{5.15}$$

Let us now present the optimal control approach so that, up to a penalisation term on the adjoint state, the mixed formulation (5.15) matches the mixed formulation associated with the optimality conditions.

The optimal control approach

Let us introduce the optimal control functional $\mathcal{J}$:

$$\inf_{g\in L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))} \mathcal{J}(g) = \frac{1}{2}\int_0^T \|u_g - f\|^2_{L^2(\Gamma)}\, dt + \frac{\epsilon}{2}\|u_g\|^2_{W(0,T)}. \tag{5.16}$$

with u_g solution in $W_0(0,T)$ of :

$$\dot{u}_g + Au_g = Bg, \text{ in } L^2(0,T;V'). \tag{5.17}$$

with $B : \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}) \to V'$ such that for all $v \in V$ and for all $g \in \tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})$:

$$\langle Bg, v\rangle_{V',V} := b(g,v) = \langle g, v\rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}),H^{\frac{1}{2}}(\tilde{\Gamma})}.$$

Proposition 5.2.12. *For $g \in L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$ the problem (5.17) has a unique solution $u_g \in W_0(0,T)$. Furthermore, there exists $C > 0$ such that:*

$$\|u_g\|_{W_0(0,T)} \leq C\|Bg\|_{L^2(0,T;V')} \leq \tilde{C}\|g\|_{L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))}.$$

Proof. The result is a direct application of [90, Theorem 4.1, Page 238] and using the continuity of the operator B. □

The following Lemma is necessary to prove that problem (5.16) admits a unique solution

Lemma 5.2.1. *For $g \in L^2(0,T,\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$ and u_g the unique solution of (5.17), it follows that there exists $C > 0$ such that:*

$$\|g\|_{L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))} \leq C\|u_g\|_{W(0,T)}.$$

Proof. Let $\varphi \in H^{\frac{1}{2}}(\tilde{\Gamma})$. Then there exits a lifting $V_\varphi \in H^1(\Omega)$ such that $V_\varphi = \varphi$ on $\tilde{\Gamma}$ and $\|V_\varphi\|_{H^1(\Omega)} \leq C\|\varphi\|_{H^{\frac{1}{2}}(\tilde{\Gamma})}$. From (5.17) we get that, that for almost all $t \in [0,T]$:

$$\langle \dot{u}(t), V_\varphi\rangle_{V',V} + a(u(t), V_\varphi) = \langle g(t), \varphi\rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}),H^{\frac{1}{2}}(\tilde{\Gamma})}$$

It follows that for almost $t \in [0,T]$:

$$\|\dot{u}(t)\|_{V'}\|V_\varphi\|_V + \|A\|\|u(t)\|_V\|V_\varphi\|_V \geq |\langle g(t), \varphi\rangle_{H^{-\frac{1}{2}}(\tilde{\Gamma}),H^{\frac{1}{2}}(\tilde{\Gamma})}|.$$

Hence, using the continuity of the lifting, for almost $t \in [0,T]$:

$$C\|\dot{u}(t)\|_{V'} + C\|A\|\|u(t)\|_V \geq \frac{|\langle g(t), \varphi\rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}),H^{\frac{1}{2}}(\tilde{\Gamma})}|}{\|\varphi\|_{H^{\frac{1}{2}}(\tilde{\Gamma})}}$$

Taking the sup on φ and taking the square, it follows that for almost all $t \in [0,T]$:

$$\|g(t)\|^2_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})} \leq 2C^2\|\dot{u}(t)\|^2_{V'} + 2C^2\|A\|^2\|u(t)\|^2_V.$$

Finally, integrating over t gets us:

$$\|g\|_{L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))} \leq \sqrt{2}C\max(1,\|A\|)\|u\|_{W(0,T)}.$$

□

Proposition 5.2.13. *The problem (5.16) admits a unique solution $g_\epsilon \in L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$.*

Proof. The functional $\mathcal{J}$ is clearly continuous and strictly convex. It remains to show that $\mathcal{J}$ is coercive on $L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$. We have:

$$\mathcal{J}(g) \geq \frac{\epsilon}{2}\|u_g\|^2_{W(0,T)}.$$

The proof is then a consequence of Lemma 5.2.1. □

Let us now introduce the following mixed formulation, inspired by the one obtained in (5.15). The main difference here being that the regularisation term $(p_\epsilon, q)_M$ has been removed. Find $(u_\epsilon, p_\epsilon) \in W_0(0,T) \times M$ such that for all $(v,q) \in W_0(0,T) \times M$,

$$\begin{cases} d(v, p_\epsilon) + \epsilon(u_\epsilon, v)_{W_0(0,T)} + (u_\epsilon, v)_{L^2(0,T;L^2(\Gamma))} = (f, v)_{L^2(0,T;L^2(\Gamma))}, \\ d(u_\epsilon, q) = 0. \end{cases} \tag{5.18}$$

Let us now prove that this mixed formulation is well-posed.

Proposition 5.2.14. *The mixed formulation (5.18) has a unique solution (u_ϵ, p_ϵ) in $W_0(0,T) \times M$.*

Proof. We have to prove that the bilinear form $j(u,v) = \epsilon(u,v)_{W_0(0,T)} + (u,v)_{L^2(0,T;L^2(\Gamma))}$ is coercive on $Ker\ b$ and that the bilinear form b satisfies an inf-sup condition. Concerning the first point, the proof is trivial. We indeed find that for $u \in W_0(0,T) \supset Ker\ b$ that:

$$j(u,u) \geq \epsilon\|u\|^2_{W_0(0,T)}.$$

Let us now consider the second point, which is equivalent to prove that there exists some $\beta > 0$ such that for all $p \in M$,

$$\sup_{u \in W_0(0,T)} \frac{d(u,p)}{\|u\|_{W_0(0,T)}} \geq \beta\|p\|_M.$$

For $p \in M$, let us introduce $u_p \in W_0(0,T)$ solution for $q \in L^2(0,T;V)$ of :

$$d(u_p, q) = \int_0^T a(p,q)\ dt \tag{5.19}$$

The solution u_p of (5.19) is solution of $\dot{u}_p + Au_p = Ap$. Since $Ap \in L^2(0,T;V')$, using Proposition 5.2.12, there exists a unique solution $u_p \in W_0(0,T)$ to (5.19) and

$$\|u_p\|_{W_0(0,T)} \leq C\|Ap\|_{L^2(0,T;V')} \leq \tilde{C}\|p\|_M.$$

since A is continuous from V to V' and the norm M is equivalent to the norm $L^2(0,T;V)$ by Poincaré inequality. We finally find that for all $p \in M$

$$\sup_{u \in W_0(0,T)} \frac{d(u,p)}{\|u\|_{W_0(0,T)}} \geq \frac{d(u_p,p)}{\|u_p\|_{W_0(0,T)}} \geq \frac{1}{\tilde{C}}\|p\|_M.$$

Which concludes the proof. □

Let us now compute the gradient of the functional $\mathcal{J}$.

Proposition 5.2.15. *The differential of $\mathcal{J}$ at point $g \in L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$ is given, for $h \in L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$ by*

$$\left\langle D_g\tilde{\mathcal{J}}(g), h\right\rangle_{(L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})))', L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))} = -\left\langle p_g, h\right\rangle_{(L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})))', L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))}. \tag{5.20}$$

where p_g is the solution in $L^2(0,T;V)$ to the problem : find $p_g \in L^2(0,T;V)$ such that for all $v \in W_0(0,T)$

$$d(v, p_g) = -\epsilon(u_g, v)_{W_0(0,T)} - (u_g, v)_{L^2(0,T;L^2(\Gamma))} + (f, v)_{L^2(0,T;L^2(\Gamma))} \tag{5.21}$$

and u_g is the solution to the problem (5.17).

Proof. Let us first verify that problem (5.21) admits a unique solution p_g in $L^2(0,T;V)$. We wish to apply the Banach-Necas-Babuska Theorem (see [58, Theorem 2.6, Page 85]). We need to show that there exists $\beta > 0$ such that

$$\inf_{p\in L^2(0,T;V)} \sup_{v\in W_0(0,T)} \frac{d(v,p)}{\|p\|_{L^2(0,T;V)}\|v\|_{W_0(0,T)}} \geq \beta, \tag{BNB1}$$

and

$$\forall v \in W_0(0,T), \quad (\forall p \in L^2(0,T;V), d(v,p) = 0) \Rightarrow (v = 0). \tag{BNB2}$$

The condition (BNB1) is a direct consequence of the inf-sup condition showed in Proposition 5.2.14. Let us prove the condition (BNB2). Let $v \in W_0(0,T)$ and let us suppose that for all $p \in L^2(0,T;V), d(v,p) = 0$, then by choosing $p = v\mathbb{1}_{[0,s]}(t)$ forall $s \in [0,T]$, we get for all $s \in [0,T]$

$$\frac{1}{2}\|v(s)\|_H^2 + \int_0^s \|\nabla v(t)\|_H^2 \, dt = 0$$

Here, we used the Green's Formula recalled in [46, Chapter XVIII, Theorem 2, Page 477]. Which finally gives us $v = 0$.
As in the previous Section, we introduce the following Lagrangian defined for $(u,p,g) \in W_0(0,T) \times L^2(0,T;V) \times L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$:

$$\mathcal{L}(u,p,g) = \frac{1}{2}\int_0^T \|u - f\|^2_{L^2(\Gamma)} \, dt + \frac{\epsilon}{2}\|u\|^2_{W(0,T)} + d(u,p) - \langle g, p\rangle_{\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}),H^{\frac{1}{2}}(\tilde{\Gamma})} \, dt.$$

To compute the gradient of the functional $\mathcal{J}$ thanks to this Lagrangian, we still observe that for $g \in L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))$:

$$\mathcal{J}(g) = \mathcal{L}(u_g, p, g), \quad \forall p \in L^2(0,T;V),$$

where u_g is the solution in $W_0(0,T)$ of (5.17). Using the chain rule, it follows that

$$\begin{aligned}\langle D_g\mathcal{J}(g), h\rangle_{(L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})))',L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))} &= \langle D_u\mathcal{L}(u_g,p,g), u_h\rangle_{(W_0(0,T))',W_0(0,T)} \\ &\quad + \langle D_g\mathcal{L}(u_g,p,g), h\rangle_{(L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})))',L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))}.\end{aligned}$$

On the one hand, we compute

$$\left\langle D_u\tilde{\mathcal{L}}(u_g,p,g), u_h\right\rangle_{(W_0(0,T))',W_0(0,T)} = d(u_h,p) + \epsilon(u_g,u_h)_{W_0(0,T)} + (u_g - f, u_h)_{L^2(0,T;L^2(\Gamma))}.$$

On the other, we easily get that

$$\langle D_g\mathcal{L}(u_g,p,g), h\rangle_{(L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})))',L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))} = -\langle p, h\rangle_{(L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma})))',L^2(0,T;\tilde{H}^{-\frac{1}{2}}(\tilde{\Gamma}))}.$$

As in Proposition 5.2.8, we wish to cancel the term $\left\langle D_u\tilde{\mathcal{L}}(u_g,p,g), u_h\right\rangle_{(W_0(0,T))',W_0(0,T)}$, to do so let us introduce as in Proposition 5.2.8 the ad hoc adjoint state p_g solution in $L^2(0,T;V)$ of (5.21). □

As a consequence of Proposition 5.2.13 and 5.2.15, the unique solution g_ϵ to the problem (5.16) satisfies

$$p_{g_\epsilon} = 0 \text{ on } \tilde{\Gamma} \times (0,T) \tag{5.22}$$

where p_{g_ϵ} is the solution to the problem (5.21), in which u_{g_ϵ} is the solution to the problem (5.17) associated with g_ϵ. With the same argument as in the previous Section, we can show that the solution $(u_{g_\epsilon}, p_{g_\epsilon})$ of (5.17) (5.21) coincide with the solution (u_ϵ, p_ϵ) of (5.18). We can also prove the same kind of convergence result as before such that $u_\epsilon \to u$ in $W_0(0,T)$ and $p_\epsilon \to 0$ in M.

It finally appears that the main difference between the two approaches is not theoretical but mainly numerical. We showed in Chapter 3 and 4 that the iterative approach is stronger when it comes to solve the problem with refined meshes. Indeed, in the approach solving the problem through a space-time mixed formulation, when the mesh is being refined, the size of the FEM matrices rapidly increase whereas in the iterative method, this size remains manageable. This allows us to consider more "realistic" set up and to get more precise results. However, the space-time mixed formulation approach allows us to manage the amplitude of the noise and to choose all the parameters with respect to δ. In the iterative method, this choice relies on a L-curve or dichotomy which is less precise.

5.3 Optimal control in a Friedrichs framework

The goal of this section is to introduce the tools needed in order to study the optimal control problem of Chapter 4 in the case where $\eta = 0$. In Chapter 4, we decided to recast the direct problem as an order 1 variational system since the regularity of the solution was handful and gave us a continuous and discrete inf-sup condition. We saw that adding the parabolic regularisation $\nu \neq 0$ was mandatory to get the inf-sup conditions and therefore to get the analysis we conducted afterwards. The question is : how to conduct such an analysis without this parabolic regularisation ν? In this section, we introduce a framework in which we manage to obtain a continuous inf-sup condition without adding a parabolic regularisation. This inf-sup condition allows us to recast an optimal control problem into a mixed formulation. For us to be able to conduct the same type of numerical analysis, as in Chapter 4, the discrete inf-sup condition is still lacking.

5.3.1 An abstract framework

Let us introduce two separable Hilbert spaces $\mathcal{H}$ and $\mathcal{G}$ and a closed operator $G : D(G) \subset \mathcal{H} \to \mathcal{G}$ such that $D(G)$ is dense in $\mathcal{H}$. In the following we will denote $\mathcal{V} = L^2(0,T;\mathcal{H}) \times L^2(0,T;\mathcal{G})$. Let us introduce $G^* : D(G^*) \subset \mathcal{G} \to \mathcal{H}$ the adjoint operator of G and let us assume that $D(G^*)$ is dense in $\mathcal{G}$. Let us introduce the following direct problem that will be studied in this section. *For $(h,f) \in \mathcal{V}$, find (u,v) such that*

$$\begin{cases} \partial_t u - G^* v = h \\ \partial_t v + Gu = f \\ u(0) = v(0) = 0. \end{cases} \tag{5.23}$$

The regularity of the solutions of (5.23) is not specified for the moment but will be described in the following. The aim of this section is to recast the problem (5.23) in a variational way such that the associated bilinear form satisfies an inf-sup condition. Let us first introduce the following unbounded operator

$$G_T : \begin{cases} D(G_T) \subset \mathcal{V} \to L^2(0,T;\mathcal{G}) \\ (u,v) \mapsto \partial_t v + Gu, \end{cases}$$

with

$$D(G_T) = L^2(0,T;D(G)) \times H^1(0,T;\mathcal{G}).$$

The following lemma establishes a density result that will be useful throughout this section.

Lemma 5.3.1. *Let Y be a separable Hilbert space and X a dense subspace of Y, then*

$$\mathcal{D}(0,T;X) \text{ is dense in } L^2(0,T;Y).$$

Proof. We have that $\mathcal{D}(0,T) \otimes X \subset \mathcal{D}(0,T;X)$. Furthermore, $\mathcal{D}(0,T) \otimes X$ is immediately dense in $L^2(0,T) \otimes Y$ which is dense in $L^2(0,T;Y)$ since Y is separable. □

Proposition 5.3.1. *$D(G_T)$ is dense in $\mathcal{V}$ and G_T is closable. The closure of this operator will be called $\overline{G_T}$ in the following.*

Proof. We have that $L^2(0,T;D(G))$ is dense in $L^2(0,T;\mathcal{H})$ and $H^1(0,T;\mathcal{G})$ is dense in $L^2(0,T;\mathcal{G})$, so that $D(G_T)$ is dense in $\mathcal{V}$. Let us prove that G_T is closable. Let $X_n = (u_n, v_n)^T \in D(G_T)$ so that

$$X_n \to 0 \text{ in } \mathcal{V},$$

and

$$G_T X_n \to Y \text{ in } L^2(0,T;\mathcal{G}).$$

Let us check that $Y = 0$, so that G_T will be closable. Let $\phi \in \mathcal{D}(0,T;D(G^*))$, we get that

$$\int_0^T (G_T X_n, \phi)_{\mathcal{G}} = \int_0^T (\partial_t v_n + Gu_n, \phi)_{\mathcal{G}} = \int_0^T (v_n, -\partial_t \phi)_{\mathcal{G}} + \int_0^T (u_n, G^*\phi)_{\mathcal{H}} \xrightarrow[n\to\infty]{} 0.$$

Finally, we have for all $\phi \in \mathcal{D}(0,T;D(G^*))$

$$\int_0^T (Y,\phi)_{\mathcal{G}} = 0.$$

Using the density of $\mathcal{D}(0,T;D(G^*))$ in $L^2(0,T;\mathcal{G})$ given by Lemma 5.3.1, we get that $Y=0$. □

Remark 5.3.1. *For details on the characterization used to prove the closability of G_T, see [97].*

It is important to notice that G_T is not a closed operator. The previous proof applied to $X_n \to X \neq 0$ in $\mathcal{V}$ would not allow us to conclude as $u \in \mathcal{H}$ the limit of u_n does not belong to $D(G)$. Let us then define

$$(G^*)_T : \begin{cases} D((G^*)_T) \subset \mathcal{V} \to L^2(0,T;\mathcal{H}) \\ (u,v) \mapsto \partial_t u - G^* v, \end{cases}$$

with

$$D((G^*)_T) = H^1(0,T;\mathcal{H}) \times L^2(0,T;D(G^*)).$$

Remark 5.3.2. *It is important to notice that in general $(G^*)_T \neq G_T^*$. In other words $(G^*)_T$ is not the adjoint of G_T.*

One can show, as in Proposition 5.3.1 that $(G^*)_T$ is closable. The closure will then be called $\overline{(G^*)_T}$. Let us define the augmented operator

$$\mathcal{A}_T \ : \ D(G_T) \cap D((G^*)_T) \subset \mathcal{V} \to L^2(0,T;\mathcal{G}) \times L^2(0,T;\mathcal{H}), \qquad \mathcal{A}_T(u,v) = (G_T(u,v), (G^*)_T(u,v)) \, .$$

Proposition 5.3.2. *$\mathcal{A}_T$ is closable. Setting $W = D(\overline{\mathcal{A}_T})$ with graph norm*

$$\|(u,v)\|_W^2 = \|(u,v)\|_{\mathcal{V}}^2 + \|\overline{G_T}(u,v)\|_{L^2(0,T;\mathcal{G})}^2 + \|\overline{(G^*)_T}(u,v)\|_{L^2(0,T;\mathcal{H})}^2,$$

$D(G_T) \cap D((G^*)_T)$ is dense in W.

Proof. If $X_n \in D(\overline{\mathcal{A}_T}), X_n \to 0$ in $\mathcal{V}$, and $\mathcal{A}_T X_n \to (Y_1, Y_2)$, then $X_n \in D(G_T), X_n \to 0, G_T X_n \to Y_1$ forces $Y_1 = 0$ by closability of G_T; likewise $Y_2 = 0$ by closability of $(G^*)_T$. Hence $\mathcal{A}_T$ is closable. Density of $D(\mathcal{A}_T) = D(G_T) \cap D((G^*))$ in $D(\overline{\mathcal{A}_T})$ for the graph norm is then by definition of the closure of $\mathcal{A}_T$. □

Since $\overline{\mathcal{A}_T}$ is by construction a closed operator, it follows that W endowed with the norm of the graph is a Hilbert space.

Remark 5.3.3. *For $X \in D(\overline{\mathcal{A}_T})$ one has $\overline{\mathcal{A}_T} X = (\overline{G_T} X, \overline{(G^*)_T} X) \in L^2(0,T;\mathcal{G}) \times L^2(0,T;\mathcal{H})$.*

We want to define a notion of initial conditions on W. Let us then introduce the following continuous linear operator:

$$\gamma_0 \ : \ \begin{cases} D(G_T) \cap D((G^*)_T) & \to \ \mathcal{H} \times \mathcal{G} \\ (u,v) & \mapsto \ (u(0), v(0)). \end{cases}$$

Proposition 5.3.3. *There exists a unique continuous operator $\tilde{\gamma_0}$ on W that extends γ_0 which will still be noted γ_0 in this section.*

Proof. Let $(u,v) \in D(G_T) \cap D((G^*)_T)$. Let us introduce

$$\begin{aligned} I &= \int_0^T (\overline{(G^*)_T}(u,v), (T-t)u)_{\mathcal{H}} \, dt + \int_0^T (\overline{G_T}(u,v), (T-t)v)_{\mathcal{G}} \, dt \\ &= \int_0^T (\partial_t u - G^* v, (T-t)u)_{\mathcal{H}} \, dt + \int_0^T (\partial_t v + Gu, (T-t)v)_{\mathcal{G}} \, dt \\ &= \int_0^T (\partial_t u, (T-t)u)_{\mathcal{H}} \, dt + \int_0^T (\partial_t v, (T-t)v)_{\mathcal{G}} \, dt. \end{aligned}$$

Furthermore, by integration by parts

$$\int_0^T (\partial_t u, (T-t)u)_{\mathcal{H}}\ dt = \int_0^T \|u(t)\|^2_{\mathcal{H}}\ dt - \int_0^T (u, (T-t)\partial_t u)_{\mathcal{H}}\ dt + \Big[(u, (T-t)u)_{\mathcal{H}}\Big]_0^T,$$

leading to

$$\int_0^T (\partial_t u, (T-t)u)_{\mathcal{H}}\ dt = \frac{1}{2}\int_0^T \|u(t)\|^2_{\mathcal{H}} - \frac{T}{2}\|u(0)\|^2_{\mathcal{H}}.$$

We get that

$$2I = \int_0^T \|u(t)\|^2_{\mathcal{H}} + \int_0^T \|v(t)\|^2_{\mathcal{G}} - T(\|u(0)\|^2_{\mathcal{H}} + \|v(0)\|^2_{\mathcal{G}}).$$

We can deduce by controlling I by the W norm of (u, v), that there exists $C > 0$ such that

$$\|\gamma_0(u, v)\|_{\mathcal{H}\times\mathcal{G}} \leq C\|(u, v)\|_W.$$

This ensures the existence and the uniqueness of a continuous extension $\tilde{\gamma_0}$ by density of $D(G_T) \cap D((G^*)_T)$ in W. □

The previous Proposition extends the notion of initial condition to elements of W so that we can finally define the Hilbert space of solutions $W_0 = \ker\gamma_0$:

$$W_0 = \left\{(u, v)^T \in D(\overline{\mathcal{A}_T}),\ (u(0), v(0)) = (0, 0)\right\},$$

endowed with the W norm. The space W_0 is still an Hilbert space as a closed subspace of W.

Remark 5.3.4. *Proposition 5.3.3 can be generalised to every $s \in [0, T]$, so that we can define $\gamma_s(u, v) = (u(s), v(s))$ as a continuous operator on W.*

Finally, we recast the problem (5.23) as : *For $(h, f) \in \mathcal{V}$, find $(u, v) \in W_0$ such that for all $(q, p) \in \mathcal{V}$:*

$$b((u, v), (q, p)) = ((h, f), (q, p))_{\mathcal{V}}, \tag{5.24}$$

with

$$b((u, v), (q, p)) = (\overline{\mathcal{A}_T}(u, v), (p, q))_{L^2(0,T,\mathcal{G})\times L^2(0,T,\mathcal{H})}.$$

In the following, we will use this more handful form for b

$$b((u, v), (q, p)) = (\overline{G_T}(u, v), p)_{L^2(0,T,\mathcal{G})} + (\overline{(G^*)_T}(u, v), q)_{L^2(0,T,\mathcal{H})}.$$

We immediately see that if (u, v) a pair of solutions of (5.24) have enough regularity, (u, v) satisfies (5.23). In other words, a variational solution with enough regularity is a strong solution. In this sense, we say that solving (5.24) amounts to solve (5.23) in a variational sense. The following proposition gives us a result of existence and uniqueness of a strong solution for the problem (5.23). This proposition will be needed in order to prove that problem (5.24) is well-posed. Before this proposition, let us describe an the spaces $D(G)$ and $D(G^*)$. We have that

$$D(G) = \{u \in \mathcal{H},\ Gu \in \mathcal{G}\}, \quad D(G^*) = \{u \in \mathcal{G},\ G^*u \in \mathcal{H}\},$$

both endowed with the norm of the graph.

Lemma 5.3.2. *Th space $D(G^*)$ endowed with the norm of the graph is a Hilbert space.*

Proof. Since $D(G)$ is dense in $\mathcal{H}$, we have that G^* is closed. It obviously follows that $D(G^*)$ is a closed space of $\mathcal{H}$. Which concludes the proof. This result is also recalled in [31, Proposition II.16, Page 28] □

We can now finally show the result of existence and uniqueness in the following proposition.

Proposition 5.3.4. *For all $(h,f) \in \mathcal{D}(0,T;\mathcal{H}) \times \mathcal{D}(0,T;\mathcal{G})$, there exists a unique pair $(u,v) \in (\mathcal{C}^1([0,T],\mathcal{H}\times\mathcal{G}) \cap \mathcal{C}^0([0,T],D(G)\times D(G^*)) \subset W$ solution of (5.23).*

Proof. Let us show that the operator $A : (u,v) \in D(A) = D(G)\times D(G^*) \subset H\times\mathcal{G} \mapsto (G^*v, -Gu) \in \mathcal{H}\times\mathcal{G}$ satisfies the two following conditions in order for us to apply Theorem 3.3.1:

- $(z, Az)_{\mathcal{H}\times\mathcal{G}} \geq 0$ for all z in $\mathcal{D}(A)$.
- $\lambda_0\mathcal{I} + A$ is onto from $\mathcal{D}(A)$ to $\mathcal{H}\times\mathcal{G}$, for all $\lambda_0 > 0$.

The first point is trivial as A is antisymmetric, we get $(z, Az)_{\mathcal{H}\times\mathcal{G}} = 0$ for all $z \in D(A)$. For the second point, let us consider $\lambda_0 > 0$ and $(h,f) \in \mathcal{H}\times\mathcal{G}$. Let us consider the following variational formulation: *Find $v \in D(G^*)$ such that for all $\tilde{v} \in D(G^*)$:*

$$\lambda_0^2(v,\tilde{v})_{\mathcal{G}} + (G^*v, G^*\tilde{v})_{\mathcal{H}} = \lambda_0(f,\tilde{v})_{\mathcal{G}} + (h, G^*\tilde{v})_{\mathcal{H}}. \tag{5.25}$$

Since $D(G^*)$ is a Hilbert space endowed with the norm of the graph by Lemma 5.3.2, we get, using Lax-Milgram theorem that there exists a unique $v \in D(G^*)$ solution of (5.25). Let $u \in \mathcal{H}$ such that

$$\lambda_0 u + G^*v = h.$$

We get from (5.25) that u satisfies for all $\tilde{v} \in D(G^*)$

$$\lambda_0(v,\tilde{v})_{\mathcal{G}} - (u, G^*\tilde{v})_{\mathcal{H}} = (f,\tilde{v})_{\mathcal{G}}.$$

It follows that $u \in D(G)$ and $\lambda v - Gu = f$. We finally get that there exists $(u,v) \in D(A) = D(G)\times D(G^*)$ such that

$$\lambda_0(u,v) + A(u,v) = (h,f).$$

The final result follows using Theorem 3.3.1. □

We have just constructed strong solutions to problem (5.23). Let us now show that the problem (5.24) admits a unique solution in W_0.

Theorem 5.3.1. *The problem (5.24) is well-posed.*

Proof. We wish to apply the Banach-Necas-Babuska Theorem (see [58, Theorem 2.6, Page 85]). We need to show that there exists $\beta > 0$ such that

$$\inf_{(u,v)\in W_0} \sup_{(q,p)\in\mathcal{V}} \frac{b((u,v),(q,p))}{\|(u,v)\|_W \|(q,p)\|_{\mathcal{V}}} \geq \beta, \tag{BNB1}$$

and

$$\forall (q,p) \in \mathcal{V}, \quad (\forall (u,v) \in W_0,\ b((u,v),(q,p)) = 0) \Rightarrow ((q,p) = 0). \tag{BNB2}$$

Let us show that the condition (BNB1) is satisfied, which is equivalent to show that for all $(u,v) \in W_0$, there exists $\beta > 0$ such that

$$\sup_{(q,p)\in\mathcal{V}} \frac{b((u,v),(q,p))}{\|(q,p)\|_{\mathcal{V}}} \geq \beta \|(u,v)\|_W.$$

Let (u_n, v_n) a sequence of $D(G_T) \cap D((G^*)_T)$ that converges towards (u,v) in W_0. Such a sequence exists by density. Let us introduce, for all $s \in [0,T]$,

$$\left| \begin{array}{l} q_n^* = \overline{(G^*)_T}(u_n, v_n) + \mathbb{1}_{[0,s]}(t) u_n \in L^2(0,T;\mathcal{H}), \\ p_n^* = \overline{G_T}(u_n, v_n) + \mathbb{1}_{[0,s]}(t) v_n \in L^2(0,T;\mathcal{G}), \end{array} \right.$$

so that

$$\begin{aligned} b((u_n,v_n),(q_n^*,p_n^*)) =& \|\overline{G_T}(u_n,v_n)\|^2_{L^2(0,T,\mathcal{G})} + \|\overline{(G^*)_T}(u_n,v_n)\|^2_{L^2(0,T,\mathcal{H})} \\ &+ (\overline{G_T}(u_n,v_n), v_n)_{L^2(0,s,\mathcal{G})} + (\overline{(G^*)_T}(u_n,v_n), u_n)_{L^2(0,s,\mathcal{H})}. \end{aligned}$$

Since $(u_n, v_n) \in D(G_T) \cap D((G^*)_T)$, we get that

$$\overline{G_T}(u_n, v_n) = \partial_t v_n + G u_n \text{ and } \overline{(G^*)_T}(u_n, v_n) = \partial_t u_n - G^* v_n,$$

so that, integrating by parts in time and using the definition of the adjoint of G, we finally get

$$b((u_n, v_n), (q_n^*, p_n^*)) = \|\overline{G_T}(u_n, v_n)\|^2_{L^2(0,T,\mathcal{G})} + \|\overline{(G^*)_T}(u_n, v_n)\|^2_{L^2(0,T,\mathcal{H})} + \frac{\|u_n(s)\|^2_{\mathcal{H}}}{2} + \frac{\|v_n(s)\|^2_{\mathcal{G}}}{2} - \left(\frac{\|u_n(0)\|^2_{\mathcal{H}}}{2} + \frac{\|v_n(0)\|^2_{\mathcal{G}}}{2}\right).$$

Furthermore, we easily get that there exists $C > 0$

$$\|(q_n^*, p_n^*)\|_{\mathcal{V}} \leq C \|(u_n, v_n)\|_W.$$

We then get for all $s \in [0, T]$

$$\frac{b((u_n, v_n), (q_n^*, p_n^*))}{\|(q_n^*, p_n^*)\|_{\mathcal{V}}} \geq \frac{1}{C\|(u_n, v_n)\|_W}\left(\|\overline{G_T}(u_n, v_n)\|^2_{L^2(0,T,\mathcal{G})} + \|\overline{(G^*)_T}(u_n, v_n)\|^2_{L^2(0,T,\mathcal{H})} + \frac{\|u_n(s)\|^2_{\mathcal{H}}}{2} + \frac{\|v_n(s)\|^2_{\mathcal{G}}}{2} - \left(\frac{\|u_n(0)\|^2_{\mathcal{H}}}{2} + \frac{\|v_n(0)\|^2_{\mathcal{G}}}{2}\right)\right).$$

Knowing that (u_n, v_n) converges towards (u, v) in W_0, we get that :

$$\begin{cases} \overline{G_T}(u_n, v_n) \to \overline{G_T}(u, v) & \text{in } L^2(0, T, \mathcal{G}) \\ \overline{(G^*)_T}(u_n, v_n) \to \overline{(G^*)_T}(u, v) & \text{in } L^2(0, T, \mathcal{H}) \\ \forall s \in [0, T], (u_n(s), v_n(s)) \to (u(s), v(s)) & \text{in } \mathcal{H} \times \mathcal{G}. \end{cases}$$

The convergence of $(u_n(s), v_n(s))$ is deduced from the continuity of the operator γ_s introduced in Remark 5.3.4. It follows that, by continuity of the bilinear form b, for all $s \in [0, T]$

$$\frac{b((u, v), (q^*, p^*))}{\|(q^*, p^*)\|_{\mathcal{V}}} \geq \frac{1}{C\|(u, v)\|_W}\left(\|\overline{G_T}(u, v)\|^2_{L^2(0,T,\mathcal{G})} + \|\overline{(G^*)_T}(u, v)\|^2_{L^2(0,T,\mathcal{H})} + \frac{\|u(s)\|^2_{\mathcal{H}}}{2} + \frac{\|v(s)\|^2_{\mathcal{G}}}{2}\right).$$

We finally get that for all $s \in [0, T]$:

$$\sup_{(q,p)\in V} \frac{b((u, v), (q, p))}{\|(q, p)\|_{\mathcal{V}}} \geq \frac{1}{C\|(u, v)\|_W}\left(\|\overline{G_T}(u, v)\|^2_{L^2(0,T,\mathcal{G})} + \|\overline{(G^*)_T}(u, v)\|^2_{L^2(0,T,\mathcal{H})} + \frac{\|u(s)\|^2_{\mathcal{H}}}{2} + \frac{\|v(s)\|^2_{\mathcal{G}}}{2}\right).$$

Now integrating over s gets us

$$T \sup_{(q,p)\in V} \frac{b((u, v), (q, p))}{\|(q, p)\|_{\mathcal{V}}} \geq \frac{1}{C\|(u, v)\|_W}\left(T\|\overline{G_T}(u, v)\|^2_{L^2(0,T,\mathcal{G})} + T\|\overline{(G^*)_T}(u, v)\|^2_{L^2(0,T,\mathcal{H})} + \frac{\|u\|^2_{L^2(0,T;\mathcal{H})}}{2} + \frac{\|v\|^2_{L^2(0,T;\mathcal{G})}}{2}\right).$$

Which gives us

$$T \sup_{(q,p)\in V} \frac{b((u, v), (q, p))}{\|(q, p)\|_{\mathcal{V}}} \geq \frac{1}{C\|(u, v)\|_W} \min\left(\frac{1}{2}, T\right) \|(u, v)\|^2_W.$$

Finally for all $(u,v) \in W_0$

$$\sup_{(q,p)\in\mathcal{V}} \frac{b((u,v),(q,p))}{\|(q,p)\|_{\mathcal{V}}} \geq \min\left(\frac{1}{8T}, \frac{1}{C}\right)\|(u,v)\|_W.$$

Let us now prove the condition (BNB2). Let $(q,p) \in \mathcal{V}$, we suppose that for all $(u,v) \in W_0$

$$b((u,v),(q,p)) = 0.$$

Using Proposition 5.3.4, it follows that for all $(\varphi_q, \varphi_p) \in \mathcal{D}(0,T;\mathcal{H}) \times \mathcal{D}(0,T;\mathcal{G})$, there exists a solution $(u,v) \in (\mathcal{C}^1([0,T],\mathcal{H}\times\mathcal{G})) \cap (\mathcal{C}^0([0,T],D(G)\times D(G^*))) \subset W_0$ such that

$$\begin{cases} \overline{G_T}(u,v) = \phi_p, \\ \overline{(G^*)_T}(u,v) = \phi_q. \end{cases}$$

It follows that for all $(\varphi_q, \varphi_p) \in \mathcal{D}(0,T;\mathcal{H}) \times \mathcal{D}(0,T;\mathcal{G})$

$$((\varphi_q,\varphi_p),(q,p))_{\mathcal{V}} = 0,$$

so that $(q,p) \in \mathcal{D}(0,T;\mathcal{H})^{\perp} \times \mathcal{D}(0,T;\mathcal{G})^{\perp}$. Using the Lemma 5.3.1, we finally get that $(q,p) = (0,0)$, which concludes the proof. □

In this section, we have introduced a useful variational framework that defines a certain type of solution in the Hilbert space W_0. The goal is now to use this framework in an optimal control approach to solve an inverse problem. The following section presents the inverse problem we want to study and shows how to generalize the work made in Chapter 4.

5.3.2 An abstract optimal control approach for solving an inverse problem

Let us now introduce the inverse problem we will focus on. The goal is to retrieve the source $F := (h,f) \in \mathcal{V}$ from a measurement y of the solution $Z_F := (u_F, v_F) \in W_0$ of problem (5.24). Thus, we introduce the measurement operator $C : \mathcal{V} \to Y$ where Y denotes the observation Hilbert space. To do so, we aim to solve the following optimal control which boils down to minimizing the following functional for $\epsilon > 0$:

$$\inf_{F\in\mathcal{V}} \mathcal{J}(F) = \int_0^T \|y(t) - CZ_F(t)\|_Y^2 \, dt + \frac{\epsilon}{2}\|F\|_{\mathcal{V}}^2. \tag{5.26}$$

Firstly, we notice that problem (5.26) admits a unique solution $F_\epsilon \in \mathcal{V}$ since $\mathcal{J}$ is clearly coercive, continuous and strictly convex. The aim is now to show that the solution F_ϵ is one of the variable involved in a mixed formulation so that studying the optimal control problem amounts to studying the behaviour of the mixed formulation, which is usually easier. The idea is to match the framework developed in Chapter 4. To prove such result, we rely on [58, Corollary 2.41, Page 103]. Let us prove the inf-sup condition hypothesis to apply this result.

Proposition 5.3.5. *The bilinear form b satisfies the following inf-sup condition*

$$\inf_{(q,p)\in\mathcal{V}} \sup_{(u,v)\in W_0} \frac{b((u,v),(q,p))}{\|(u,v)\|_W \|(q,p)\|_{\mathcal{V}}} \geq \beta,$$

Proof. We need to show that there exists $\beta > 0$ such that for all $(p,q) \in \mathcal{V}$:

$$\sup_{(u,v)\in W_0} \frac{b((u,v),(q,p))}{\|(u,v)\|_W} \geq \beta\|(q,p)\|_{\mathcal{V}}.$$

For all $(q,p) \in V$, let us introduce the unique solution $(u^*, v^*) \in W_0$ to the weak formulation

$$b((u^*,v^*),(\tilde{q},\tilde{p})) = ((q,p),(\tilde{q},\tilde{p}))_{\mathcal{V}}, \quad \forall (\tilde{q},\tilde{p}) \in \mathcal{V} \tag{5.27}$$

Thanks to Theorem 5.3.1, we know that there exists a unique such pair (u^*, v^*) and there exists $C > 0$ such that:

$$\|(u^*,v^*)\|_W \leq C\|(q,p)\|_{\mathcal{V}}. \tag{5.28}$$

Using (5.27) and (5.28) we get that b satisfies the previous inf-sup condition for $\beta = 1/C$. □

Remark 5.3.5. *It is important to notice that the inf-sup condition of the previous proposition is different from the one proved in Theorem 1.3.4 as the spaces have been swapped in the inf-sup. In this sense, we call the inf-sup condition of Proposition 5.3.5 an inverse inf-sup condition.*

With the additional coercivity of the functional $\mathcal{J}$, the Proposition 5.3.5 allows us to show, as in Chapter 4, that seeking a solution F_ϵ of (5.26) amounts to seek F in the following mixed formulation: *Find $Z = (u,v,F) \in W_0 \times \mathcal{V}$ and $P = (p,q) \in \mathcal{V}$ such that for all $\tilde{Z} \in W_0 \times \mathcal{V}$ and $\tilde{P} \in \mathcal{V}$:*

$$\begin{cases} j(Z,\tilde{Z}) + d(\tilde{Z},P) = \ell(\tilde{Z}) \\ d(Z,\tilde{P}) = 0, \end{cases} \tag{5.29}$$

with

$$\begin{cases} j(Z,\tilde{Z}) = \epsilon(F,\tilde{F})_{\mathcal{V}} + \displaystyle\int_0^T (C(u,v),C(\tilde{u},\tilde{v}))_Y \, dt \\ d(Z,P) = b((u,v),(p,q)) - (F,(p,q))_{\mathcal{V}} \\ l(Z) = \displaystyle\int_0^T (y,C(u,v))_Y \, dt. \end{cases}$$

To prove the equivalence between the optimal control problem and the mixed formulation, we would proceed similarly as in Chapter 4. To do so, we would need to justify the existence and uniqueness of the adjoint state P. We would have proceeded as in Chapter 4, here we only gave details on the original part, namely the inverse inf-sup condition.

5.3.3 The complete model as an hyperbolic Friedrichs system

The goal of this section is now to show that the complete model can be recast in the previously presented abstract framework. We then solve the optimal control problem of source identification within this framework, with a particular focus on the numerical implementation. The main theoretical result of the section is to show that solving the optimal control problem amounts to solving a mixed variational formulation so that the same type of argumentation as in Chapter 4 could be done. We recall here all the operators and functional spaces introduced in [50] to recast the problem (1.1) in a Friedrichs system as presented previously. Let us first introduce two Hilbert spaces:

$$\mathcal{H} = L^2(\Omega)^d, \quad \mathcal{G} = L^2(\Omega) \times L^2(\Gamma_0),$$

where d is the dimension of the space. The space $\mathcal{G}$ is endowed with the following norm

$$\|(\varphi,\gamma)\|_{\mathcal{G}}^2 = \frac{1}{c^2}\|\varphi\|_{L^2(\Omega)}^2 + \frac{1}{g}\|\gamma\|_{L^2(\Gamma_0)}^2.$$

and $\mathcal{H}$ is endowed with the canonical L^2 norm. Let us recall that $\mathcal{V} = L^2(0,T;\mathcal{H}) \times L^2(0,T;\mathcal{G})$. Let us now introduce

$$D(G) = \left\{ U \in H(\mathrm{div},\Omega) \mid \mathrm{U} \cdot \nu|_{\Gamma_0} \in \mathrm{L}^2(\Gamma_0) \right\},$$

and

$$\forall U \in D(G), \; GU = (c^2 \mathrm{div} U, -gU \cdot \nu|_{\Gamma_0})^T,$$

where ν represents the outward unit normal to the domain Ω. We have that $D(G)$ is dense in $\mathcal{H}$ and G is closed. One can show, as in [50], that

$$D(G^*) = \left\{ V \in (\varphi, \gamma)^T \in \mathcal{G},\ \varphi \in H^1(\Omega), \gamma = \varphi|_{\Gamma_0} \right\},$$

and

$$\forall V = (\varphi, \gamma)^T \in D(G^*),\ G^*V = -\nabla\varphi.$$

In the same way, $D(G^*)$ is dense in $\mathcal{G}$. We then consider the problem: *For* $\chi \in L^2(0,T; H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))$, *find* $(U,V) \in \mathcal{V}$ *such that*

$$\begin{cases} \partial_t U - G^*V = 0, \\ \partial_t V + GU = B\displaystyle\int_0^t \chi(s)\ ds, \\ U(0) = V(0) = 0. \end{cases} \tag{5.30}$$

with $B\chi = GG^*(R_{-H}\chi, 0)^T + c^2(R_{-H}\chi, 0)^T \in \mathcal{G}$ for all $\chi \in H^{-\frac{1}{2}}(\Gamma_{-H+\theta})$ and R_{-H} the lifting operator introduced in (3.10). This lifting operator B is a slight adaptation of the one introduced in Chapter 3 in the semigroup framework. Let us now show that if $V = (\varphi, \gamma)^T$ is a solution of (5.30) then φ is a solution of (1.1). The proof of the converse would work the same way. Let us consider $(U,V) \in \mathcal{V}$ solution of (5.30). Let us first notice that $V \in D(G^*)$. Taking the time derivative of the second line of (5.30) gives us that V satisfies for almost all $t \in (0,T)$:

$$\partial_t^2 V + GG^*V = B\chi. \tag{5.31}$$

From (5.31), we get that for all $\tilde{V} = (\tilde{\varphi}, \tilde{\gamma})^T \in D(G^*)$ and for almost all $t \in (0,T)$

$$\left(\partial_t^2 V, \tilde{V}\right)_{\mathcal{G}} + \left(G^*V, G^*\tilde{V}\right)_{\mathcal{H}} = (B\chi, \tilde{V})_{\mathcal{G}}. \tag{5.32}$$

Then, we get that for all $\tilde{\varphi} \in H^1(\Omega)$ and for almost all $t \in (0,T)$

$$\frac{1}{c^2}\left(\partial_t^2\varphi, \tilde{\varphi}\right)_{L^2(\Omega)} + \frac{1}{g}\left(\partial_t^2\varphi|_{\Gamma_0}, \tilde{\varphi}|_{\Gamma_0}\right)_{L^2(\Gamma_0)} + (\nabla\varphi, \nabla\tilde{\varphi})_{L^2(\Omega)} = (\nabla R_{-H}\chi, \nabla\tilde{\varphi})_{L^2(\Omega)} + (R_{-H}\chi, \tilde{\varphi})_{L^2(\Omega)}.$$

It follows, using the definition of $R_{-H}\chi$, that for all $\tilde{\varphi} \in H^1(\Omega)$ and for almost all $t \in (0,T)$

$$\frac{1}{c^2}\left(\partial_t^2\varphi, \tilde{\varphi}\right)_{L^2(\Omega)} + \frac{1}{g}\left(\partial_t^2\varphi|_{\Gamma_0}, \tilde{\varphi}|_{\Gamma_0}\right)_{L^2(\Gamma_0)} + (\nabla\varphi, \nabla\tilde{\varphi})_{L^2(\Omega)} = \langle \chi, \tilde{\varphi} \rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})}.$$

Choosing the test function $\tilde{\varphi}$ in the right way gets us immediately that φ satisfies (1.1). One can also easily deduce that $V(0) = \partial_t V(0) = 0$, so that $\varphi(0) = \partial_t\varphi(0) = 0$. We have finally managed to recast the problem (1.1) in the problem (5.30). Finally, using the previous abstract framework, there exists a unique solution $(U,V) \in W_0$ such that, for all $(q,p) \in \mathcal{V}$,

$$b((U,V),(q,p)) = \left(B\int_0^t \chi(s)\ ds, p\right)_{L^2(0,T;\mathcal{G})}, \tag{5.33}$$

with b being the bilinear form introduced in the previous section.

Remark 5.3.6. *It is important to notice that in (5.33), the right hand side cannot be written as*

$$\left\langle \int_0^t \chi(s)\ ds, p \right\rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})},$$

since the test function $p \in L^2(\Omega)$. *This more handful form will be used when it comes to discretising the equation (5.33) as the tests functions will then have enough regularities.*

Let us now specify the optimal control approach used to solve the tsunami inverse problem. To simplify the future implementation of the method we consider a measurement $y \in L^2(0,T;L^2(\Gamma_0))$ of $\varphi|_{\Gamma_0}$ which corresponds to the antiderivative of the free surface s, that is

$$\varphi|_{\Gamma_0} = -g \int_0^t s(\tau)\ d\tau.$$

We introduce the observation operator $C : \mathcal{H} \times \mathcal{G} \to L^2(\Gamma_0)$ such that for all $(U,V) = ((\psi_1,\psi_2),(\varphi,\gamma)) \in \mathcal{H}\times\mathcal{G},\ C(U,V) = \gamma$. We aim to solve the following problem, for $\epsilon > 0$

$$\inf_{\chi \in L^2(0,T;L^2(\Gamma_{-H+\theta}))} \mathcal{J}(\chi) := \int_0^T \frac{1}{2}\|y(t) - CZ(t)\|^2_{L^2(\Gamma_0)} + \frac{\epsilon}{2}\|\chi(t)\|^2_{L^2(\Gamma_{-H+\theta})}\ dt, \tag{5.34}$$

where $Z = (U,V)$ is the unique solution in W_0 of (5.33). The previous problem admits a unique solution χ_ϵ since $\mathcal{J}$ is continuous, strictly convex and coercive. The previous analysis in the abstract case shows immediately that seeking χ_ϵ the solution to (5.34) amounts to find χ solution of the mixed formulation:
Find $Z = (U,V,\chi) \in W_0 \times L^2(0,T;L^2(\Gamma_{-H+\theta}))$ *and* $P = (q,p) \in \mathcal{V}$ *such that for all* $\tilde{Z} \in W_0 \times L^2(0,T;L^2(\Gamma_{-H+\theta}))$ *and* $\tilde{P} \in \mathcal{V}$*:*

$$\begin{cases} j(Z,\tilde{Z}) + d(\tilde{Z},P) = \ell(\tilde{Z}) \\ d(Z,\tilde{P}) = 0. \end{cases} \tag{5.35}$$

with

$$\begin{cases} j(Z,\tilde{Z}) = \displaystyle\int_0^T \epsilon(\chi,\tilde{\chi})_{L^2(\Gamma_{-H+\theta})} + (C(U,V),C(\tilde{U},\tilde{V}))_{L^2(\Gamma_0)}\ dt \\ d(Z,P) = b((U,V),(q,p)) - (B\displaystyle\int_0^t \chi(s)\ ds, p)_{L^2(0,T;\mathcal{G})} \\ l(Z) = \displaystyle\int_0^T (y, C(U,V))_{L^2(\Gamma_0)}\ dt. \end{cases}$$

Remark 5.3.7. *Here by simplicity, we have considered that the operator C corresponds to the observation of γ on the whole surface Γ_0. In the following section, we will also consider partial measurements on the surface such that C gives γ only on a subset of Γ_0. This setup is more realistic as it is difficult to get a measurement on the whole surface of the ocean*

5.3.4 Discretization issues and chosen approach

The last part of this section is now dedicated to numerical experiments. Let us first present the numerical schemes used to solve the previous optimal control problem. In this section, we present the method that we actually implemented to solve such an optimal control problem and more particularly the discretisation we chose, based on a discretize-then-optimize approach. If the implementation is rather straightforward, the analysis of such an approach is quite technical as seen in Chapter 4. This numerical analysis will not be conducted in this thesis as it lacks a fundamental result, which is the discrete inf-sup condition within the previous framework. The difficulty stems from the lack of a discrete energy identity within the right norms. Let us just present the discretisation we chose for the discretize-then-optimize approach. On the one hand, the time discretisation is based on a classical finite difference scheme, where the time interval $(0,T)$ is divided into N sub intervals $(n\tau,(n+1)\tau)$, with $\tau = T/N$ and $n = 0,\cdots,N-1$. On the other hand, we introduce some finite conformal dimensional spaces : $U_h \subset L^2(\Gamma_{-H+\theta}), Y_h \subset L^2(\Gamma_0)$ and $W_h \subset D(G)\times D(G^*)$ with h being the space step of the mesh. Based on such discretisation, instead of χ, Z and y we consider $\chi_h^\tau = (\chi_h^n)_{n=1\cdots N} \in U_h^N, y_h^\tau = (y_h^n)_{n=1\cdots N} \in Y_h^N$ and $Z_h^\tau = (Z_h^n)_{n=0\cdots N} \in W_h^{N+1}$. In regard of the approach proposed in Chapter 4, the following backward Euler scheme is the one we should implement to obtain the same numerical analysis results.

$$\inf_{\chi_h^\tau \in U_h^N} \mathcal{J}_h^\tau(\chi_h^\tau) = \sum_{n=1}^N \frac{\tau}{2}\|y_h^n - C_h Z_h^n\|^2_{L^2(\Gamma_0)} + \frac{\tau\epsilon}{2}\|\chi_h^n\|^2_{L^2(\Gamma_{-H+\theta})},$$

with $Z_h^n = (U_h^n, V_h^n)$ solution in W_h^{N+1} of:

$$\begin{cases} \dfrac{U_h^n - U_h^{n-1}}{\tau} - G_h^* V_h^n = 0, \quad n = 1, \cdots, N \\ \dfrac{V_h^n - V_h^{n-1}}{\tau} + G_h U_h^n = -B\zeta_h^n, \quad n = 1, \cdots, N \\ U_h^0 = V_h^0 = 0, \end{cases}$$

with $\dfrac{\zeta_h^n - \zeta_h^{n-1}}{\tau} = -\chi_h^n, n = 1, \cdots, N$. Here, G_h corresponds to the restriction of G to the discrete spaces. To bound the domain, we have imposed homogeneous Dirichlet conditions on $\Sigma_{\pm R}$ with a sufficiently large R. Adding Dirichlet conditions allows us not to add an operator on the transverse boundaries which would not fit the previous abstract framework. The equivalence between problem (5.34) and the mixed formulation (5.35) given by [58, Corollary 2.41, Page 103] is the key to get a rigorous numerical analysis of the optimal control problem as in Chapter 4. The remaining work would be to get the error estimate, which would require the discrete inf-sup condition and the estimate of the projection operator P_h introduced in Chapter 4.
In practice, the previous discretisation scheme tends to be too expensive to implement. Indeed, leading to very large inversion in realistic case. It would be preferable to use state of the art solvers based on $\mathbb{Q}_k$ finite elements and inversion free approach with the fast explicit library OndoMathX. This implies to use a different scheme which unfortunately cannot enter the previous framework. we choose to implement a different time scheme. This cannot be analysed within the presented framework but allows for realistic setups. Let us consider the resulting discretization:

$$\inf_{\chi_h^\tau \in U_h^N} \mathcal{J}_h^\tau(\chi_h^\tau) = \sum_{n=1}^{N} \frac{\tau}{2} \|y_h^n - C_h Z_h^n\|^2_{L^2(\Gamma_0)} + \sum_{n=1}^{N-1} \frac{\tau\epsilon}{2} \|\chi_h^n\|^2_{L^2(\Gamma_{-H+\theta})}, \tag{5.36}$$

with Z_h^n being the solution to the problem

$$\begin{cases} \dfrac{U_h^{n+\frac{1}{2}} - U_h^{n-\frac{1}{2}}}{\tau} - G_h^* V_h^n = 0, \quad n = 1, \cdots, N \\ \dfrac{V_h^n - V_h^{n-1}}{\tau} + G_h U_h^{n-\frac{1}{2}} + D_h V_h^{n-\frac{1}{2}} = -B_h \zeta_h^n, \quad n = 1, \cdots, N \\ U_h^0 = V_h^0 = 0, \end{cases} \tag{5.37}$$

with $U_h^{n+\frac{1}{2}} = \frac{U_h^{n+1} + U_h^n}{2}$. In order to numerically solve this problem, we have added artificial viscosity conditions on transverse boundaries $\Sigma_{\pm R}$ through the operator D_h. While the scheme (5.37) is still implicit, as it forces to invert $Id + D_h$, we do not have to invert G_h and G_h^* anymore.It is readily seen that an equivalent scheme to (5.36)(5.37), which is easier to implement in practice, is:

$$\inf_{\chi_h^\tau \in U_h^N} \mathcal{J}_h^\tau(\chi_h^\tau) = \sum_{n=1}^{N} \frac{\tau}{2} \|y_h^n - \gamma_{\Gamma_0} \varphi_h^n\|^2_{L^2(\Gamma_0)} + \sum_{n=1}^{N-1} \frac{\tau\epsilon}{2} \|\chi_h^n\|^2_{L^2(\Gamma_{-H+\theta})},$$

with V_h^n is the solution to the problem

$$\begin{cases} \dfrac{V_h^{n+1} - 2V_h^n + V_h^{n-1}}{\tau^2} + A_h V_h^n + D_h \dfrac{V_h^{n+1} - V_h^{n-1}}{2\tau} = B_h \chi_h^n, \quad n = 0, \cdots, N-1 \\ V_h^0 = V_h^{-1} = 0, \end{cases} \tag{5.38}$$

with $A_h = G_h G_h^*$ and $V_h^n = (\varphi_h^n, \gamma_{\Gamma_0} \varphi_h^n)^T$. Introducing $\overrightarrow{\varphi}_h^n$, $\overrightarrow{\chi}_h^n$, the finite elements degree of freedom vectors at time $t = n\tau$, let us present the actual scheme we implement at the discrete level for all $n = 0, \cdots, N-1$:

$$\begin{cases} (\mathbb{M} + \mathbb{M}_{\Gamma_0}) \dfrac{\overrightarrow{\varphi}_h^{n+1} - 2\overrightarrow{\varphi}_h^n + \overrightarrow{\varphi}_h^{n-1}}{\tau^2} + \mathbb{K} \overrightarrow{\varphi}_h^n + \mathbb{M}_{\Sigma_{\pm R}} \dfrac{\overrightarrow{\varphi}_h^{n+1} - \overrightarrow{\varphi}_h^{n-1}}{2\tau} = \mathbb{M}_{\Gamma_{-H+\theta}} \overrightarrow{\chi}_h^n, \\ \overrightarrow{\varphi}_h^0 = \overrightarrow{\varphi}_h^{-1} = 0, \end{cases}$$

with $\mathbb{M}, \mathbb{M}_{\Gamma_0}, \mathbb{M}_{\Gamma_{-H+\theta}}, \mathbb{M}_{\Sigma_{\pm R}}$ being the mass matrices on $\Omega, \Gamma_0, \Gamma_{-H+\theta}, \Sigma_{\pm R}$ respectively and $\mathbb{K}$ being the rigidity matrix. In the library OndoMathX, we use quadratures rule to compute the mass matrices so that these matrices are diagonal. Therefore, there is no matrix inversion to compute $\overrightarrow{\varphi}_h^{n+1}$. This scheme is therefore explicit. To find the minimizer χ_h^τ of the previous functional, we use the same approach as in Chapter 4. We compute the exact gradient using a discrete adjoint, we finally apply a L-BFGS minimization algorithm. We now specify the adjoint state p_h^τ associated with the functional $\mathcal{J}_h^\tau$ and the expression of the gradient. Introducing $\overrightarrow{p}_h^n$ in the same way as before, we get that the adjoint state satisfies for all $n = 2, \cdots, N$

$$\begin{cases} (\mathbb{M} + \mathbb{M}_{\Gamma_0}) \dfrac{\overrightarrow{p}_h^{n+1} - 2\overrightarrow{p}_h^n + \overrightarrow{p}_h^{n-1}}{\tau^2} + \mathbb{K}\overrightarrow{p}_h^n - \mathbb{M}_{\Sigma_{\pm R}} \dfrac{\overrightarrow{p}_h^{n+1} - \overrightarrow{p}_h^{n-1}}{2\tau} = \mathbb{M}_{\Gamma_0}(\overrightarrow{y}_h^n - \gamma_{\Gamma_0}\overrightarrow{\varphi}_h^n), \\ \overrightarrow{p}_h^N = \overrightarrow{p}_h^{N+1} = 0. \end{cases}$$

Finally, we can compute the gradient of $\mathcal{J}_h^\tau$ as

$$\nabla \mathcal{J}_h^\tau(\overrightarrow{\chi}_h^n) = \tau\left(\epsilon\overrightarrow{\chi}_h^n - \gamma_{\Gamma_{-H+\theta}}\overrightarrow{p}_h^n\right), \quad n = 1, \cdots, N-1,$$

Remark 5.3.8. *We notice that the choice of the discretisation scheme is such that the adjoint state is almost identical as the forward state. This result is particularly useful in practice as it speeds up the computation in the library OndoMathX. Indeed, both schemes are explicit schemes which allow us to solve each iteration without having to invert any matrices except diagonal ones.*

5.3.5 Numerical experiments

This section presents numerical results in a "realistic" domain using the OndoMathX library. To do so, we choose a source χ and we compute the associated solution using the direct problem. Our goal is now to retrieve this known source χ from the only knowledge of this surface measurement and to compare to the exact value of χ. Let us choose

$$\chi(x,t) = f(x)g(t),$$

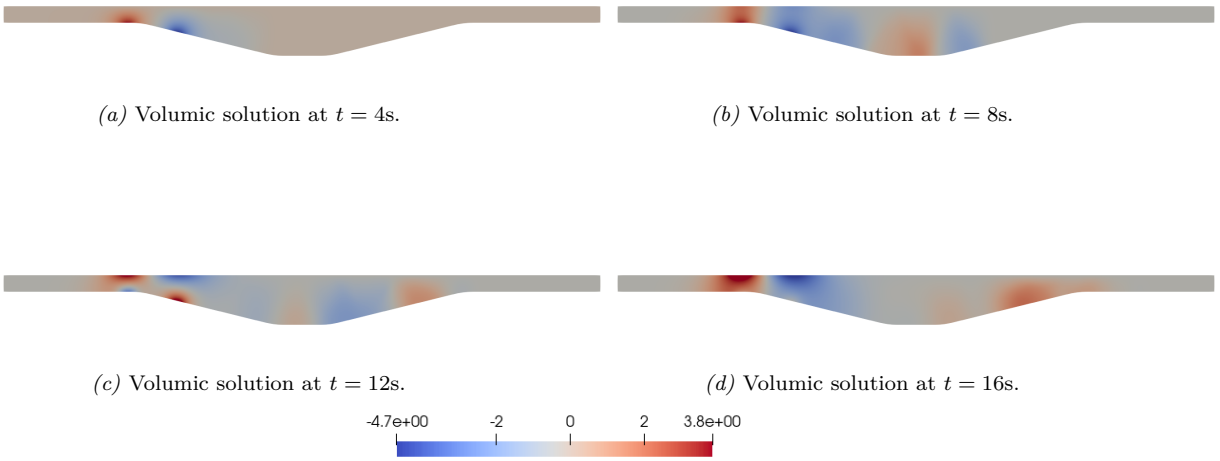
with

$$g(t) = 0.016t(2 - (0.04t^2 - 1))e^{-0.02t^2},$$

and

$$f(x) = e^{\frac{10}{(x-5)^2-1}+10}\mathbb{1}_{]4,6[}(x) - e^{\frac{10}{(x-7)^2-1}+10}\mathbb{1}_{]6,8[}(x),$$

so that for all t, χ has a zero mean in space. The constants are being chosen with respect to the constant of the problem. Let us emphasize that we implement a dimensionless version of the problem such that one unity of space corresponds to $1.5km$ in reality. Let us first present results of simulation for the direct problem. We use the previous source term χ over a time window of 10 minutes for the direct problem, that is $T = 600s$. The domain ranges from $x = 0$ to $x = 24$ which corresponds to $36km$. The depth is 2, which corresponds to $3km$. The shape of the domain is made clear on Figure 5.2 as we present the propagation of waves in the whole fluid. Figure 5.2 and 5.1 shows the resulting volumic propagation associated with the previous source χ in the ocean. The chosen domain represents what could happen in reality: the earthquake occurs next to an oceanic rift, the tsunami propagates on a deep ocean until hitting shallower depths. This domain is compatible with the observation of both the acoustic and gravity effects.
As presented earlier, the complete model accounts for two different phenomena. In Figure 5.2 and Figure 5.1, we highlight these two phenomena numerically. On Figure 5.2, we clearly see wave packets propagating on the surface of the domain corresponding to the gravity waves. On Figure 5.1, we see at an early stage after the earthquake, the propagation in the volume of fast acoustic waves. To put the emphasis on these two phenomena, we need a really refined mesh to account for both high frequency acoustic waves and high wavelength gravity waves. Thanks to the specialised OndoMathX library, we used highly refined meshes for this domain and high order interpolation polynoms $\mathbb{Q}_4$ in system (5.37). On Figure 5.3, we present snapshots of the free surface of the ocean, which allow us to put the emphasis even more on the wave packets propagating at the surface. As we lack some real measurements, the free surface presented on Figure 5.3 will represent our artificial



(a) Volumic solution at $t = 4$s.

(b) Volumic solution at $t = 8$s.

(c) Volumic solution at $t = 12$s.

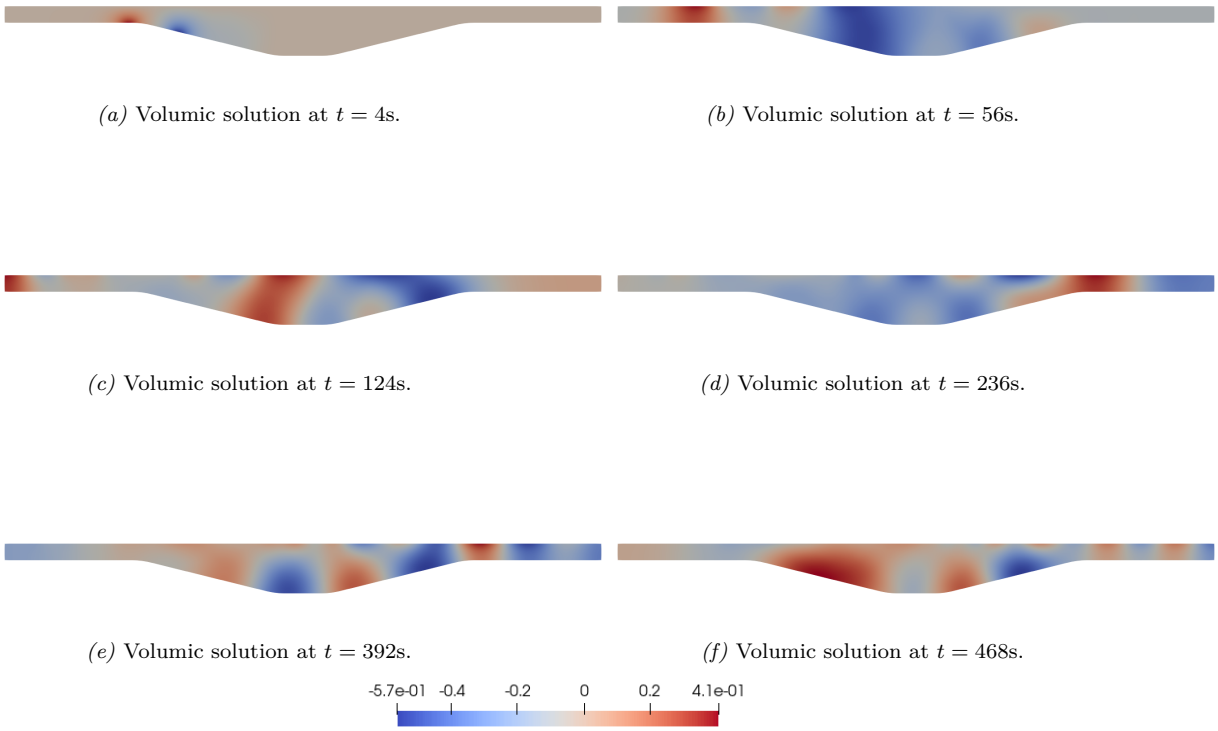
(d) Volumic solution at $t = 16$s.

Figure 5.1: Volumic solution at early stages after the earthquake simulated with the direct problem.

data for the inverse problems. This measurement will be eventually contaminated with noise using the same principle as in the previous chapters. Let us then present the parameters used for the inverse problems and let us present numerical results.

To limit the size of the inverse problem, we choose to retrieve the source term χ on a smaller time window $T = 200s$. The Figure 5.4 and Figure 5.5 show snapshots of the reconstruction without noise and with 1% of noise, respectively. Even if the reconstruction deteriorates as the noise on the observation increases, the results are still really satisfactory in regard of the size of the domain. It is important to notice that these two first reconstruction experiments have been done taking into account the observation of all the surface at any time. Considering that such observations are obviously unrealistic, in the following we present reconstruction of the source term χ only using partial observations on the surface. It is also essential to notice that we consider satellites observations, so that we have access to the "continuous" data on the surface and not only a discrete sample given by buoys. In Figure 5.6, we have reconstructed the source χ on the bottom by a partial measurement on the surface. Instead of getting a measurement on the whole surface from 0 to 24, we had only considered the measurement from 3 to 17. The efficient reconstruction made in Figure 5.6 shows that in a case that accounts for "real life" measurements, our method manage to reconstruct the deformation of the bottom. As said before, these "real life" measurements consider that we are able to retrieve continuous observations from a satellite on a large interval of the sea surface. In reality, the noise contamination on observations would probably be larger and the observations would have to be reinterpolated a posteriori.

Finally, since our goal is to predict the behaviour of the tsunami in the future we present a result of reconstruction of the free surface on Figure 5.7 with no noise and on Figure 5.8 with 1% noise. As previously said, the reconstruction of the bottom deformations χ have been made on a smaller time frame of $T = 200$s. On Figure 5.7 and Figure 5.8, we show the free surface at later times until 600s from the reconstruction of χ showed in Figure 5.4. The satisfying result on Figure 5.7 and Figure 5.8 shows that we are able to predict the behavior of a tsunami in the future. These differents numerical experiments, while still being made in a simplified framework (no real data, knowledge of the support of the source χ and knowledge of the bathymetry), present an encouraging perspective in this domain of predicting tsunami propagation. It is indeed quite impressive to notice that even with noise on the measurements the prediction of the surface waves is almost perfect. However, these first results are still made in an optimal setup, as the noise on the measurements is still quite low.



(a) Volumic solution at $t = 4$s.

(b) Volumic solution at $t = 56$s.

(c) Volumic solution at $t = 124$s.

(d) Volumic solution at $t = 236$s.

(e) Volumic solution at $t = 392$s.

(f) Volumic solution at $t = 468$s.

Figure 5.2: Volumic solution over time simulated with the direct problem.

A new framework has finally been introduced, allowing the analysis of the optimal control approach without any additional parabolic regularisation. This Friedrich's framework has then been used numerically, without proper justification, to tackle large oceanic domains. While the mathematical analysis for this framework remains to be completed, the numerical results have been very satisfactory and have paved the way for future real-life applications. The rigorous analysis derived for the parabolic regularisation will need to be carried out within the Friedrich's framework to justify the obtained numerical results. We did not make this analysis by lack of time, however although the analysis is very technical and we still don't know exactly how to proceed, we have ideas we will need to investigate. These numerical results would also need to be complemented with simulations using real satellite observations. Furthermore, in the simulation we carried out in Friedrich's framework, we assume that the bathymetry of the ocean floor was already known. It would be interesting to couple the two inverse problems studied in this thesis: first solving for the ocean bathymetry, and secondly solving for the source at the bottom.

The models we have chosen in this thesis were linear and therefore since we assumed that the free surface measurements were located close to the support of the bottom displacement. In reality, the propagation of a tsunami is highly non-linear, especially as the waves approach the coast. Solving the same inverse problem using a non-linear model such as the Saint-Venant equations, and employing the same approaches, would be a real challenge. Beyond the study of the inverse problem, the non-linearity of the modelling would make the analysis of the direct problem quite technical. It is important to note that in the Saint-Venant equations the source term is played by an initial condition. In Chapter 1, we derived a logarithmic stability inequality for an optimal control problem. However, the norms used in this criterion do not coincide with those employed in Friedrich's previous framework, so this stability inequality is not useful in practice. To reconcile this inequality with the optimal control problem used in practice, one could devise another criterion

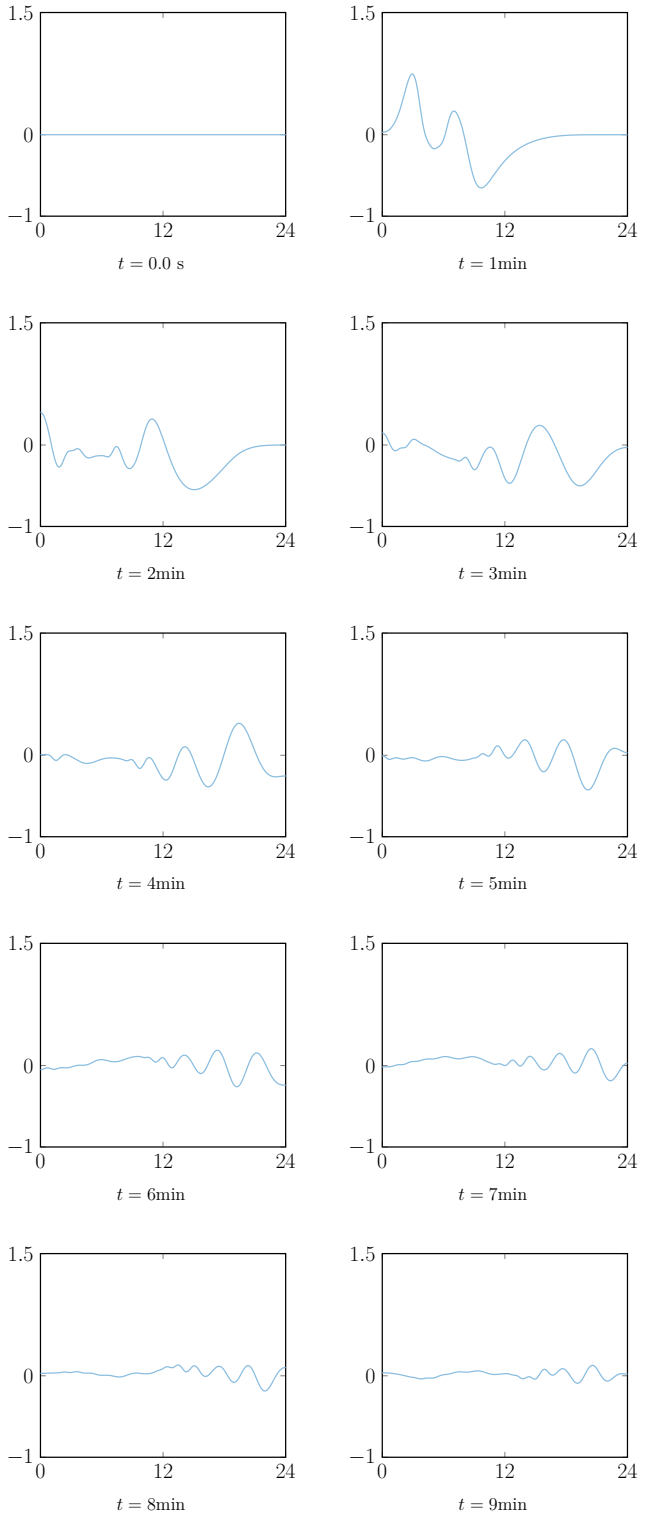


Figure 5.3: Measurement of the free surface on Γ_0.

that matches the inequality. Another possibility, using interpolation results on Sobolev spaces, would be to weaken the stability inequality so that it matches the criterion already used in practice.

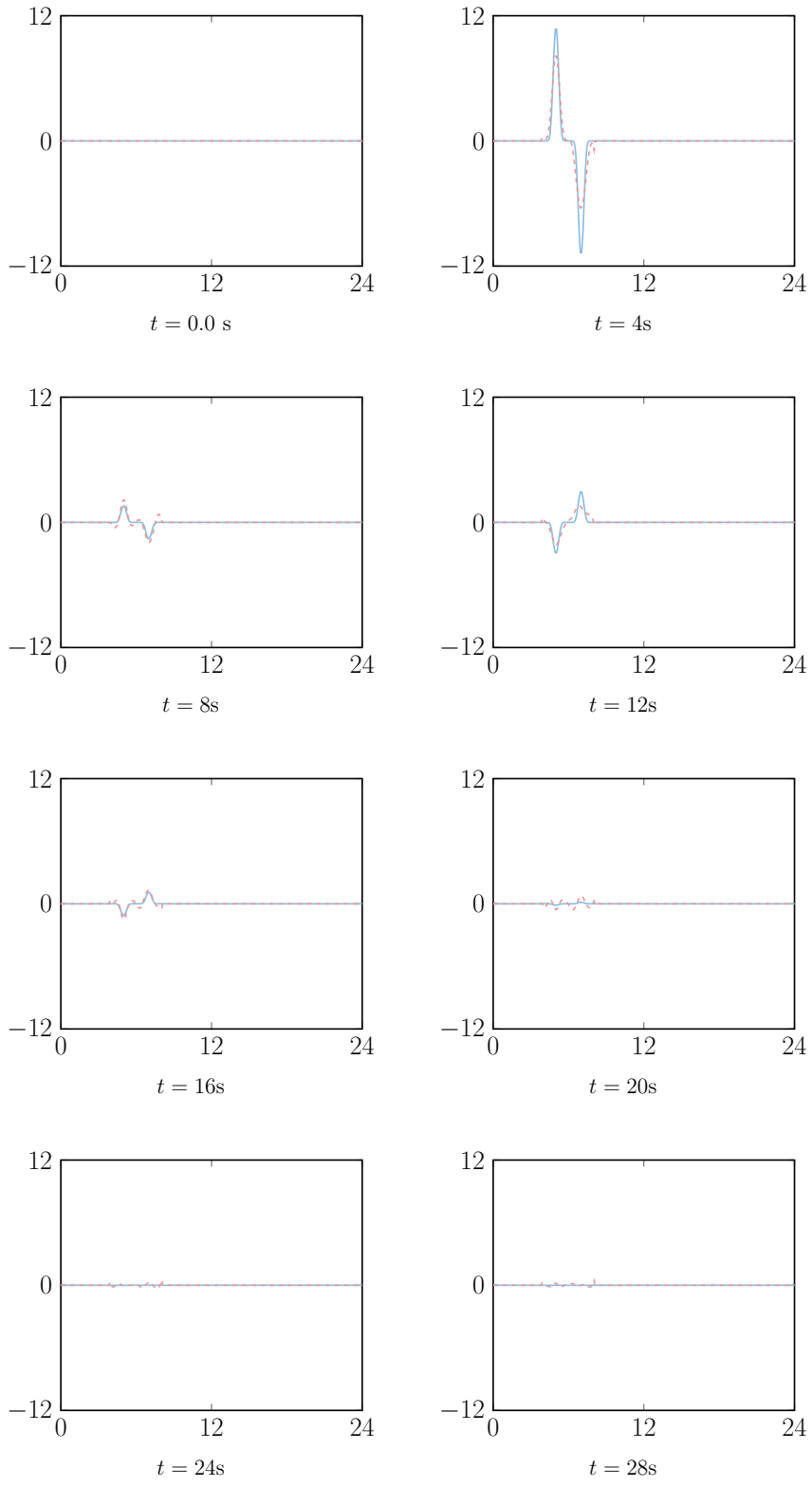


Figure 5.4: Reconstructed χ at different times. Continuous line : true χ. Dashed line : reconstructed χ.

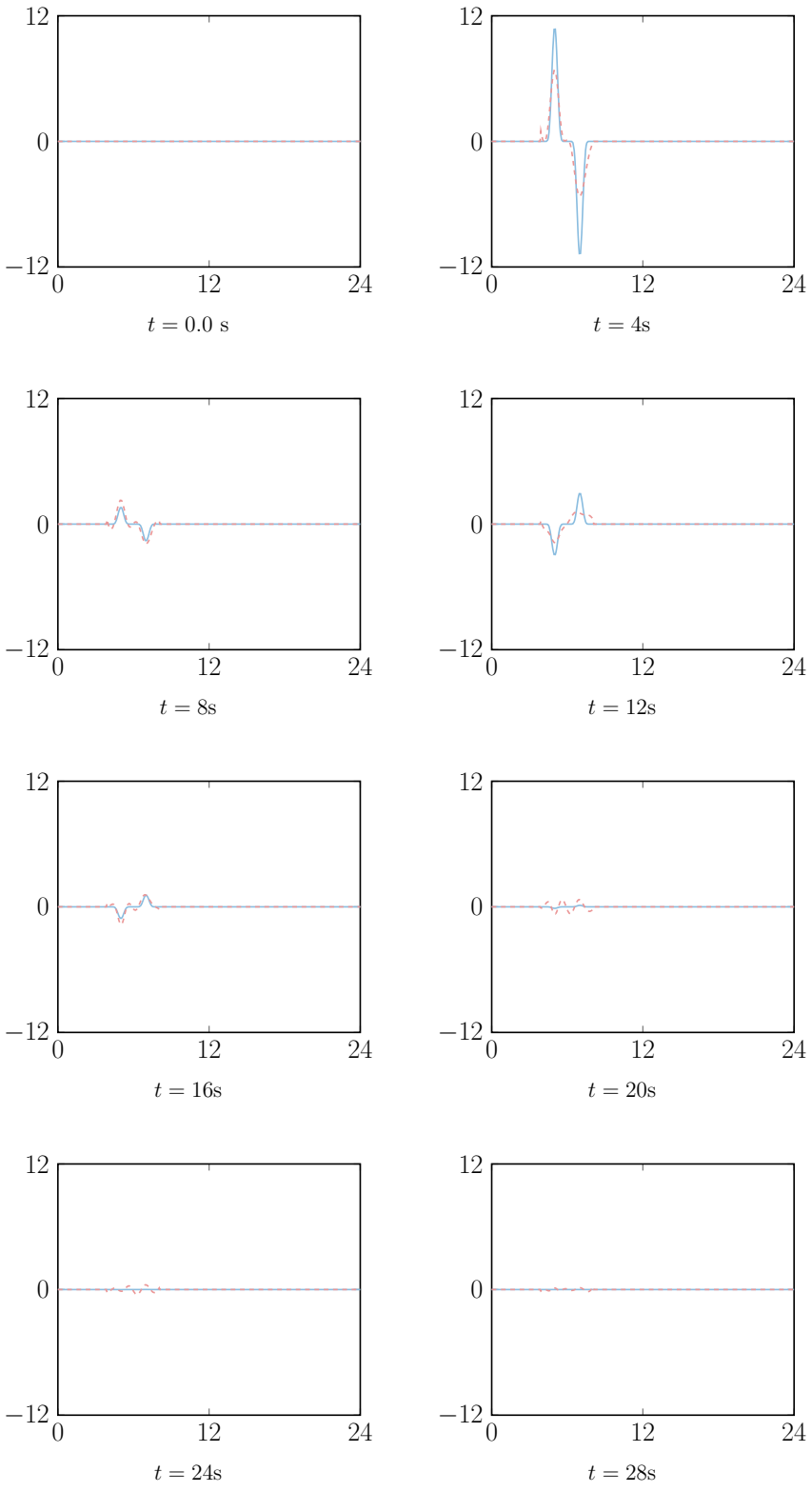


Figure 5.5: Reconstructed χ at different times with 1% of noise. Continuous line : true χ. Dashed line : reconstructed χ.

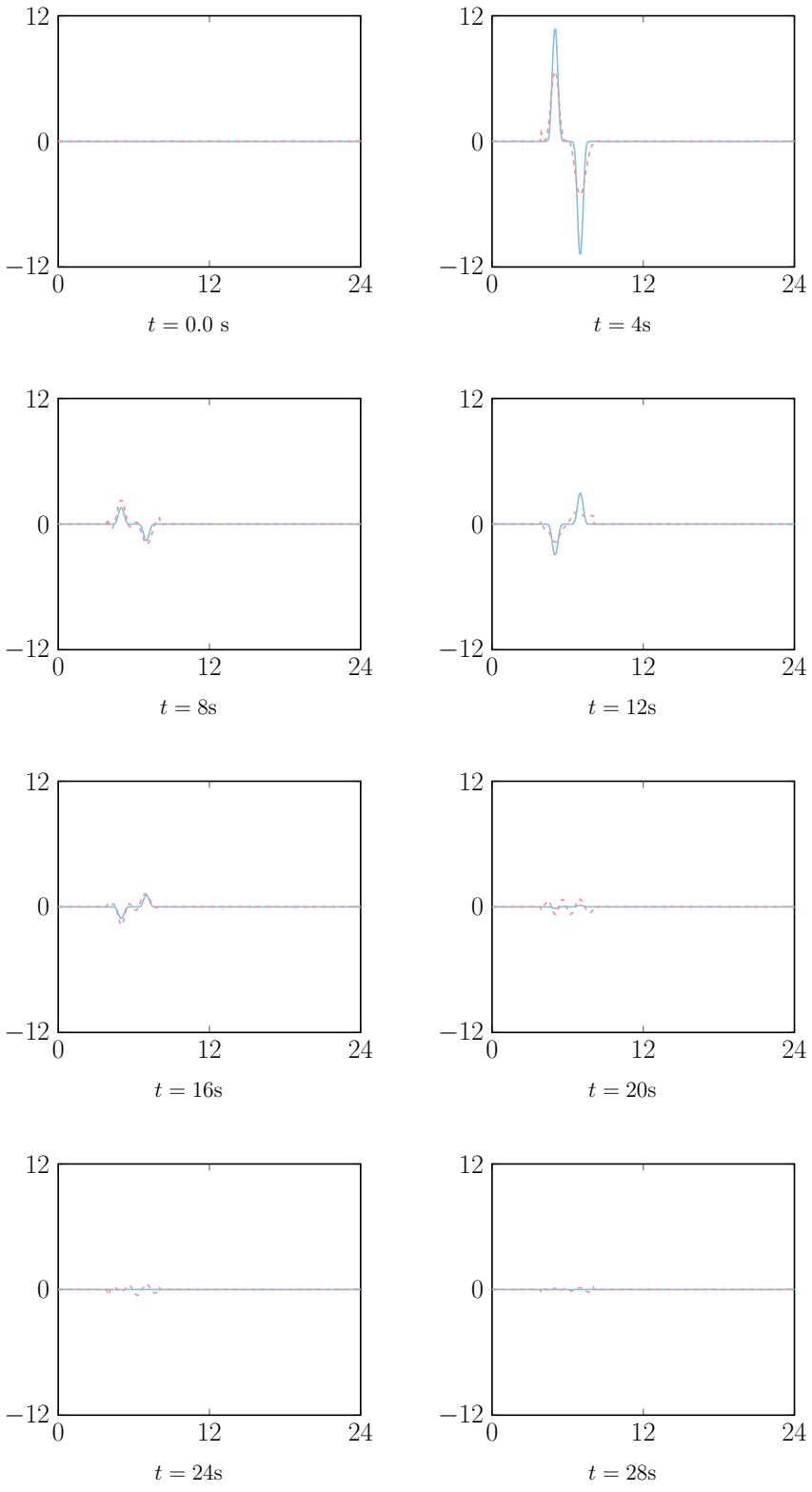


Figure 5.6: Reconstructed χ at different times with partial measurements. Continuous line : true χ. Dashed line : reconstructed χ.

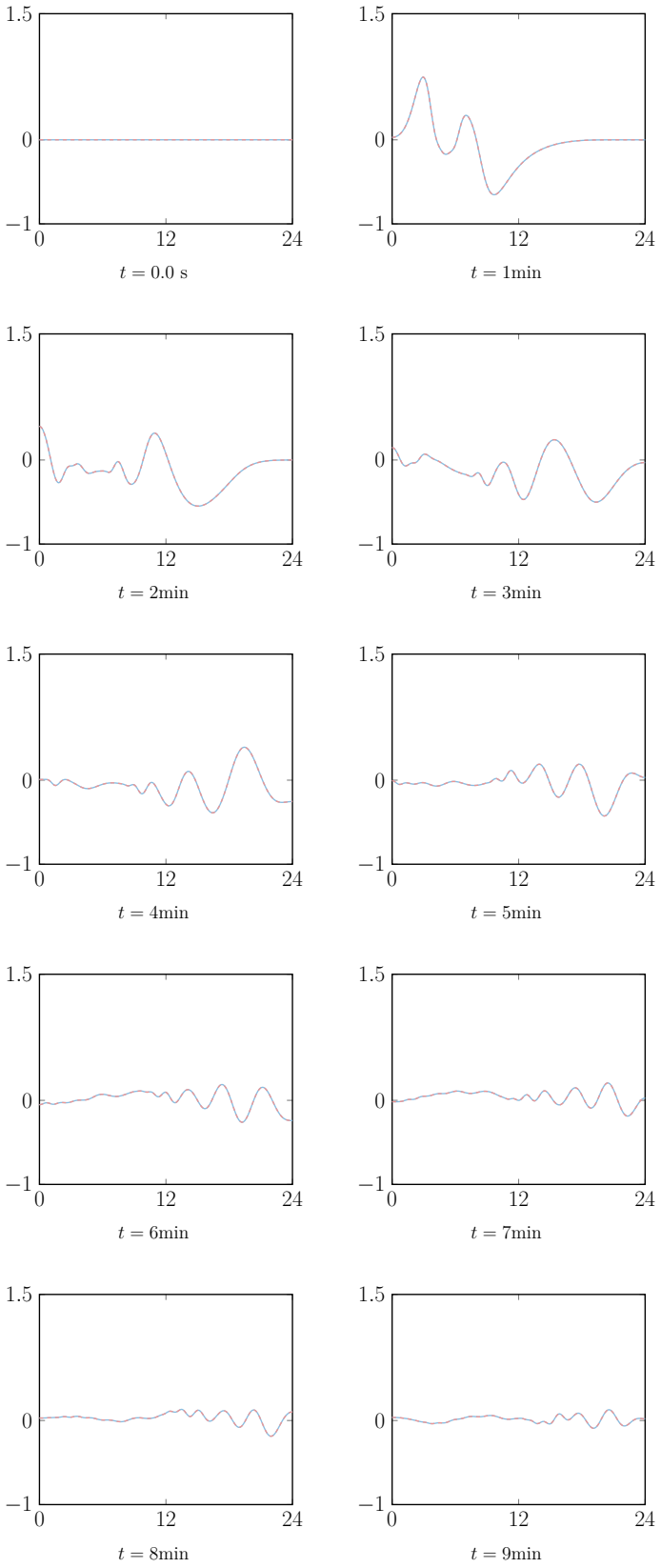


Figure 5.7: Reconstructed free surface at different times. Continuous line : measurement of the free surface. Dashed line : reconstructed free surface.

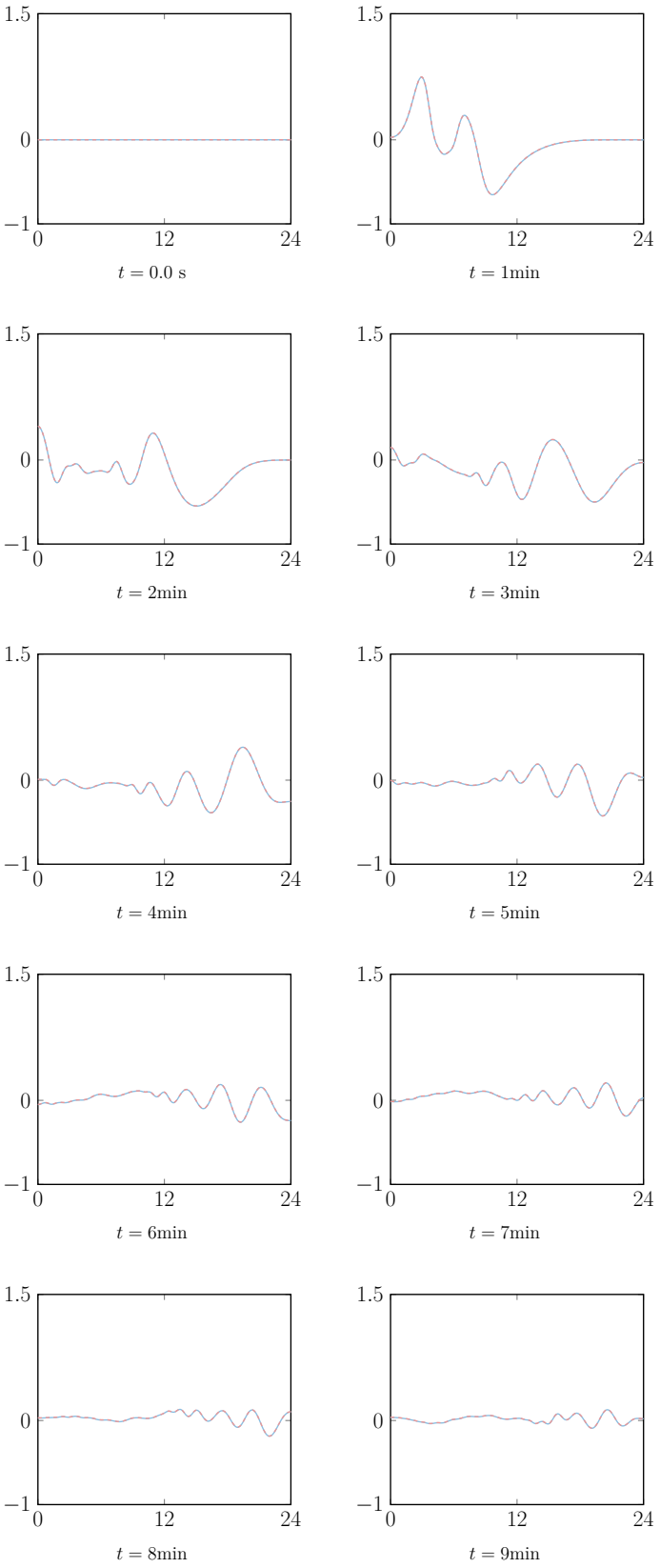


Figure 5.8: Reconstructed free surface at different times with 1% of noise. Continuous line : measurement of the free surface. Dashed line : reconstructed free surface.

Appendix A

Construction of first order variational solutions

A.1 Variational solutions for the tsunami problem

In Chapter 3, we derived two types of solutions for the tsunami problem : strong solutions in the semigroup sense and order 2 variational solutions. The strong solution is regular but requires high regularity on the source term χ whereas the order 2 variational solutions requires less regularity but the framework is less handful when it comes to optimal control. For the sake of completeness and to develop variational solutions that looks like the one describes in Chapter 4 but without parabolic regularisation, in the following Proposition, we introduce a third type of solution: the so-called first order variational solutions [90], which we construct from the previously defined strong solutions. The idea here is to build a solution that facilitates numerical implementation through a variational formulation while relaxing the regularity we ask for the source χ. We will finally show that even this new framework, that looks like the parabolic framework introduced in Chapter 4, does not match with the optimal control framework with which we worked. Let us first recall two Hilbert spaces previously introduced

$$\mathcal{H} = H^1(\Omega_\theta) \times L^2(\Omega_\theta) \times L^2(\Gamma_0), \quad \mathcal{V} = \left\{ (\varphi, \psi, s)^T \in H^1(\Omega_\theta) \times H^1(\Omega_\theta) \times L^2(\Gamma_0), \psi|_{\Gamma_0} + gs = 0 \right\}.$$

Proposition A.1.1. *Suppose that $\chi \in H^1(0,T; H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))$ and $\chi(0) = 0$. Then there exists a unique variational solution $Z = (\varphi, \psi, s)^T \in \mathcal{C}^0([0,T], \mathcal{H})$ such that*

$$Z(0) = 0,$$

that is, such that for all $\tilde{s} \in L^2(\Gamma_0)$:

$$\frac{d}{dt}(\varphi|_{\Gamma_0}(t), \tilde{s})_{L^2(\Gamma_0)} = -g(s(t), \tilde{s})_{L^2(\Gamma_0)} \text{ in } \mathcal{D}'(0,T), \tag{A.1}$$

and the following equation is satisfied in $\mathcal{D}'(0,T)$ for all $\tilde{Z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s})^T \in \mathcal{V}$:

$$\frac{d}{dt}(Z(t), \tilde{Z})_{\tilde{\mathcal{H}}} + \tilde{a}(Z(t), \tilde{Z}) = \left\langle \chi(t), \tilde{\psi} \right\rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})}, \tag{A.2}$$

with

$$(Z, \tilde{Z})_{\tilde{\mathcal{H}}} = (\varphi, \tilde{\varphi})_{L^2(\Omega_\theta)} + \frac{1}{c^2}(\psi, \tilde{\psi})_{L^2(\Omega_\theta)} + g(s, \tilde{s})_{L^2(\Gamma_0)},$$

and

$$\tilde{a}(Z, \tilde{Z}) = -(\psi, \tilde{\varphi})_{L^2(\Omega_\theta)} + (\nabla\varphi, \nabla\tilde{\psi})_{L^2(\Omega_\theta)}.$$

Remark A.1.1. *Solving (A.1) and (A.2) amounts to solve (1.1) in a weak sense. Indeed, if Z has enough regularity the two formulations are equivalent. In this sense, we call those solutions "variational".*

Proof. The proof relies on the construction of strong solutions carried out previously, together with a density argument. Let χ_n be a sequence of functions in $H^2([0,T], H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))$ converging to χ in $H^1([0,T], H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))$ such that $\chi_n(0) = 0$. Such a sequence exists by density. Then, according to Theorem 3.2.1, there exists a unique strong solution $z_n = (\phi_n, \psi_n, s_n)^T$ in $\mathcal{C}^1([0,T], \mathcal{H}) \cap \mathcal{C}^0([0,T], X)$ satisfying:

$$\begin{cases} \dot{z}_n = Az_n - D_{-H}\dot{\chi}_n + c^2 E D_{-H}\chi_n, \\ z_n(0) = 0. \end{cases} \tag{A.3}$$

By taking the inner product of the first equation of (A.3) with $\tilde{Z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s})^T \in \mathcal{V}$, we obtain in $\mathcal{D}'(0,T)$:

$$\begin{aligned}(\dot{\phi}_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} + \frac{1}{c^2}(\dot{\psi}_n(t), \tilde{\psi})_{L^2(\Omega_\theta)} + g(\dot{s}_n(t), \tilde{s})_{L^2(\Gamma_0)} &= (\psi_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} - (\nabla\phi_n(t), \nabla\tilde{\psi})_{L^2(\Omega_\theta)} \\ &\quad - (R_{-H}\dot{\chi}_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} + (R_{-H}\chi_n(t), \tilde{\psi})_{L^2(\Omega_\theta)}.\end{aligned}$$

It follows that, with $\varphi_n = \phi_n + R_{-H}\chi_n$, for all $\tilde{Z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s})^T \in \mathcal{V}$, we have in $\mathcal{D}'(0,T)$:

$$\begin{aligned}(\dot{\varphi}_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} + \frac{1}{c^2}(\dot{\psi}_n(t), \tilde{\psi})_{L^2(\Omega_\theta)} + g(\dot{s}_n(t), \tilde{s})_{L^2(\Gamma_0)} &= (\psi_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} - (\nabla\varphi_n(t), \nabla\tilde{\psi})_{L^2(\Omega_\theta)} \\ &\quad + \left\langle \chi_n(t), \tilde{\psi} \right\rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})}.\end{aligned} \tag{A.4}$$

We now aim to pass to the limit in equation (A.4). Let $\psi_{n,m} = \psi_n - \psi_m$, and we adopt the same notation for the other quantities. Applying the result of Proposition 1.3.2, we obtain the following energy identity for all $t \in [0,T]$:

$$\begin{aligned}\frac{1}{2c^2}\int_{\Omega_\theta} |\psi_{n,m}(t)|^2 \, dx + \frac{1}{2}\int_{\Omega_\theta} |\nabla\varphi_{n,m}(t)|^2 \, dx + \frac{1}{2}\int_{\Omega_\theta} \varphi_{n,m}(t)^2 \, dx + \frac{g}{2}\int_{\Gamma_0} |s_{n,m}(t)|^2 \, d\sigma \\ \leq K\|\chi_{n,m}\|^2_{H^1([0,T], H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))}.\end{aligned}$$

Since χ_n converges to χ in $H^1(0,T, H^{-\frac{1}{2}}(\Gamma_\theta))$, it follows that $(\chi_n)n \in \mathbb{N}$ is a Cauchy sequence in the same space. We then deduce that (φ_n) is a Cauchy sequence in $\mathcal{C}^0([0,T], H^1(\Omega_\theta))$, (ψ_n) in $\mathcal{C}^0([0,T], L^2(\Omega_\theta))$, and (s_n) in $\mathcal{C}^0([0,T], L^2(\Gamma_0))$. Hence, by completeness of the spaces involved, there exist $\varphi \in \mathcal{C}^0([0,T], H^1(\Omega_\theta))$, $\psi \in \mathcal{C}^0([0,T], L^2(\Omega_\theta))$, and $s \in \mathcal{C}^0([0,T], L^2(\Gamma_0))$ such that

$$\varphi_n \to \varphi \text{ in } \mathcal{C}^0([0,T], H^1(\Omega_\theta)), \quad \psi_n \to \psi \text{ in } \mathcal{C}^0([0,T], L^2(\Omega_\theta)), \quad s_n \to s \text{ in } \mathcal{C}^0([0,T], L^2(\Gamma_0))$$

The convergence of the inertial terms (i.e., the terms in $\dot{\varphi}_n, \dot{\psi}_n$, and $\dot{s}_n$) in (A.4) is established via integration in the sense of distributions: for $\Psi \in \mathcal{D}(0,T)$, we have, for example,

$$\begin{aligned}\int_0^T \int_{\Omega_\theta} \partial_t\psi_n(t,x)\tilde{\psi}(x)\Psi(t) \, dxdt &= -\int_0^T \int_{\Omega_\theta} \psi_n(t,x)\tilde{\psi}(x)\dot{\Psi}(t) \, dxdt \\ &\xrightarrow[n\to\infty]{} -\int_0^T \int_{\Omega_\theta} \psi(t,x)\tilde{\psi}(x)\dot{\Psi}(t) \, dxdt.\end{aligned}$$

Let us notice that

$$\partial_t\varphi_n = \psi_n. \tag{A.5}$$

Before passing to the limit in (A.4), we need to determine the limit of the following term:

$$(\psi_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} = (\psi_n(t), \tilde{\varphi})_{L^2(\Omega_\theta)} + (\nabla\psi_n(t), \nabla\tilde{\varphi})_{L^2(\Omega_\theta)}$$

$$= (\psi_n(t), \tilde{\varphi})_{L^2(\Omega_\theta)} + (\nabla \partial_t \varphi_n(t), \nabla \tilde{\varphi})_{L^2(\Omega_\theta)}.$$

This formulation thus allows us to pass to the limit and conclude that in $\mathcal{D}'(0,T)$:

$$(\psi_n(t), \tilde{\varphi})_{H^1(\Omega_\theta)} \xrightarrow[n\to\infty]{} (\psi(t), \tilde{\varphi})_{L^2(\Omega_\theta)} + \frac{d}{dt}(\nabla\varphi(t), \nabla\tilde{\varphi})_{L^2(\Omega_\theta)}.$$

The limit of the term involving $\partial_t \varphi_n$ is treated in the same way as the inertial terms. We then obtain the limit using the convergence of φ_n in $H^1(\Omega_\theta)$. Thus, by passing to the limit in (A.4), we find that the functions φ, ψ, s satisfy, for all $(\tilde{\varphi}, \tilde{\psi}, \tilde{s})^T \in \mathcal{V}$:

$$\begin{aligned}\frac{d}{dt}(\varphi(t), \tilde{\varphi})_{H^1(\Omega_\theta)} + \frac{d}{dt}\frac{1}{c^2}(\psi(t), \tilde{\psi})_{L^2(\Omega_\theta)} + \frac{d}{dt} g(s(t), \tilde{s})_{L^2(\Gamma_0)} = (\psi(t), \tilde{\varphi})_{L^2(\Omega_\theta)} + \frac{d}{dt}(\nabla\varphi(t), \nabla\tilde{\varphi})_{L^2(\Omega_\theta)} \\ - (\nabla\varphi(t), \nabla\tilde{\psi})_{L^2(\Omega_\theta)} + \left\langle \chi(t), \tilde{\psi} \right\rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})}.\end{aligned} \tag{A.6}$$

It follows that φ, ψ, s satisfy, for all $(\tilde{\varphi}, \tilde{\psi}, \tilde{s})^T \in \mathcal{V}$:

$$\begin{aligned}\frac{d}{dt}(\varphi(t), \tilde{\varphi})_{L^2(\Omega_\theta)} + \frac{d}{dt}\frac{1}{c^2}(\psi(t), \tilde{\psi})_{L^2(\Omega_\theta)} + \frac{d}{dt} g(s(t), \tilde{s})_{L^2(\Gamma_0)} = (\psi(t), \tilde{\varphi})_{L^2(\Omega_\theta)} - (\nabla\varphi(t), \nabla\tilde{\psi})_{L^2(\Omega_\theta)} \\ + \left\langle \chi(t), \tilde{\psi} \right\rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})}.\end{aligned}$$

Which is equivalent to the following equation being satisfied for $Z = (\varphi, \psi, s)^T \in \mathcal{C}^0([0,T]; \mathcal{H})$ for all $\tilde{Z} = (\tilde{\varphi}, \tilde{\psi}, \tilde{s})^T \in \mathcal{V}$:

$$\frac{d}{dt}(Z(t), \tilde{Z})_{\tilde{\mathcal{H}}} + \tilde{a}(Z(t), \tilde{Z}) = \left\langle \chi(t), \tilde{\psi} \right\rangle_{H^{-\frac{1}{2}}(\Gamma_{-H+\theta}), H^{\frac{1}{2}}(\Gamma_{-H+\theta})}, \qquad \text{in } \mathcal{D}'(0,T).$$

Passing to the limit in (A.5), we also get that in $\mathcal{D}'(0,T)$, for all $\tilde{s} \in L^2(\Gamma_0)$:

$$\frac{d}{dt}(\varphi|_{\Gamma_0}(t), \tilde{s})_{L^2(\Gamma_0)} = -g(s(t), \tilde{s})_{L^2(\Gamma_0)}$$

The intial conditions on φ, ψ and s easily pass to the limit, so that $Z(0) = 0$. We have just proved the existence of a solution. Let us now prove the uniqueness of the solution for a $\chi \in H^1(0,T; H^{-\frac{1}{2}}(\Gamma_{-H+\theta}))$ given. The uniqueness comes down to prove that if $\chi = 0$ then Z solution of (A.2) and (A.1) is also such that $Z = 0$. From (A.2), we deduce that for all $\Psi \in \mathcal{D}(0,T)$ and for all $\tilde{Z} \in \mathcal{V}$:

$$-\int_0^T (Z(t), \tilde{Z})_{\tilde{\mathcal{H}}} \dot{\Psi}(t)\, dt + \int_0^T \tilde{a}(Z(t), \tilde{Z}) \Psi(t)\, dt = 0.$$

From this expression, we get that there exists $C > 0$ such that

$$\left| \int_0^T (Z(t), \tilde{Z})_{\tilde{\mathcal{H}}} \dot{\Psi}(t)\, dt \right| \leq C \|\Psi\|_{L^2(0,T)} \|\tilde{Z}\|_{\tilde{\mathcal{V}}}$$

with $\tilde{\mathcal{V}} = L^2(\Omega_\theta) \times H^1(\Omega_\theta) \times L^2(\Gamma_0)$ endowed with its canonic norm. Introducing the Gelfand triplet $\tilde{\mathcal{V}} \hookrightarrow \tilde{H} \equiv \tilde{H}' \hookrightarrow \tilde{\mathcal{V}}'$, we get that $\dot{Z} \in L^2(0,T; \tilde{\mathcal{V}}')$. Here we can easily obtain that $\tilde{\mathcal{V}}' = L^2(\Omega_\theta) \times (H^1(\Omega_\theta))' \times L^2(\Gamma_0)$. From the density of $\mathcal{D}(0,T) \otimes \mathcal{V}$ in $L^2(0,T; \tilde{\mathcal{V}})$, it then follows that (A.2) implies that Z satisfies for all $\tilde{Z} \in L^2(0,T; \tilde{\mathcal{V}})$:

$$\int_0^T \left\langle \dot{Z}(t), \tilde{Z}(t) \right\rangle_{\tilde{\mathcal{V}}', \tilde{\mathcal{V}}} + \tilde{a}(Z(t), \tilde{Z}(t))\, dt = 0. \tag{A.7}$$

Let us introduce the following Ladyzhenskaya tests functions [82] for all $\tau \in [0,T]$

$$\tilde{Z}(t) = \left(\begin{cases} -\int_t^\tau \varphi(s)\, ds, & \tau \geq t \\ 0 \end{cases}, \begin{cases} -\int_t^\tau \varphi(s)\, ds, & \tau \geq t \\ 0 \end{cases}, 0 \right) \in L^2(0,T; \tilde{\mathcal{V}}).$$

It is important here to notice that the previous test function is admissible, in other words $\tilde{Z} \in L^2(0,T;\tilde{V})$. First, let us notice that since $\frac{d}{dt}(\varphi,\tilde{\varphi})_{L^2(\Omega_\theta)} = (\psi,\tilde{\varphi})_{L^2(\Omega_\theta)}, \forall \tilde{\varphi} \in H^1(\Omega_\theta)$ in $\mathcal{D}'(0,T)$, we get that for all $t \in [0,T]$

$$\varphi(t) = \int_0^t \psi(s)\ ds. \tag{A.8}$$

The equation (A.7) then becomes for all $\tau \in [0,T]$

$$\int_0^\tau \left(\dot{\varphi}(t), -\int_t^\tau \varphi(s)\ ds\right)_{L^2(\Omega_\theta)} + \left\langle \dot{\psi}, -\int_t^\tau \varphi(s)\ ds \right\rangle_{(H^1(\Omega_\theta))',H^1(\Omega_\theta)} - \left(\psi, -\int_t^\tau \varphi(s)\ ds\right)_{L^2(\Omega_\theta)} + \left(\nabla\varphi, \nabla\left(-\int_t^\tau \varphi(s)\ ds\right)\right)_{L^2(\Omega_\theta)} dt = 0.$$

Integrating by parts, we get for all $\tau \in [0,T]$

$$\int_0^\tau \|\varphi\|^2_{L^2(\Omega_\theta)}\ dt + \frac{1}{2}\left\| \int_0^\tau \psi \right\|^2_{L^2(\Omega_\theta)} - \int_0^\tau \left\| \int_0^t \psi \right\|^2_{L^2(\Omega_\theta)} dt + \frac{1}{2}\left\| \nabla\left(\int_0^\tau \varphi \right) \right\|^2_{L^2(\Omega_\theta)} = 0.$$

It follows that for all $\tau \in [0,T]$

$$\frac{1}{2}\left\| \int_0^\tau \psi \right\|^2_{L^2(\Omega_\theta)} + \frac{1}{2}\left\| \nabla\left(\int_0^\tau \varphi \right) \right\|^2_{L^2(\Omega_\theta)} = 0.$$

Which gives us finally for all $\tau \in [0,T]$

$$\frac{1}{2}\|\varphi(\tau)\|^2_{L^2(\Omega_\theta)} + \frac{1}{2}\left\| \nabla\left(\int_0^\tau \varphi \right) \right\|^2_{L^2(\Omega_\theta)} = 0.$$

Which gives us that $\varphi = 0$. Finally, using (A.8), we get that $\psi = 0$. Finally, using the link between φ and s on Γ_0, we get that $s = 0$. We then get that $Z = 0$, which conclude the proof. □

We have thus constructed two types of solutions: strong solutions, which require very restrictive regularity assumptions on the source term but are highly regular themselves and variational solutions that we have just established. As expected, the variational formulation is better suited for numerical discretization. We showed in Chapter 4 that up to a parabolic regularization, we could define variational solutions in a sense defined in [90]. In this appendix, we have defined different kind of variational solutions but without adding any type of regularization. In Chapter 4, we have chosen variational solutions of a regularized problem since we managed to obtain an inf-sup condition in the spaces of solutions. The variational solutions obtained in this appendix are not well suited for such an inf-sup condition, which prevented us from doing the same analysis than in Chapter 4. Indeed, the regularity of the first order variational solution is such that the adjoint state in the inf-sup condition does not belong to the energy state, so that we cannot obtain a control of it within the right norm.

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ECOLE
DOCTORALE
DE MATHEMATIQUES
HADAMARD

Titre: Un problème inverse en océanographie : Identification des déformations du fond marin par mesure sur la surface libre

Mots clés: Problèmes inverses, equation d'ondes, analyse numérique, formulations mixtes, contrôle optimal, condition inf-sup

Résumé: Cette thèse concerne l'étude d'un problème inverse d'océanographie. A la suite d'un tsunami causé par un séisme sous-marin, peut-on par la mesure de la déformation de la surface libre en déduire les caractéristiques du séisme au fond de l'océan? Les modèles mathématiques décrivant la propagation d'un tsunami sont nombreux et pour la plupart non linéaires. Au temps court, on peut modéliser ce phénomène par un modèle linéaire qui sera par simplicité utilisé dans toute la thèse. Ce problème inverse mal posé sera étudié au travers de deux cadres bien distincts. Nous proposons tout d'abord une approche, tant dans le domaine fréquentiel que temporel, par formulation mixte espace/temps associée à une régularisation de Tikhonov. La prise en compte du bruit est ici un point délicat sur lequel nous développons une approche rigoureuse pour fixer le paramètre de régularisation en fonction de l'amplitude du bruit. Cette méthode bien qu'efficace et ne nécessitant qu'une unique itération souffre cependant d'un cout de stockage numérique important lié notamment à la taille des matrices utilisées. Pour pallier ce problème, une seconde approche itérative par contrôle optimal est utilisée. On minimise alors une fonctionnelle moindres carrés en construisant un état adjoint et en utilisant l'algorithme de descente L-BFGS. On propose également l'analyse numérique rigoureuse de cette méthode numérique en utilisant la condition d'optimalité vue comme une formulation mixte. On obtient ainsi dans un cadre abstrait une estimation d'erreur entre la solution discrétisée et la solution continue du problème de contrôle optimal. Cependant ce cadre fonctionnel dans lequel notre analyse est réalisée n'existe qu'au prix d'une régularisation parabolique de notre modèle de départ, ce qui correspond à l'ajout d'un terme de viscosité cinématique. Nous présentons ainsi finalement un nouveau cadre fonctionnel permettant l'analyse numérique sans ajout de terme de régularisation. L'analyse reste à faire ici mais le cadre est précisée rigoureusement. Des simulations numériques de reconstruction utilisant des paramètres réalistes sont finalement présentés.

Title: An inverse problem in oceanography: Identification of the bottom deformations from measurements on the free surface

Keywords: Inverse problems, waves equation, numerical analysis, mixed formulations, optimal control, inf-sup condition

Abstract: This thesis is devoted to the study of an inverse problem in oceanography. Following a tsunami generated by an underwater earthquake, can the characteristics of the seismic event at the ocean floor be inferred from measurements of the deformation of the free surface? Numerous mathematical models have been developed to describe tsunami propagation, most of which are nonlinear. Over short time scales, however, this phenomenon can be accurately represented by a linear model, which is adopted throughout this thesis for the sake of simplicity.
This ill-posed inverse problem is investigated within two distinct frameworks. First, we propose an approach in both the frequency and time domains based on a space–time mixed formulation combined with Tikhonov regularization. The treatment of measurement noise is a delicate issue, and a rigorous methodology is developed to determine the regularization parameter as a function of the noise amplitude. Although this method is effective and requires only a single iteration, it suffers from a significant computational storage cost, primarily due to the size of the matrices involved. To overcome this limitation, a second iterative approach based on optimal control theory is considered. The method consists in minimizing a least-squares functional by constructing an adjoint state and employing the L-BFGS descent algorithm. A rigorous numerical analysis of this approach is also developed by interpreting the optimality condition as a mixed formulation.
Within an abstract framework, we derive an error estimate between the discrete solution and the continuous solution of the optimal control problem. However, the functional setting in which this analysis is carried out is only available through a parabolic regularization of the original model, corresponding to the addition of a kinematic viscosity term. We therefore introduce a new functional framework that enables numerical analysis without the inclusion of any regularization term. While the corresponding analysis remains to be completed, the framework itself is established rigorously.
Finally, numerical reconstruction experiments based on realistic physical parameters are presented.

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